

Model Analysis

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1 Data analysis /Dataframe import and filters

1.1 DataFrame import and filters

The data has been imported with the following(s) function(s):

```

def build_df():
    if flag_IR:
        df, features_engineered = import_data_IR()
        if flag_IR_maskP:
            df = mask_period_IR(df)
        if flag_IR_balR:
            df = mask_balanceHC_random_IR(df, n_samples=None)
    if flag_ER:
        df, features_engineered = import_data_ER()

    cols_to_check = target_col + features_X
    mask = np.isfinite(df[cols_to_check]).all(axis=1)
    df = df[mask]

    cols = list(dict.fromkeys(target_col + features_X +
                              feats_to_balance + feats_grouping +
                              feats_reference+ feats_postproc))

    df = df[cols]
    #df['campaign'] = "ALL"

    return df

```

```

def import_data_IR():
    if target_col[0] == "i_NH3":
        #file_path = "csv/innova/switched/df_signal_decomposition_NH3_current_db2.csv"
        file_path = "csv/innova/notswitched/df_signal_decomposition_current.csv"
    elif target_col[0] == "i_CO2":
        file_path = "csv/innova/switched/df_signal_decomposition_CO2_current.csv"
    else:
        raise ValueError("Target is not supported")

    df = pd.read_csv(file_path, sep=";")
    df["datetime"] = pd.to_datetime(df["datetime"])
    df.sort_values("datetime").reset_index(drop=True)

    if target_col[0] == "i_NH3":
        df["campaign"] = df["datetime"].apply(assign_campaign)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv/innova/notswitched/features_decomposition_list.json", "r") as f:

```

```

        dict_features = json.load(f)
elif target_col[0] == "i_CO2":
    df["campaign"] = df["datetime"].apply(assign_campaign_iCO2)
    df = df[df["campaign"] != "other"].copy()
    df["innova_point"] = df.apply(assign_innova_point_iCO2, axis=1)
    df = df[df["innova_point"] != "no"].copy()
    with open("csv/innova/switched/features_decomposition_CO2_list.json", "r") as f:
        dict_features = json.load(f)
else:
    raise ValueError("The target col do not match any existing function")

df["hot"] = df.apply(assign_hot_season,axis=1)
df["umidity_range"] = df.apply(assign_humidity, axis = 1)
#Removing of rabbit 4
df["column"] = df.apply(assign_colonna, axis = 1)
df['height'] = df.apply(assign_altezza, axis = 1)

return df,dict_features

```

```

def assign_innova_point(row):
    camp = row["campaign"]
    rid = row["id_rabbit"]

    # if camp == "2025_Jun":
    #     mapping = {2: "sl11", 4: "sl5", 6: "sl4", 1: "sl3", 5: "no"}
    # elif camp == "2025_Sep":
    #     mapping = {2: "sl11", 4: "sl5", 3: "sl4", 6: "sl3", 5: "no"}
    if camp == "2025_Apr":
        #mapping = {1: "no", 2: "no", 4: "no", 6: "sl5"}
        #mapping = {1: "no", 2: "sl5", 4: "no", 6: "sl5"}
        mapping = {1: "no", 2: "no", 4: "no", 6: "no"}
    elif camp == "2025_Jun":
        mapping = {2: "sl11", 4: "sl5", 1: "sl3", 6: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 1: "sl4", 6: "sl3", 5: "no"}
    elif camp == "2025_Sep":
        mapping = {2: "sl11", 4: "sl5", 6: "sl3", 3: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 6: "sl4", 3: "sl3", 5: "no"}
    elif camp == "2026_Gen":
        #mapping = {1: "no", 4: "no", 5: "sl11", 6: "no"}

```

```

    #mapping = {1: "sl11", 4: "no", 5: "sl11", 6: "sl11"}
    mapping = {1: "no", 4: "no", 5: "no", 6: "no"}
else:
    mapping = {}

return mapping.get(rid, "no")

```

Function print :

1.2 Metadata

Nombre de lignes df	8168
Nombre de lignes df clean	8168
Nombre de features	5
% de NaN	0.0
Taille dataset entraînement	5717
Taille dataset test	2450
Métrique utilisée	r2

Column target : i_NH3

Features_X : r_NH3, r_temperature, r_umidity, r_CO2, r_THI

feats_to_balance :

feats_grouping : id_rabbit, id_channel

feats_postproc : datetime, id_channel, id_rabbit, campaign

feats_ref :

id_cols : campaign

Grouping : id_rabbit, id_channel

2 Model(s) analysis

2.1 Explanation legend tables

Columns

- rank: ranking regarding cross validation test

- train_cv: mean score train of the CV
- test_cv: mean score test of the CV
- test: score on the testing dataset
- diff: test-test_cv
- r_test: test/test_cv
- r_train: train_cv/test_cv

Colors:

[Orange] $r_test < 0.95$ (under performance of the test)

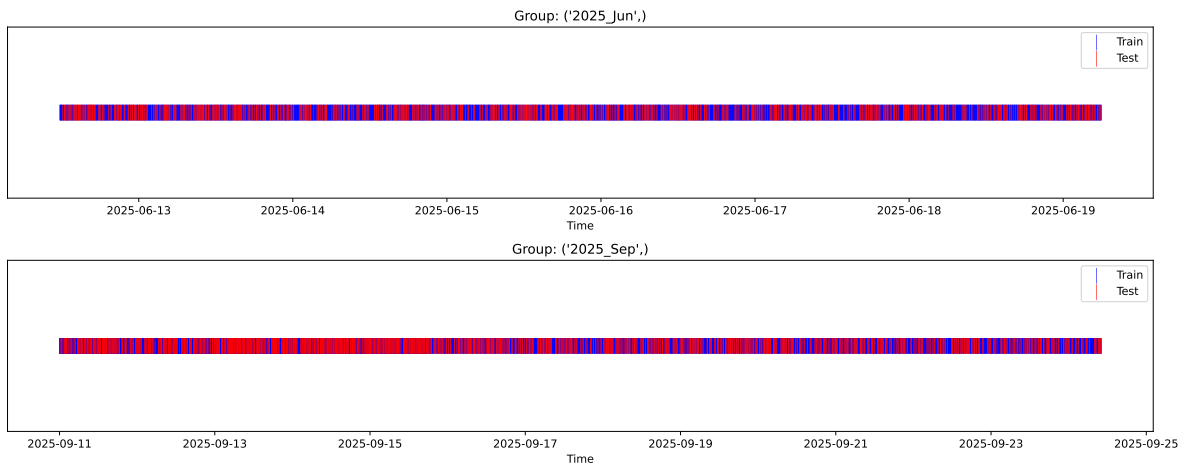
[Green] $r_test > 1.05$ (over performance of the test)

[Red] r_train not in $[0.95, 1.05]$ (gap between train/test)

2.2 Repartition of train and test

2.2.1 External Split: X_train vs X_test

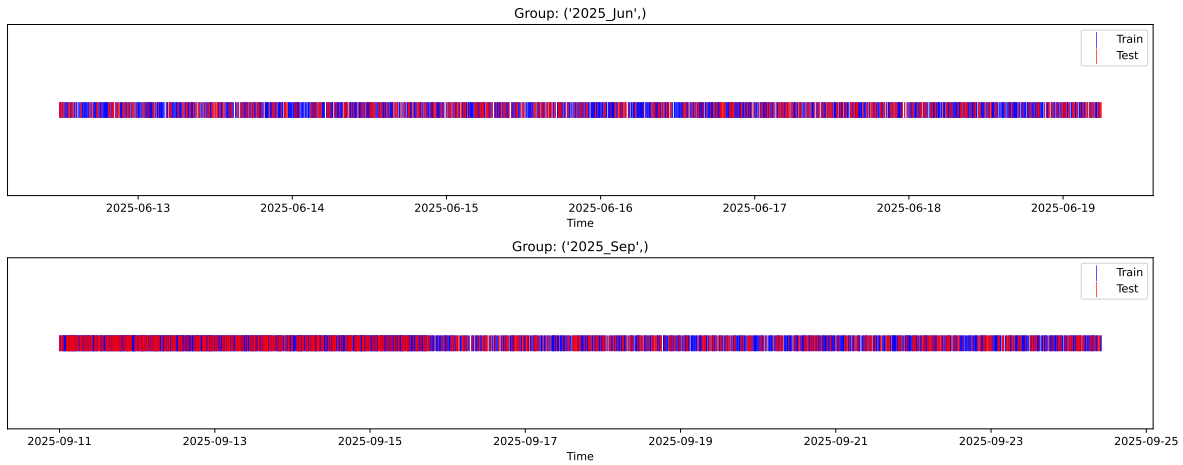
External Split: Train: 5717 samples Test: 2451 samples



2.2.2 Internal CV Folds

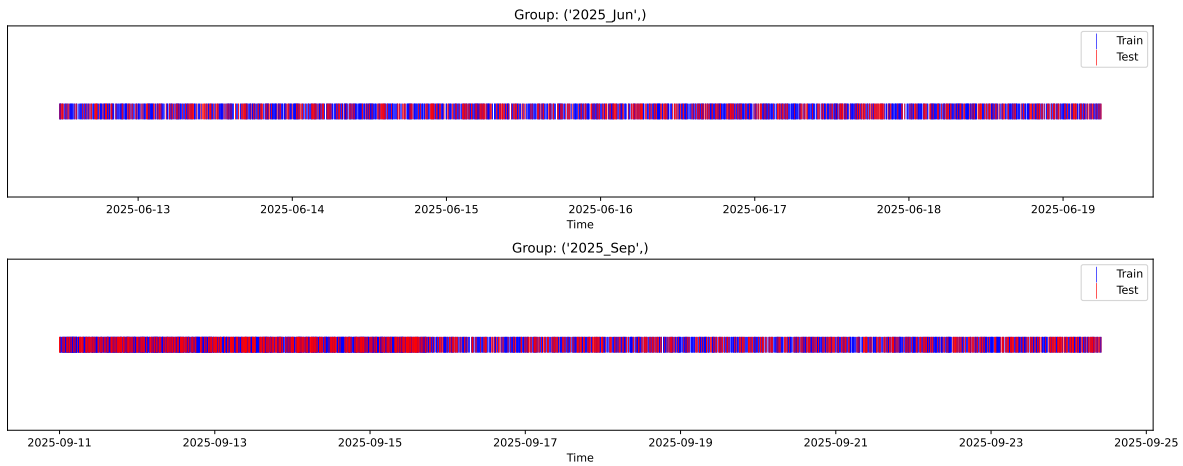
Fold 1/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



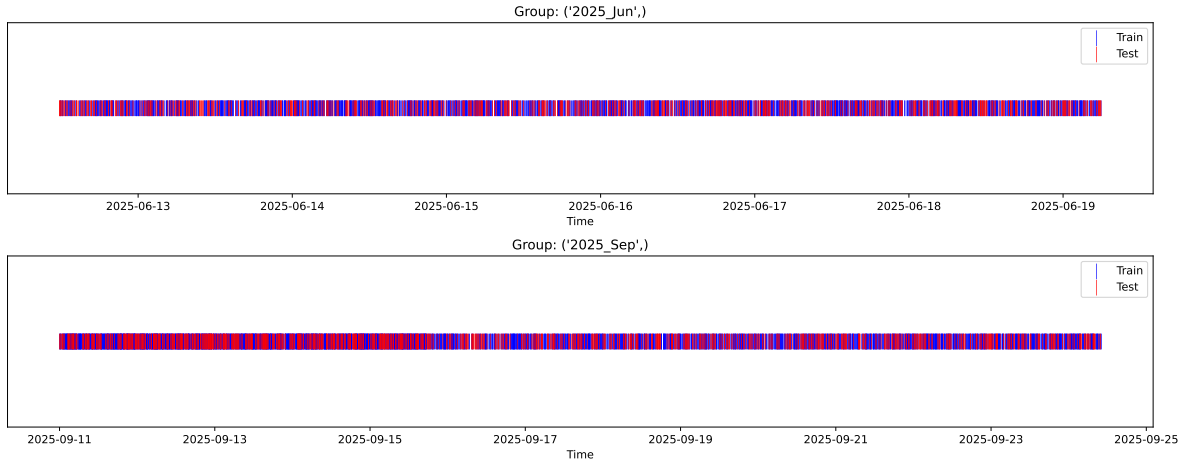
Fold 2/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



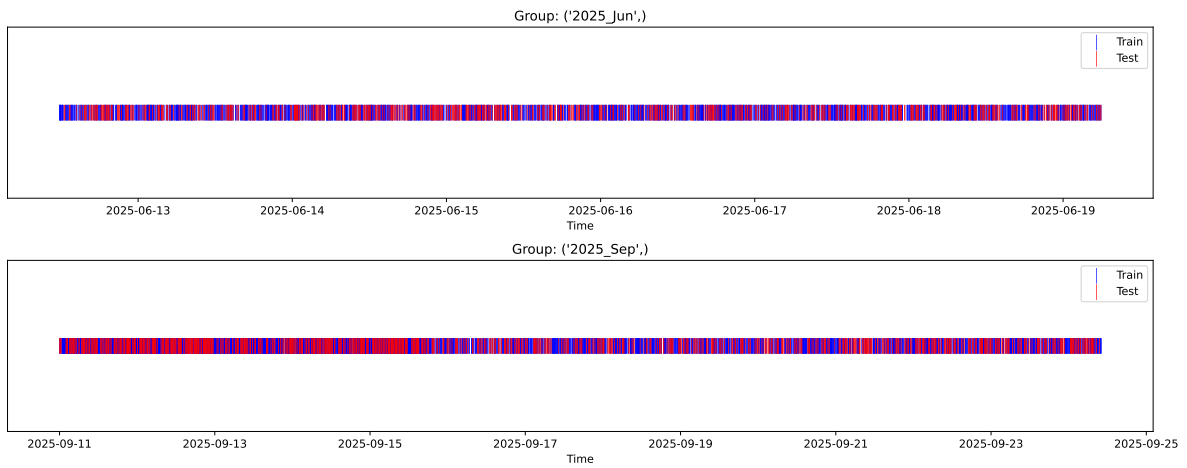
Fold 3/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



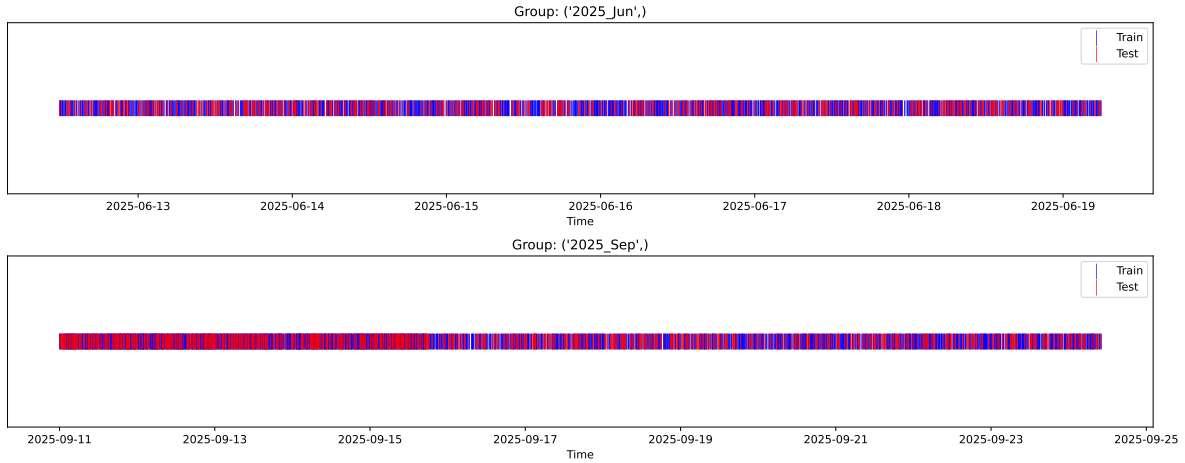
Fold 4/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



Fold 5/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



2.3 Model : Linear

2.3.1 Gridsearch — Group: 1_3

Len(group) : 795, Len(training) : 557, Len(test) : 238

Distribution y_train vs y_test: y_train: mean=2.597, std=0.579, min=1.437, max=4.986
y_test: mean=2.492, std=0.625, min=1.466, max=5.447

Fold 0 — y_test: mean=2.57, std=0.62, y_train: mean=2.61, std=0.56

Fold 1 — y_test: mean=2.53, std=0.55, y_train: mean=2.63, std=0.59

Fold 2 — y_test: mean=2.65, std=0.55, y_train: mean=2.58, std=0.59

Fold 3 — y_test: mean=2.64, std=0.59, y_train: mean=2.58, std=0.57

Fold 4 — y_test: mean=2.56, std=0.54, y_train: mean=2.61, std=0.59

train scores CV: [0.56730715 0.5334636 0.56784944 0.59353423 0.55417991]

test scores CV: [0.5431311 0.6179077 0.53621024 0.4830173 0.57408525]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.5633	0.5509	0.0445	0.4979	0.5003	-0.0505	0.9082	1.0225

Coefficients:

$\text{coef_r_CO2} = -0.0079$

$\text{coef_r_NH3} = 0.0524$

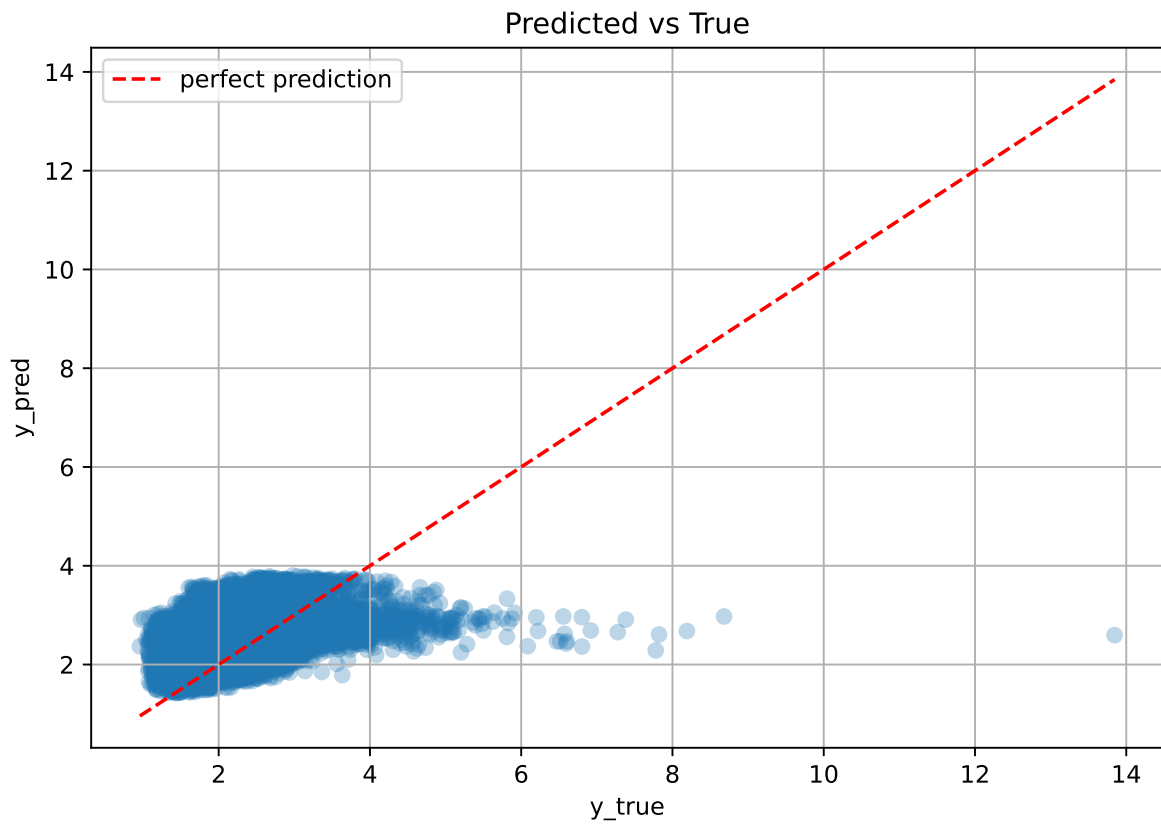
$\text{coef_r_THI} = 0.5946$

$\text{coef_r_temperature} = -0.3581$

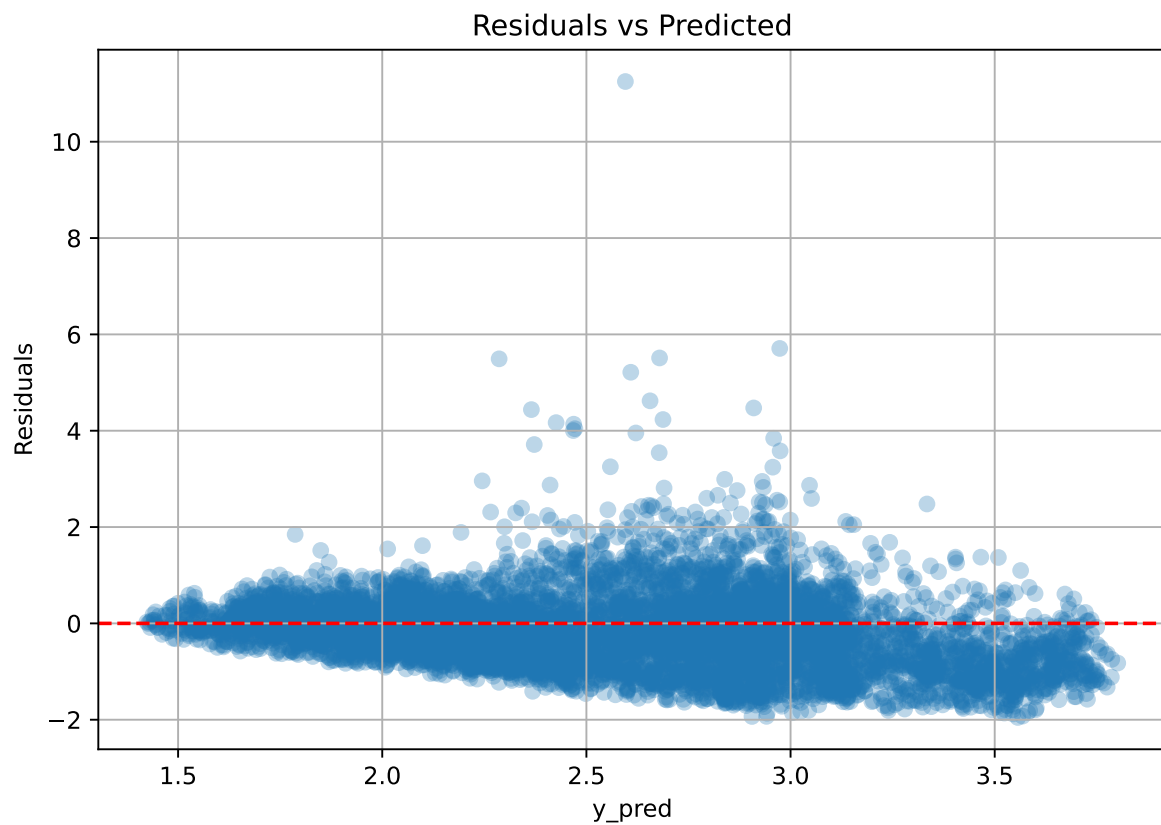
$\text{coef_r_umidity} = -0.2680$

$\text{intercept} = 2.5975$

2.3.2 Scatter



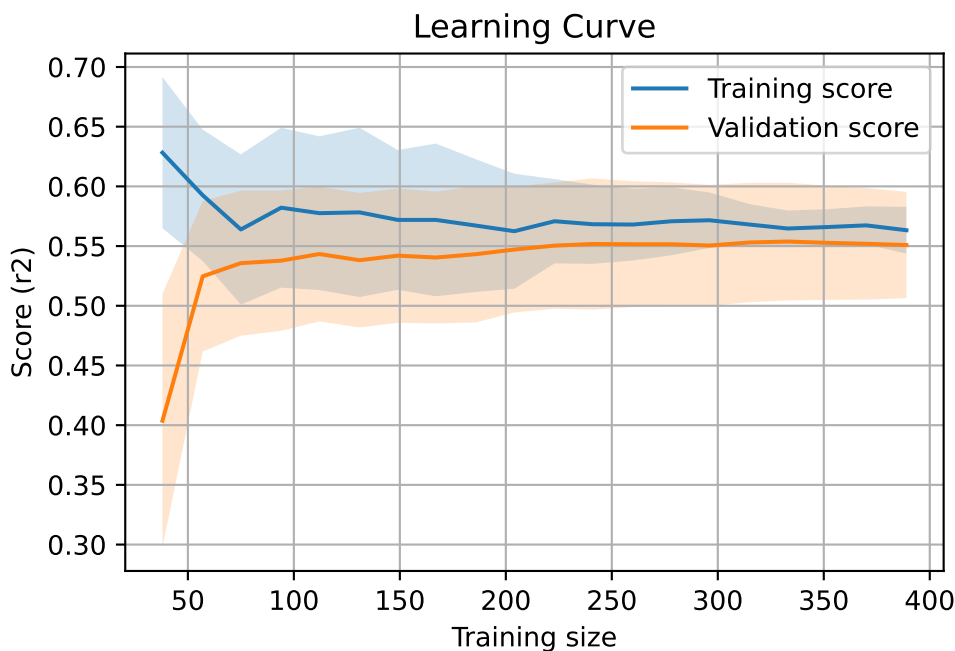
2.3.3 Scatter residuals



2.3.4 Learning curve — Group: 1_3

`Len(group)` : 795, `Len(training)` : 557,

`Len(training_real)` : 390, `Len(test_of_training)` : 167, `Len(test)` : 238



2.3.5 Gridsearch — Group: 2_11

Len(group) : 2726, Len(training) : 1909, Len(test) : 817

Distribution y_train vs y_test: y_train: mean=2.540, std=0.859, min=1.150, max=8.682
y_test: mean=2.561, std=0.947, min=1.120, max=13.845

Fold 0 — y_test: mean=2.58, std=0.86, y_train: mean=2.52, std=0.86

Fold 1 — y_test: mean=2.56, std=0.84, y_train: mean=2.53, std=0.87

Fold 2 — y_test: mean=2.50, std=0.84, y_train: mean=2.55, std=0.87

Fold 3 — y_test: mean=2.53, std=0.87, y_train: mean=2.55, std=0.85

Fold 4 — y_test: mean=2.53, std=0.82, y_train: mean=2.54, std=0.87

train scores CV: [0.46497694 0.46580479 0.46495948 0.46844345 0.45091902]

test scores CV: [0.45757013 0.45504757 0.45666765 0.44978339 0.49647352]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.463	0.4631	0.0169	0.3942	0.3945	-0.0686	0.8519	0.9998

Coefficients:

coef_r_CO2 = 0.0009

coef_r_NH3 = 0.0485

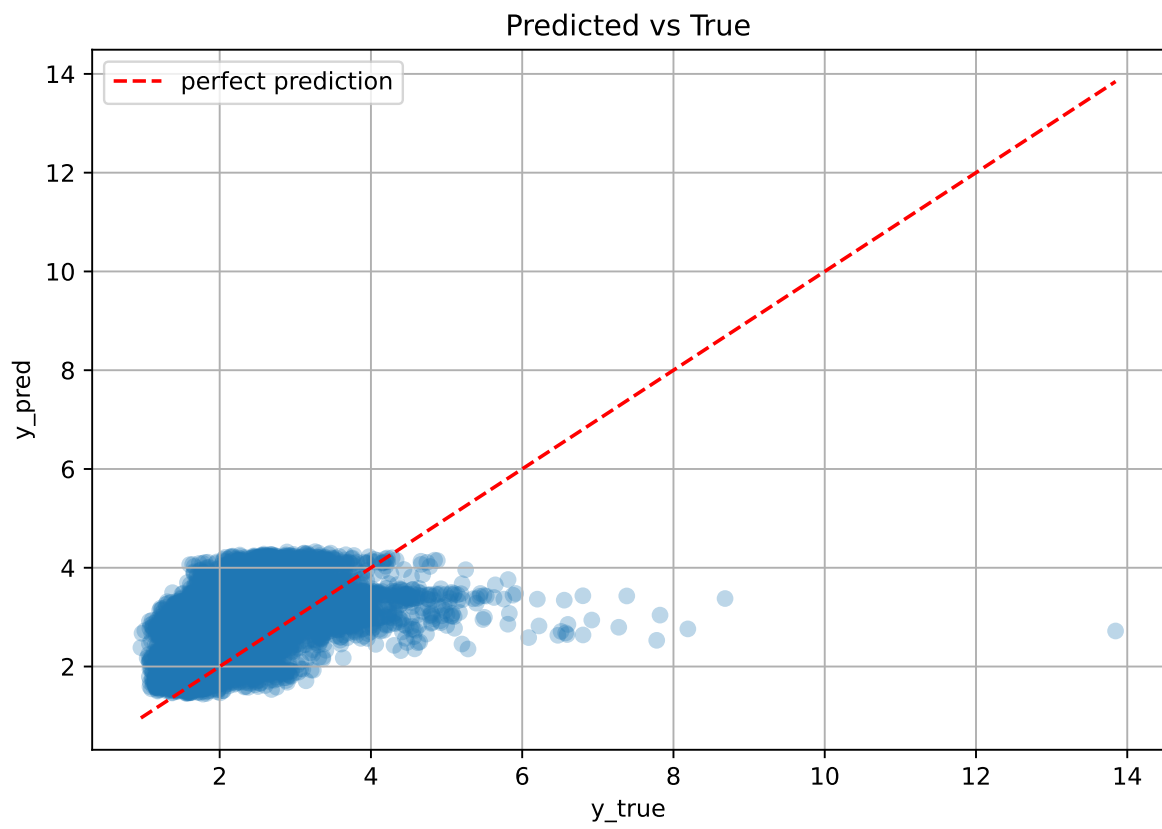
coef_r_THI = 0.8323

coef_r_temperature = -0.9731

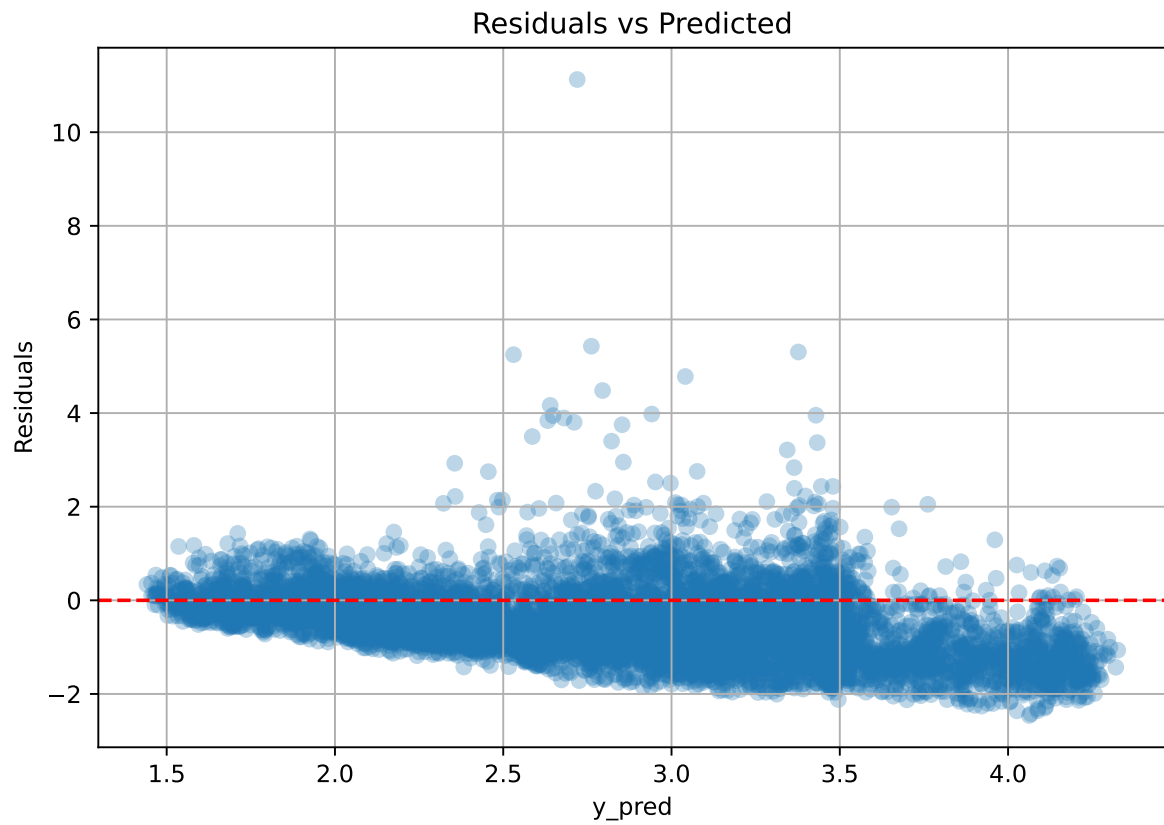
coef_r_umidity = -0.7852

intercept = 2.5398

2.3.6 Scatter



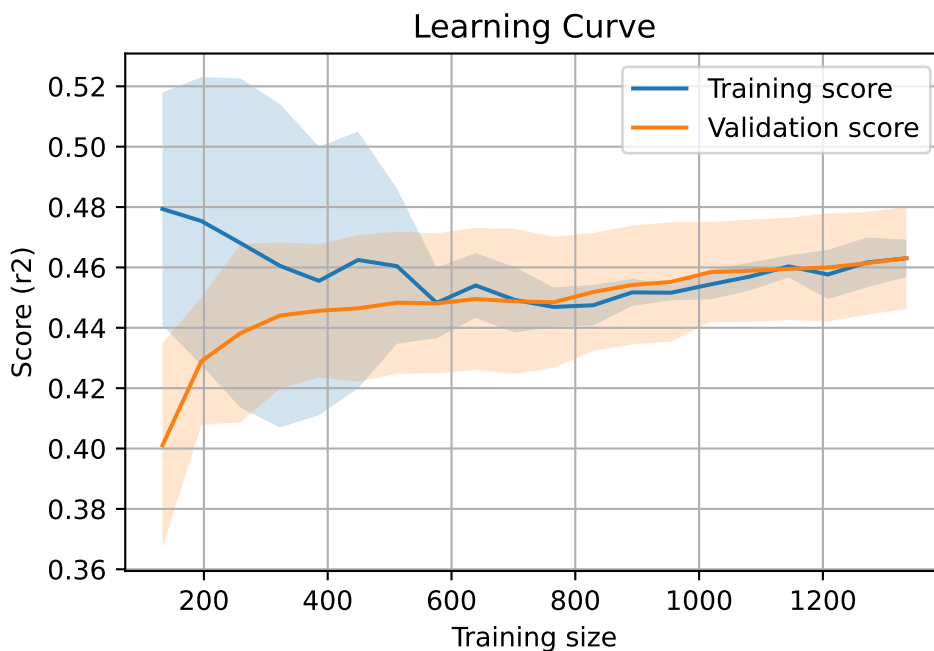
2.3.7 Scatter residuals



2.3.8 Learning curve — Group: 2_11

`Len(group)` : 2726, `Len(training)` : 1909,

`Len(training_real)` : 1337, `Len(test_of_training)` : 572, `Len(test)` : 817



2.3.9 Gridsearch — Group: 3_4

Len(group) : 568, Len(training) : 398, Len(test) : 170

Distribution y_train vs y_test: y_train: mean=1.928, std=0.816, min=0.959, max=6.804
y_test: mean=1.833, std=0.729, min=1.098, max=7.822

Fold 0 — y_test: mean=1.93, std=0.75, y_train: mean=1.93, std=0.84

Fold 1 — y_test: mean=1.85, std=0.67, y_train: mean=1.96, std=0.87

Fold 2 — y_test: mean=2.00, std=0.92, y_train: mean=1.90, std=0.76

Fold 3 — y_test: mean=1.92, std=0.88, y_train: mean=1.93, std=0.78

Fold 4 — y_test: mean=1.92, std=0.86, y_train: mean=1.93, std=0.80

train scores CV: [0.3155034 0.29364485 0.29854522 0.32073505 0.30855581]

test scores CV: [0.17929864 0.27024886 0.27717852 0.24665152 0.26532812]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.3074	0.2477	0.0357	0.1869	0.2027	-0.045	0.8182	1.2408

Coefficients:

$\text{coef_r_CO2} = 0.1066$

$\text{coef_r_NH3} = -0.2625$

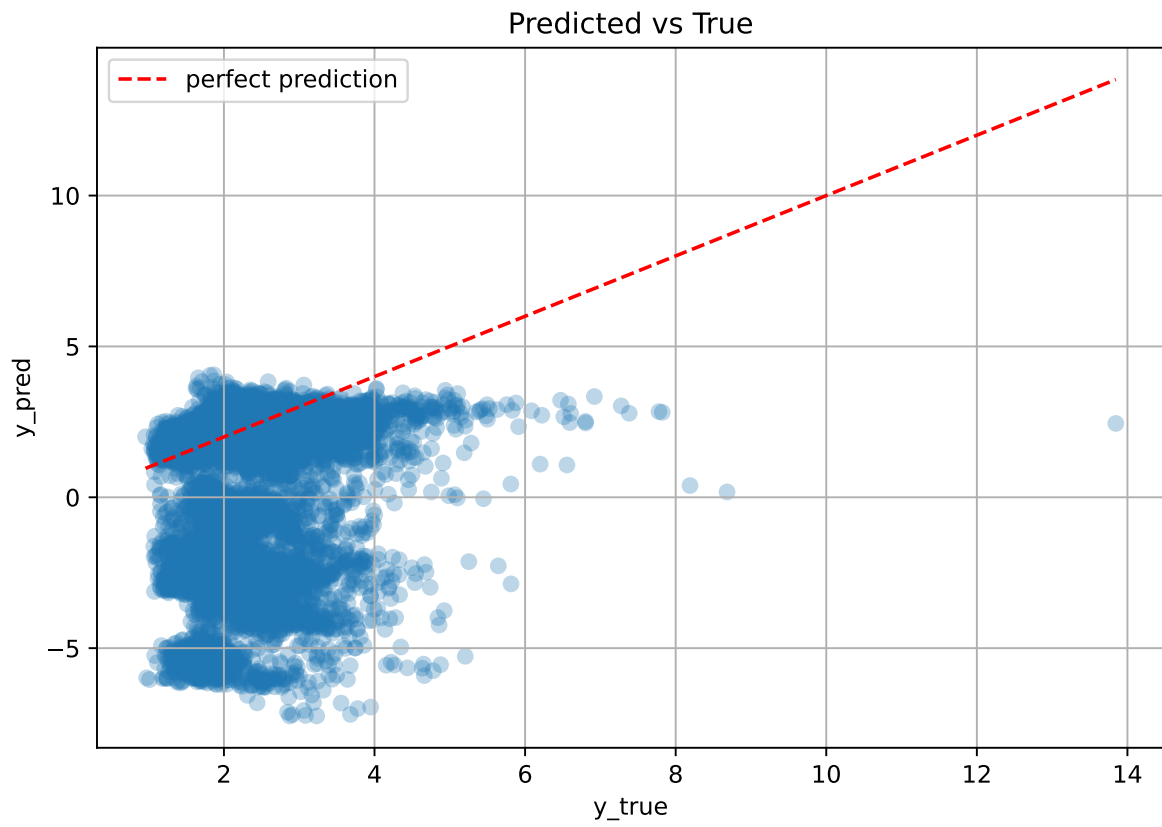
$\text{coef_r_THI} = 2.6048$

$\text{coef_r_temperature} = -3.7070$

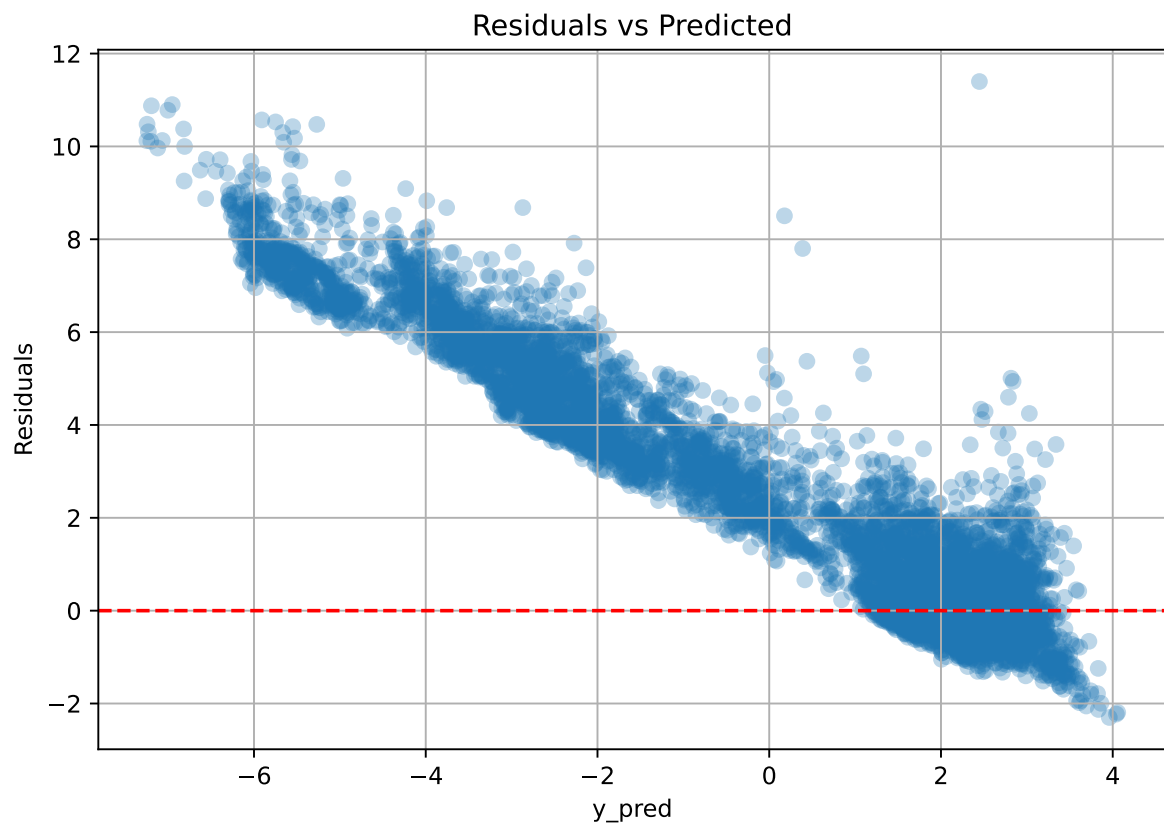
$\text{coef_r_umidity} = -1.4591$

$\text{intercept} = 1.9278$

2.3.10 Scatter



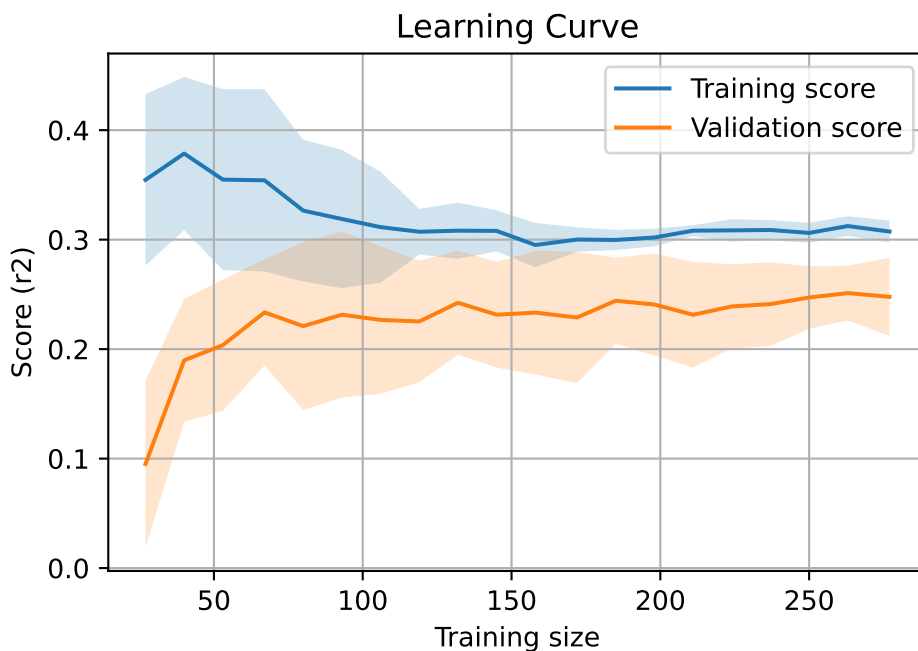
2.3.11 Scatter residuals



2.3.12 Learning curve — Group: 3_4

`Len(group)` : 568, `Len(training)` : 398,

`Len(training_real)` : 279, `Len(test_of_training)` : 119, `Len(test)` : 170



2.3.13 Gridsearch — Group: 4_5

Len(group) : 2726, Len(training) : 1909, Len(test) : 817

Distribution y_train vs y_test: y_train: mean=2.238, std=0.681, min=0.973, max=5.813
y_test: mean=2.212, std=0.659, min=1.071, max=4.858

Fold 0 — y_test: mean=2.27, std=0.68, y_train: mean=2.22, std=0.68

Fold 1 — y_test: mean=2.25, std=0.71, y_train: mean=2.23, std=0.67

Fold 2 — y_test: mean=2.20, std=0.65, y_train: mean=2.25, std=0.69

Fold 3 — y_test: mean=2.21, std=0.65, y_train: mean=2.25, std=0.70

Fold 4 — y_test: mean=2.22, std=0.63, y_train: mean=2.24, std=0.70

train scores CV: [0.4277706 0.44021497 0.41172833 0.43714743 0.44005866]

test scores CV: [0.41299685 0.39069188 0.45040747 0.3878951 0.37549212]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.4314	0.4035	0.0264	0.4566	0.4572	0.0537	1.1332	1.0691

Coefficients:

$\text{coef_r_CO2} = -0.0952$

$\text{coef_r_NH3} = -0.1254$

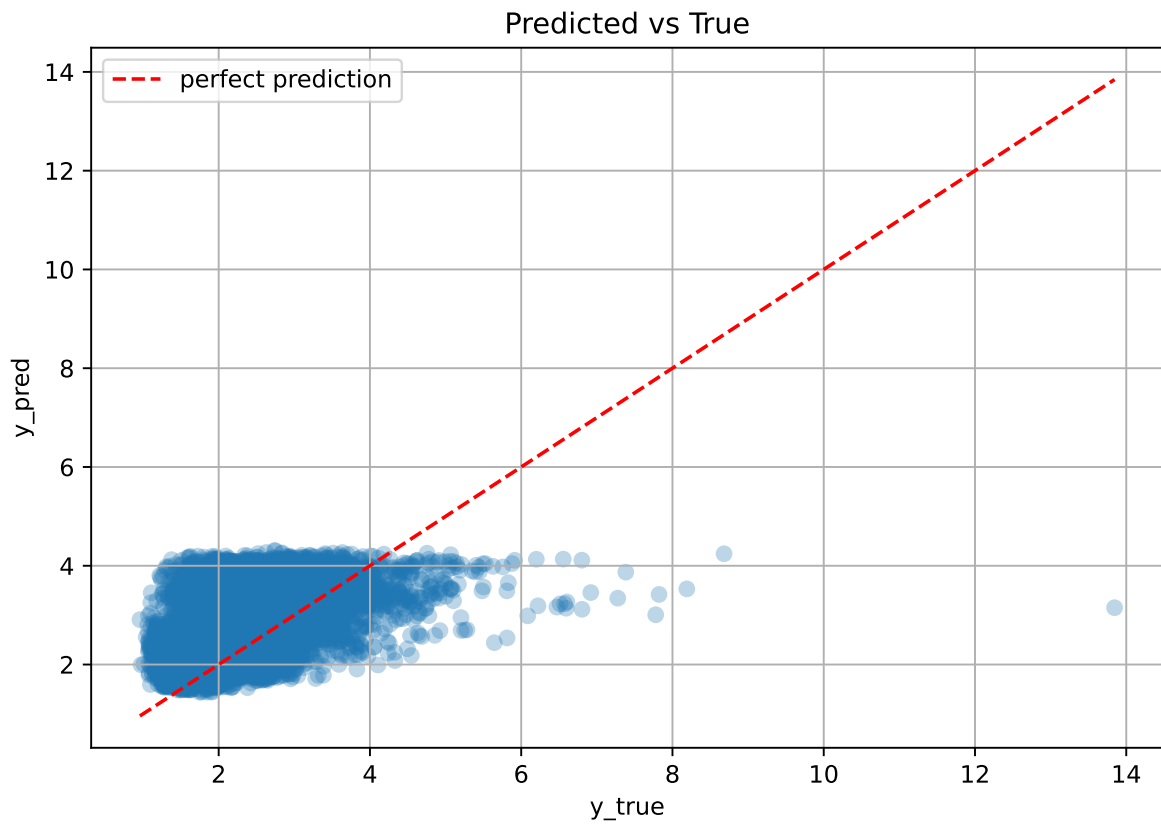
$\text{coef_r_THI} = 0.0994$

$\text{coef_r_temperature} = 0.0362$

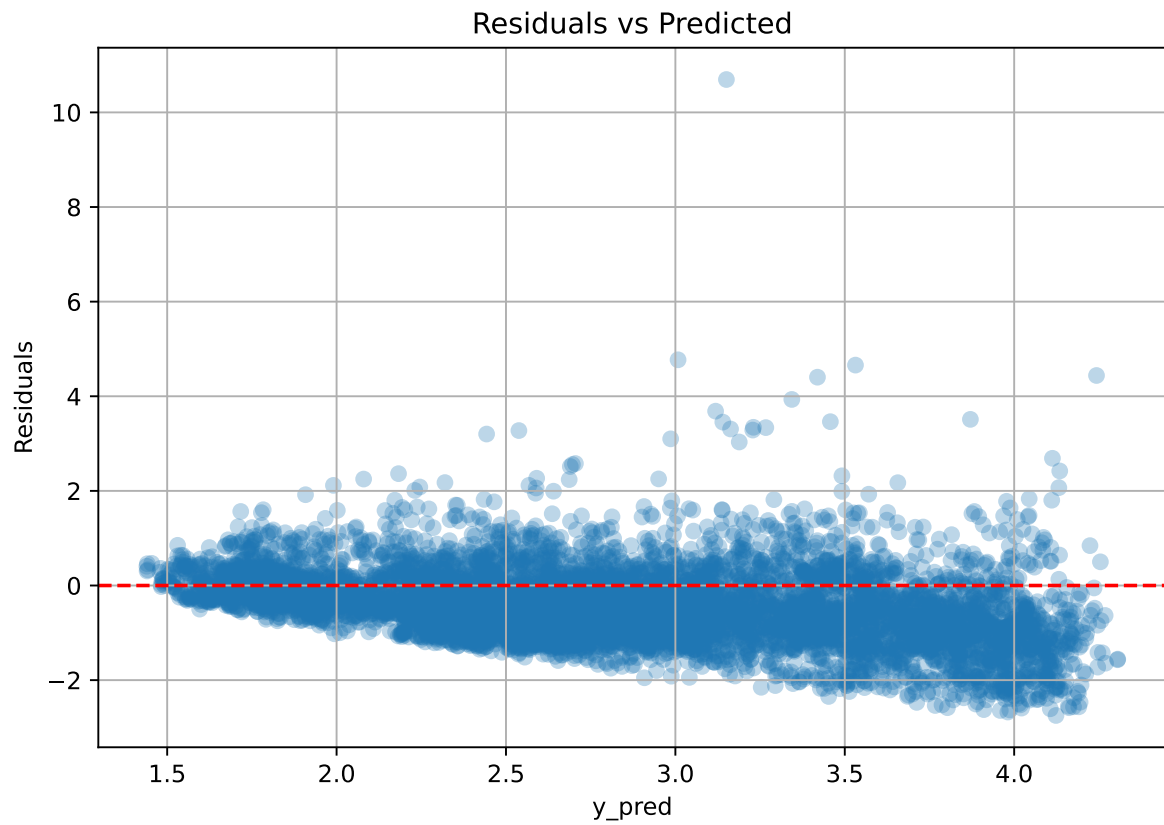
$\text{coef_r_umidity} = -0.4502$

$\text{intercept} = 2.2379$

2.3.14 Scatter



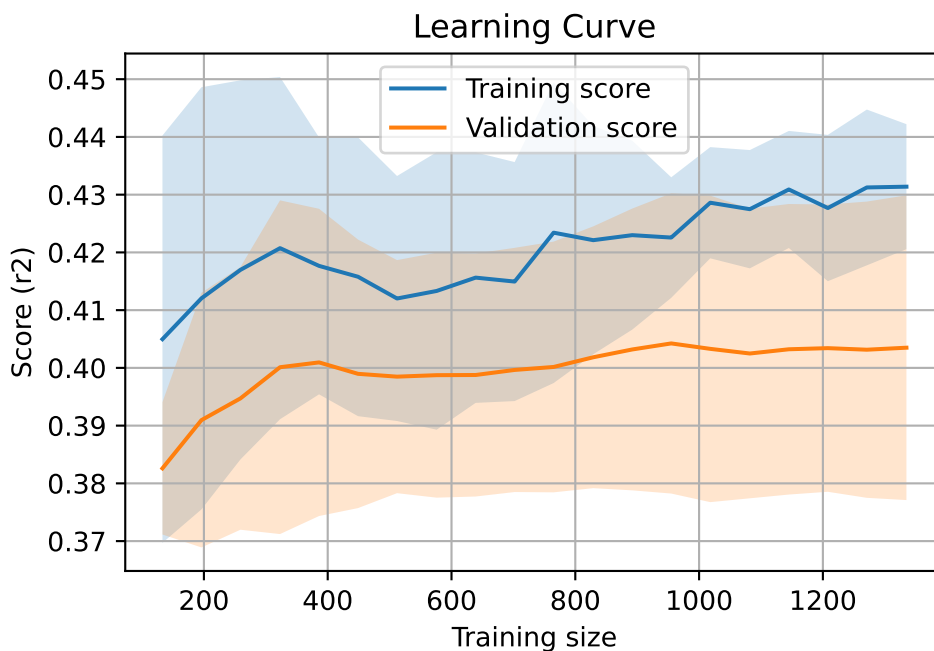
2.3.15 Scatter residuals



2.3.16 Learning curve — Group: 4_5

`Len(group)` : 2726, `Len(training)` : 1909,

`Len(training_real)` : 1337, `Len(test_of_training)` : 572, `Len(test)` : 817



2.3.17 Gridsearch — Group: 6_3

Len(group) : 561, Len(training) : 393, Len(test) : 168

Distribution y_train vs y_test: y_train: mean=1.935, std=0.720, min=1.041, max=6.086
y_test: mean=2.031, std=0.744, min=1.078, max=4.912

Fold 0 — y_test: mean=1.85, std=0.64, y_train: mean=1.97, std=0.75

Fold 1 — y_test: mean=1.98, std=0.75, y_train: mean=1.91, std=0.70

Fold 2 — y_test: mean=1.91, std=0.77, y_train: mean=1.95, std=0.70

Fold 3 — y_test: mean=1.92, std=0.68, y_train: mean=1.94, std=0.73

Fold 4 — y_test: mean=1.99, std=0.74, y_train: mean=1.91, std=0.71

train scores CV: [0.60490696 0.59617869 0.68479321 0.61316376 0.62706164]

test scores CV: [0.74052864 0.71853853 0.55240443 0.71109031 0.66169535]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.6252	0.6769	0.0674	0.6159	0.6233	-0.0536	0.9209	0.9237

Coefficients:

$\text{coef_r_CO2} = -0.0551$

$\text{coef_r_NH3} = -0.2103$

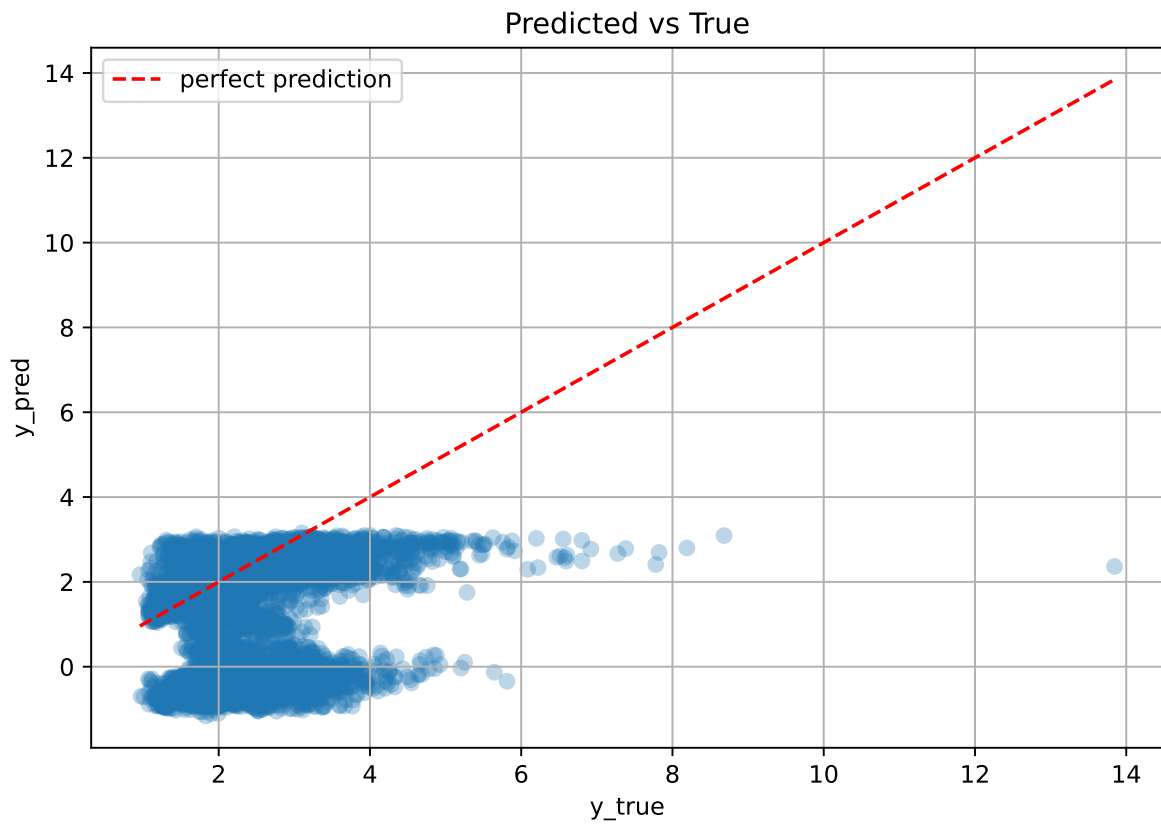
$\text{coef_r_THI} = 1.2572$

$\text{coef_r_temperature} = -1.5809$

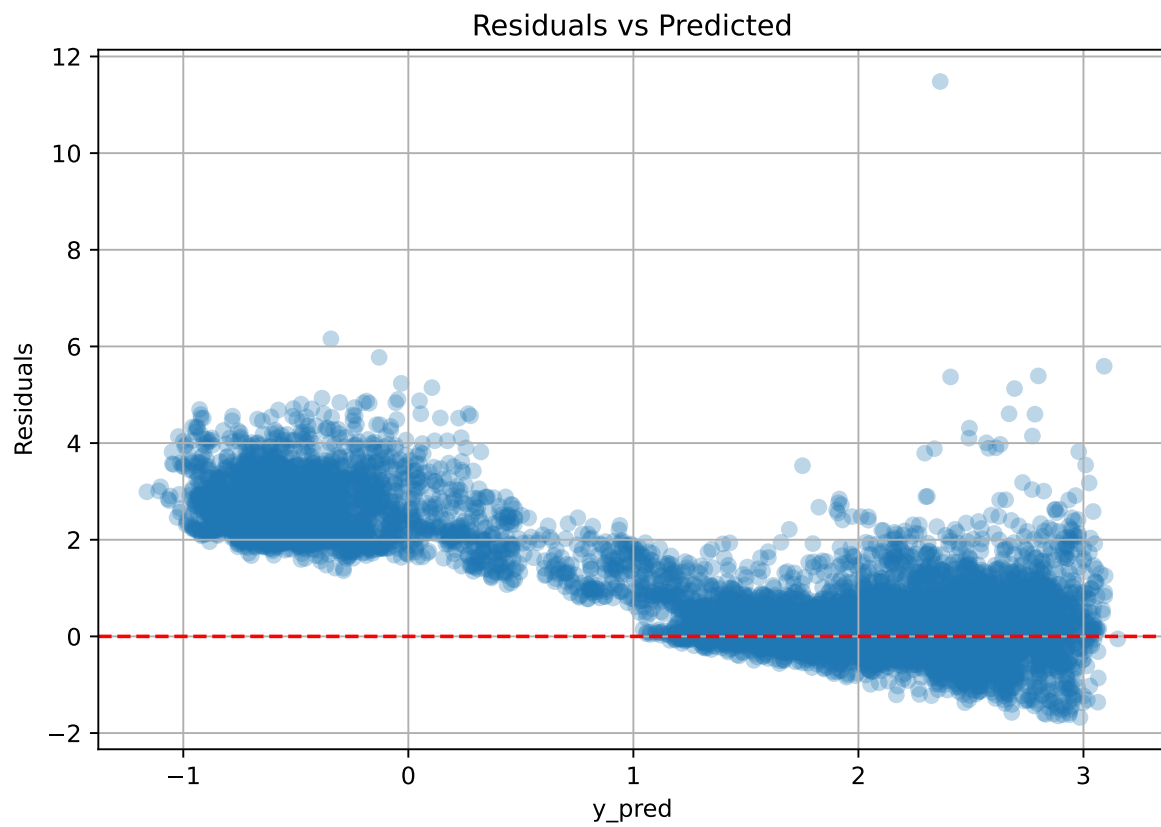
$\text{coef_r_umidity} = -0.8280$

$\text{intercept} = 1.9349$

2.3.18 Scatter



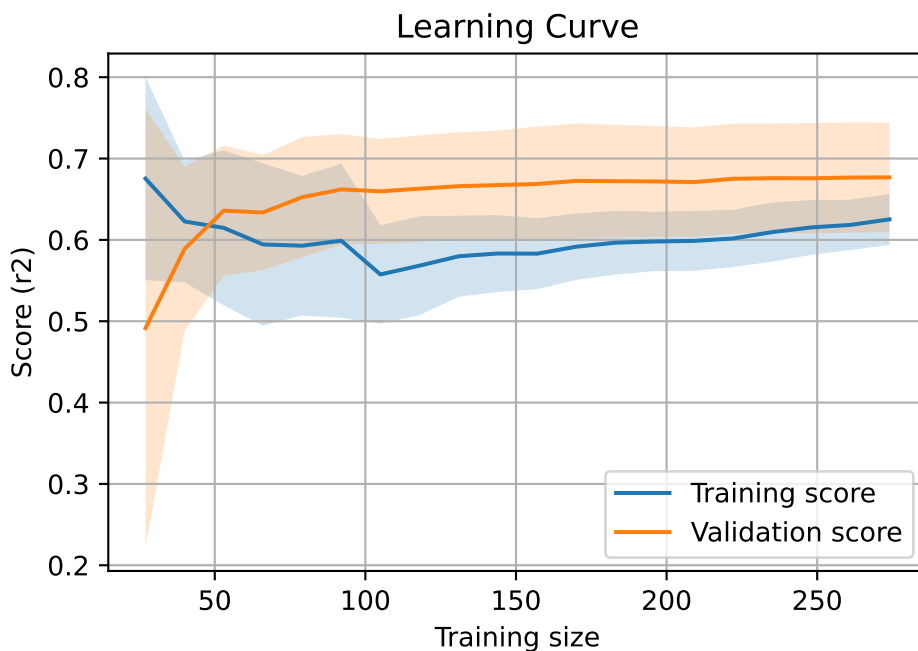
2.3.19 Scatter residuals



2.3.20 Learning curve — Group: 6_3

`Len(group)` : 561, `Len(training)` : 393,

`Len(training_real)` : 276, `Len(test_of_training)` : 117, `Len(test)` : 168



2.3.21 Gridsearch — Group: 6_4

Len(group) : 792, Len(training) : 555, Len(test) : 237

Distribution y_train vs y_test: y_train: mean=2.486, std=0.782, min=1.253, max=7.383
y_test: mean=2.410, std=0.685, min=1.398, max=5.144

Fold 0 — y_test: mean=2.46, std=0.72, y_train: mean=2.50, std=0.81

Fold 1 — y_test: mean=2.48, std=0.72, y_train: mean=2.49, std=0.81

Fold 2 — y_test: mean=2.47, std=0.79, y_train: mean=2.49, std=0.78

Fold 3 — y_test: mean=2.55, std=0.82, y_train: mean=2.46, std=0.76

Fold 4 — y_test: mean=2.50, std=0.78, y_train: mean=2.48, std=0.78

train scores CV: [0.15506482 0.17197221 0.15800044 0.17447249 0.20192166]

test scores CV: [0.23114143 0.18901692 0.21005333 0.16444206 0.10988493]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.1723	0.1809	0.0418	0.2601	0.2639	0.083	1.4589	0.9523

Coefficients:

$\text{coef_r_CO2} = -0.1789$

$\text{coef_r_NH3} = 0.0279$

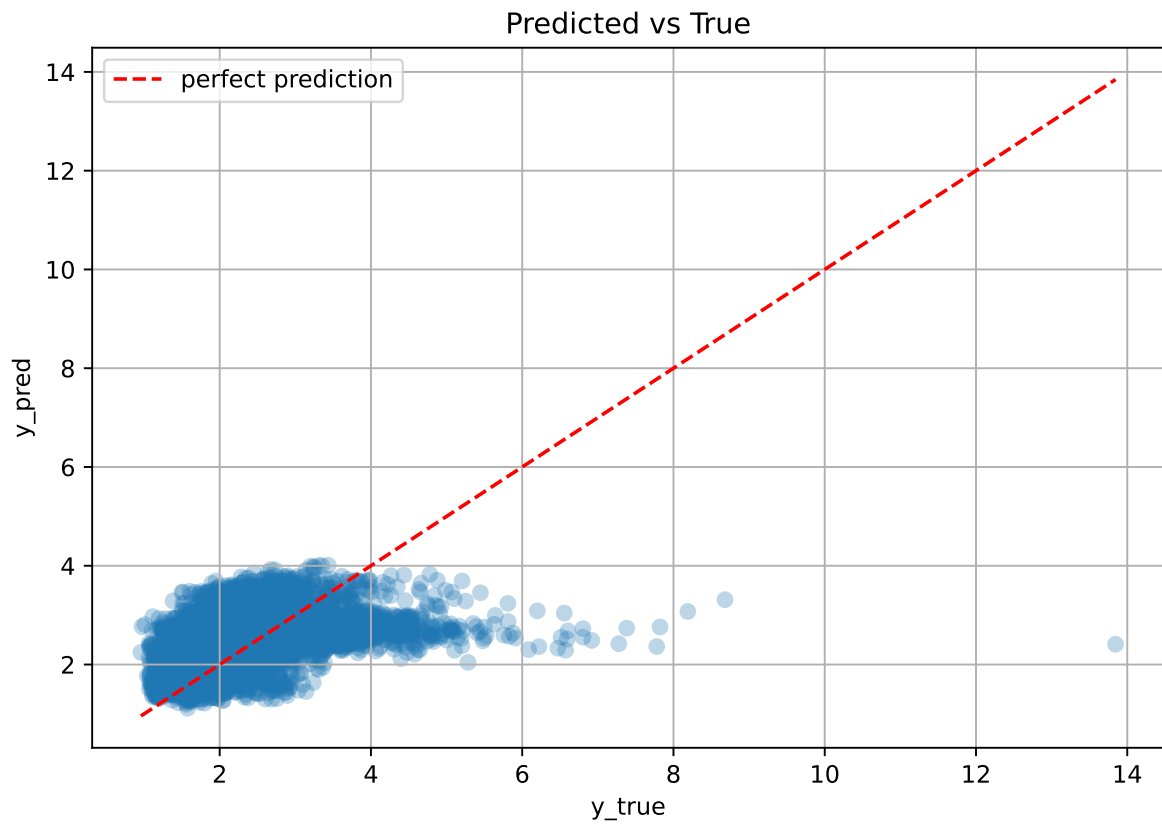
$\text{coef_r_THI} = 1.6225$

$\text{coef_r_temperature} = -2.4188$

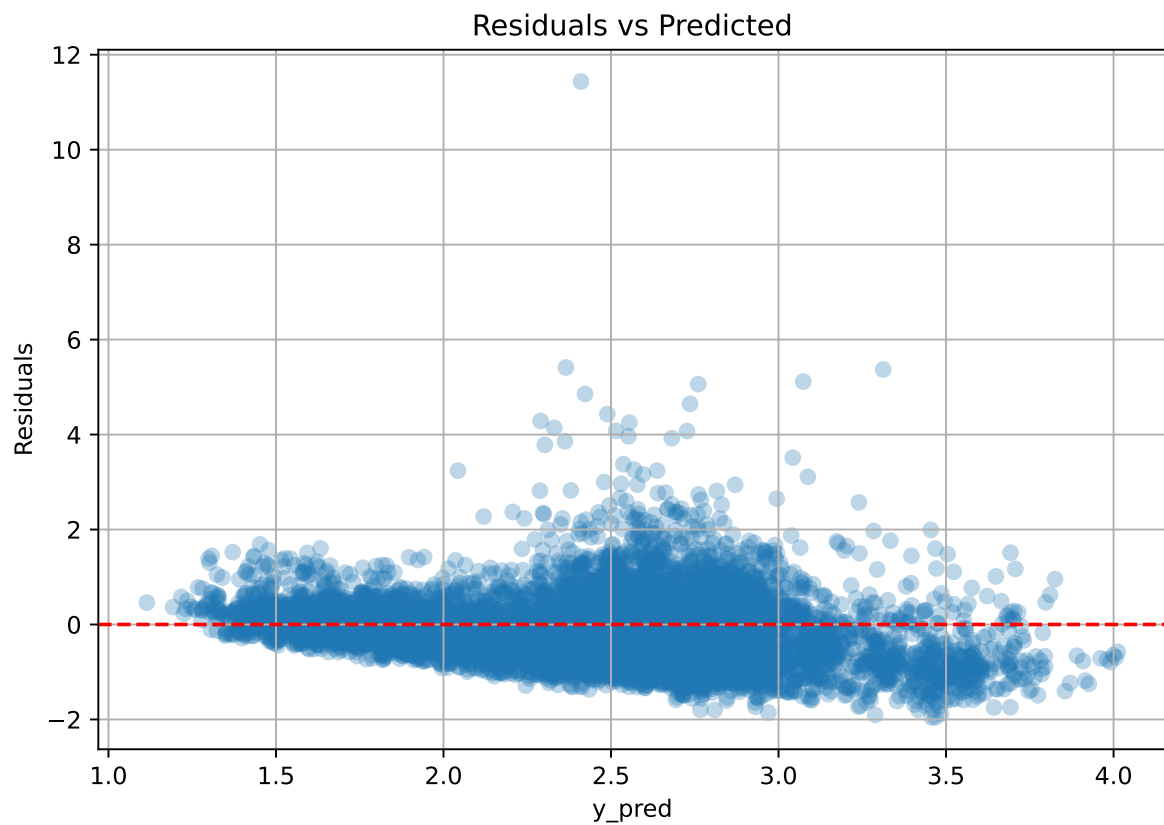
$\text{coef_r_umidity} = -1.1321$

$\text{intercept} = 2.4860$

2.3.22 Scatter



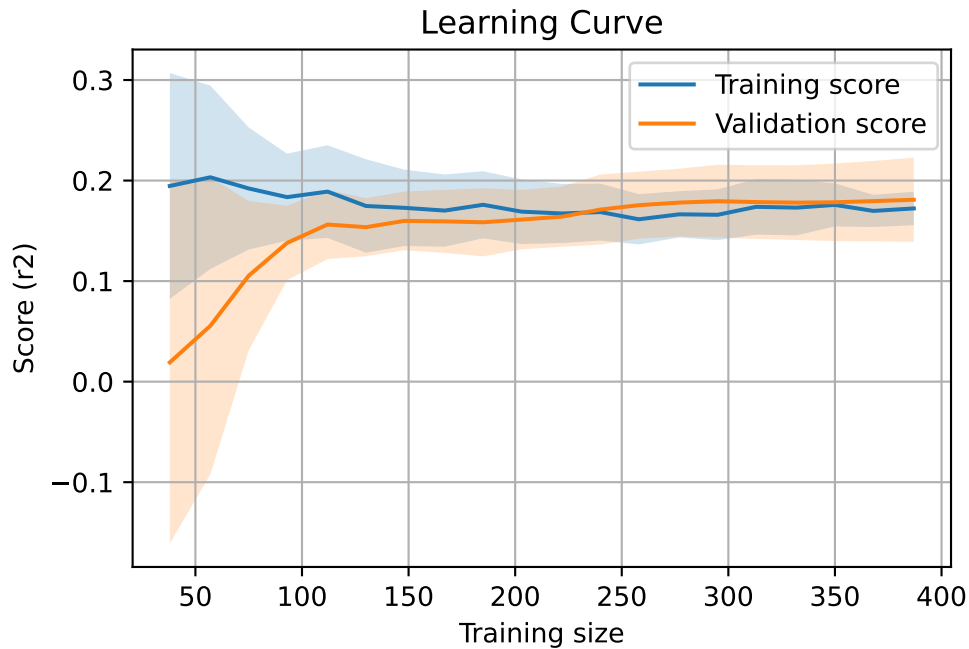
2.3.23 Scatter residuals



2.3.24 Learning curve — Group: 6_4

`Len(group)` : 792, `Len(training)` : 555,

`Len(training_real)` : 389, `Len(test_of_training)` : 166, `Len(test)` : 237



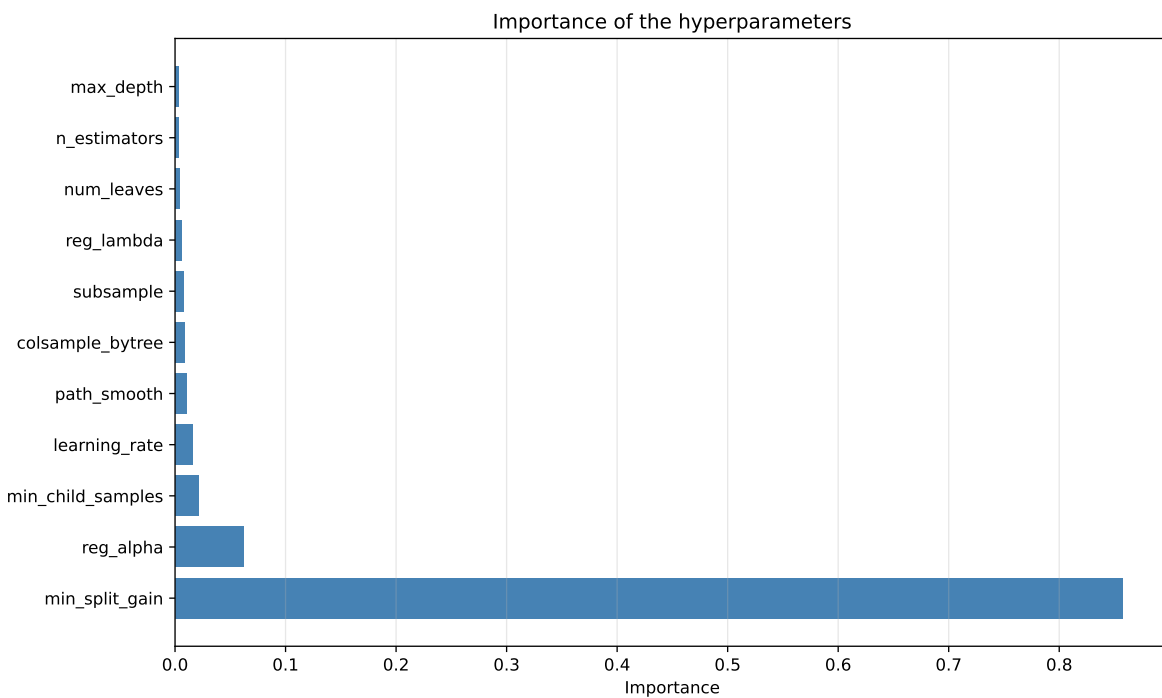
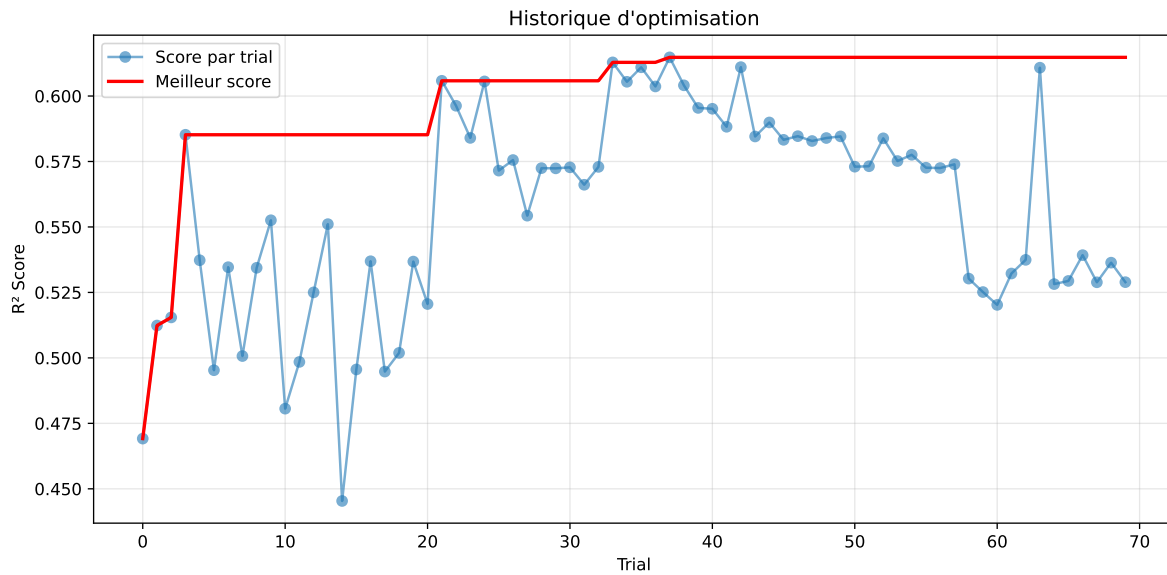
2.4 Model : LGBM

2.4.1 Gridsearch — Group: 1_3

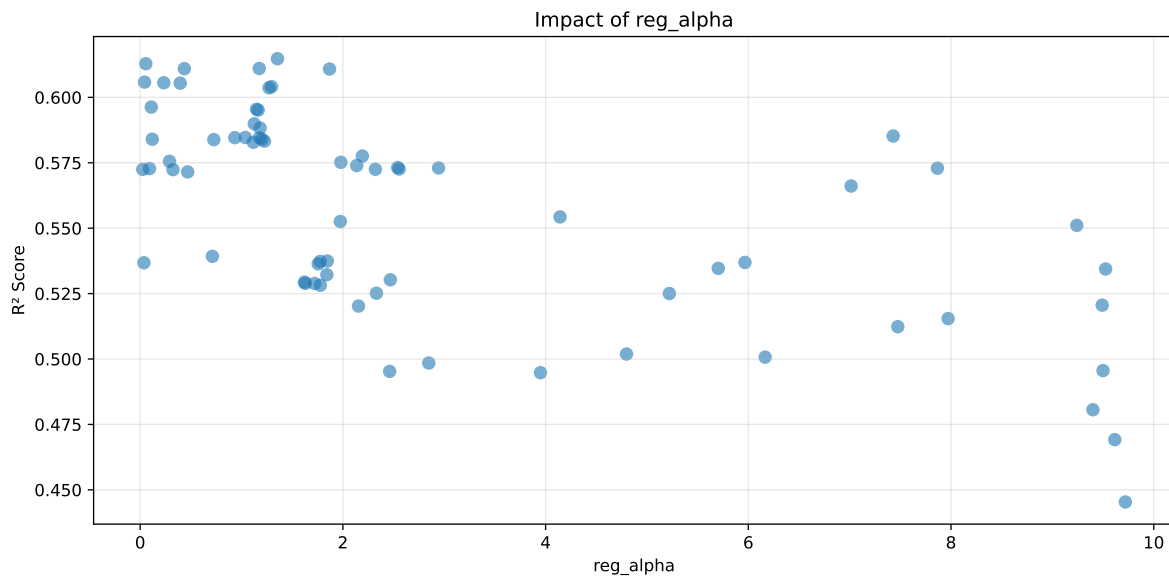
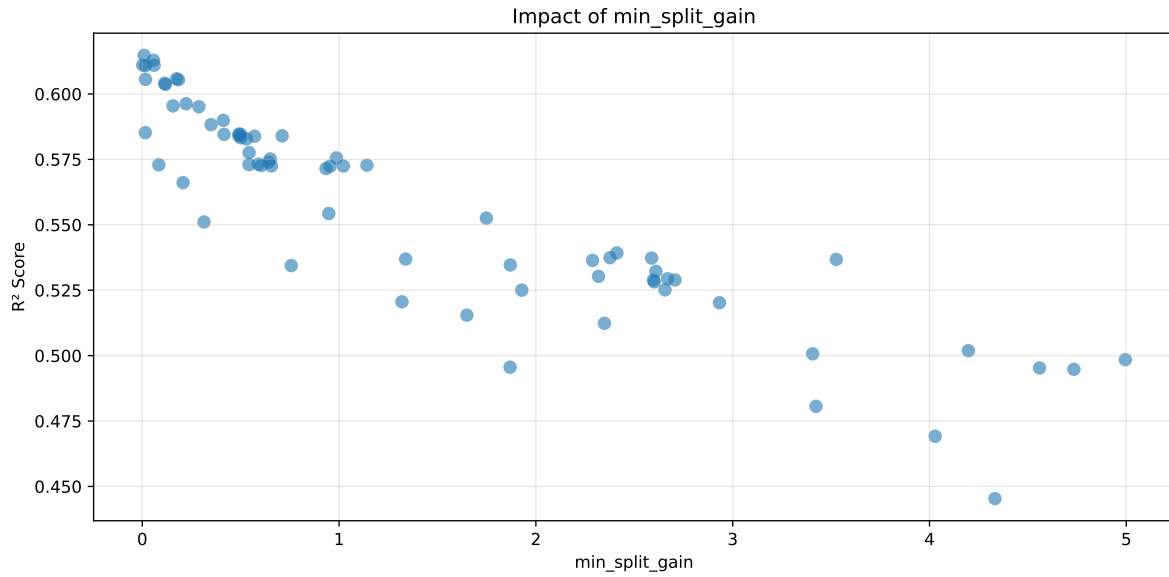
Len(group) : 795, Len(training) : 557, Len(test) : 238

Distribution y_train vs y_test: y_train: mean=2.597, std=0.579, min=1.437, max=4.986
y_test: mean=2.492, std=0.625, min=1.466, max=5.447

===== TOP Importance FEATURES =====
feature importance r_temperature 85 r_umidity 49 r_CO2 30 r_THI 10 r_NH3 3

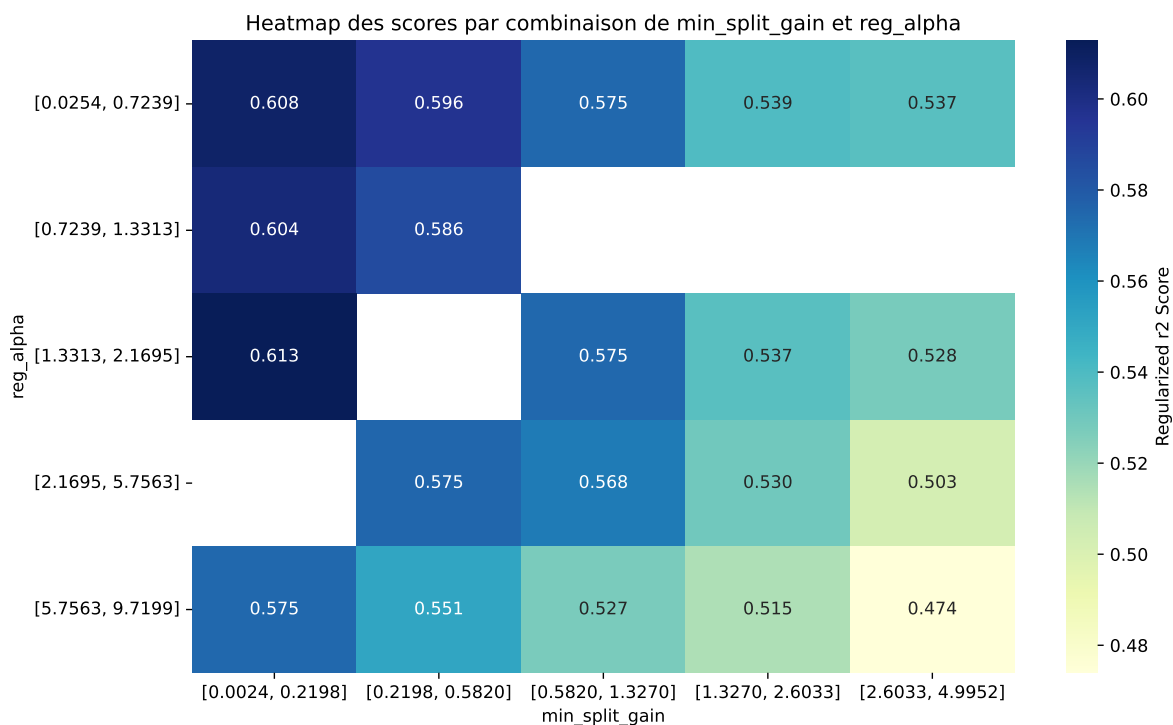


Paramètres avec >5% d'importance : ['min_split_gain', 'reg_alpha']



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The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.6125	0.5725	0.54	0.0431	0.4522	0.5588	-0.0136	0.9762	1.07
2	0.6126	0.5724	0.5399	0.0437	0.4517	0.5603	-0.012	0.979	1.0703
3	0.6118	0.5715	0.5395	0.0429	0.4506	0.5579	-0.0136	0.9762	1.0705
4	0.589	0.5543	0.5159	0.0418	0.4501	0.5393	-0.015	0.9729	1.0627
5	0.6143	0.5728	0.5445	0.0445	0.4482	0.5599	-0.0129	0.9776	1.0725
6	0.6192	0.5756	0.5478	0.0445	0.4447	0.5622	-0.0134	0.9768	1.0758
7	0.6162	0.5732	0.5423	0.0431	0.4442	0.5596	-0.0136	0.9763	1.075
8	0.6177	0.5739	0.5422	0.0429	0.4428	0.5617	-0.0122	0.9787	1.0762
9	0.5892	0.5526	0.5181	0.0451	0.4427	0.5374	-0.0152	0.9725	1.0662
10	0.6194	0.5752	0.5446	0.0417	0.4424	0.561	-0.0142	0.9753	1.0769
---	---	---	---	---	---	---	---	---	---
70	0.7864	0.6056	0.6471	0.0586	0.0631	0.6516	0.046	1.0759	1.2986

Hyperparameters:

Rank 1: {'reg__n_estimators': 600, 'reg__max_depth': 4, 'reg__learning_rate': 0.04216088444071403, 'reg__num_leaves': 15, 'reg__subsample': 0.5105574206484402, 'reg__colsample_bytree': 0.8949244548579935, 'reg__min_child_samples': 22, 'reg__reg_alpha': 0.02541928948538441, 'reg__reg_lambda': 47.42011290873358, 'reg__min_split_gain': 1.0216464216479086, 'reg__path_smooth': 3.2006907093899373}

Rank 2: {'reg__n_estimators': 600, 'reg__max_depth': 4, 'reg__learning_rate': 0.011031281891493733, 'reg__num_leaves': 16, 'reg__subsample': 0.50297629083647, 'reg__colsample_bytree': 0.8985656275630216, 'reg__min_child_samples': 20, 'reg__reg_alpha': 0.32404768574835074, 'reg__reg_lambda': 47.31364583195327, 'reg__min_split_gain': 0.9539603640767714, 'reg__path_smooth': 3.6125220196310552}

Rank 3: {'reg__n_estimators': 600, 'reg__max_depth': 4, 'reg__learning_rate': 0.09496543467173472, 'reg__num_leaves': 16, 'reg__subsample': 0.5103531236662526, 'reg__colsample_bytree': 0.5199594408636008, 'reg__min_child_samples': 21, 'reg__reg_alpha': 0.468407275657273, 'reg__reg_lambda': 49.15126574711327, 'reg__min_split_gain': 0.9336726023669835, 'reg__path_smooth': 3.279901387195308}

Rank 4: {'reg__n_estimators': 600, 'reg__max_depth': 4, 'reg__learning_rate': 0.04263705225215113, 'reg__num_leaves': 16, 'reg__subsample': 0.503524131729034, 'reg__colsample_bytree': 0.5028554553555874, 'reg__min_child_samples': 21, 'reg__reg_alpha': 4.142657752086015, 'reg__reg_lambda': 49.978253631878275, 'reg__min_split_gain': 0.9477500951976406, 'reg__path_smooth': 3.1306341732716136}

Rank 5: {'reg__n_estimators': 600, 'reg__max_depth': 4, 'reg__learning_rate': 0.010659523859829411, 'reg__num_leaves': 15, 'reg__subsample': 0.5018771136500397, 'reg__colsample_bytree': 0.8825444080932013, 'reg__min_child_samples': 24, 'reg__reg_alpha': 0.09227179078341408, 'reg__reg_lambda': 27.838543592738215, 'reg__min_split_gain': 1.141476598395874, 'reg__path_smooth': 3.4574651296333005}

Rank 6: {'reg__n_estimators': 600, 'reg__max_depth': 4, 'reg__learning_rate': 0.09779696227259908, 'reg__num_leaves': 17, 'reg__subsample': 0.6075912398878351, 'reg__colsample_bytree': 0.8379453355657491, 'reg__min_child_samples': 22, 'reg__reg_alpha': 0.2894730089235562, 'reg__reg_lambda': 32.40758769050149, 'reg__min_split_gain': 0.9866394647778929, 'reg__path_smooth': 3.251513567166265}

Rank 7: {'reg__n_estimators': 500, 'reg__max_depth': 3, 'reg__learning_rate': 0.057228617035578, 'reg__num_leaves': 20, 'reg__subsample': 0.6693190024366751, 'reg__colsample_bytree': 0.8087672433237421, 'reg__min_child_samples': 27, 'reg__reg_alpha': 2.540484251135565, 'reg__reg_lambda': 39.15738141501163, 'reg__min_split_gain': 0.5896122776516741, 'reg__path_smooth': 0.6355867552014358}

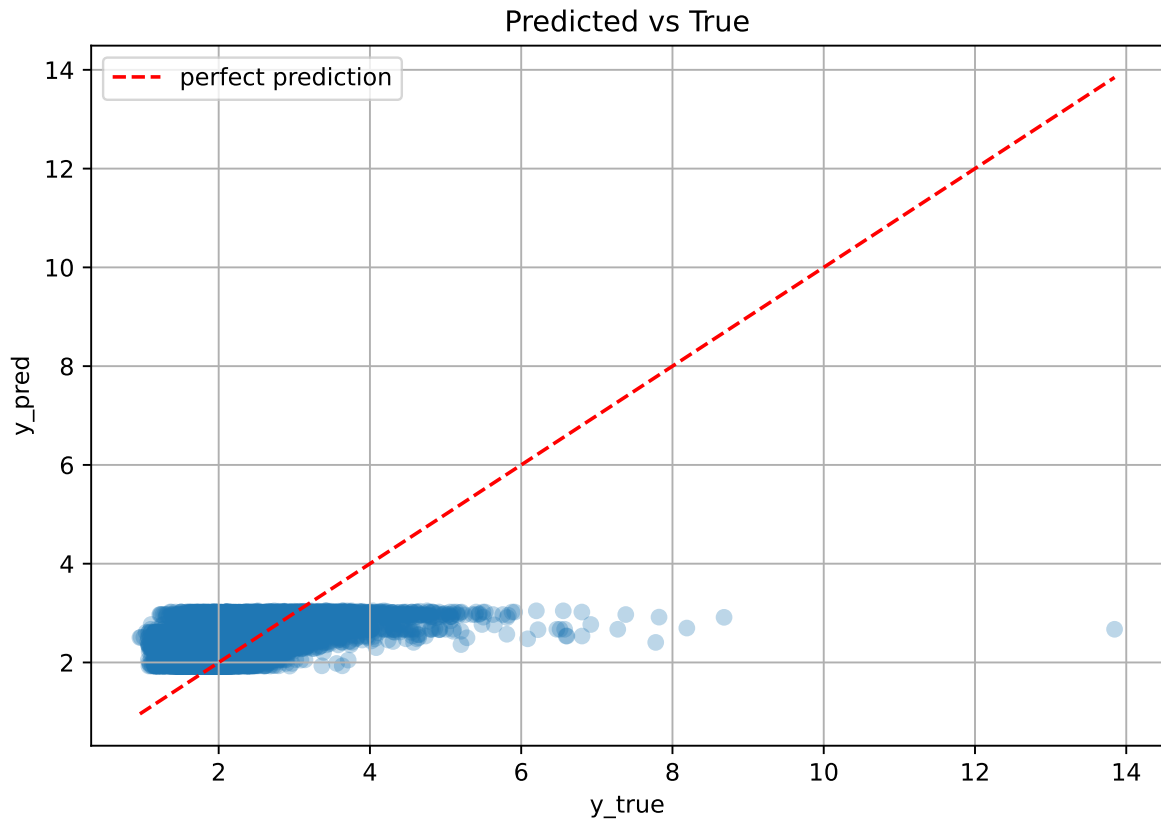
Rank 8: {'reg__n_estimators': 800, 'reg__max_depth': 3, 'reg__learning_rate': 0.052891468369436055, 'reg__num_leaves': 23, 'reg__subsample': 0.7019291687697476, 'reg__colsample_bytree': 0.8184147579825743, 'reg__min_child_samples': 17, 'reg__reg_alpha': 2.136698135562604, 'reg__reg_lambda': 39.56967096252523, 'reg__min_split_gain': 0.642464855488065, 'reg__path_smooth': 0.5762084418961995}

Rank 9: {'reg__n_estimators': 600, 'reg__max_depth': 3, 'reg__learning_rate': 0.06562290633732172, 'reg__num_leaves': 25, 'reg__subsample': 0.5164883875531059,

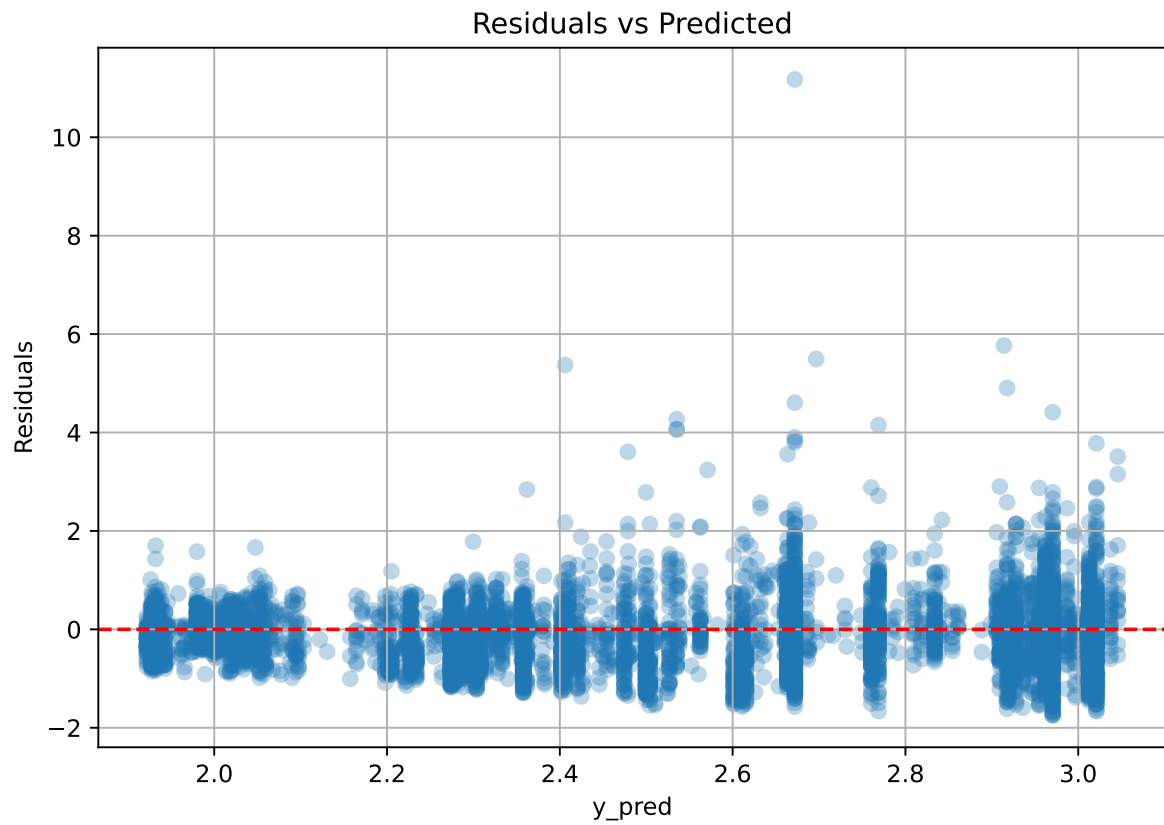
'reg__colsample_bytree': 0.7655110320162272, 'reg__min_child_samples': 10, 'reg__reg_alpha': 1.9747599687226813, 'reg__reg_lambda': 21.058883073731543, 'reg__min_split_gain': 1.7484360847957614, 'reg__path_smooth': 3.4230434754380727}

Rank 10: {'reg__n_estimators': 800, 'reg__max_depth': 3, 'reg__learning_rate': 0.08666899634100304, 'reg__num_leaves': 21, 'reg__subsample': 0.6639637858053002, 'reg__colsample_bytree': 0.8143618936552222, 'reg__min_child_samples': 17, 'reg__reg_alpha': 1.979701549468876, 'reg__reg_lambda': 39.073621612554426, 'reg__min_split_gain': 0.6501998185540712, 'reg__path_smooth': 0.850605413193549}

2.4.2 Scatter



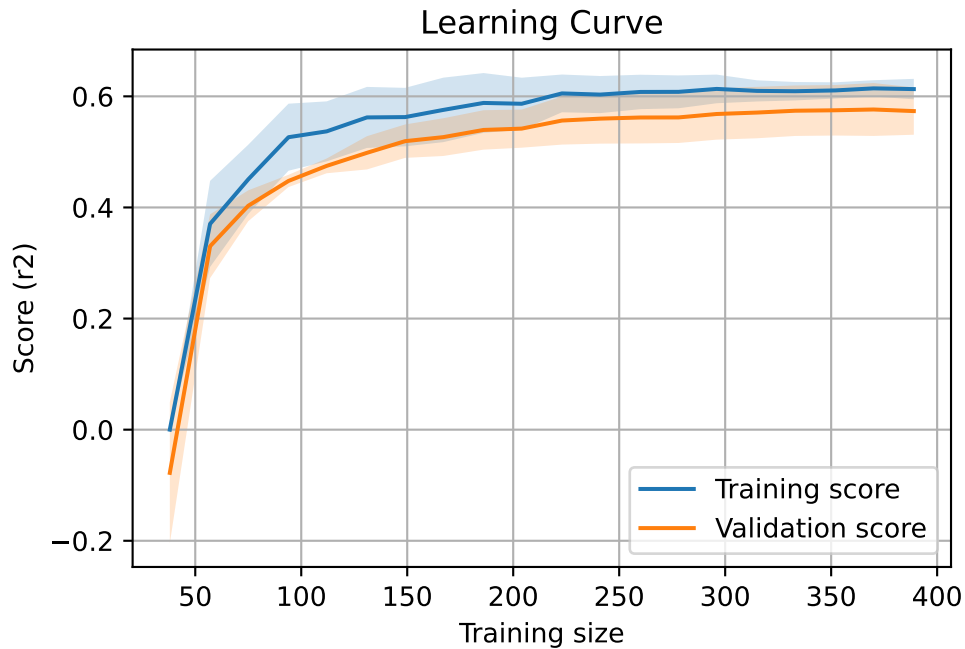
2.4.3 Scatter residuals



2.4.4 Learning curve — Group: 1_3

`Len(group)` : 795, `Len(training)` : 557,

`Len(training_real)` : 390, `Len(test_of_training)` : 167, `Len(test)` : 238

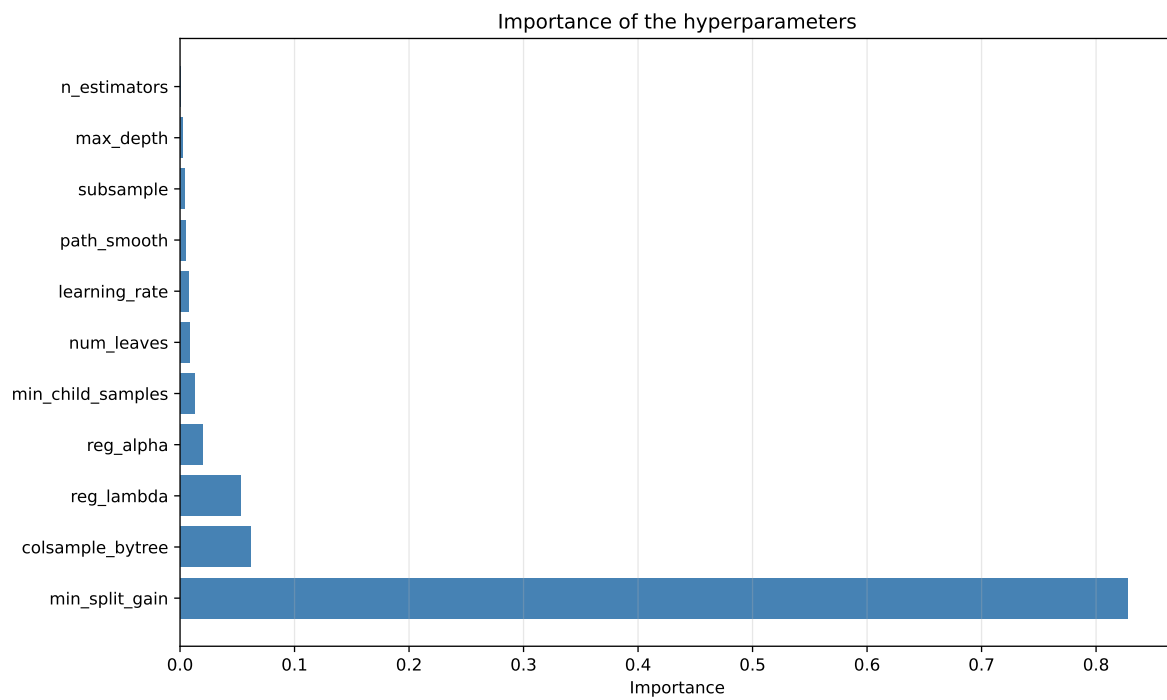
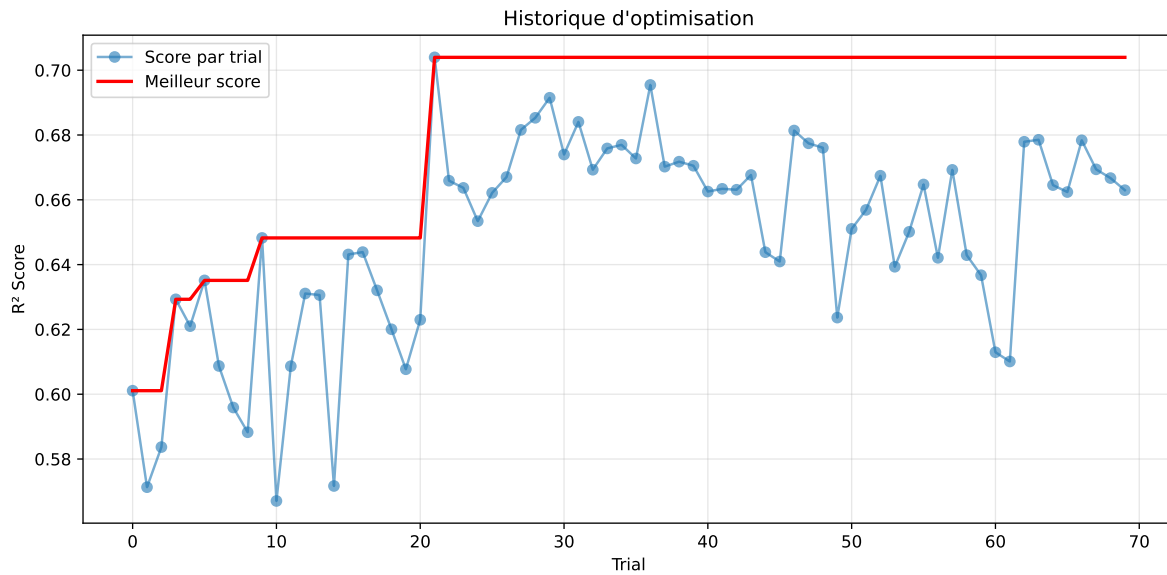


2.4.5 Gridsearch — Group: 2_11

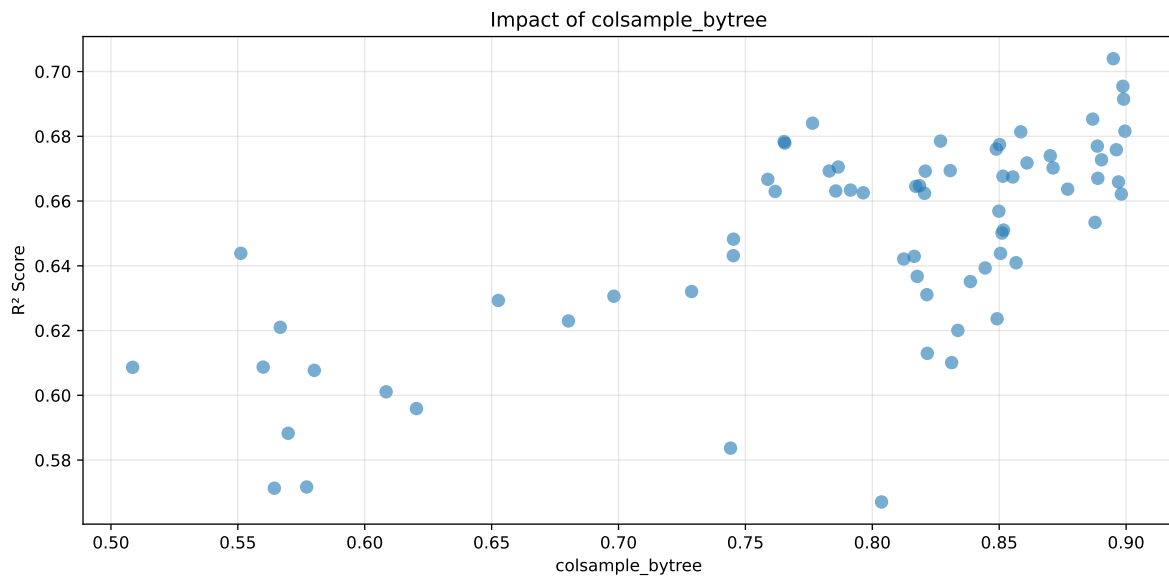
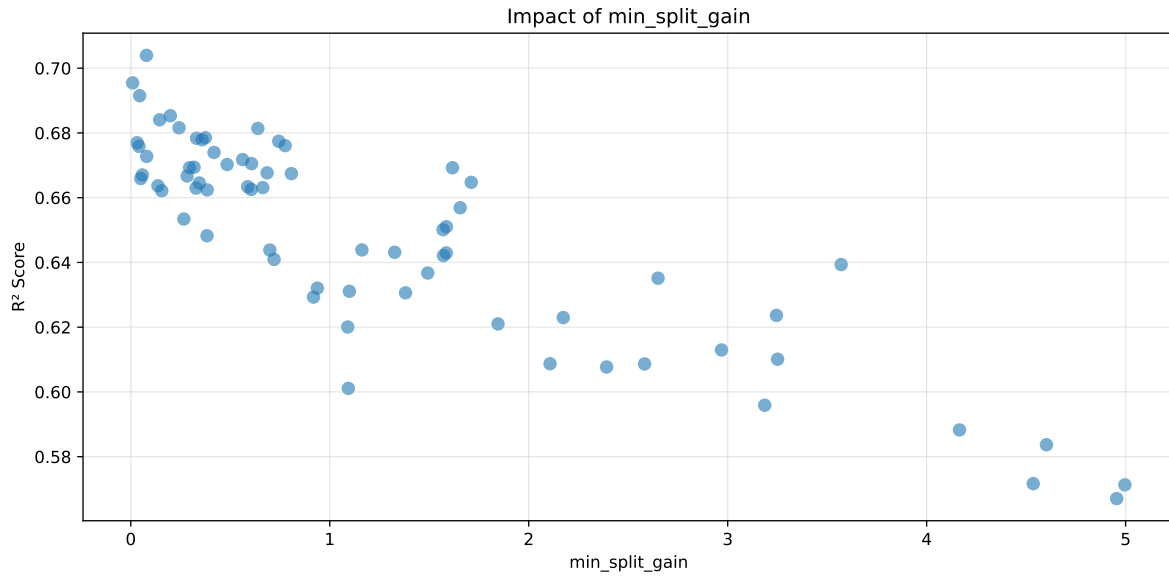
Len(group) : 2726, **Len(training)** : 1909, **Len(test)** : 817

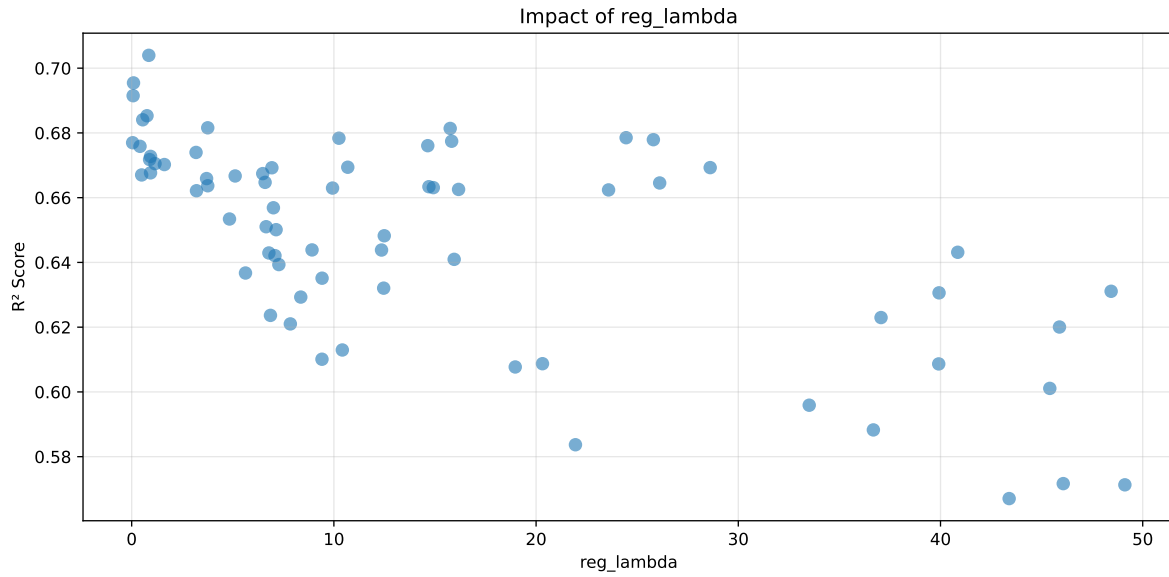
Distribution y_train vs y_test: y_train: mean=2.540, std=0.859, min=1.150, max=8.682
y_test: mean=2.561, std=0.947, min=1.120, max=13.845

===== TOP Importance FEATURES =====
feature importance r_NH3 242 r_umidity 212 r_CO2 162 r_temperature 91 r_THI 45



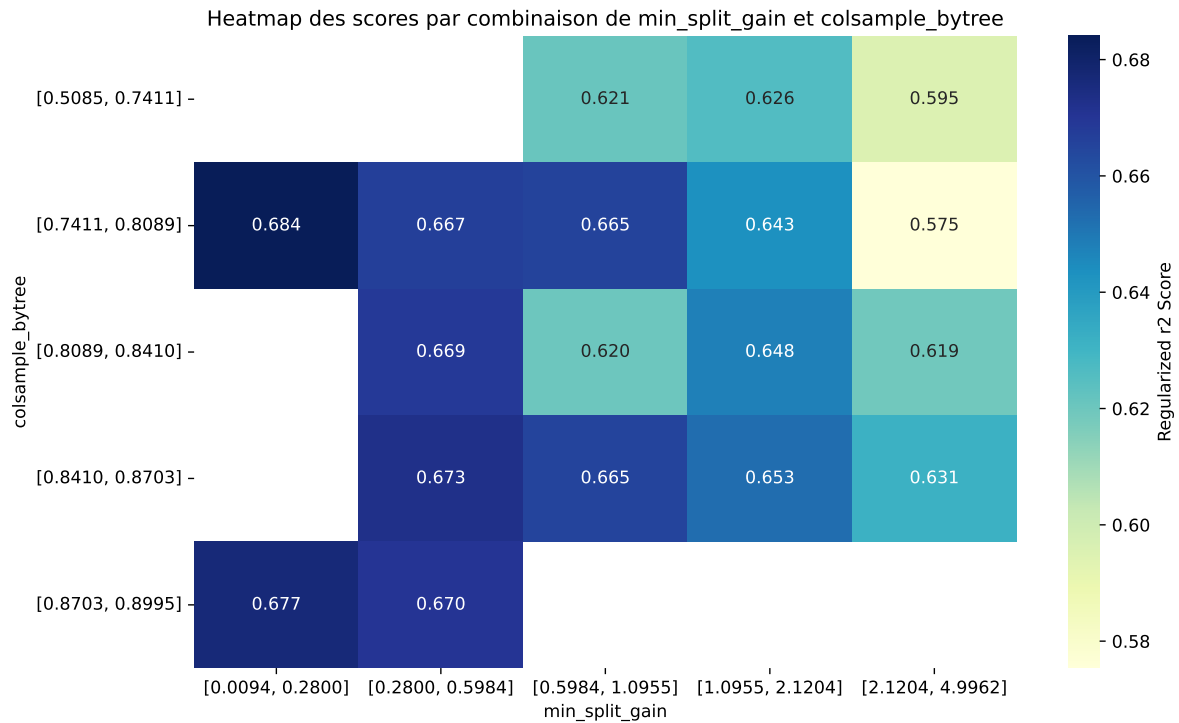
Paramètres avec >5% d'importance : ['min_split_gain', 'colsample_bytree', 'reg_lambda']





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The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.682	0.651	0.5512	0.0137	0.651	0.5687	-0.0824	0.8735	1.0476
2	0.6815	0.6501	0.5506	0.014	0.6501	0.5672	-0.0829	0.8724	1.0482
3	0.6715	0.6439	0.5517	0.0146	0.6439	0.5668	-0.0771	0.8803	1.043
4	0.673	0.6438	0.5483	0.0122	0.6438	0.5662	-0.0776	0.8795	1.0454
5	0.6698	0.6431	0.5484	0.0133	0.6431	0.5643	-0.0788	0.8775	1.0414
6	0.6701	0.6429	0.5452	0.0137	0.6429	0.5644	-0.0785	0.8778	1.0422
7	0.67	0.6421	0.545	0.013	0.6421	0.5622	-0.0799	0.8756	1.0435
8	0.6696	0.6409	0.5468	0.0135	0.6409	0.5639	-0.077	0.8798	1.0447
9	0.6627	0.6393	0.5396	0.0136	0.6393	0.5605	-0.0788	0.8767	1.0366
10	0.6593	0.6367	0.5424	0.0145	0.6367	0.5585	-0.0782	0.8772	1.0354
---	---	---	---	---	---	---	---	---	---
70	0.8153	0.704	0.5875	0.0111	0.37	0.6026	-0.1014	0.856	1.1581

Hyperparameters:

Rank 1: {'reg__n_estimators': 500, 'reg__max_depth': 6, 'reg__learning_rate': 0.03787970456512325, 'reg__num_leaves': 23, 'reg__subsample': 0.6032084814760359, 'reg__colsample_bytree': 0.85169194469598, 'reg__min_child_samples': 33, 'reg__reg_alpha': 3.348379756888436, 'reg__reg_lambda': 6.649844282617125, 'reg__min_split_gain': 1.5862810002692738, 'reg__path_smooth': 0.6118777701148851}

Rank 2: {'reg__n_estimators': 800, 'reg__max_depth': 5, 'reg__learning_rate': 0.06053201085712836, 'reg__num_leaves': 43, 'reg__subsample': 0.590845626406677, 'reg__colsample_bytree': 0.851111283554411, 'reg__min_child_samples': 31, 'reg__reg_alpha': 3.515264491550764, 'reg__reg_lambda': 7.142479940618003, 'reg__min_split_gain': 1.5694307084232415, 'reg__path_smooth': 4.185894295740189}

Rank 3: {'reg__n_estimators': 800, 'reg__max_depth': 3, 'reg__learning_rate': 0.08083631738119192, 'reg__num_leaves': 38, 'reg__subsample': 0.5288521437275644, 'reg__colsample_bytree': 0.5511718818603067, 'reg__min_child_samples': 20, 'reg__reg_alpha': 2.6093681287306616, 'reg__reg_lambda': 8.9193919692525, 'reg__min_split_gain': 1.161657604458413, 'reg__path_smooth': 2.974182860065847}

Rank 4: {'reg__n_estimators': 700, 'reg__max_depth': 4, 'reg__learning_rate': 0.03835785730129544, 'reg__num_leaves': 26, 'reg__subsample': 0.6613061594363632, 'reg__colsample_bytree': 0.8504983859987089, 'reg__min_child_samples': 29, 'reg__reg_alpha': 8.084904823669982, 'reg__reg_lambda': 12.356384177443825, 'reg__min_split_gain': 0.698957488313468, 'reg__path_smooth': 0.5524806583101605}

Rank 5: {'reg__n_estimators': 600, 'reg__max_depth': 5, 'reg__learning_rate': 0.029549244595798545, 'reg__num_leaves': 35, 'reg__subsample': 0.6247156217668879,

'reg__colsample_bytree': 0.745243861735963, 'reg__min_child_samples': 36, 'reg__reg_alpha': 2.5383154656022064, 'reg__reg_lambda': 40.852527750045894, 'reg__min_split_gain': 1.3263076719511995, 'reg__path_smooth': 0.9536760996263549}

Rank 6: {'reg__n_estimators': 800, 'reg__max_depth': 5, 'reg__learning_rate': 0.05286761889796594, 'reg__num_leaves': 43, 'reg__subsample': 0.6017641693213994, 'reg__colsample_bytree': 0.8165208811702136, 'reg__min_child_samples': 26, 'reg__reg_alpha': 5.53132981442985, 'reg__reg_lambda': 6.785862065347048, 'reg__min_split_gain': 1.585828216336536, 'reg__path_smooth': 4.178216816542681}

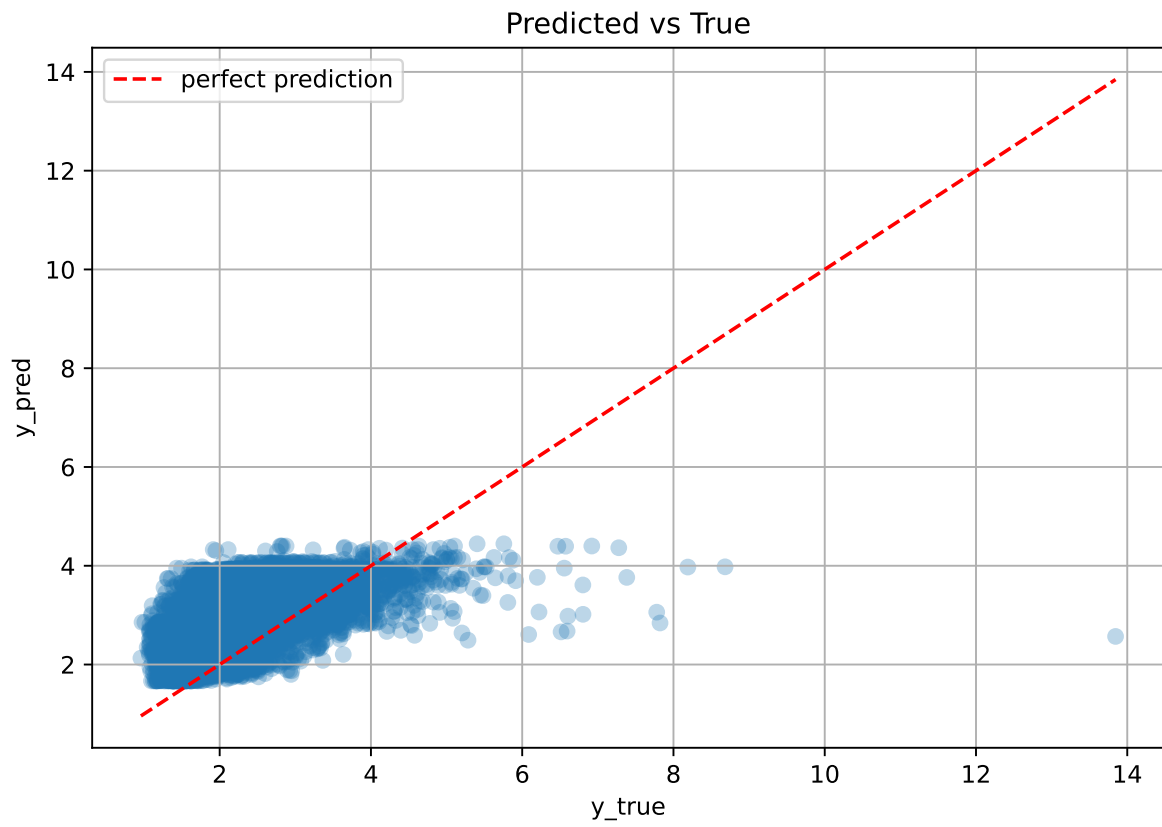
Rank 7: {'reg__n_estimators': 800, 'reg__max_depth': 5, 'reg__learning_rate': 0.05286881087903922, 'reg__num_leaves': 44, 'reg__subsample': 0.6026691319610006, 'reg__colsample_bytree': 0.8123772600580836, 'reg__min_child_samples': 25, 'reg__reg_alpha': 5.719033131428771, 'reg__reg_lambda': 7.086070645904692, 'reg__min_split_gain': 1.5715566162085337, 'reg__path_smooth': 4.184043359682066}

Rank 8: {'reg__n_estimators': 700, 'reg__max_depth': 4, 'reg__learning_rate': 0.03870882826481671, 'reg__num_leaves': 24, 'reg__subsample': 0.6634654567118112, 'reg__colsample_bytree': 0.8566682999558561, 'reg__min_child_samples': 32, 'reg__reg_alpha': 8.074787603923514, 'reg__reg_lambda': 15.948940902924974, 'reg__min_split_gain': 0.7202589568377578, 'reg__path_smooth': 0.5773102836244206}

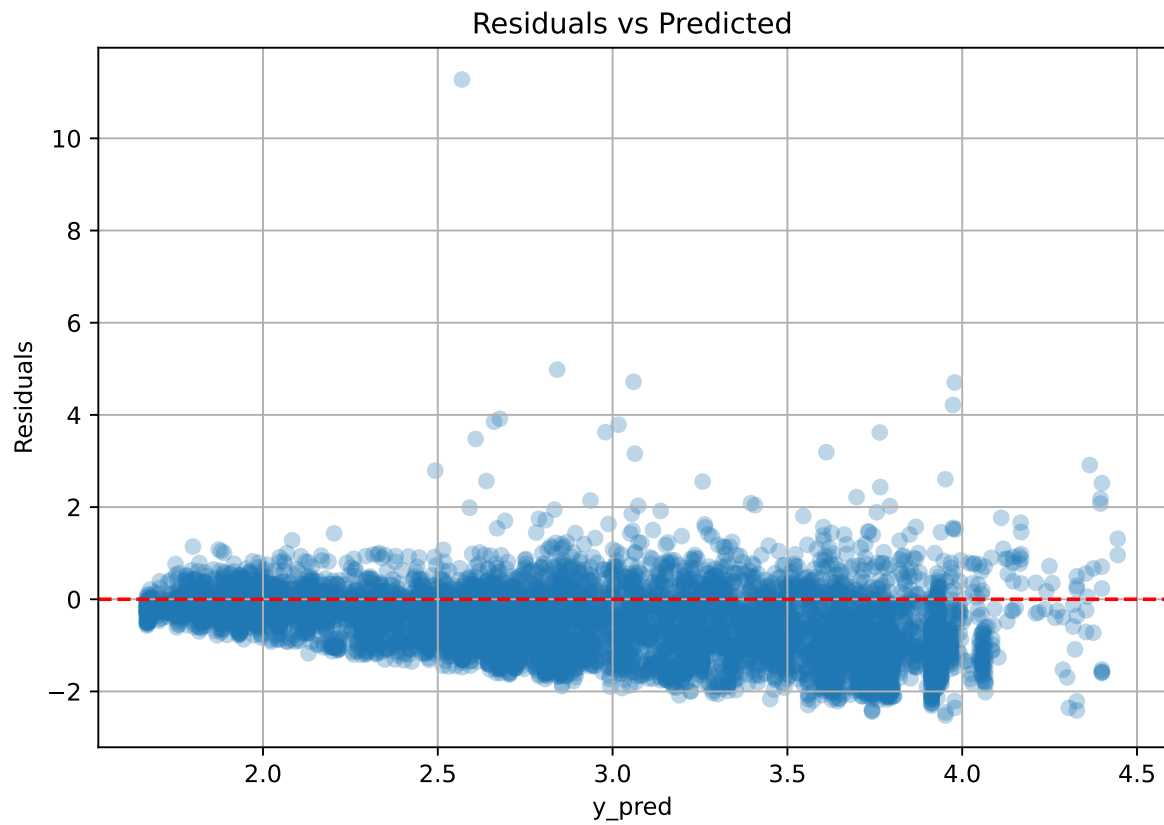
Rank 9: {'reg__n_estimators': 800, 'reg__max_depth': 6, 'reg__learning_rate': 0.062373654475971554, 'reg__num_leaves': 42, 'reg__subsample': 0.6043728538043468, 'reg__colsample_bytree': 0.8444648553339273, 'reg__min_child_samples': 34, 'reg__reg_alpha': 0.030305882114484817, 'reg__reg_lambda': 7.278689133947419, 'reg__min_split_gain': 3.5703422167078136, 'reg__path_smooth': 4.3926048028223805}

Rank 10: {'reg__n_estimators': 800, 'reg__max_depth': 3, 'reg__learning_rate': 0.013262058710910768, 'reg__num_leaves': 46, 'reg__subsample': 0.6064423060321079, 'reg__colsample_bytree': 0.8176952349314709, 'reg__min_child_samples': 25, 'reg__reg_alpha': 5.885006534158755, 'reg__reg_lambda': 5.628658304524644, 'reg__min_split_gain': 1.4925712160377405, 'reg__path_smooth': 4.044004385248877}

2.4.6 Scatter



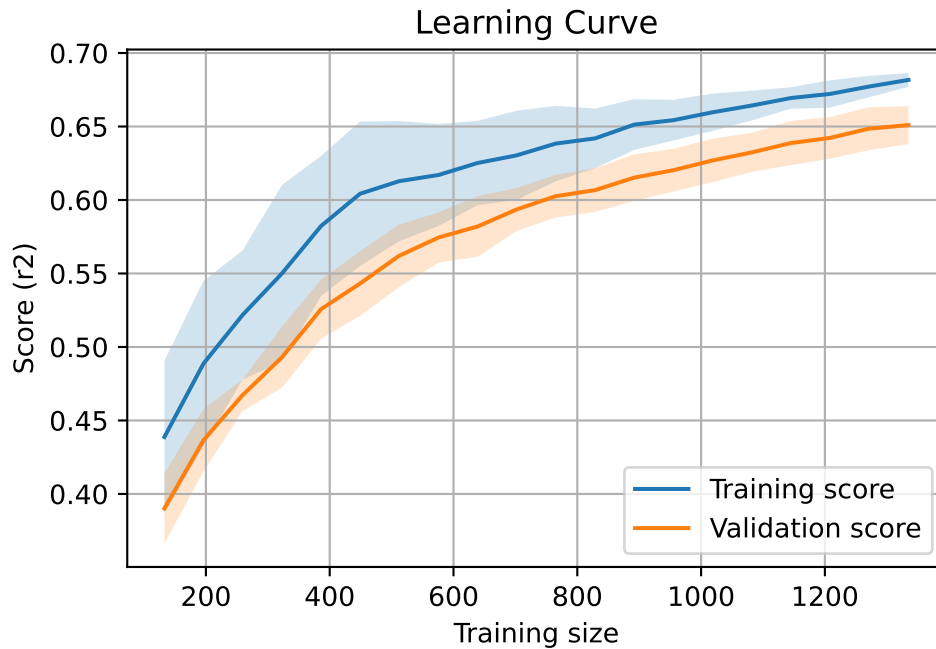
2.4.7 Scatter residuals



2.4.8 Learning curve — Group: 2_11

`Len(group)` : 2726, `Len(training)` : 1909,

`Len(training_real)` : 1337, `Len(test_of_training)` : 572, `Len(test)` : 817

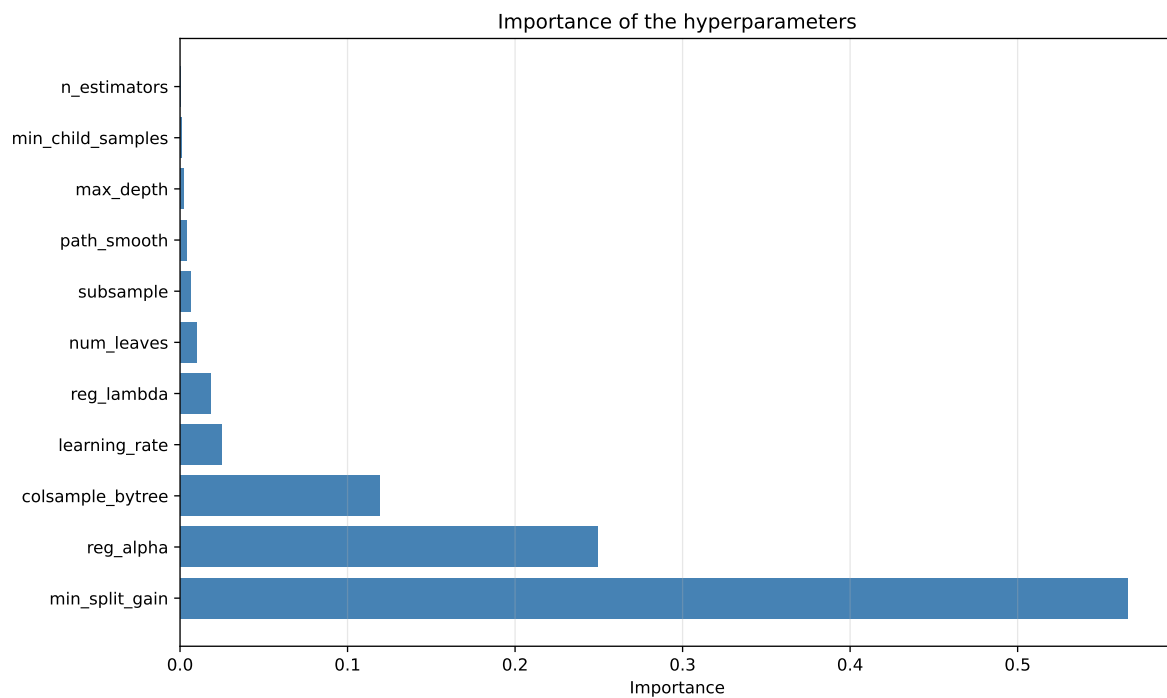
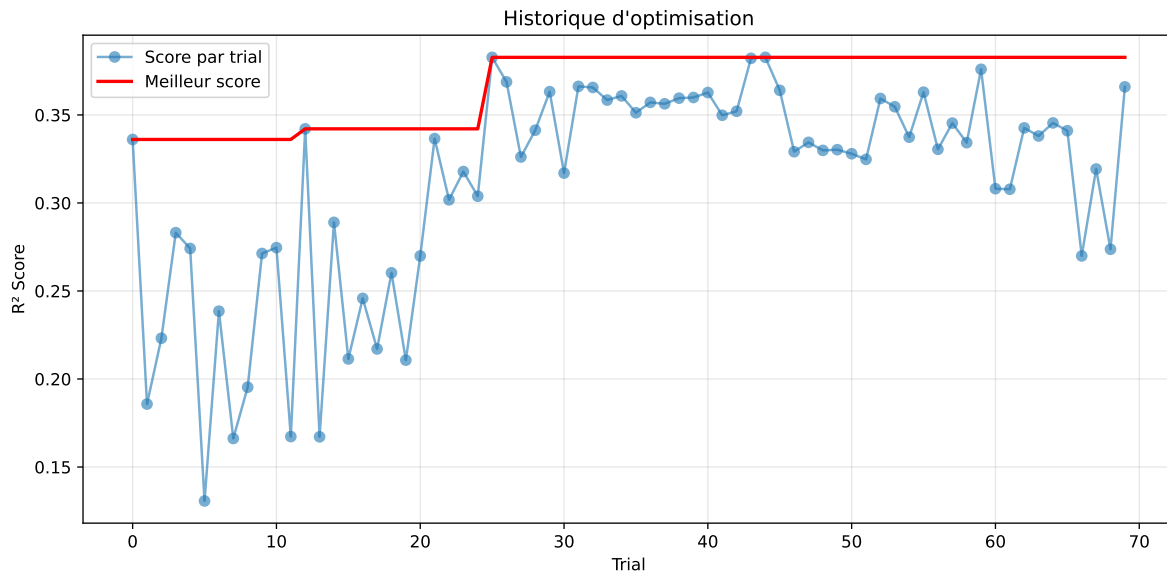


2.4.9 Gridsearch — Group: 3_4

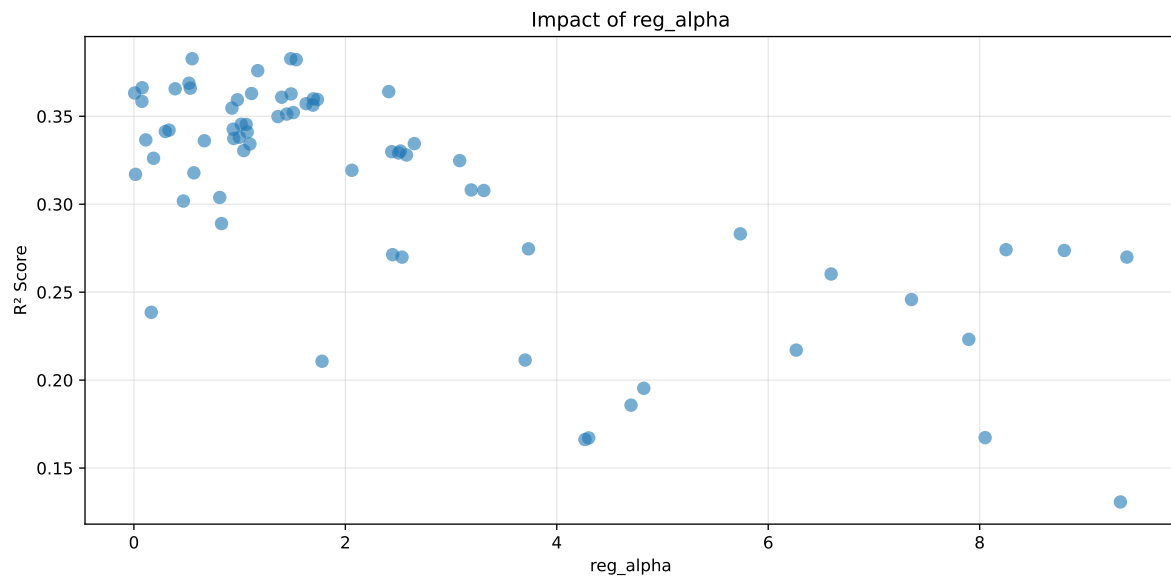
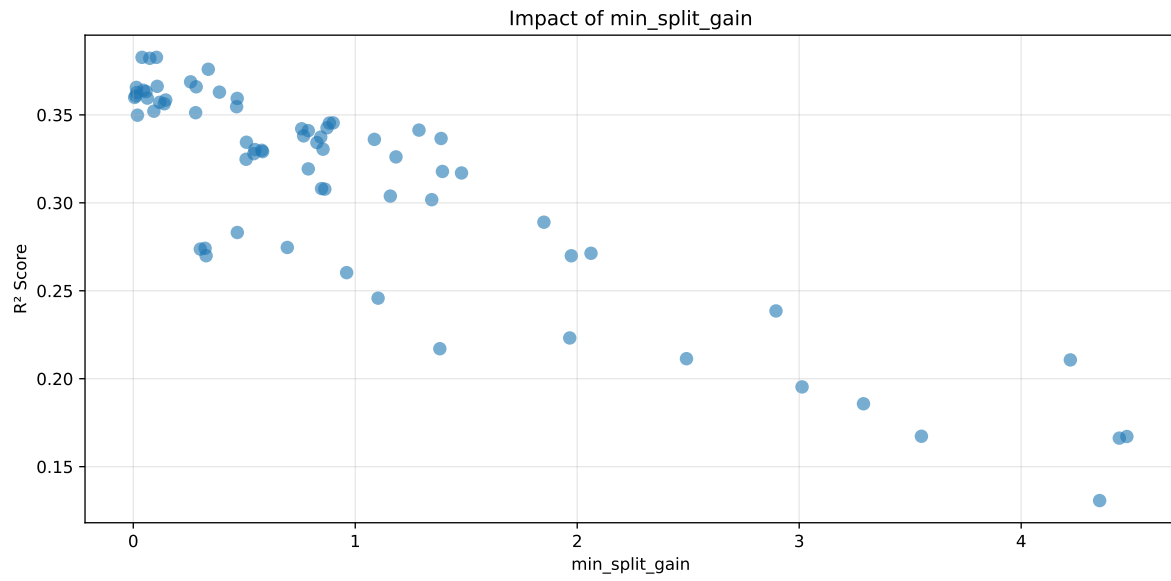
Len(group) : 568, Len(training) : 398, Len(test) : 170

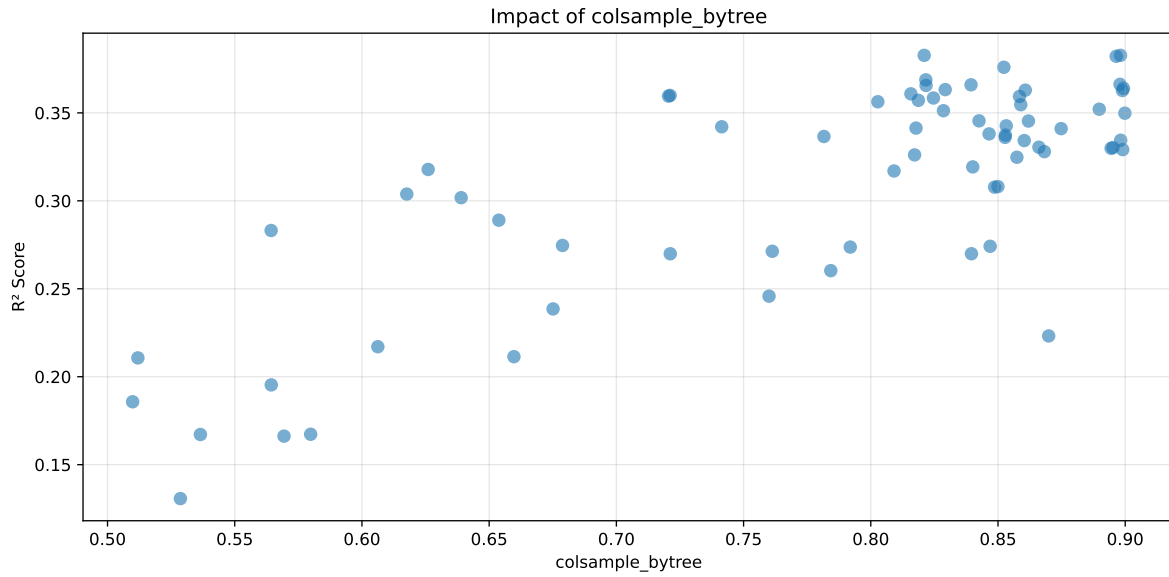
Distribution y_train vs y_test: y_train: mean=1.928, std=0.816, min=0.959, max=6.804
y_test: mean=1.833, std=0.729, min=1.098, max=7.822

===== TOP Importance FEATURES =====
feature importance r_umidity 25 r_NH3 12 r_THI 10 r_temperature 4 r_CO2 4



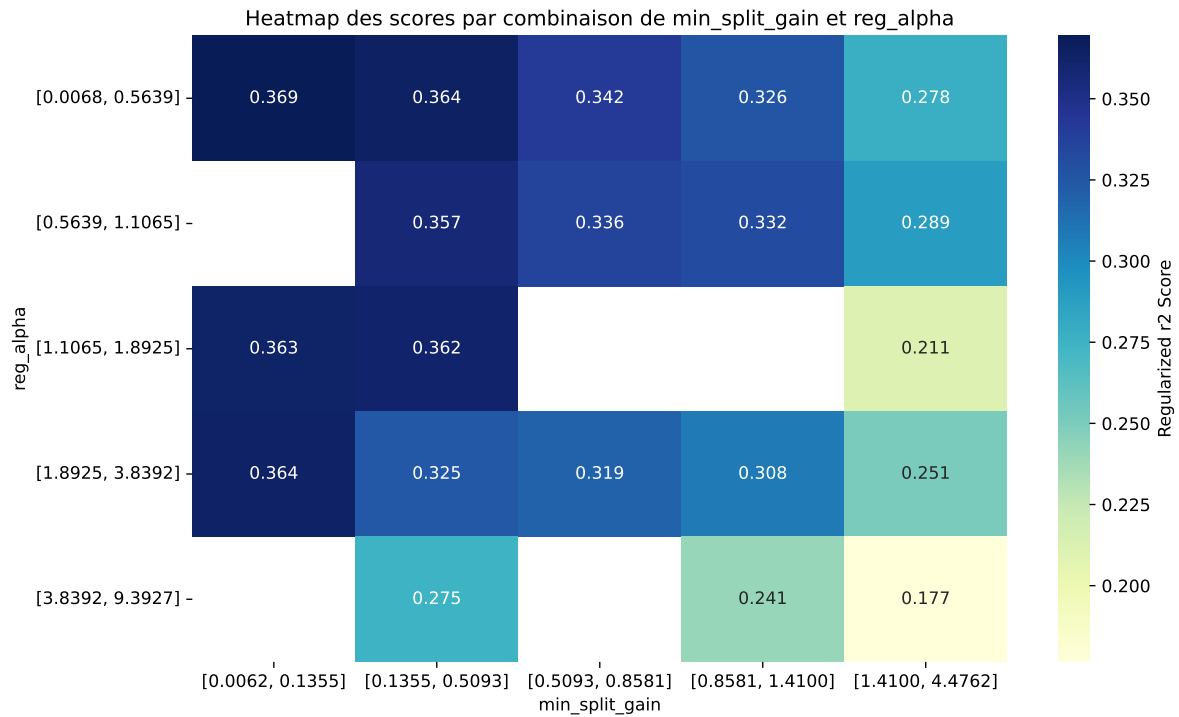
Paramètres avec >5% d'importance : ['min_split_gain', 'reg_alpha', 'colsample_bytree']





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The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.2961	0.2458	0.1001	0.0336	0.0948	0.1436	-0.1022	0.5842	1.2048
2	0.2166	0.1858	0.0697	0.0357	0.0932	0.1084	-0.0773	0.5838	1.1662
3	0.1948	0.1662	0.0629	0.035	0.0804	0.0874	-0.0788	0.5257	1.172
4	0.1493	0.1307	0.0464	0.03	0.0748	0.0739	-0.0569	0.565	1.1425
5	0.2571	0.2114	0.0784	0.0352	0.0743	0.1257	-0.0857	0.5944	1.2162
6	0.202	0.1672	0.0566	0.0307	0.0625	0.0913	-0.0758	0.5462	1.2086
7	0.279	0.2232	0.0943	0.0292	0.0558	0.1266	-0.0965	0.5675	1.25
8	0.2441	0.1953	0.0743	0.0283	0.0492	0.1187	-0.0767	0.6074	1.2494
9	0.3058	0.2385	0.1002	0.0333	0.0366	0.1399	-0.0987	0.5864	1.2821
10	0.2114	0.1673	0.0625	0.0254	0.0348	0.093	-0.0743	0.5561	1.2639
---	---	---	---	---	---	---	---	---	---
70	0.726	0.3632	0.1829	0.0837	-0.7251	0.1899	-0.1733	0.5229	1.9988

Hyperparameters:

Rank 1: {'reg__n_estimators': 200, 'reg__max_depth': 4, 'reg__learning_rate': 0.09995208640275326, 'reg__num_leaves': 54, 'reg__subsample': 0.608310559340913, 'reg__colsample_bytree': 0.7600119676917894, 'reg__min_child_samples': 47, 'reg__reg_alpha': 7.355155059754807, 'reg__reg_lambda': 32.84183856400163, 'reg__min_split_gain': 1.1029357834397613, 'reg__path_smooth': 3.3880902411197527}

Rank 2: {'reg__n_estimators': 300, 'reg__max_depth': 8, 'reg__learning_rate': 0.03631596362676469, 'reg__num_leaves': 53, 'reg__subsample': 0.8388720304693287, 'reg__colsample_bytree': 0.5098692555324766, 'reg__min_child_samples': 42, 'reg__reg_alpha': 4.702204921480728, 'reg__reg_lambda': 46.61244310458119, 'reg__min_split_gain': 3.289581946134809, 'reg__path_smooth': 2.3568120327832753}

Rank 3: {'reg__n_estimators': 800, 'reg__max_depth': 6, 'reg__learning_rate': 0.010606026084510514, 'reg__num_leaves': 21, 'reg__subsample': 0.7743480503754068, 'reg__colsample_bytree': 0.569376715061128, 'reg__min_child_samples': 20, 'reg__reg_alpha': 4.266773756387182, 'reg__reg_lambda': 49.40994480365103, 'reg__min_split_gain': 4.442054890208807, 'reg__path_smooth': 0.6206369866430134}

Rank 4: {'reg__n_estimators': 400, 'reg__max_depth': 4, 'reg__learning_rate': 0.04576402337089281, 'reg__num_leaves': 47, 'reg__subsample': 0.8976147180647855, 'reg__colsample_bytree': 0.5286411673137645, 'reg__min_child_samples': 38, 'reg__reg_alpha': 9.331816240129772, 'reg__reg_lambda': 45.37079892964628, 'reg__min_split_gain': 4.353442195638506, 'reg__path_smooth': 0.5181072877867099}

Rank 5: {'reg__n_estimators': 800, 'reg__max_depth': 8, 'reg__learning_rate': 0.07014755533603848, 'reg__num_leaves': 44, 'reg__subsample': 0.5544585926652388,

'reg__colsample_bytree': 0.6597388987502024, 'reg__min_child_samples': 44, 'reg__reg_alpha': 3.7008170137903775, 'reg__reg_lambda': 37.40871269249278, 'reg__min_split_gain': 2.492269135563246, 'reg__path_smooth': 3.721766150115002}

Rank 6: {'reg__n_estimators': 800, 'reg__max_depth': 6, 'reg__learning_rate': 0.06840916897359546, 'reg__num_leaves': 41, 'reg__subsample': 0.7875285986432077, 'reg__colsample_bytree': 0.5365049035862894, 'reg__min_child_samples': 37, 'reg__reg_alpha': 4.302038532180406, 'reg__reg_lambda': 26.593201886608327, 'reg__min_split_gain': 4.4761774026268615, 'reg__path_smooth': 1.4182425745129956}

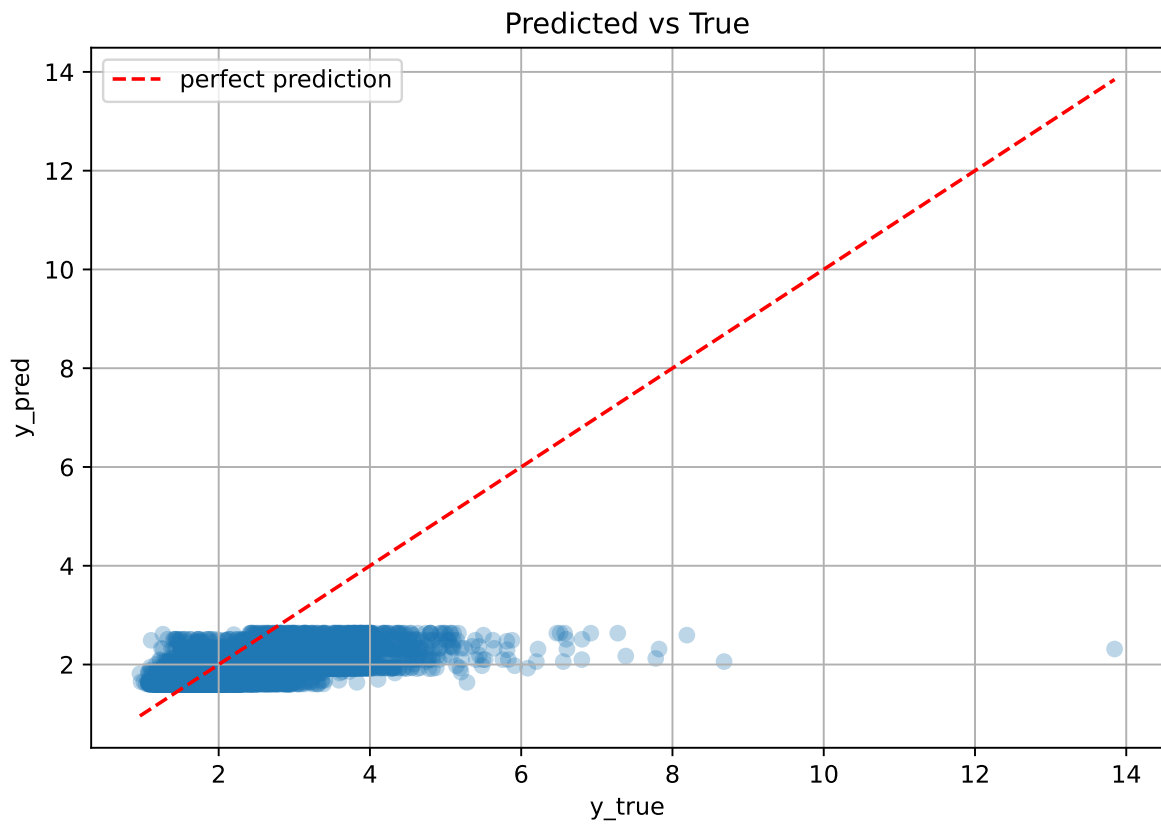
Rank 7: {'reg__n_estimators': 200, 'reg__max_depth': 6, 'reg__learning_rate': 0.07663832886634563, 'reg__num_leaves': 54, 'reg__subsample': 0.7547643893593303, 'reg__colsample_bytree': 0.8699176274203255, 'reg__min_child_samples': 15, 'reg__reg_alpha': 7.89762158434238, 'reg__reg_lambda': 17.967170678171072, 'reg__min_split_gain': 1.9663543174692273, 'reg__path_smooth': 0.6531091634010883}

Rank 8: {'reg__n_estimators': 500, 'reg__max_depth': 6, 'reg__learning_rate': 0.05418400149671304, 'reg__num_leaves': 45, 'reg__subsample': 0.7966212339538978, 'reg__colsample_bytree': 0.5643602430522505, 'reg__min_child_samples': 34, 'reg__reg_alpha': 4.822904858194269, 'reg__reg_lambda': 24.606258746929573, 'reg__min_split_gain': 3.0129436778074083, 'reg__path_smooth': 3.1733784536986436}

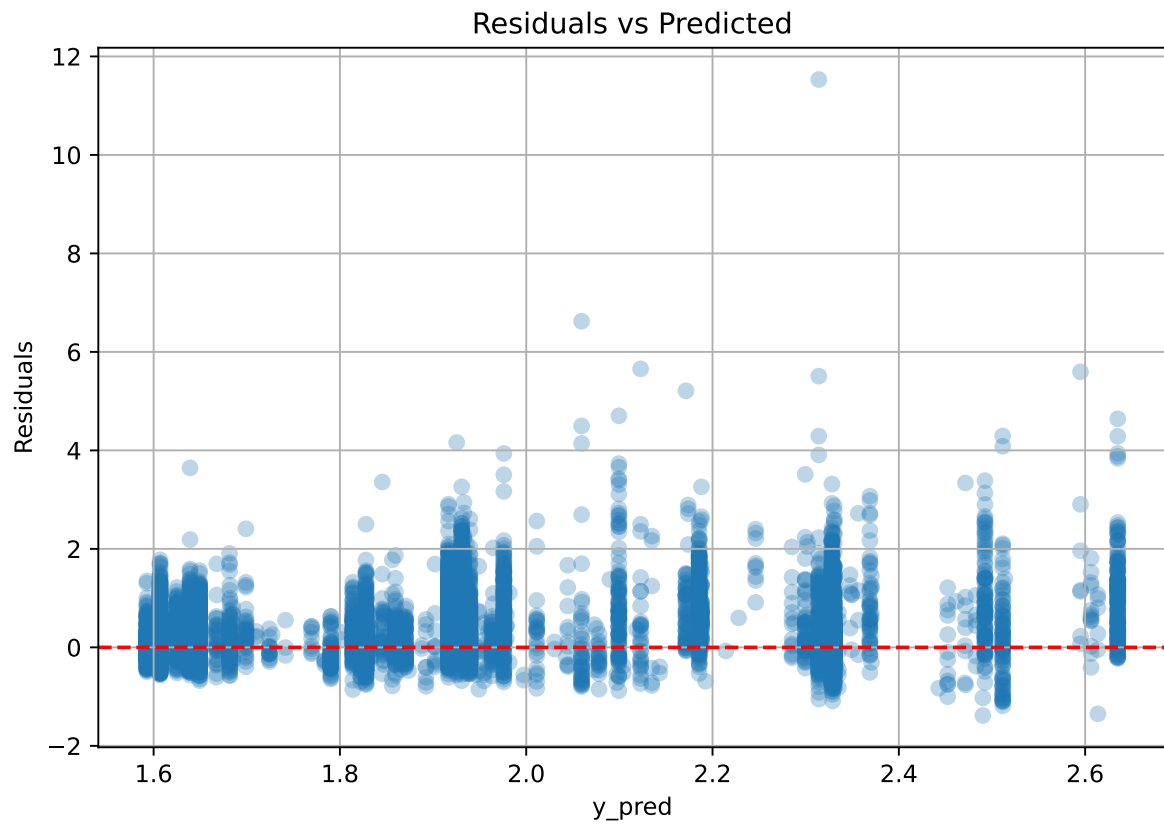
Rank 9: {'reg__n_estimators': 700, 'reg__max_depth': 7, 'reg__learning_rate': 0.06036574647627586, 'reg__num_leaves': 27, 'reg__subsample': 0.6726984000915742, 'reg__colsample_bytree': 0.6751167015038402, 'reg__min_child_samples': 37, 'reg__reg_alpha': 0.16399002230852888, 'reg__reg_lambda': 39.6711873979759, 'reg__min_split_gain': 2.895480619883247, 'reg__path_smooth': 3.030252549836718}

Rank 10: {'reg__n_estimators': 200, 'reg__max_depth': 6, 'reg__learning_rate': 0.079766565356118, 'reg__num_leaves': 28, 'reg__subsample': 0.5929045574050403, 'reg__colsample_bytree': 0.5798394155904252, 'reg__min_child_samples': 34, 'reg__reg_alpha': 8.051864015110963, 'reg__reg_lambda': 4.674784113145236, 'reg__min_split_gain': 3.5506509339380754, 'reg__path_smooth': 2.1528759167245015}

2.4.10 Scatter



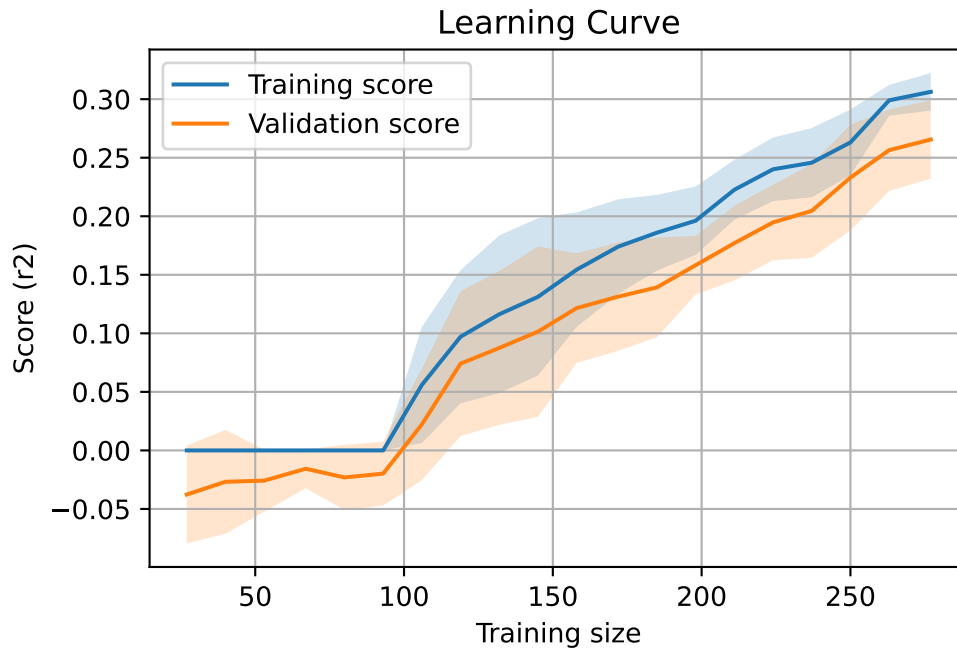
2.4.11 Scatter residuals



2.4.12 Learning curve — Group: 3_4

`Len(group)` : 568, `Len(training)` : 398,

`Len(training_real)` : 279, `Len(test_of_training)` : 119, `Len(test)` : 170

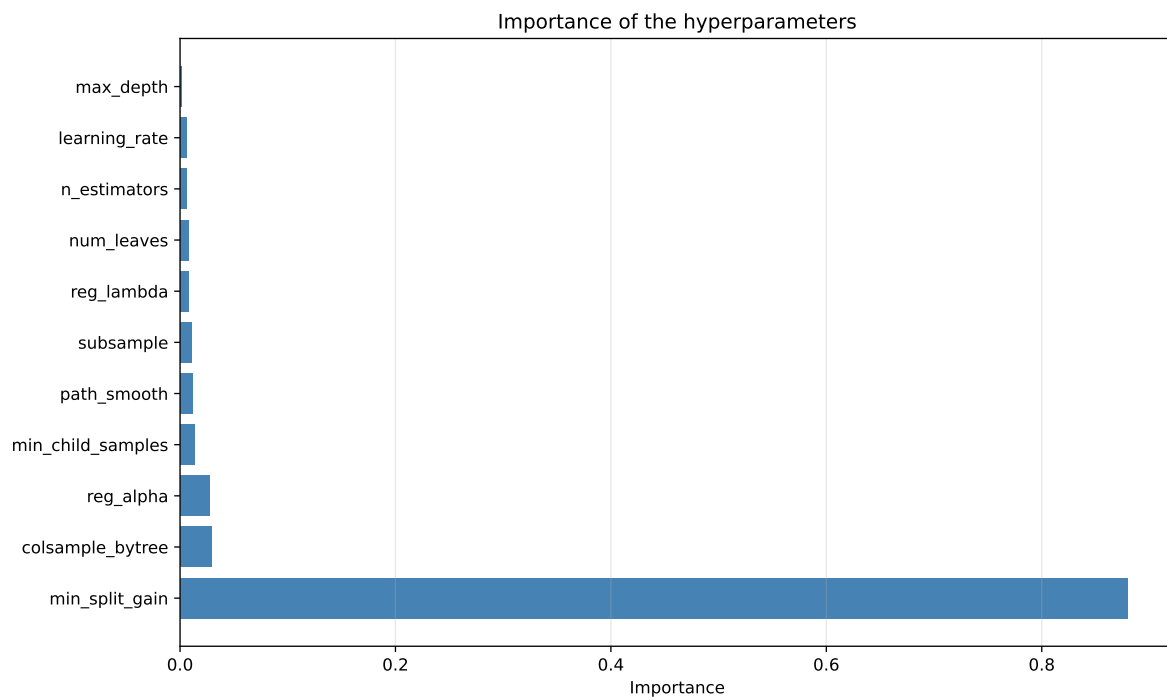
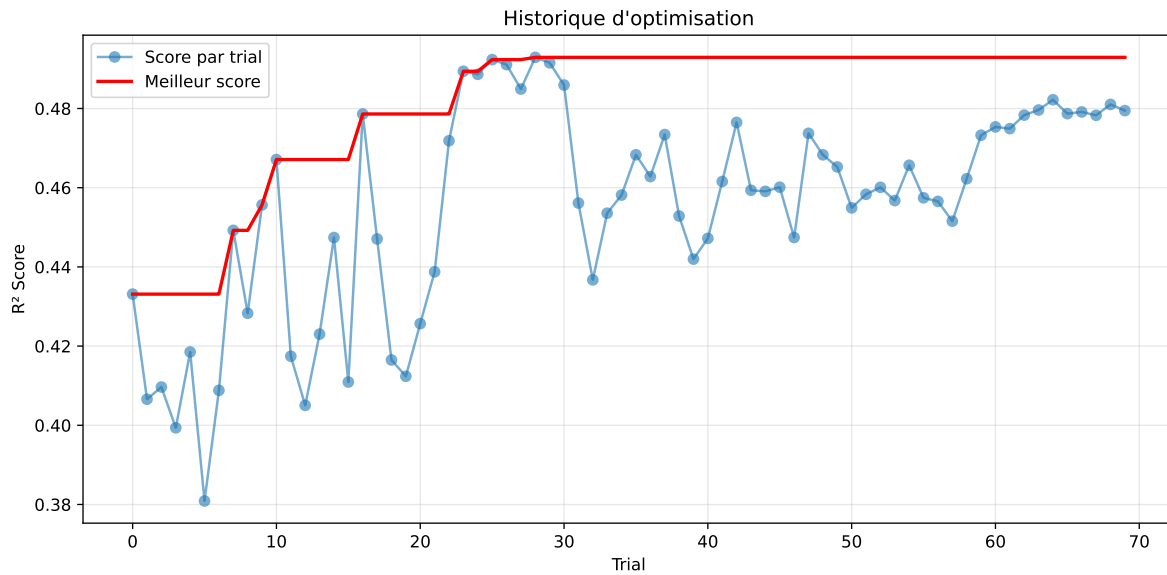


2.4.13 Gridsearch — Group: 4_5

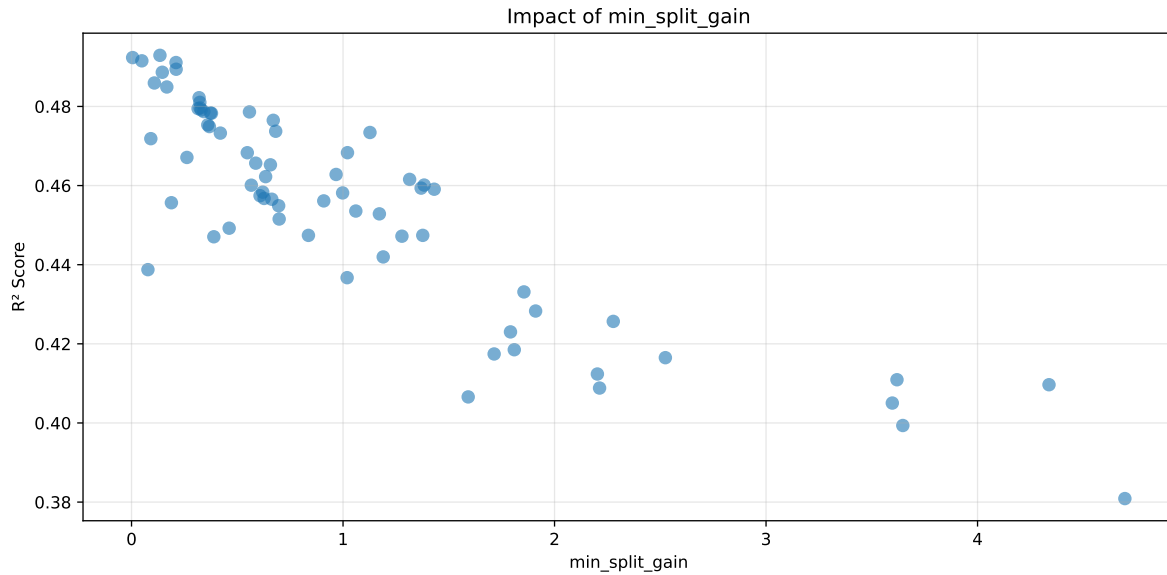
Len(group) : 2726, Len(training) : 1909, Len(test) : 817

Distribution y_train vs y_test: y_train: mean=2.238, std=0.681, min=0.973, max=5.813
y_test: mean=2.212, std=0.659, min=1.071, max=4.858

===== TOP Importance FEATURES =====
feature importance r_umidity 64 r_NH3 24 r_CO2 17 r_temperature 16 r_THI 6



Paramètres avec >5% d'importance : ['min_split_gain']



Pas assez de paramètres avec >5% d'importance pour une heatmap 2D.

rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.4533	0.4124	0.471	0.0242	0.2897	0.4944	0.0821	1.199	1.0992
2	0.4593	0.4165	0.4753	0.0249	0.2882	0.4961	0.0796	1.1911	1.1027
3	0.4462	0.4066	0.4691	0.0231	0.2878	0.4903	0.0837	1.2058	1.0974
4	0.4609	0.4174	0.4756	0.0246	0.2872	0.4987	0.0813	1.1948	1.104
5	0.4379	0.3994	0.4637	0.0257	0.2837	0.485	0.0857	1.2146	1.0965
6	0.4505	0.4088	0.474	0.0242	0.2837	0.4939	0.085	1.208	1.1021
7	0.4133	0.3809	0.4427	0.0243	0.2835	0.4647	0.0838	1.22	1.0852
8	0.4642	0.4185	0.4822	0.0255	0.2814	0.4969	0.0784	1.1874	1.1092
9	0.4774	0.4283	0.4878	0.025	0.2811	0.5069	0.0786	1.1836	1.1146
10	0.4478	0.405	0.4708	0.026	0.2769	0.4905	0.0855	1.2111	1.1055
---	---	---	---	---	---	---	---	---	---
70	0.6711	0.4923	0.5679	0.033	-0.044	0.5888	0.0965	1.1959	1.3631

Hyperparameters:

Rank 1: {'reg__n_estimators': 300, 'reg__max_depth': 6, 'reg__learning_rate': 0.07956110718660912, 'reg__num_leaves': 57, 'reg__subsample': 0.8070838774432845, 'reg__colsample_bytree': 0.8676055050483982, 'reg__min_child_samples': 34, 'reg__reg_alpha': 7.432502609864827, 'reg__reg_lambda': 23.86908780412933, 'reg__min_split_gain': 2.2025850177196222, 'reg__path_smooth': 2.1403079343083586}

Rank 2: {'reg__n_estimators': 200, 'reg__max_depth': 7, 'reg__learning_rate': 0.04198906373676476, 'reg__num_leaves': 37, 'reg__subsample': 0.7874702981185386,

'reg__colsample_bytree': 0.852004456666967, 'reg__min_child_samples': 38, 'reg__reg_alpha': 3.357353315649992, 'reg__reg_lambda': 42.24971181897498, 'reg__min_split_gain': 2.5240079300821163, 'reg__path_smooth': 3.1739951186766318}

Rank 3: {'reg__n_estimators': 400, 'reg__max_depth': 7, 'reg__learning_rate': 0.015152737298236844, 'reg__num_leaves': 19, 'reg__subsample': 0.5434167337288756, 'reg__colsample_bytree': 0.5129993195496966, 'reg__min_child_samples': 41, 'reg__reg_alpha': 9.157163860385365, 'reg__reg_lambda': 39.343477273489455, 'reg__min_split_gain': 1.5917811822862304, 'reg__path_smooth': 0.11960380708912755}

Rank 4: {'reg__n_estimators': 800, 'reg__max_depth': 8, 'reg__learning_rate': 0.06788948856671713, 'reg__num_leaves': 62, 'reg__subsample': 0.5762979035644082, 'reg__colsample_bytree': 0.8045761992652745, 'reg__min_child_samples': 30, 'reg__reg_alpha': 8.881706257848895, 'reg__reg_lambda': 19.042590531056824, 'reg__min_split_gain': 1.7144281276847546, 'reg__path_smooth': 1.8688917604475175}

Rank 5: {'reg__n_estimators': 400, 'reg__max_depth': 7, 'reg__learning_rate': 0.06270048211761503, 'reg__num_leaves': 22, 'reg__subsample': 0.6440110993833447, 'reg__colsample_bytree': 0.6011620588993537, 'reg__min_child_samples': 45, 'reg__reg_alpha': 3.558671800670301, 'reg__reg_lambda': 34.53754137977565, 'reg__min_split_gain': 3.6466518883073418, 'reg__path_smooth': 0.2549765946985305}

Rank 6: {'reg__n_estimators': 600, 'reg__max_depth': 3, 'reg__learning_rate': 0.07017570329525734, 'reg__num_leaves': 58, 'reg__subsample': 0.6179288947984658, 'reg__colsample_bytree': 0.6591626784936925, 'reg__min_child_samples': 35, 'reg__reg_alpha': 4.907644057032736, 'reg__reg_lambda': 49.9794925169133, 'reg__min_split_gain': 2.21292088867112, 'reg__path_smooth': 0.21458964924287338}

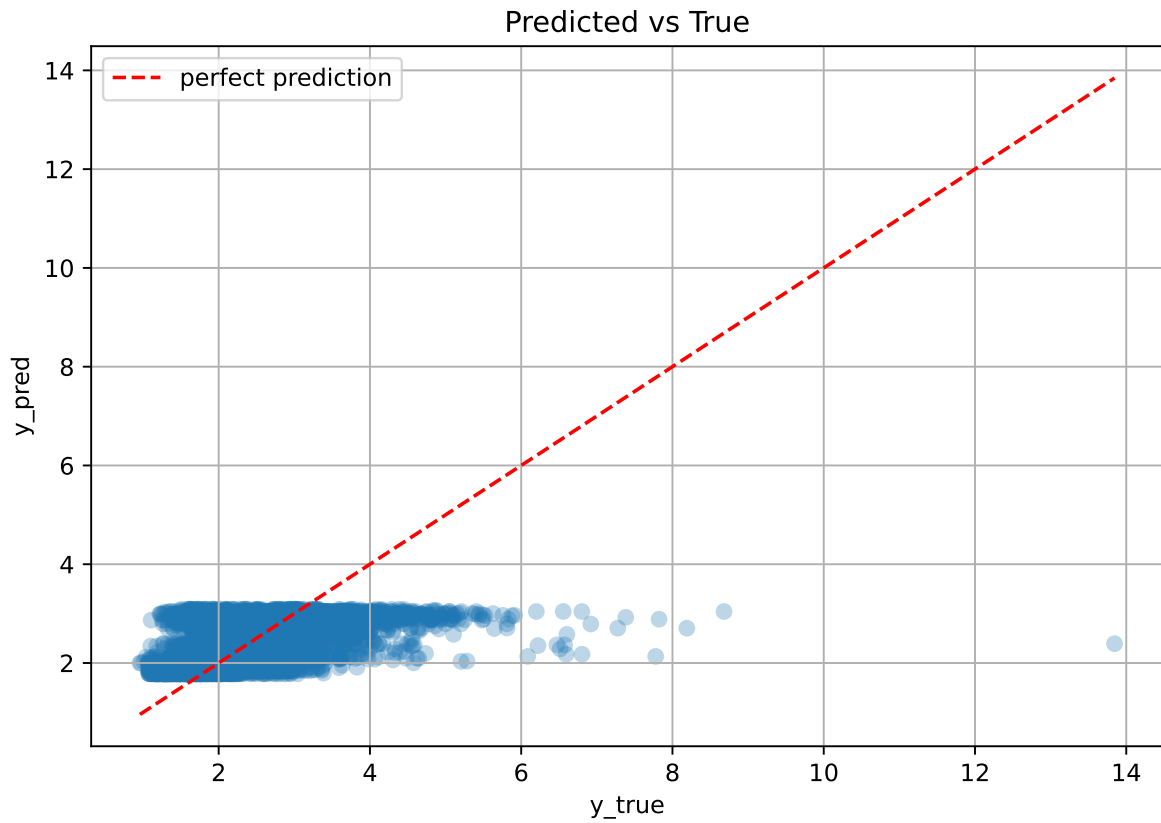
Rank 7: {'reg__n_estimators': 200, 'reg__max_depth': 8, 'reg__learning_rate': 0.06492085357120088, 'reg__num_leaves': 31, 'reg__subsample': 0.7005211425612292, 'reg__colsample_bytree': 0.5444452713051624, 'reg__min_child_samples': 50, 'reg__reg_alpha': 5.911977476855899, 'reg__reg_lambda': 44.53229012672016, 'reg__min_split_gain': 4.696573084340056, 'reg__path_smooth': 4.413259399828792}

Rank 8: {'reg__n_estimators': 300, 'reg__max_depth': 4, 'reg__learning_rate': 0.024811776081323723, 'reg__num_leaves': 53, 'reg__subsample': 0.7731420496504191, 'reg__colsample_bytree': 0.5039899480061353, 'reg__min_child_samples': 11, 'reg__reg_alpha': 7.833573637116661, 'reg__reg_lambda': 12.507879459397836, 'reg__min_split_gain': 1.8096494212523428, 'reg__path_smooth': 4.782826243058464}

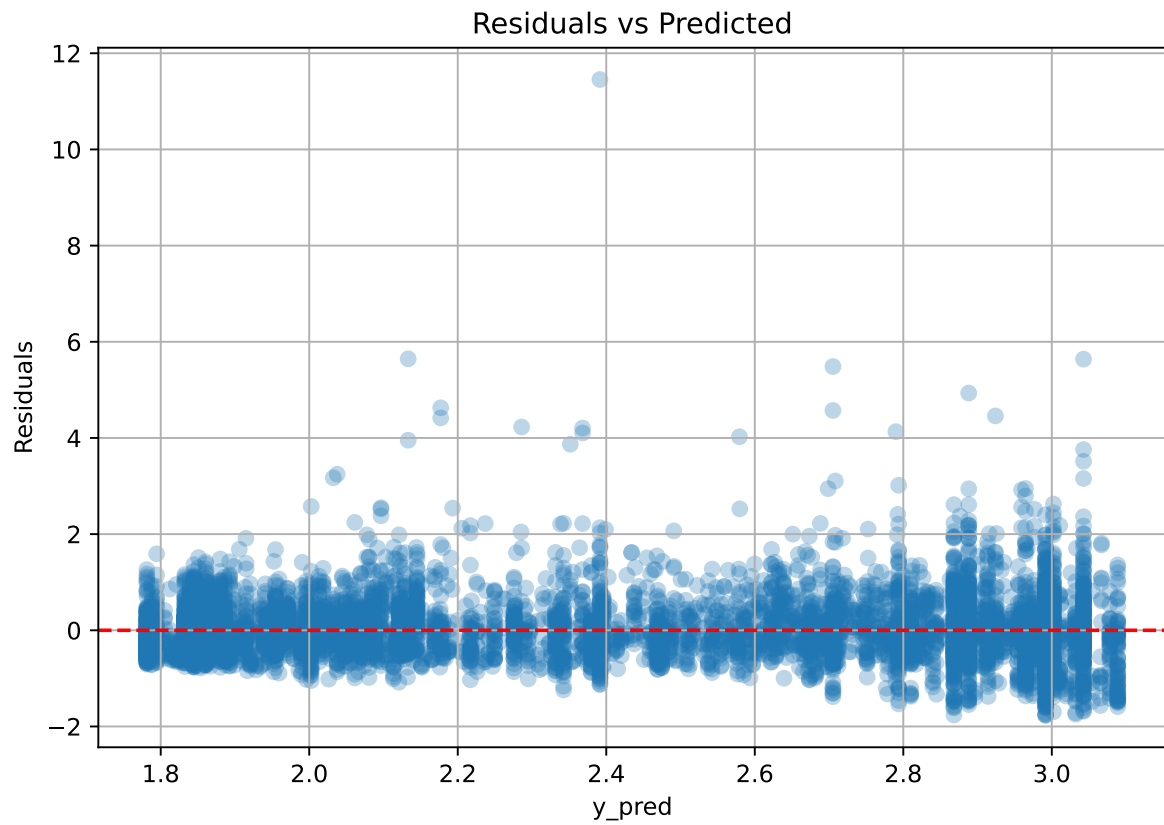
Rank 9: {'reg__n_estimators': 700, 'reg__max_depth': 7, 'reg__learning_rate': 0.03289414561964485, 'reg__num_leaves': 62, 'reg__subsample': 0.7919854549372631, 'reg__colsample_bytree': 0.710117297279161, 'reg__min_child_samples': 25, 'reg__reg_alpha': 3.1961759602454998, 'reg__reg_lambda': 39.9147510708009, 'reg__min_split_gain': 1.9105462461760485, 'reg__path_smooth': 3.5624512397410673}

Rank 10: {'reg__n_estimators': 300, 'reg__max_depth': 6, 'reg__learning_rate': 0.04693512417425347, 'reg__num_leaves': 28, 'reg__subsample': 0.6946662704111946, 'reg__colsample_bytree': 0.6760917381729867, 'reg__min_child_samples': 14, 'reg__reg_alpha': 4.4804201147659475, 'reg__reg_lambda': 14.562023348952208, 'reg__min_split_gain': 3.5969857575074657, 'reg__path_smooth': 1.1060516636033746}

2.4.14 Scatter



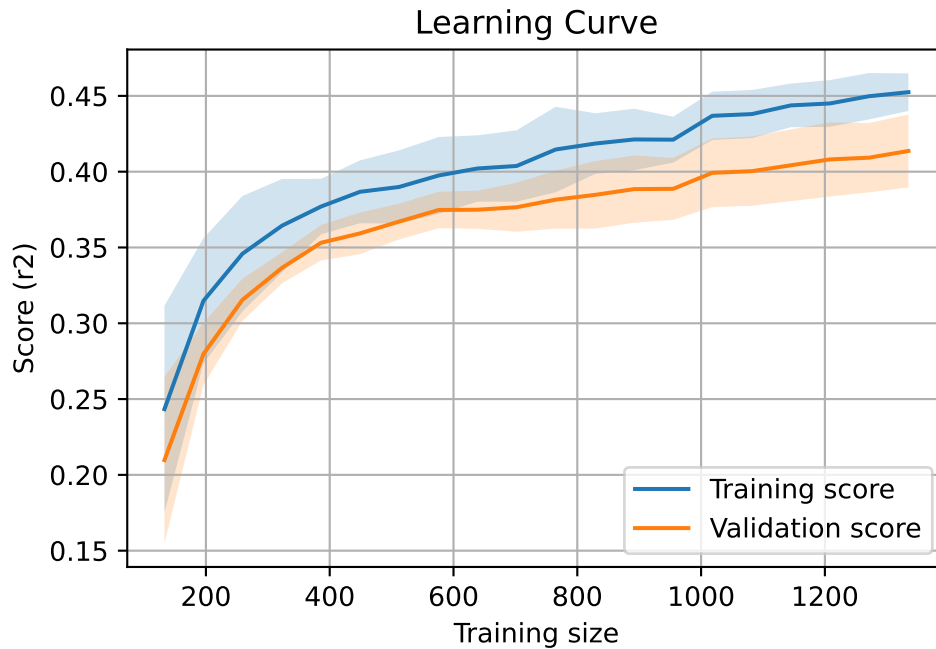
2.4.15 Scatter residuals



2.4.16 Learning curve — Group: 4_5

`Len(group)` : 2726, `Len(training)` : 1909,

`Len(training_real)` : 1337, `Len(test_of_training)` : 572, `Len(test)` : 817

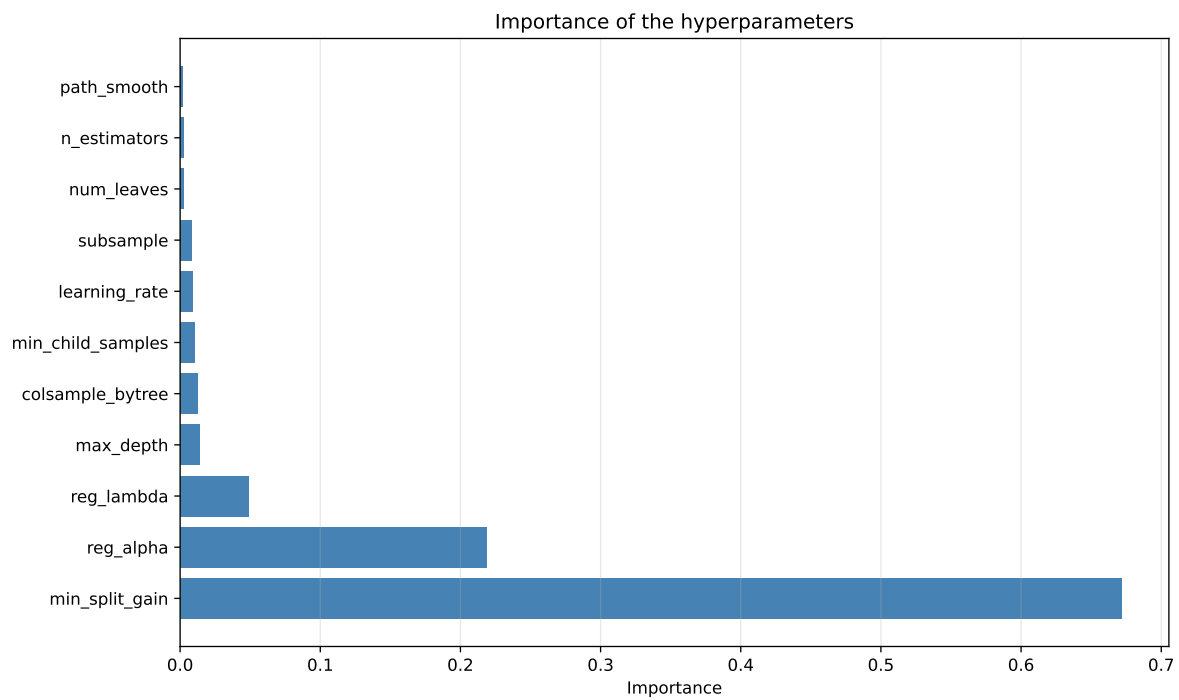
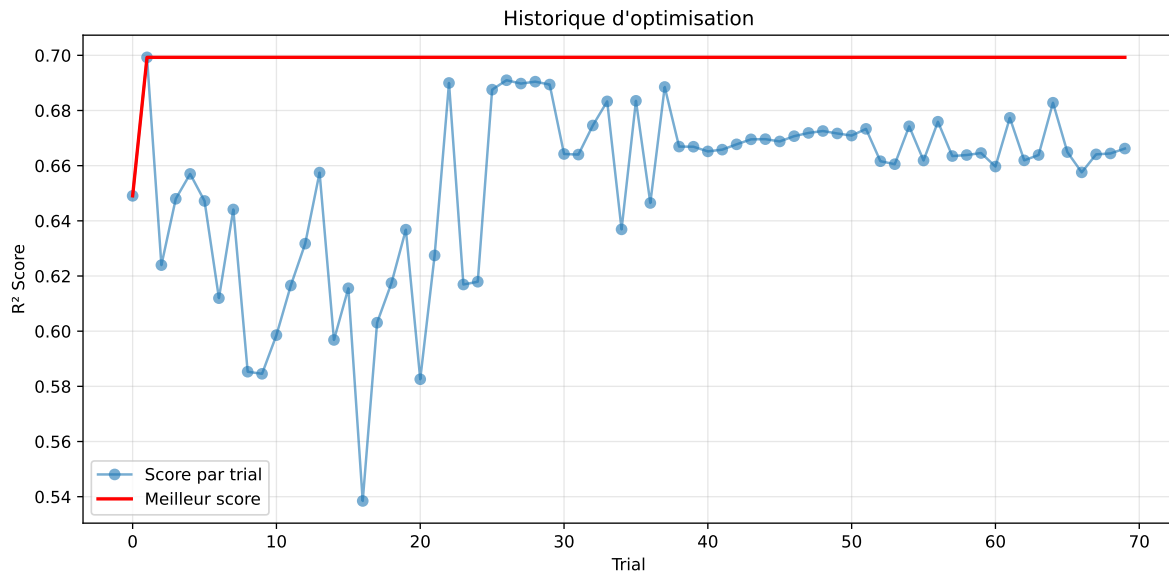


2.4.17 Gridsearch — Group: 6_3

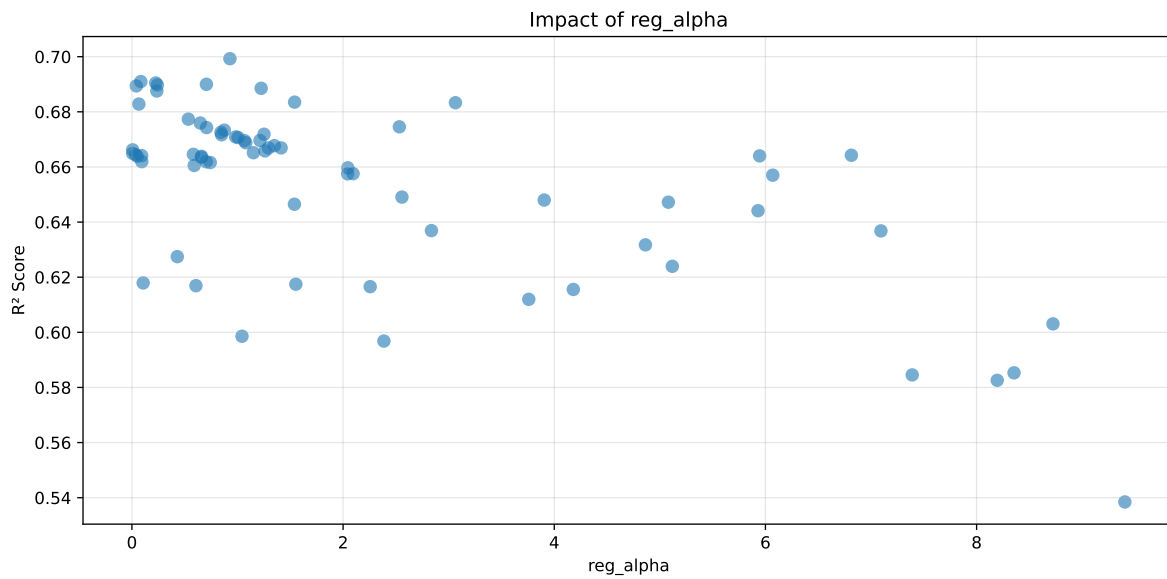
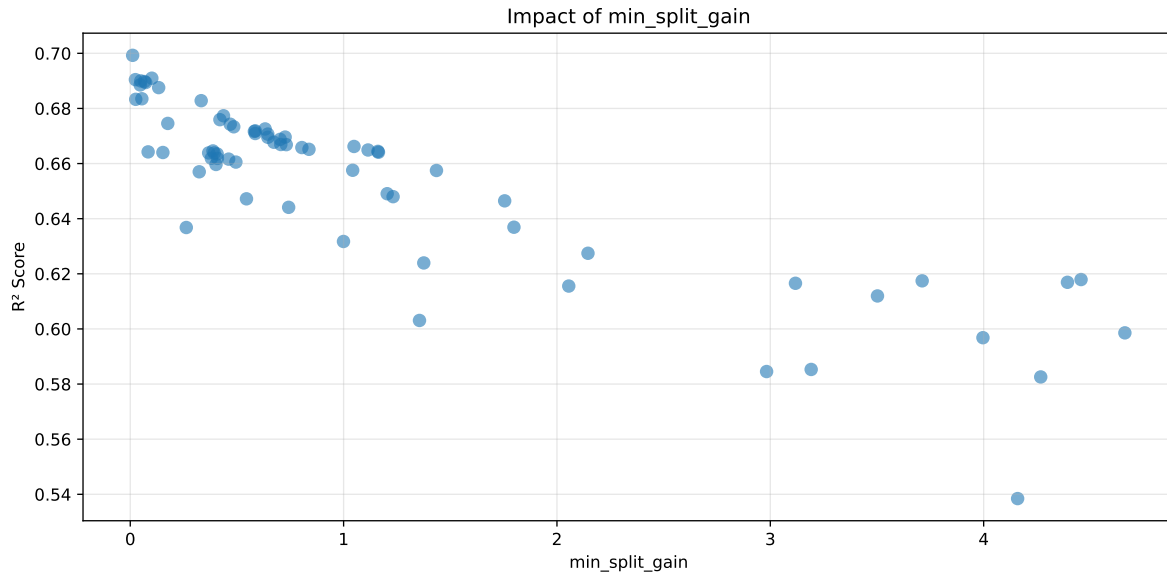
Len(group) : 561, Len(training) : 393, Len(test) : 168

Distribution y_train vs y_test: y_train: mean=1.935, std=0.720, min=1.041, max=6.086
y_test: mean=2.031, std=0.744, min=1.078, max=4.912

===== TOP Importance FEATURES =====
feature importance r_umidity 111 r_THI 63 r_NH3 62 r_temperature 41 r_CO2 37

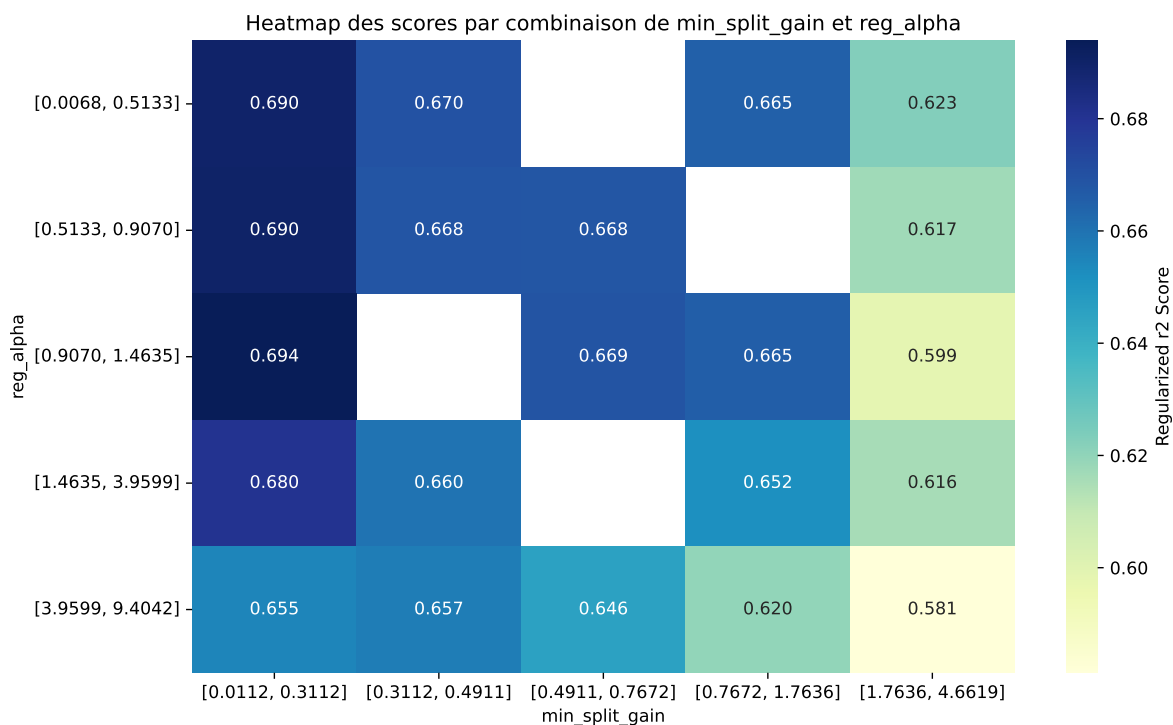


Paramètres avec >5% d'importance : ['min_split_gain', 'reg_alpha']



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The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.7175	0.691	0.6363	0.054	0.691	0.678	-0.013	0.9812	1.0384
2	0.7123	0.6885	0.6383	0.0557	0.6885	0.6759	-0.0126	0.9817	1.0345
3	0.6986	0.6876	0.6241	0.0573	0.6876	0.6658	-0.0217	0.9684	1.0161
4	0.6962	0.6835	0.6231	0.0583	0.6835	0.6563	-0.0273	0.9601	1.0185
5	0.7037	0.6833	0.6279	0.0596	0.6833	0.6761	-0.0072	0.9895	1.0298
6	0.6876	0.6828	0.617	0.0594	0.6828	0.6559	-0.0269	0.9606	1.007
7	0.6724	0.6773	0.6025	0.0597	0.6773	0.637	-0.0403	0.9405	0.9927
8	0.6655	0.6759	0.5929	0.0606	0.6759	0.613	-0.0629	0.9069	0.9845
9	0.6674	0.6746	0.5939	0.0614	0.6746	0.6103	-0.0642	0.9048	0.9894
10	0.6598	0.6743	0.5877	0.0594	0.6743	0.6022	-0.072	0.8932	0.9786
---	---	---	---	---	---	---	---	---	---
70	0.823	0.6993	0.7016	0.0555	0.328	0.7662	0.0669	1.0957	1.177

Hyperparameters:

Rank 1: {'reg__n_estimators': 600, 'reg__max_depth': 3, 'reg__learning_rate': 0.08458439084213326, 'reg__num_leaves': 17, 'reg__subsample': 0.5104470887553797, 'reg__colsample_bytree': 0.8980914341240254, 'reg__min_child_samples': 50, 'reg__reg_alpha': 0.08446030064033194, 'reg__reg_lambda': 38.4971143338706, 'reg__min_split_gain': 0.10110564841644587, 'reg__path_smooth': 4.939921778955668}

Rank 2: {'reg__n_estimators': 700, 'reg__max_depth': 3, 'reg__learning_rate': 0.09554745648326475, 'reg__num_leaves': 15, 'reg__subsample': 0.5710796335855343, 'reg__colsample_bytree': 0.851018096426981, 'reg__min_child_samples': 41, 'reg__reg_alpha': 1.2240595801786143, 'reg__reg_lambda': 39.79941015711939, 'reg__min_split_gain': 0.046603036886087586, 'reg__path_smooth': 2.8364629394255427}

Rank 3: {'reg__n_estimators': 800, 'reg__max_depth': 3, 'reg__learning_rate': 0.07997722697656792, 'reg__num_leaves': 16, 'reg__subsample': 0.5327106899658682, 'reg__colsample_bytree': 0.8950812304840172, 'reg__min_child_samples': 50, 'reg__reg_alpha': 0.23608441064389574, 'reg__reg_lambda': 34.794135573113344, 'reg__min_split_gain': 0.13317663407048067, 'reg__path_smooth': 0.1219726042366478}

Rank 4: {'reg__n_estimators': 700, 'reg__max_depth': 3, 'reg__learning_rate': 0.05803281818160028, 'reg__num_leaves': 23, 'reg__subsample': 0.582598600667018, 'reg__colsample_bytree': 0.849235852238124, 'reg__min_child_samples': 42, 'reg__reg_alpha': 1.5407300997368008, 'reg__reg_lambda': 43.10134788122731, 'reg__min_split_gain': 0.05440143766925987, 'reg__path_smooth': 2.662423748557655}

Rank 5: {'reg__n_estimators': 500, 'reg__max_depth': 3, 'reg__learning_rate': 0.06069424579379728, 'reg__num_leaves': 58, 'reg__subsample': 0.5710282253360414, 'reg__colsample_bytree': 0.774781421609963, 'reg__min_child_samples': 24, 'reg__reg_alpha': 3.0641245299392255, 'reg__reg_lambda': 49.71483195556224, 'reg__min_split_gain': 0.025645888754236346, 'reg__path_smooth': 2.0838120303602277}

Rank 6: {'reg__n_estimators': 800, 'reg__max_depth': 7, 'reg__learning_rate': 0.07228296508557426, 'reg__num_leaves': 19, 'reg__subsample': 0.5193976738251992, 'reg__colsample_bytree': 0.8048385358614994, 'reg__min_child_samples': 38, 'reg__reg_alpha': 0.06567007474167186, 'reg__reg_lambda': 9.383495580320467, 'reg__min_split_gain': 0.3324791002784457, 'reg__path_smooth': 0.5404224484546334}

Rank 7: {'reg__n_estimators': 600, 'reg__max_depth': 7, 'reg__learning_rate': 0.033662070234740236, 'reg__num_leaves': 19, 'reg__subsample': 0.5448300833140111, 'reg__colsample_bytree': 0.8156608658874694, 'reg__min_child_samples': 38, 'reg__reg_alpha': 0.5342191400970535, 'reg__reg_lambda': 8.68577257759678, 'reg__min_split_gain': 0.4370456492472479, 'reg__path_smooth': 1.7501524456952693}

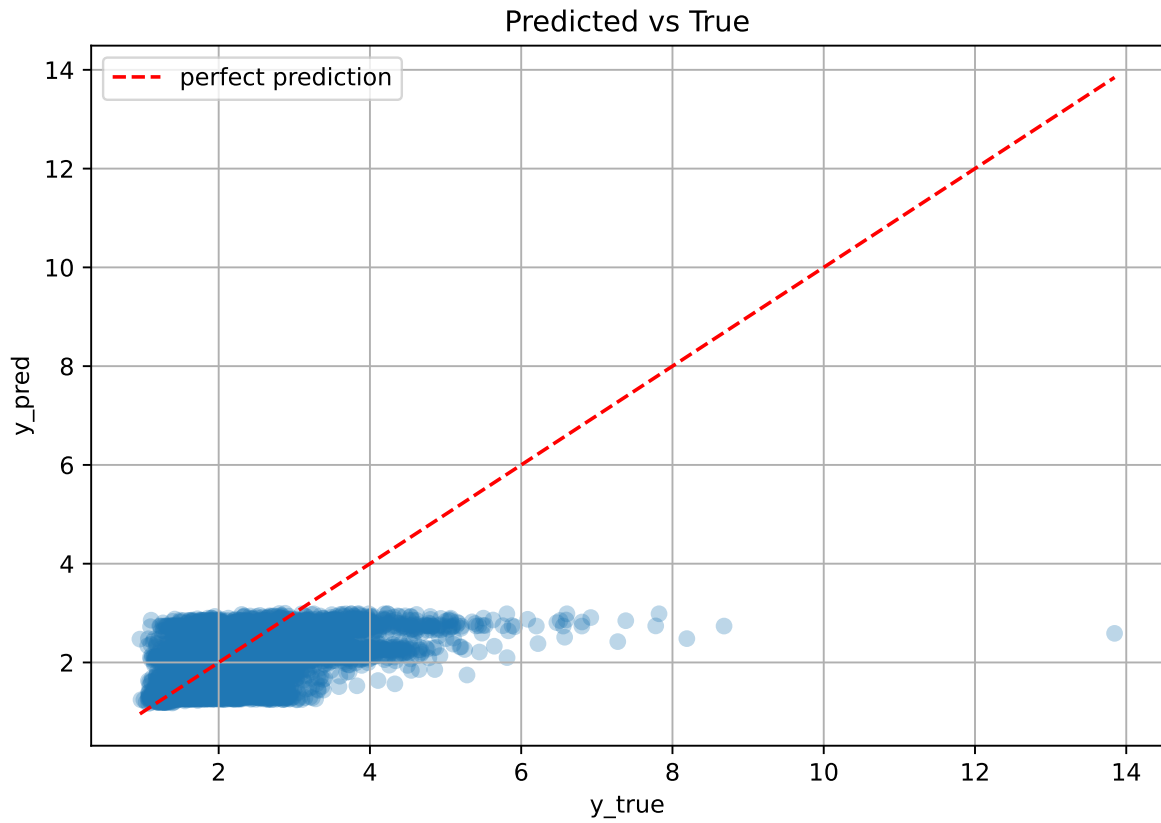
Rank 8: {'reg__n_estimators': 600, 'reg__max_depth': 4, 'reg__learning_rate': 0.02761379759369719, 'reg__num_leaves': 19, 'reg__subsample': 0.5421595215828766, 'reg__colsample_bytree': 0.7249979947763352, 'reg__min_child_samples': 48, 'reg__reg_alpha': 0.6481852582383079, 'reg__reg_lambda': 15.232121308065288, 'reg__min_split_gain': 0.42021728506861916, 'reg__path_smooth': 0.46496458144229674}

Rank 9: {'reg__n_estimators': 500, 'reg__max_depth': 3, 'reg__learning_rate': 0.07294063223250336, 'reg__num_leaves': 58, 'reg__subsample': 0.5705394576615009,

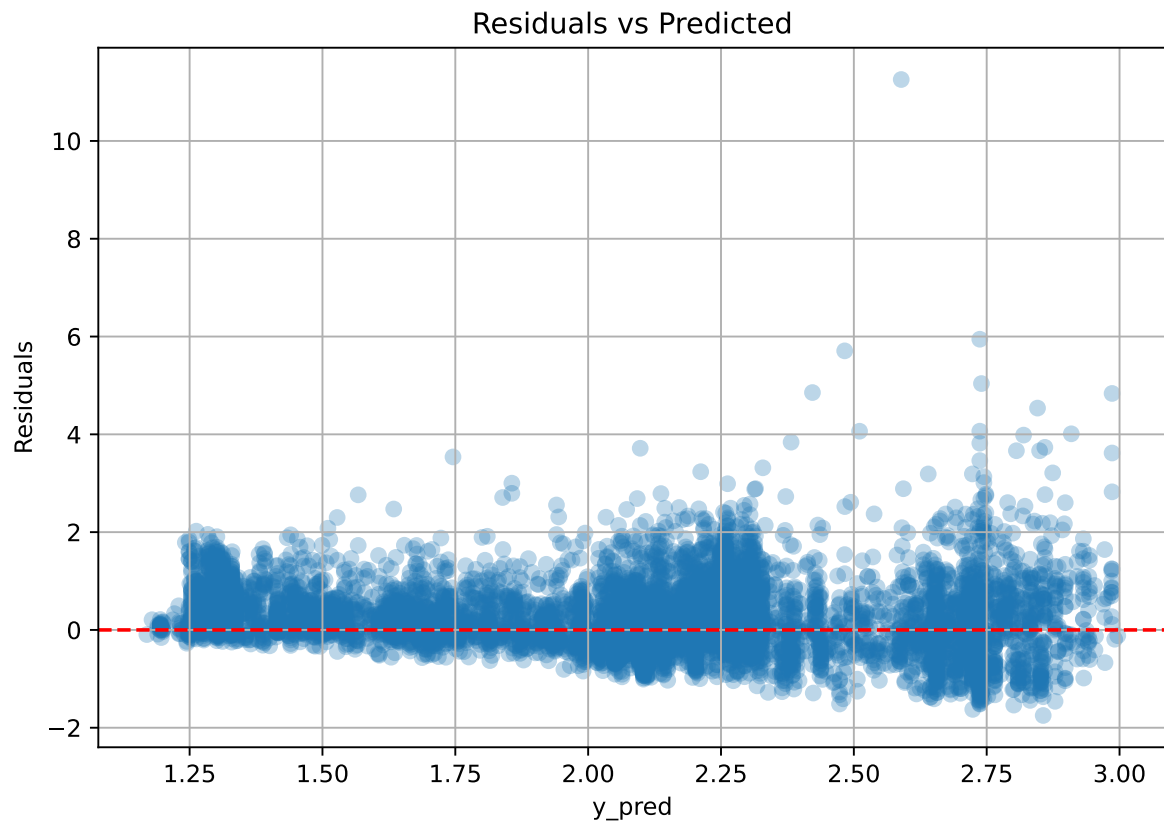
'reg__colsample_bytree': 0.8841867633892386, 'reg__min_child_samples': 26, 'reg__reg_alpha': 2.532774256091468, 'reg__reg_lambda': 49.61585317218634, 'reg__min_split_gain': 0.17592644905190147, 'reg__path_smooth': 0.15484520137054636}

Rank 10: {'reg__n_estimators': 800, 'reg__max_depth': 4, 'reg__learning_rate': 0.03148857651968543, 'reg__num_leaves': 22, 'reg__subsample': 0.5398148203914005, 'reg__colsample_bytree': 0.717631364413984, 'reg__min_child_samples': 46, 'reg__reg_alpha': 0.7072246575948631, 'reg__reg_lambda': 30.858011815407732, 'reg__min_split_gain': 0.46865610644356415, 'reg__path_smooth': 0.4791254406778065}

2.4.18 Scatter



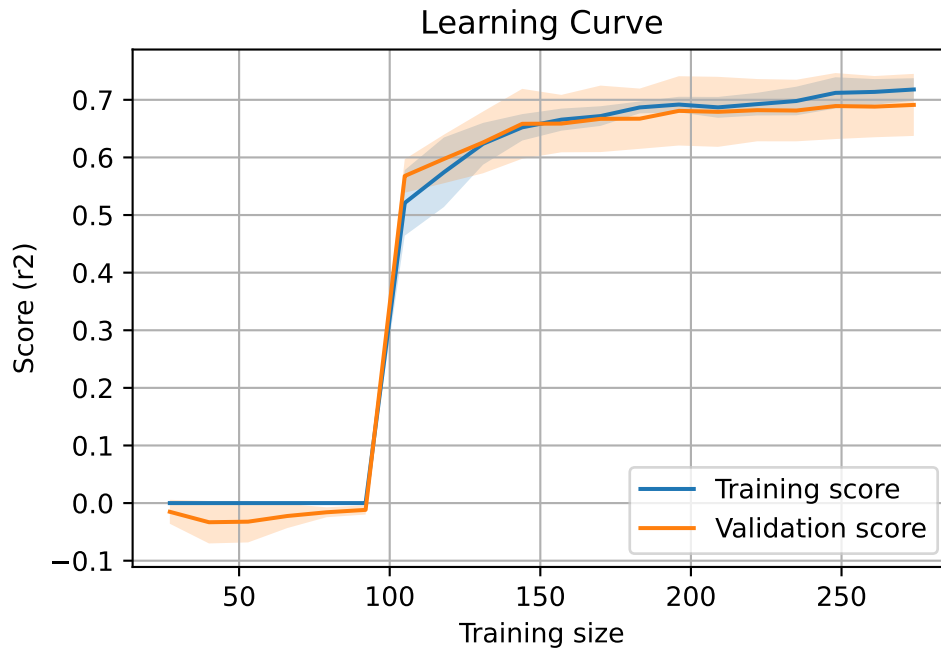
2.4.19 Scatter residuals



2.4.20 Learning curve — Group: 6_3

`Len(group)` : 561, `Len(training)` : 393,

`Len(training_real)` : 276, `Len(test_of_training)` : 117, `Len(test)` : 168

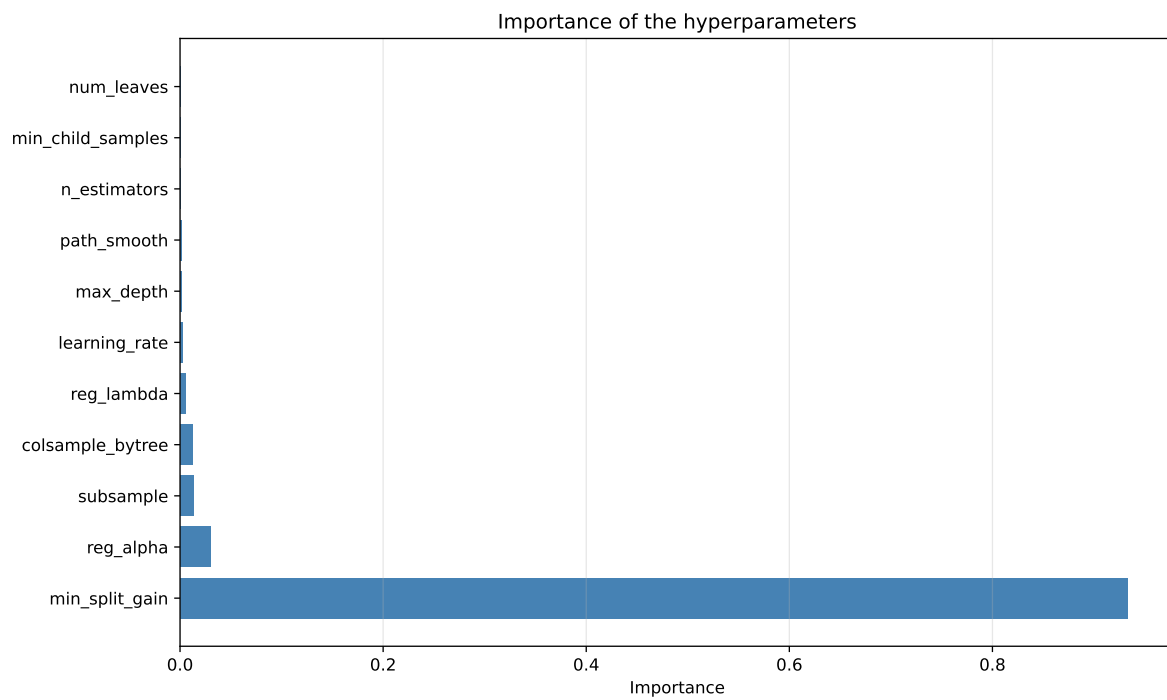
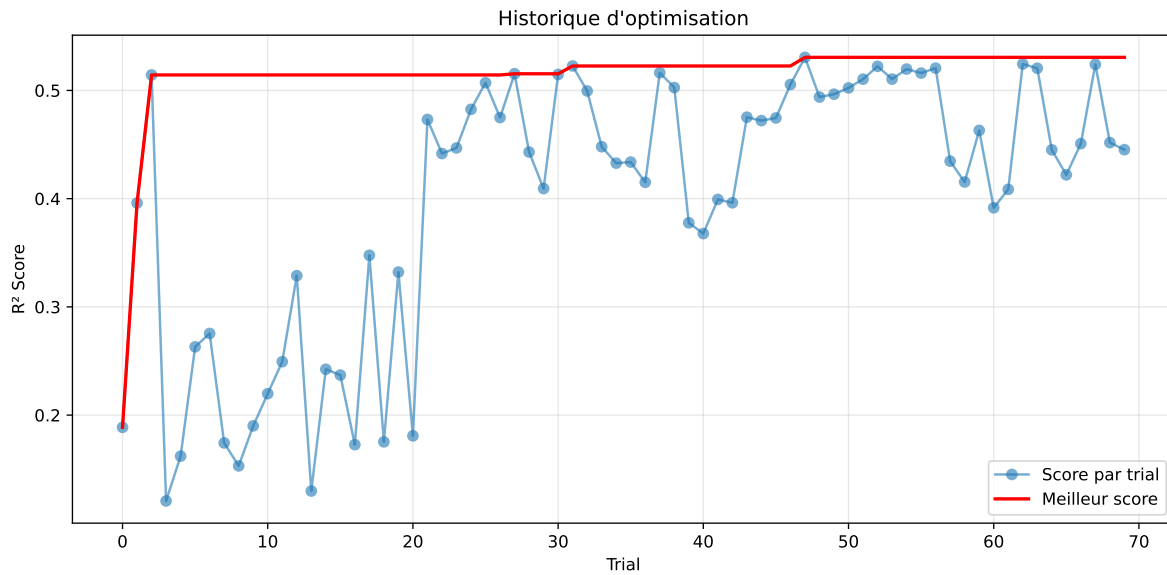


2.4.21 Gridsearch — Group: 6_4

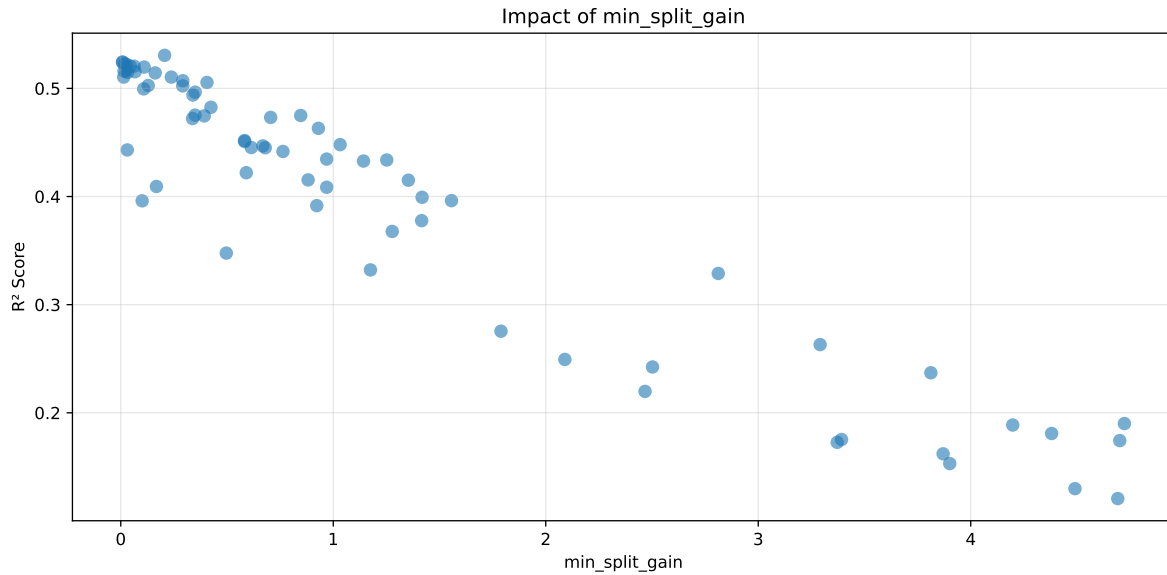
Len(group) : 792, Len(training) : 555, Len(test) : 237

Distribution y_train vs y_test: y_train: mean=2.486, std=0.782, min=1.253, max=7.383
y_test: mean=2.410, std=0.685, min=1.398, max=5.144

===== TOP Importance FEATURES =====
feature importance r_THI 47 r_NH3 46 r_umidity 42 r_temperature 23 r_CO2 14



Paramètres avec >5% d'importance : ['min_split_gain']



Pas assez de paramètres avec >5% d'importance pour une heatmap 2D.

rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.5125	0.4328	0.5178	0.029	0.1938	0.5578	0.125	1.2889	1.184
2	0.5296	0.4453	0.5327	0.0187	0.1923	0.5734	0.1281	1.2877	1.1894
3	0.4269	0.3677	0.4417	0.0381	0.1902	0.5117	0.144	1.3915	1.1609
4	0.5379	0.4509	0.5414	0.0188	0.1897	0.581	0.1302	1.2887	1.1931
5	0.5392	0.4518	0.5412	0.0182	0.1895	0.5836	0.1318	1.2918	1.1935
6	0.4911	0.4151	0.4933	0.0272	0.1871	0.5399	0.1248	1.3008	1.1831
7	0.5268	0.4417	0.5238	0.0257	0.1864	0.5823	0.1406	1.3183	1.1927
8	0.5338	0.4468	0.5353	0.0231	0.1859	0.5944	0.1476	1.3303	1.1947
9	0.529	0.4431	0.5164	0.0328	0.1852	0.563	0.1199	1.2706	1.194
10	0.472	0.3993	0.4752	0.0405	0.1811	0.5311	0.1318	1.3302	1.1821
---	---	---	---	---	---	---	---	---	---
70	0.8022	0.5105	0.6343	0.032	-0.3647	0.6478	0.1373	1.269	1.5715

Hyperparameters:

Rank 1: {'reg__n_estimators': 700, 'reg__max_depth': 4, 'reg__learning_rate': 0.09715082596013144, 'reg__num_leaves': 31, 'reg__subsample': 0.5047537225318536, 'reg__colsample_bytree': 0.7764859661522394, 'reg__min_child_samples': 24, 'reg__reg_alpha': 0.18433678595540304, 'reg__reg_lambda': 46.32190945028451, 'reg__min_split_gain': 1.1423881749476172, 'reg__path_smooth': 2.1531938598333866}

Rank 2: {'reg__n_estimators': 600, 'reg__max_depth': 8, 'reg__learning_rate': 0.02284310412404127, 'reg__num_leaves': 41, 'reg__subsample': 0.8876910765709008,

'reg__colsample_bytree': 0.8159557097806637, 'reg__min_child_samples': 29, 'reg__reg_alpha': 1.9024465942433877, 'reg__reg_lambda': 49.84443037165895, 'reg__min_split_gain': 0.6143376955027323, 'reg__path_smooth': 1.4980895595311088}

Rank 3: {'reg__n_estimators': 600, 'reg__max_depth': 4, 'reg__learning_rate': 0.05425811556854865, 'reg__num_leaves': 51, 'reg__subsample': 0.6309884605831213, 'reg__colsample_bytree': 0.881885232719323, 'reg__min_child_samples': 31, 'reg__reg_alpha': 2.6829085412873965, 'reg__reg_lambda': 47.50640908256052, 'reg__min_split_gain': 1.2774755734416634, 'reg__path_smooth': 2.5978246447479965}

Rank 4: {'reg__n_estimators': 600, 'reg__max_depth': 8, 'reg__learning_rate': 0.023939012005992986, 'reg__num_leaves': 36, 'reg__subsample': 0.5242874644447403, 'reg__colsample_bytree': 0.7217796709748567, 'reg__min_child_samples': 29, 'reg__reg_alpha': 1.7752362901459098, 'reg__reg_lambda': 49.816939751906865, 'reg__min_split_gain': 0.5829377500002693, 'reg__path_smooth': 2.898528543763062}

Rank 5: {'reg__n_estimators': 600, 'reg__max_depth': 8, 'reg__learning_rate': 0.034285533141191205, 'reg__num_leaves': 41, 'reg__subsample': 0.5571488537333397, 'reg__colsample_bytree': 0.7354816379587079, 'reg__min_child_samples': 29, 'reg__reg_alpha': 1.715358344812064, 'reg__reg_lambda': 49.59132293530384, 'reg__min_split_gain': 0.5822365414081755, 'reg__path_smooth': 3.058823971771262}

Rank 6: {'reg__n_estimators': 700, 'reg__max_depth': 4, 'reg__learning_rate': 0.09672862235382083, 'reg__num_leaves': 30, 'reg__subsample': 0.509013693897434, 'reg__colsample_bytree': 0.7666540584598751, 'reg__min_child_samples': 23, 'reg__reg_alpha': 0.20277341747663952, 'reg__reg_lambda': 49.75742241787524, 'reg__min_split_gain': 1.3535501342477754, 'reg__path_smooth': 2.3699086975844383}

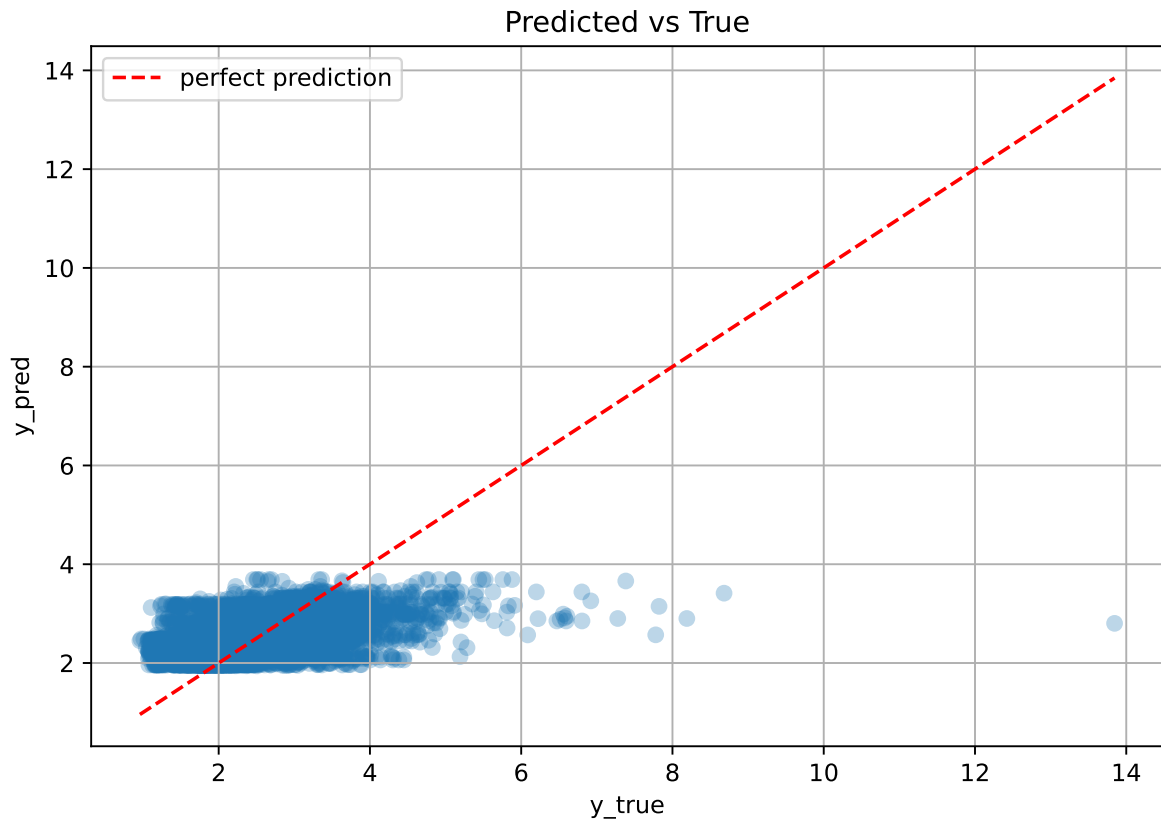
Rank 7: {'reg__n_estimators': 500, 'reg__max_depth': 5, 'reg__learning_rate': 0.03773146926467207, 'reg__num_leaves': 63, 'reg__subsample': 0.5691666248826267, 'reg__colsample_bytree': 0.8122199800654211, 'reg__min_child_samples': 46, 'reg__reg_alpha': 1.609337902674607, 'reg__reg_lambda': 2.533497630820566, 'reg__min_split_gain': 0.7632700124815486, 'reg__path_smooth': 1.6770962216114296}

Rank 8: {'reg__n_estimators': 600, 'reg__max_depth': 5, 'reg__learning_rate': 0.0419571781242129, 'reg__num_leaves': 60, 'reg__subsample': 0.5722787702172097, 'reg__colsample_bytree': 0.8001983017609957, 'reg__min_child_samples': 49, 'reg__reg_alpha': 1.479667570743083, 'reg__reg_lambda': 1.0778267292626822, 'reg__min_split_gain': 0.668741748203936, 'reg__path_smooth': 1.933097684658136}

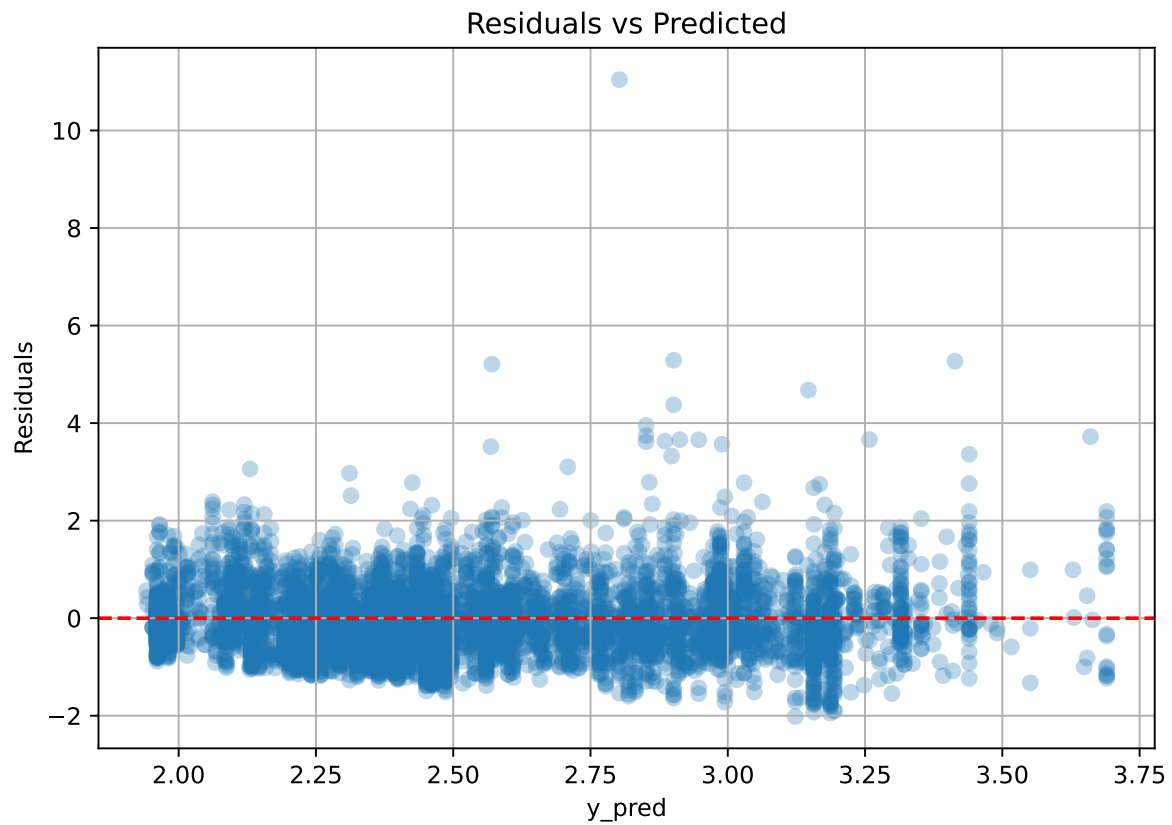
Rank 9: {'reg__n_estimators': 800, 'reg__max_depth': 7, 'reg__learning_rate': 0.03773224469050269, 'reg__num_leaves': 30, 'reg__subsample': 0.5859533186157176, 'reg__colsample_bytree': 0.7932999601950568, 'reg__min_child_samples': 27, 'reg__reg_alpha': 9.989976789875929, 'reg__reg_lambda': 0.12955574133343006, 'reg__min_split_gain': 0.030644378121098548, 'reg__path_smooth': 2.072897973446271}

Rank 10: {'reg__n_estimators': 700, 'reg__max_depth': 6, 'reg__learning_rate': 0.09398283997665266, 'reg__num_leaves': 51, 'reg__subsample': 0.6402381092216513, 'reg__colsample_bytree': 0.7568746447642202, 'reg__min_child_samples': 31, 'reg__reg_alpha': 2.4760992053322703, 'reg__reg_lambda': 12.098089162282706, 'reg__min_split_gain': 1.4180434515134535, 'reg__path_smooth': 2.4494855988517648}

2.4.22 Scatter



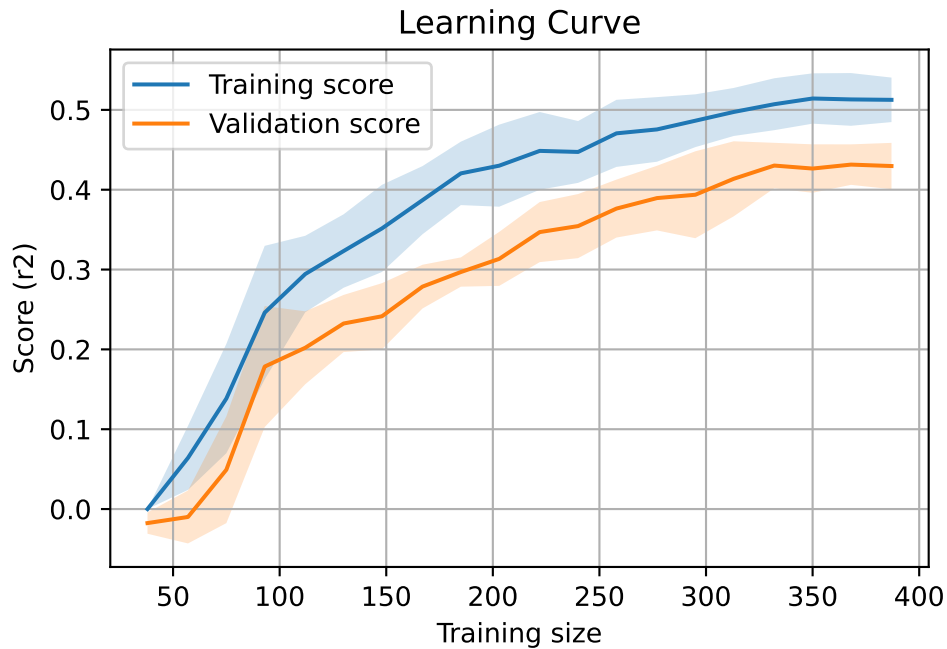
2.4.23 Scatter residuals



2.4.24 Learning curve — Group: 6_4

`Len(group)` : 792, `Len(training)` : 555,

`Len(training_real)` : 389, `Len(test_of_training)` : 166, `Len(test)` : 237



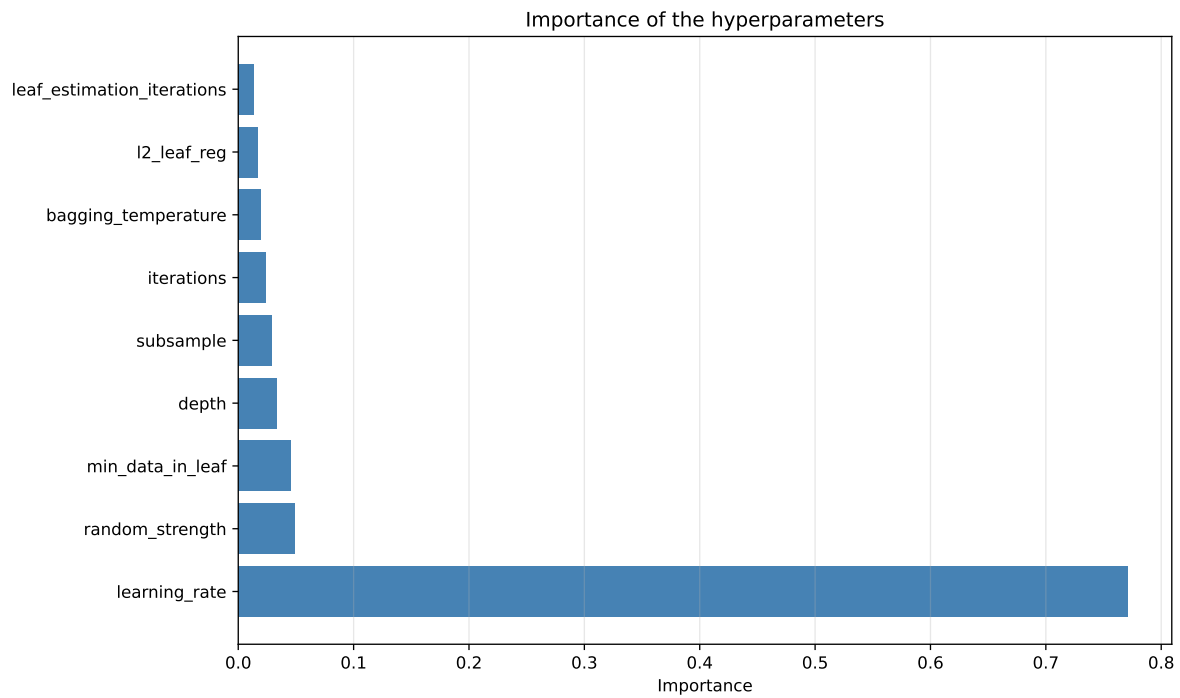
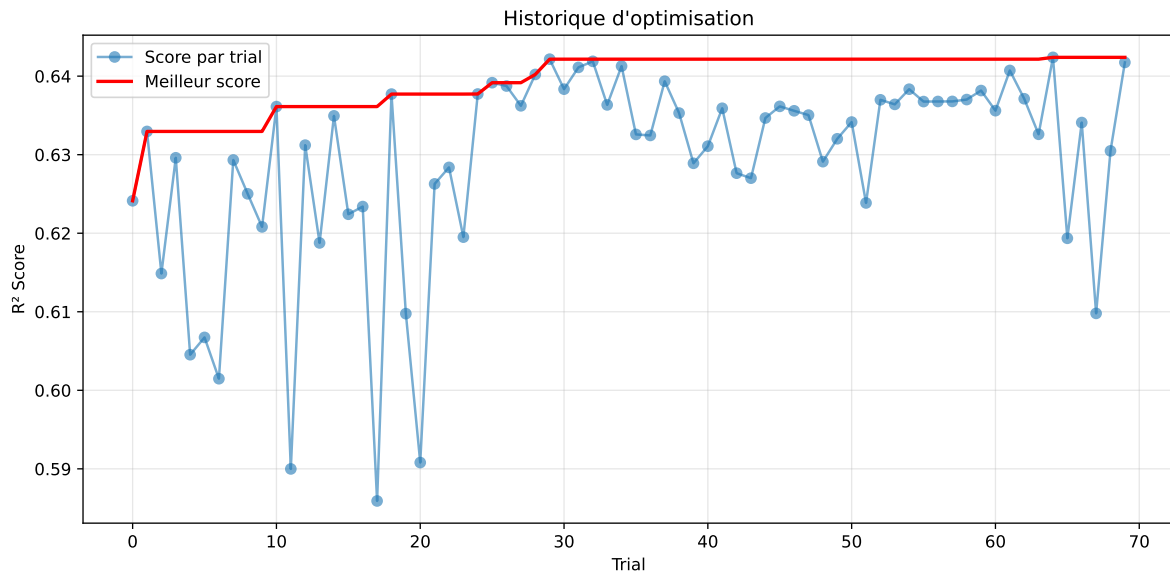
2.5 Model : CatBoost

2.5.1 Gridsearch — Group: 1_3

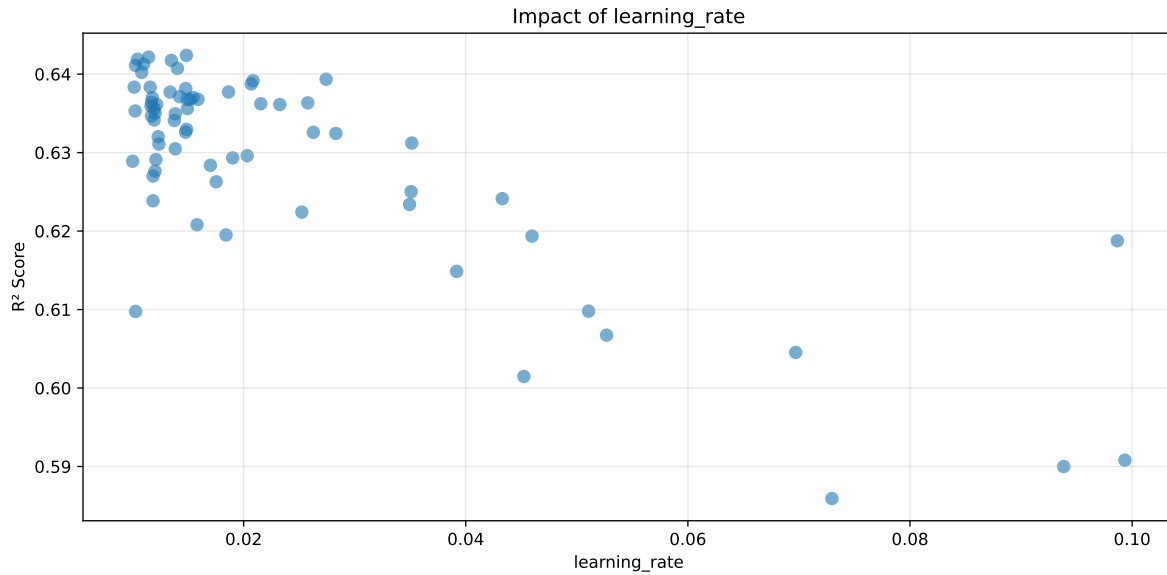
Len(group) : 795, Len(training) : 557, Len(test) : 238

Distribution y_train vs y_test: y_train: mean=2.597, std=0.579, min=1.437, max=4.986
y_test: mean=2.492, std=0.625, min=1.466, max=5.447

===== TOP Importance FEATURES =====
feature importance r_temperature 27.831125 r_umidity 25.477012 r_THI 20.210212 r_CO2
15.601361 r_NH3 10.880290



Paramètres avec >5% d'importance : ['learning_rate']



Pas assez de paramètres avec >5% d'importance pour une heatmap 2D.

rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.6554	0.6098	0.5829	0.0451	0.4727	0.5988	-0.011	0.982	1.0749
2	0.6835	0.6234	0.6096	0.0513	0.4432	0.6136	-0.0098	0.9843	1.0964
3	0.7105	0.6312	0.6258	0.0457	0.3933	0.634	0.0028	1.0044	1.1256
4	0.7171	0.633	0.6242	0.0465	0.3804	0.6323	-0.0007	0.9989	1.133
5	0.7204	0.6289	0.6177	0.0454	0.3544	0.6246	-0.0043	0.9932	1.1455
6	0.7364	0.635	0.6275	0.0457	0.331	0.6359	0.0009	1.0013	1.1596
7	0.7424	0.6353	0.6253	0.0438	0.3139	0.6364	0.0011	1.0017	1.1686
8	0.7423	0.6347	0.6298	0.0446	0.3119	0.638	0.0034	1.0053	1.1695
9	0.7447	0.6359	0.6326	0.0449	0.3096	0.6356	-0.0003	0.9995	1.1711
10	0.7451	0.6362	0.6325	0.0445	0.3092	0.6387	0.0025	1.004	1.1713
---	---	---	---	---	---	---	---	---	---
70	0.9903	0.5859	0.6527	0.0621	-0.6273	0.6335	0.0476	1.0813	1.6902

Hyperparameters:

Rank 1: {'reg__iterations': 600, 'reg__depth': 5, 'reg__learning_rate': 0.010273467057771879, 'reg__l2_leaf_reg': 43.91225475795349, 'reg__bagging_temperature': 1.8245365260314288, 'reg__random_strength': 2.515282363708394, 'reg__subsample': 0.5819239880048545, 'reg__min_data_in_leaf': 26, 'reg__leaf_estimation_iterations': 1}

Rank 2: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.03493661896913573, 'reg__l2_leaf_reg': 15.506369657314968, 'reg__bagging_temperature': 1.7176850242071529, 'reg__random_strength': 4.236887396734121, 'reg__subsample': 0.5398293955690805, 'reg__min_data_in_leaf': 29, 'reg__leaf_estimation_iterations': 8}

Rank 3: {'reg__iterations': 200, 'reg__depth': 4, 'reg__learning_rate': 0.03513947555989072, 'reg__l2_leaf_reg': 41.51172137127661, 'reg__bagging_temperature': 4.750606781303062, 'reg__random_strength': 3.133424802453054, 'reg__subsample': 0.5870643659114487, 'reg__min_data_in_leaf': 50, 'reg__leaf_estimation_iterations': 7}

Rank 4: {'reg__iterations': 300, 'reg__depth': 5, 'reg__learning_rate': 0.014870904297572074, 'reg__l2_leaf_reg': 26.429650308383206, 'reg__bagging_temperature': 0.5746347905368243, 'reg__random_strength': 1.8824174311168163, 'reg__subsample': 0.522049564149818, 'reg__min_data_in_leaf': 39, 'reg__leaf_estimation_iterations': 8}

Rank 5: {'reg__iterations': 300, 'reg__depth': 7, 'reg__learning_rate': 0.010007445315225932, 'reg__l2_leaf_reg': 32.69932737662701, 'reg__bagging_temperature': 2.3143960799582746, 'reg__random_strength': 1.6404130059177462, 'reg__subsample': 0.8271895540738308, 'reg__min_data_in_leaf': 22, 'reg__leaf_estimation_iterations': 9}

Rank 6: {'reg__iterations': 300, 'reg__depth': 7, 'reg__learning_rate': 0.012034949372664968, 'reg__l2_leaf_reg': 39.556575136147664, 'reg__bagging_temperature': 2.1149219600889824, 'reg__random_strength': 1.609275188108226, 'reg__subsample': 0.8676342864799433, 'reg__min_data_in_leaf': 10, 'reg__leaf_estimation_iterations': 9}

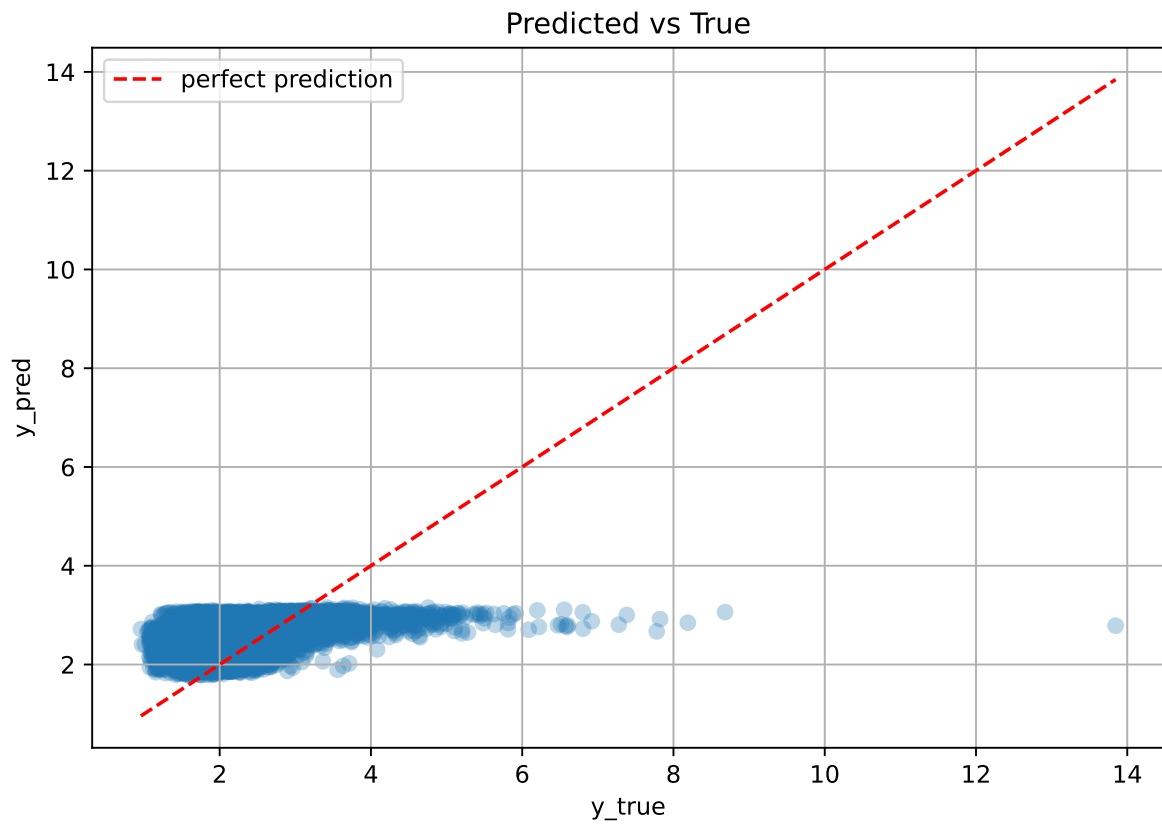
Rank 7: {'reg__iterations': 300, 'reg__depth': 8, 'reg__learning_rate': 0.010231450922110683, 'reg__l2_leaf_reg': 31.107495962849402, 'reg__bagging_temperature': 2.353134368329093, 'reg__random_strength': 1.5420409542957598, 'reg__subsample': 0.8984150895882435, 'reg__min_data_in_leaf': 21, 'reg__leaf_estimation_iterations': 9}

Rank 8: {'reg__iterations': 300, 'reg__depth': 7, 'reg__learning_rate': 0.011702759312382032, 'reg__l2_leaf_reg': 29.264205802644867, 'reg__bagging_temperature': 2.276630142598911, 'reg__random_strength': 1.6054505654326086, 'reg__subsample': 0.877697767534836, 'reg__min_data_in_leaf': 10, 'reg__leaf_estimation_iterations': 9}

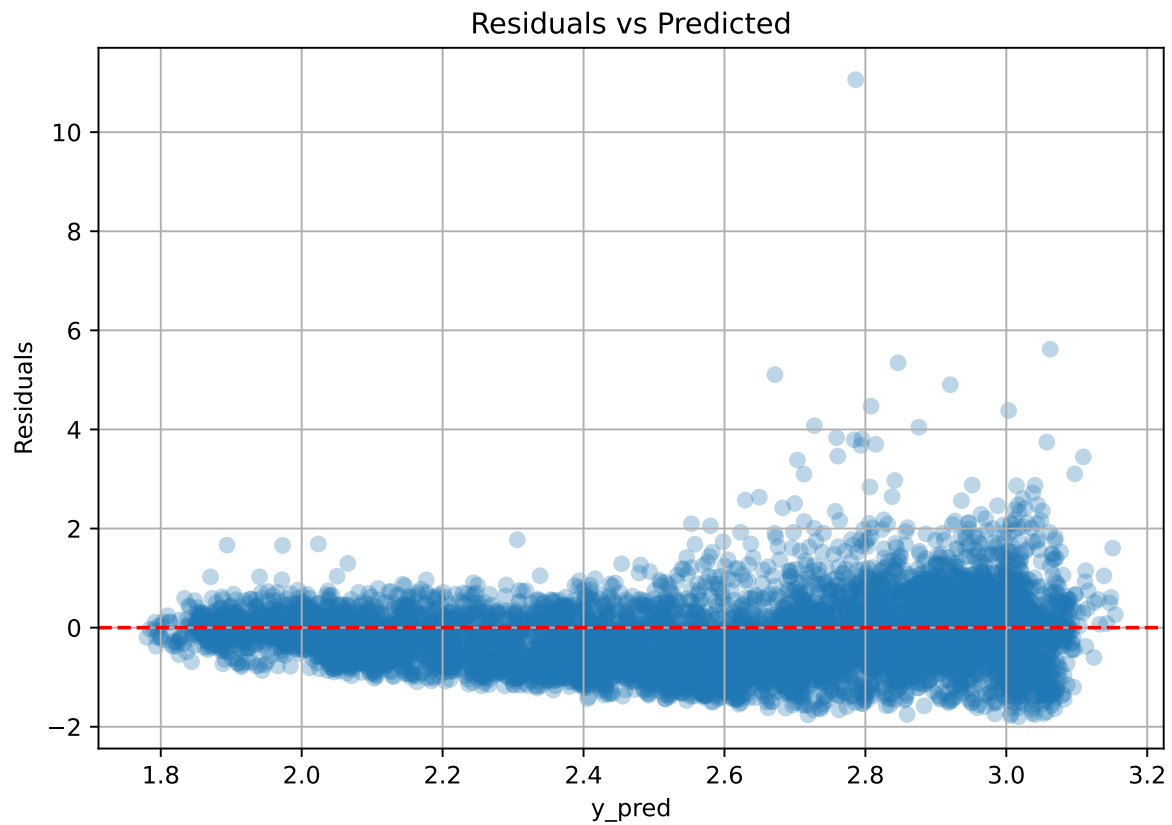
Rank 9: {'reg__iterations': 300, 'reg__depth': 7, 'reg__learning_rate': 0.01166534996397246, 'reg__l2_leaf_reg': 28.533868705046356, 'reg__bagging_temperature': 2.206666039833838, 'reg__random_strength': 1.512416526315748, 'reg__subsample': 0.8145023083318517, 'reg__min_data_in_leaf': 10, 'reg__leaf_estimation_iterations': 9}

Rank 10: {'reg__iterations': 300, 'reg__depth': 7, 'reg__learning_rate': 0.01216856744388358, 'reg__l2_leaf_reg': 29.67334746045567, 'reg__bagging_temperature': 2.195716248047531, 'reg__random_strength': 1.7046748553444164, 'reg__subsample': 0.8715187208182072, 'reg__min_data_in_leaf': 11, 'reg__leaf_estimation_iterations': 9}

2.5.2 Scatter



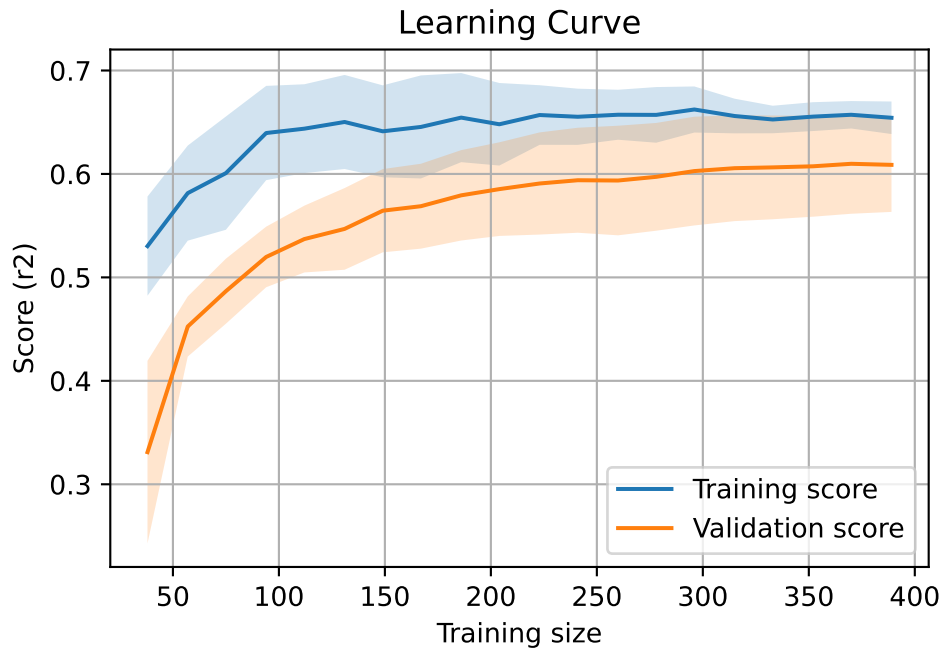
2.5.3 Scatter residuals



2.5.4 Learning curve — Group: 1_3

`Len(group)` : 795, `Len(training)` : 557,

`Len(training_real)` : 390, `Len(test_of_training)` : 167, `Len(test)` : 238

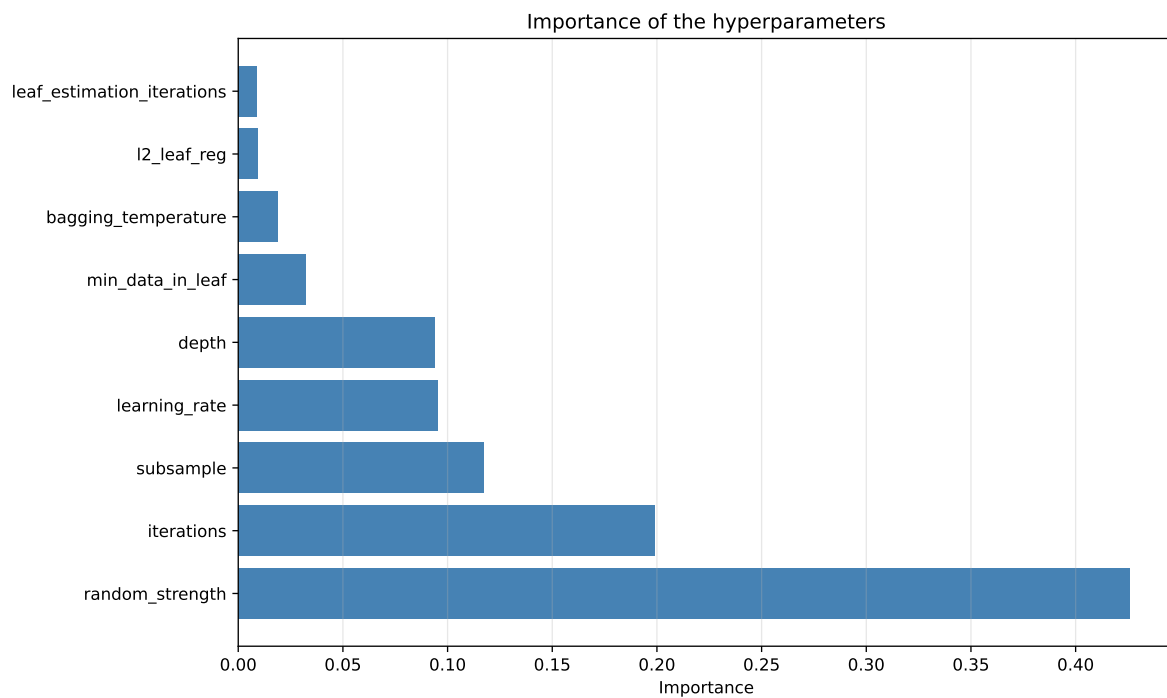
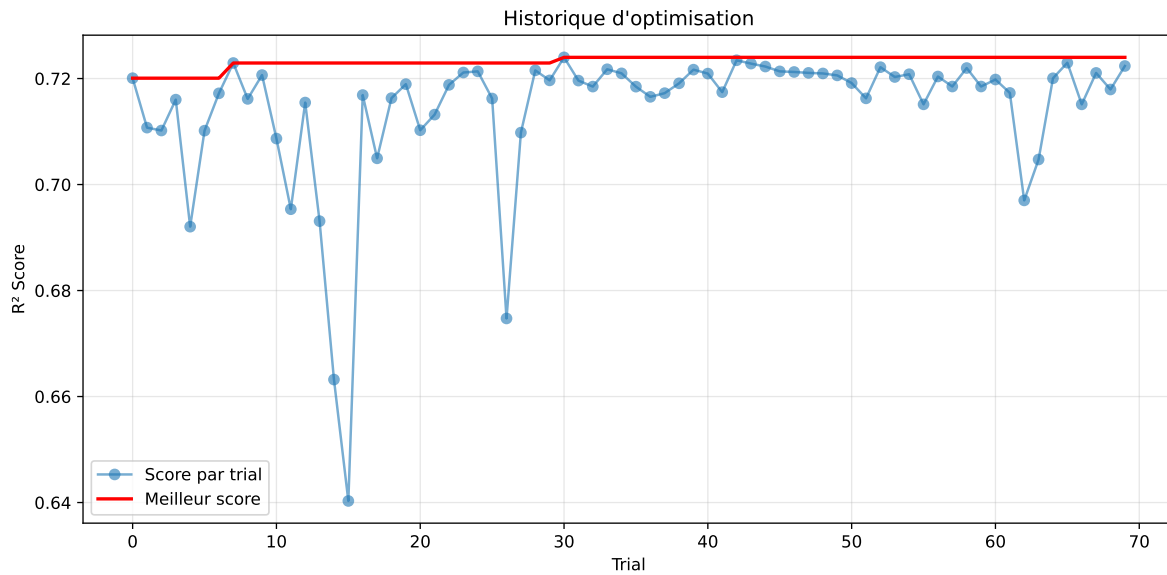


2.5.5 Gridsearch — Group: 2_11

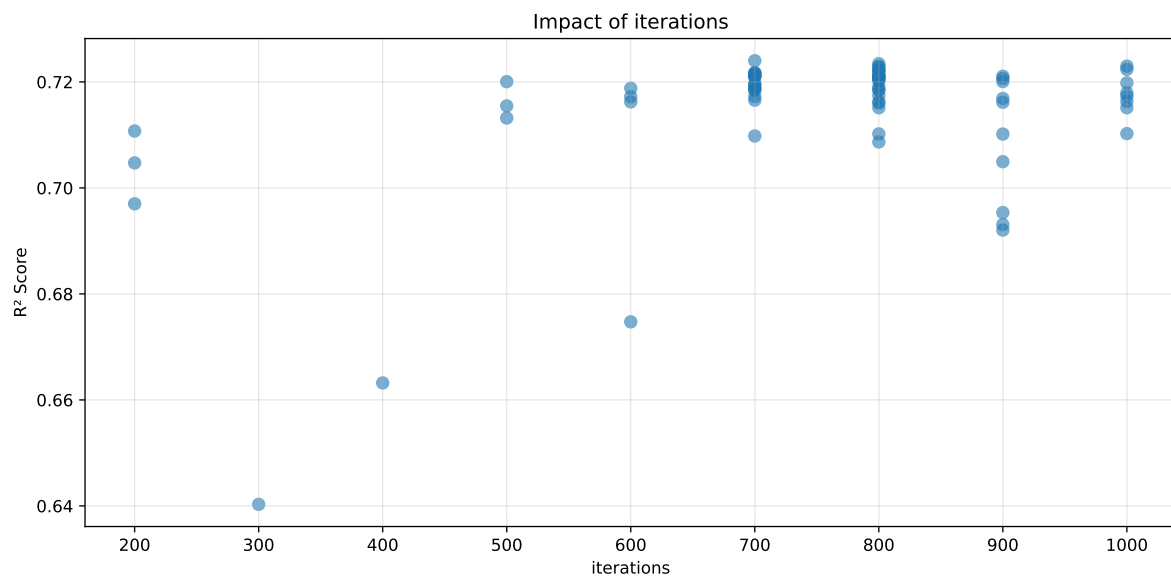
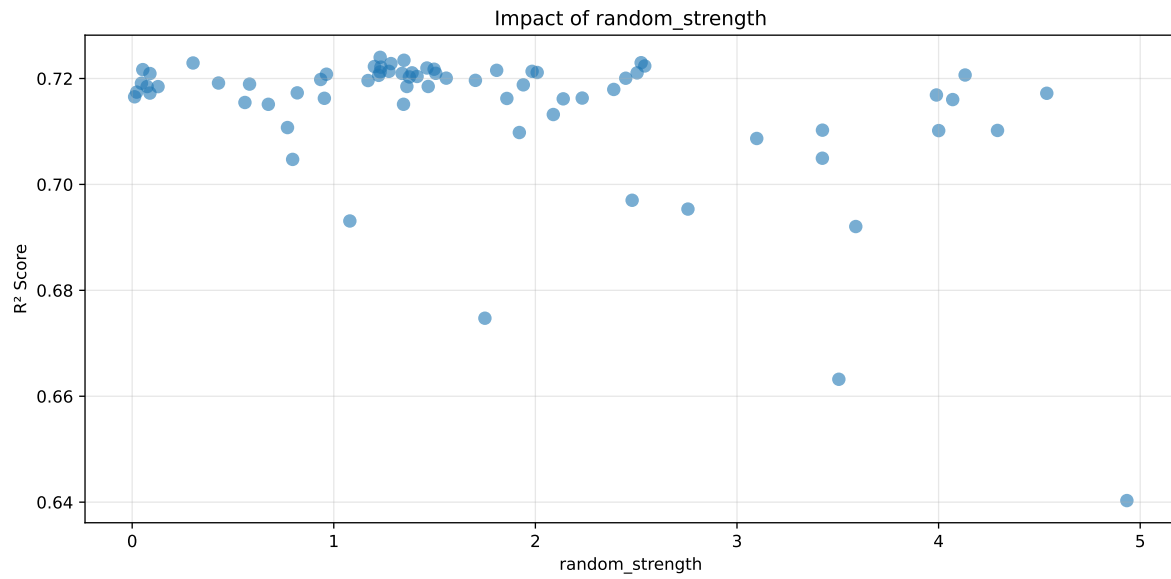
Len(group) : 2726, Len(training) : 1909, Len(test) : 817

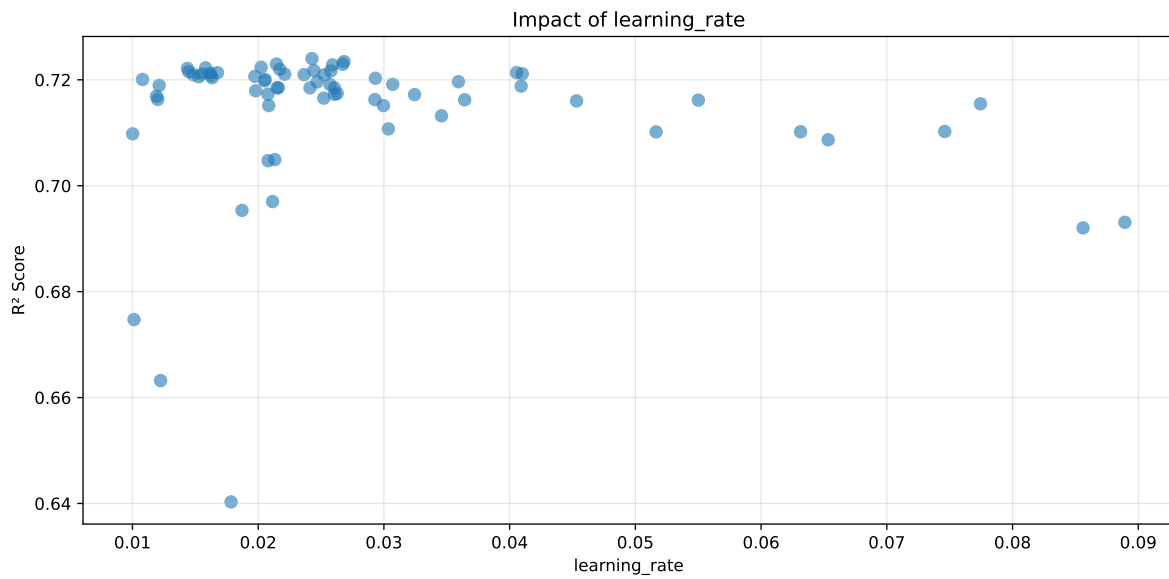
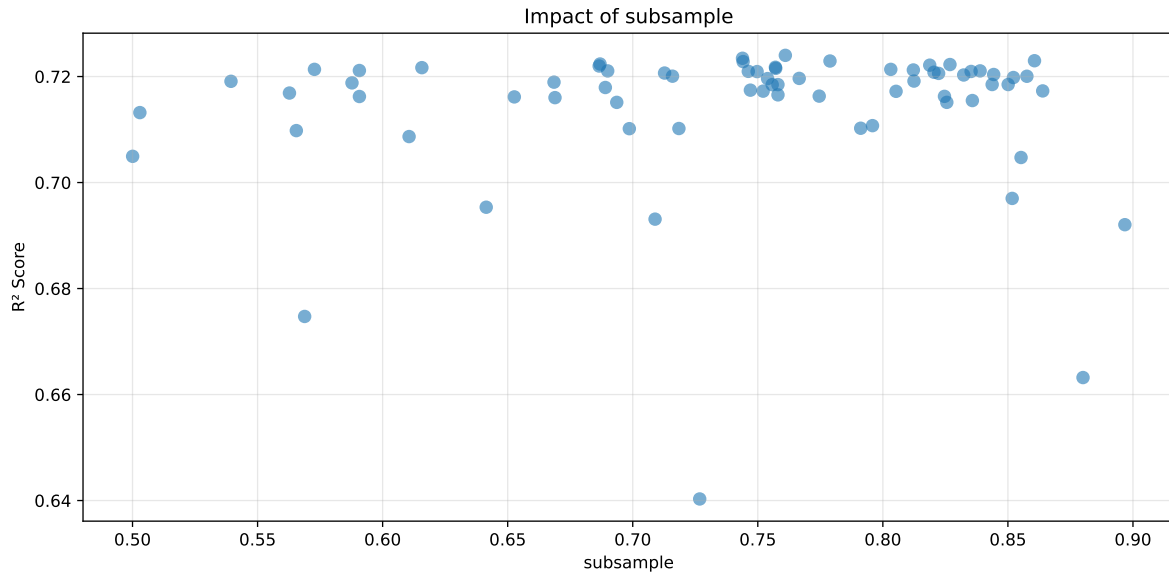
Distribution y_train vs y_test: y_train: mean=2.540, std=0.859, min=1.150, max=8.682
y_test: mean=2.561, std=0.947, min=1.120, max=13.845

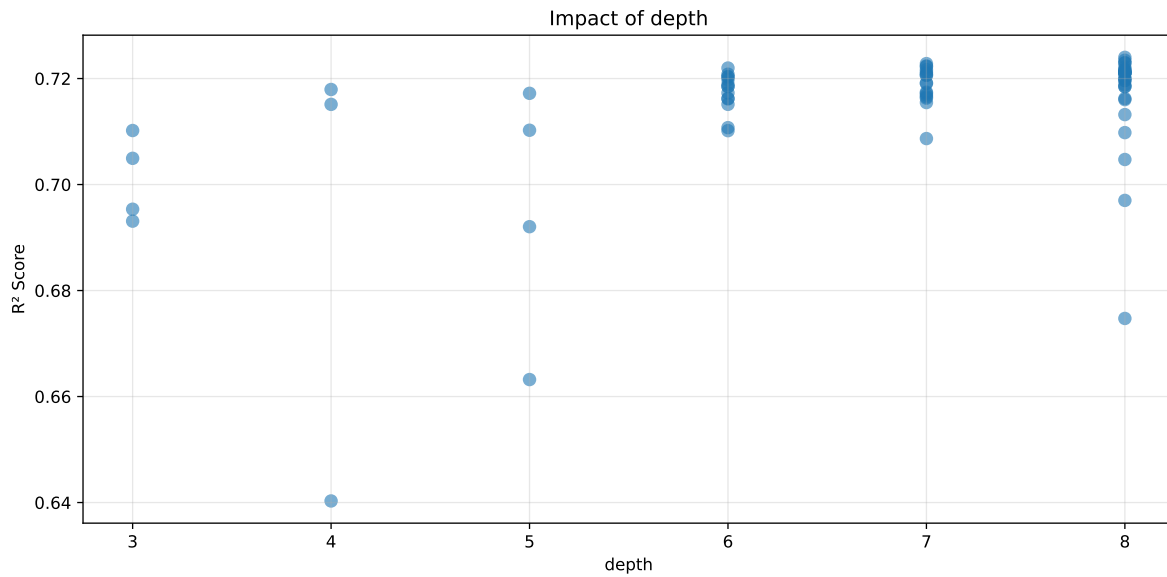
===== TOP Importance FEATURES =====
feature importance r_umidity 35.828560 r_NH3 34.073233 r_CO2 14.031609 r_temperature
9.131845 r_THI 6.934754



Paramètres avec >5% d'importance : ['random_strength', 'iterations', 'subsample', 'learning_rate', 'depth']

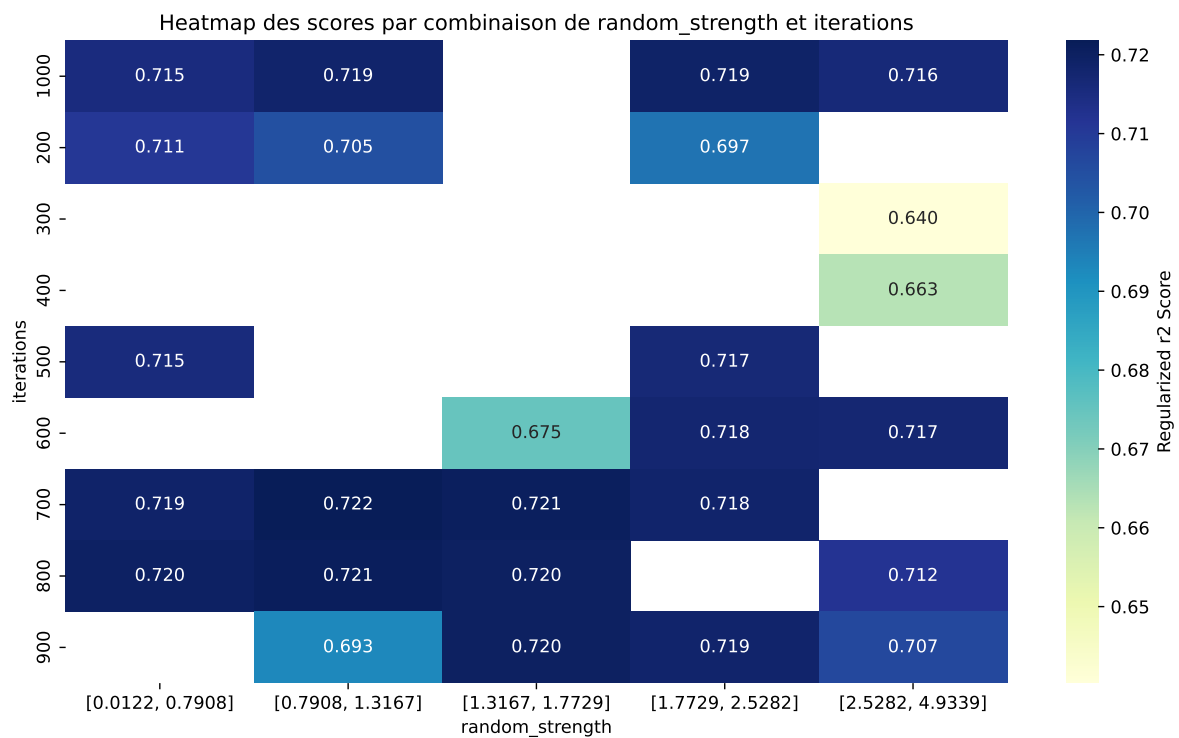






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The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.681	0.6632	0.5729	0.0126	0.6632	0.582	-0.0812	0.8776	1.0268
2	0.6485	0.6403	0.5527	0.012	0.6403	0.5613	-0.079	0.8766	1.0129
3	0.7356	0.6953	0.5986	0.0196	0.5744	0.5988	-0.0965	0.8612	1.058
4	0.7162	0.6747	0.5772	0.0123	0.5503	0.59	-0.0847	0.8745	1.0615
5	0.7584	0.7049	0.605	0.0185	0.5444	0.6047	-0.1003	0.8578	1.0759
6	0.7589	0.697	0.5991	0.0159	0.5114	0.6085	-0.0885	0.8731	1.0888
7	0.7836	0.7107	0.6015	0.0161	0.4921	0.6093	-0.1015	0.8572	1.1026
8	0.7866	0.7098	0.609	0.0139	0.4793	0.6134	-0.0964	0.8641	1.1082
9	0.7977	0.7163	0.6077	0.0164	0.472	0.6111	-0.1052	0.8532	1.1137
10	0.7952	0.7132	0.6091	0.0151	0.4672	0.6136	-0.0996	0.8604	1.115
---	---	---	---	---	---	---	---	---	---
70	0.9661	0.692	0.5725	0.0129	-0.13	0.5858	-0.1063	0.8464	1.396

Hyperparameters:

Rank 1: {'reg__iterations': 400, 'reg__depth': 5, 'reg__learning_rate': 0.012235857751134813, 'reg__l2_leaf_reg': 43.27437814975455, 'reg__bagging_temperature': 1.7205646091386562, 'reg__random_strength': 3.5052036956185235, 'reg__subsample': 0.8800279996018991, 'reg__min_data_in_leaf': 50, 'reg__leaf_estimation_iterations': 6}

Rank 2: {'reg__iterations': 300, 'reg__depth': 4, 'reg__learning_rate': 0.017843662585602663, 'reg__l2_leaf_reg': 32.819244304646645, 'reg__bagging_temperature': 0.5282565254888127, 'reg__random_strength': 4.933925777698461, 'reg__subsample': 0.7267713838306024, 'reg__min_data_in_leaf': 40, 'reg__leaf_estimation_iterations': 3}

Rank 3: {'reg__iterations': 900, 'reg__depth': 3, 'reg__learning_rate': 0.01871105816443942, 'reg__l2_leaf_reg': 38.680853810457045, 'reg__bagging_temperature': 2.5714531349752336, 'reg__random_strength': 2.7567430433843496, 'reg__subsample': 0.6414186535787567, 'reg__min_data_in_leaf': 38, 'reg__leaf_estimation_iterations': 3}

Rank 4: {'reg__iterations': 600, 'reg__depth': 8, 'reg__learning_rate': 0.010121603190611889, 'reg__l2_leaf_reg': 20.842348901034665, 'reg__bagging_temperature': 2.043955046622875, 'reg__random_strength': 1.749580087558376, 'reg__subsample': 0.5688207294961103, 'reg__min_data_in_leaf': 31, 'reg__leaf_estimation_iterations': 1}

Rank 5: {'reg__iterations': 900, 'reg__depth': 3, 'reg__learning_rate': 0.0213207494732443, 'reg__l2_leaf_reg': 42.27041811794936, 'reg__bagging_temperature': 3.0346234811861343, 'reg__random_strength': 3.4235401265886054, 'reg__subsample': 0.5000457113375703, 'reg__min_data_in_leaf': 48, 'reg__leaf_estimation_iterations': 5}

Rank 6: {'reg__iterations': 200, 'reg__depth': 8, 'reg__learning_rate': 0.021143469558699467, 'reg__l2_leaf_reg': 16.823094344572382, 'reg__bagging_temperature': 0.9191828448848316, 'reg__random_strength': 2.4798812005010062, 'reg__subsample': 0.8517109834139996, 'reg__min_data_in_leaf': 38, 'reg__leaf_estimation_iterations': 10}

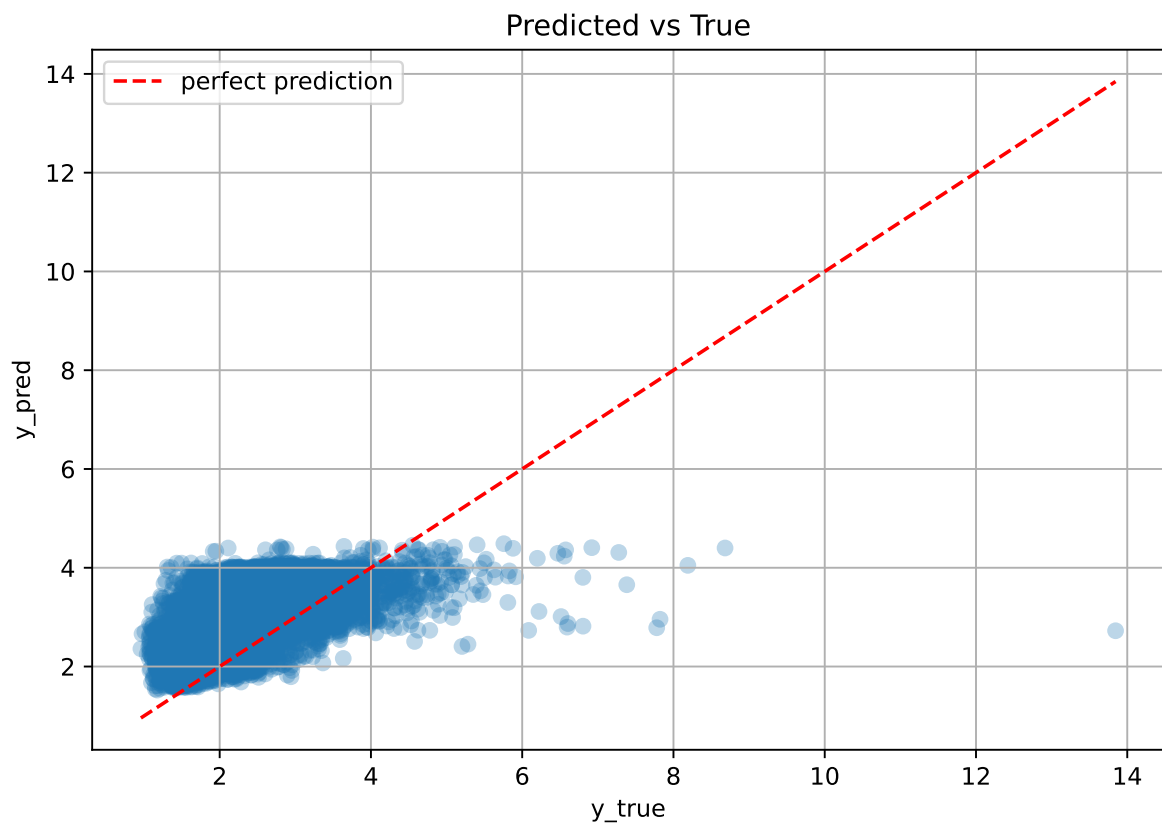
Rank 7: {'reg__iterations': 200, 'reg__depth': 6, 'reg__learning_rate': 0.03035186956539708, 'reg__l2_leaf_reg': 12.879256773431242, 'reg__bagging_temperature': 3.8597449649838262, 'reg__random_strength': 0.7703786458341727, 'reg__subsample': 0.7958563145967742, 'reg__min_data_in_leaf': 14, 'reg__leaf_estimation_iterations': 9}

Rank 8: {'reg__iterations': 700, 'reg__depth': 8, 'reg__learning_rate': 0.010014229397299148, 'reg__l2_leaf_reg': 49.39275434178347, 'reg__bagging_temperature': 1.84028090940035, 'reg__random_strength': 1.9204371507725462, 'reg__subsample': 0.565497178678088, 'reg__min_data_in_leaf': 26, 'reg__leaf_estimation_iterations': 10}

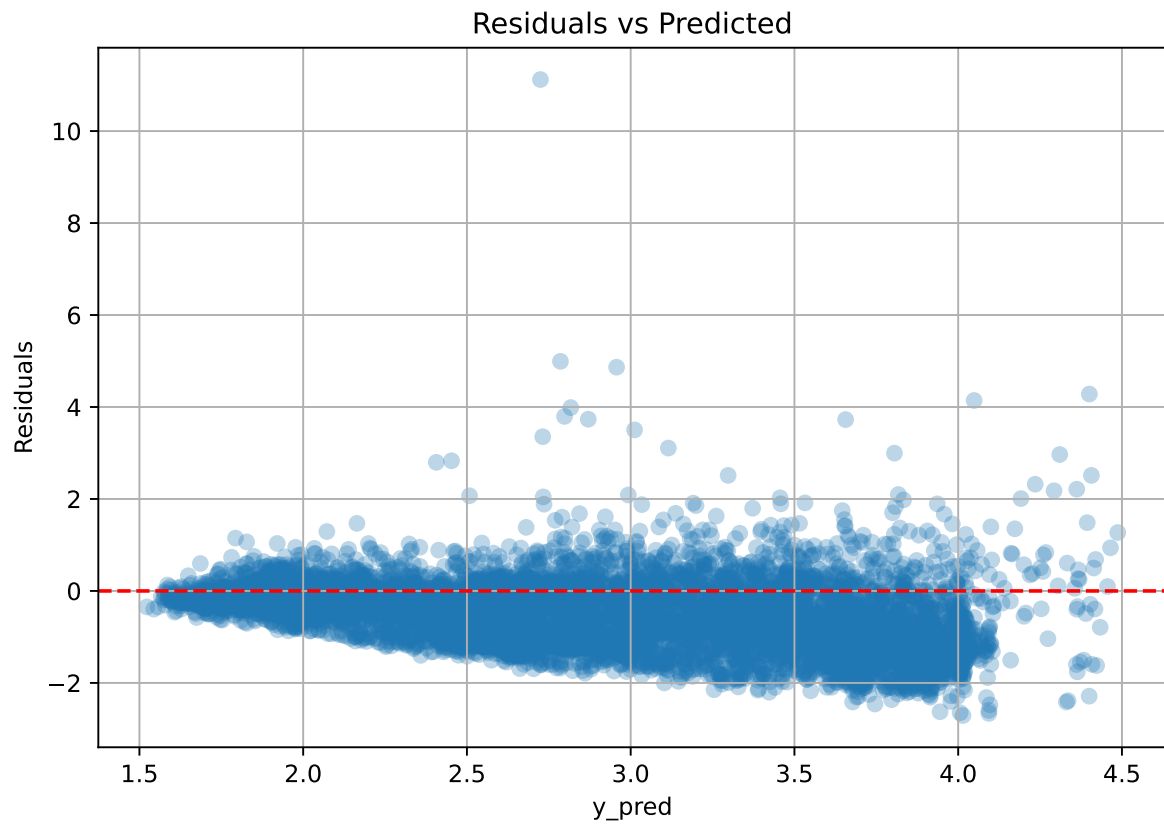
Rank 9: {'reg__iterations': 1000, 'reg__depth': 7, 'reg__learning_rate': 0.012011463741962413, 'reg__l2_leaf_reg': 6.9707381853733885, 'reg__bagging_temperature': 1.6965386847784736, 'reg__random_strength': 2.2322802256186924, 'reg__subsample': 0.7745545964598803, 'reg__min_data_in_leaf': 21, 'reg__leaf_estimation_iterations': 1}

Rank 10: {'reg__iterations': 500, 'reg__depth': 8, 'reg__learning_rate': 0.034581519055985756, 'reg__l2_leaf_reg': 21.045965680508075, 'reg__bagging_temperature': 2.723376305049338, 'reg__random_strength': 2.0889206546481764, 'reg__subsample': 0.5029393279266664, 'reg__min_data_in_leaf': 30, 'reg__leaf_estimation_iterations': 1}

2.5.6 Scatter



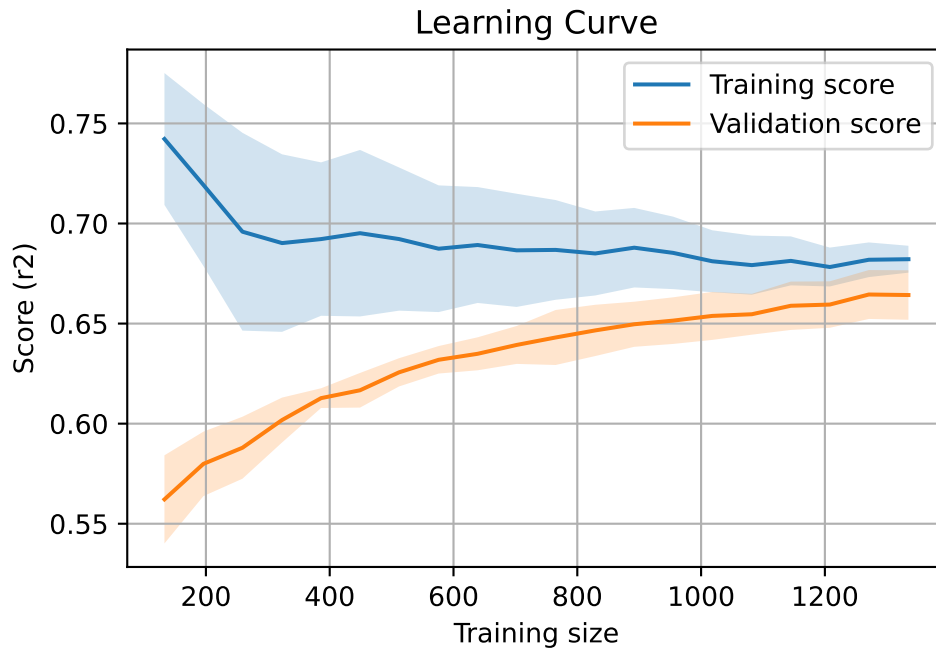
2.5.7 Scatter residuals



2.5.8 Learning curve — Group: 2_11

`Len(group)` : 2726, `Len(training)` : 1909,

`Len(training_real)` : 1337, `Len(test_of_training)` : 572, `Len(test)` : 817

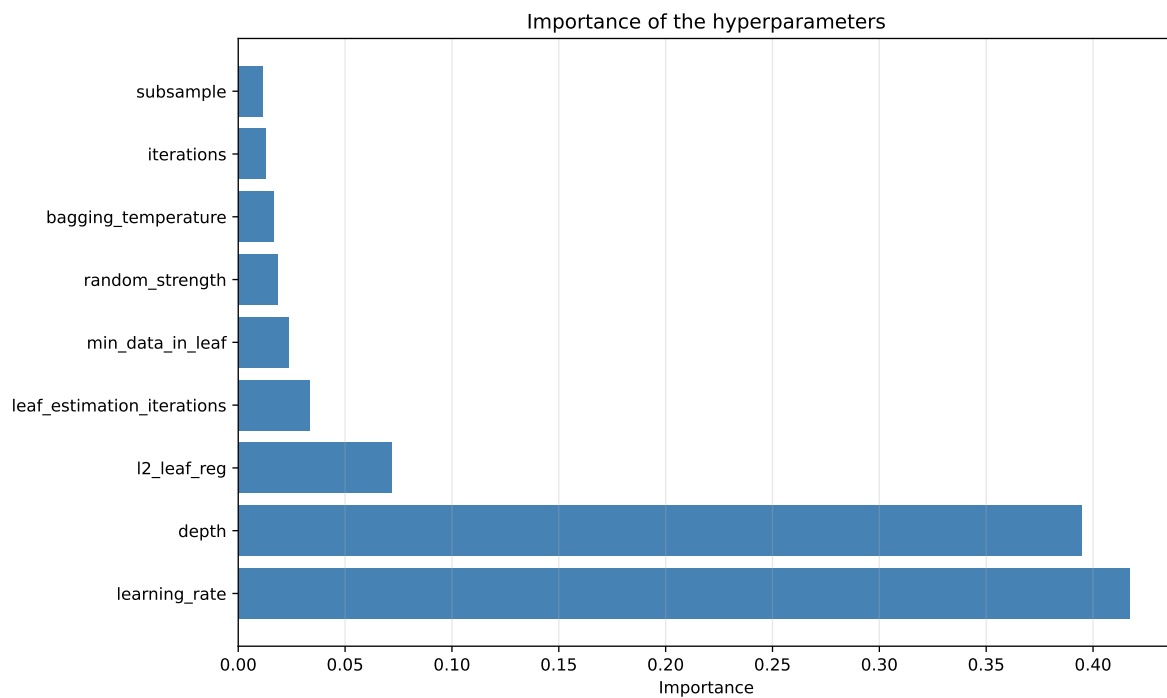
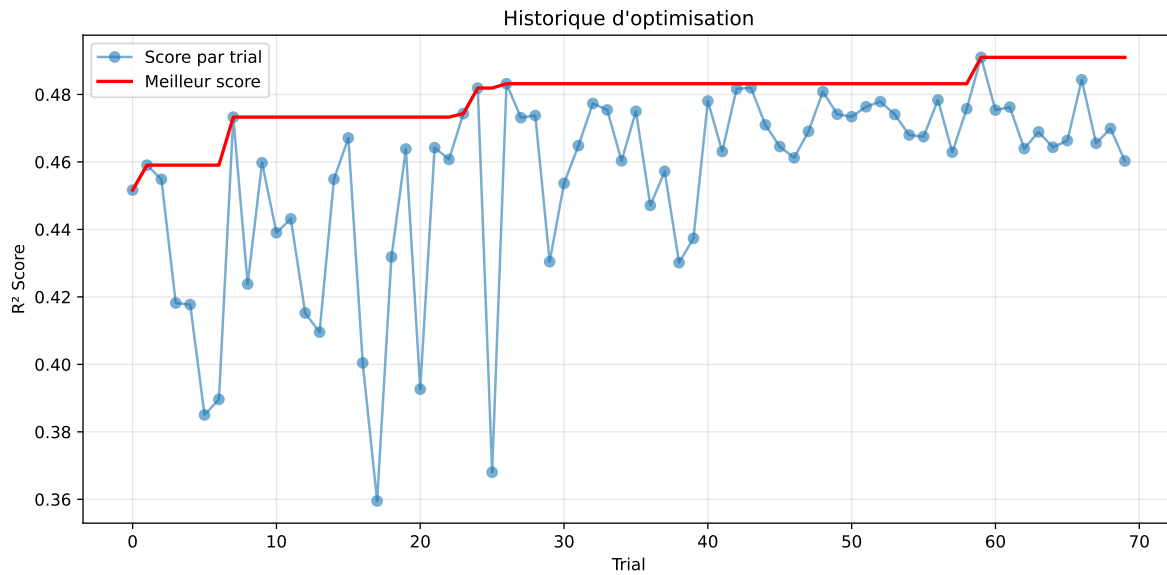


2.5.9 Gridsearch — Group: 3_4

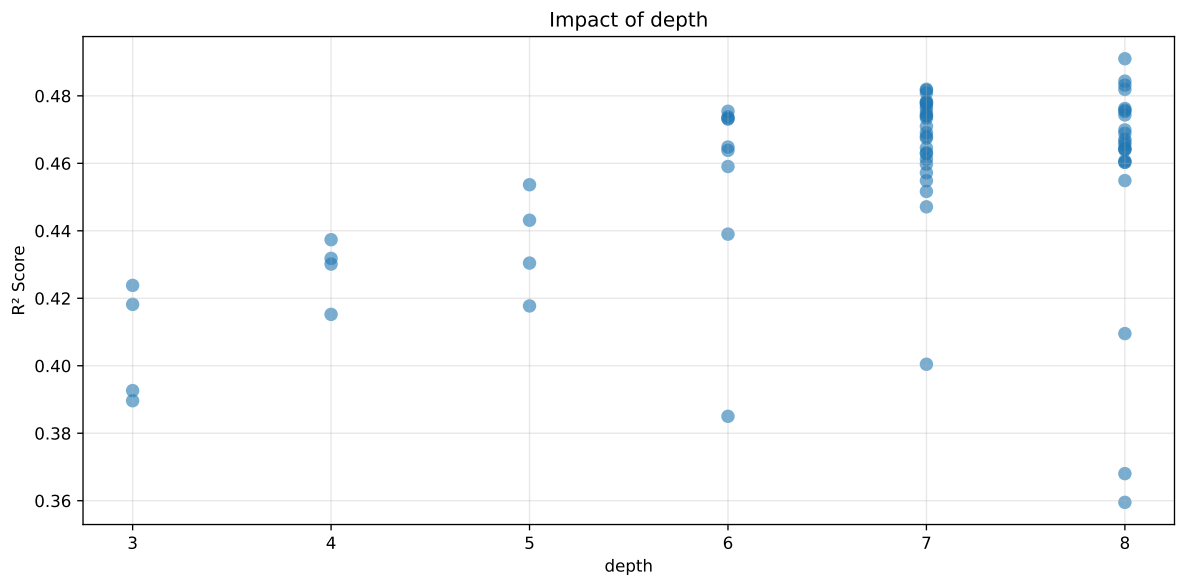
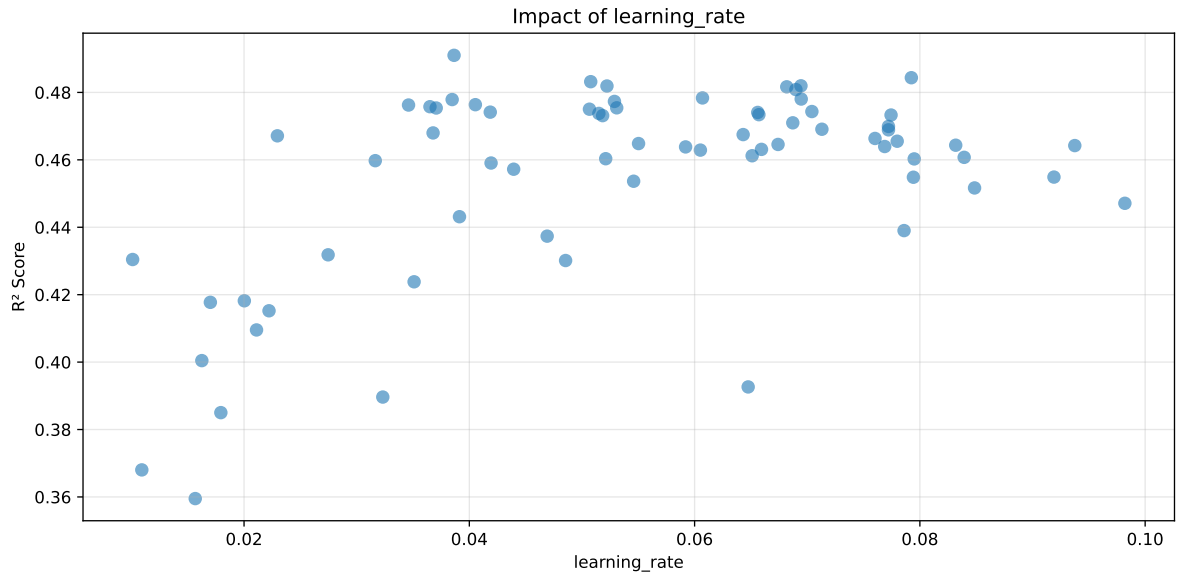
Len(group) : 568, Len(training) : 398, Len(test) : 170

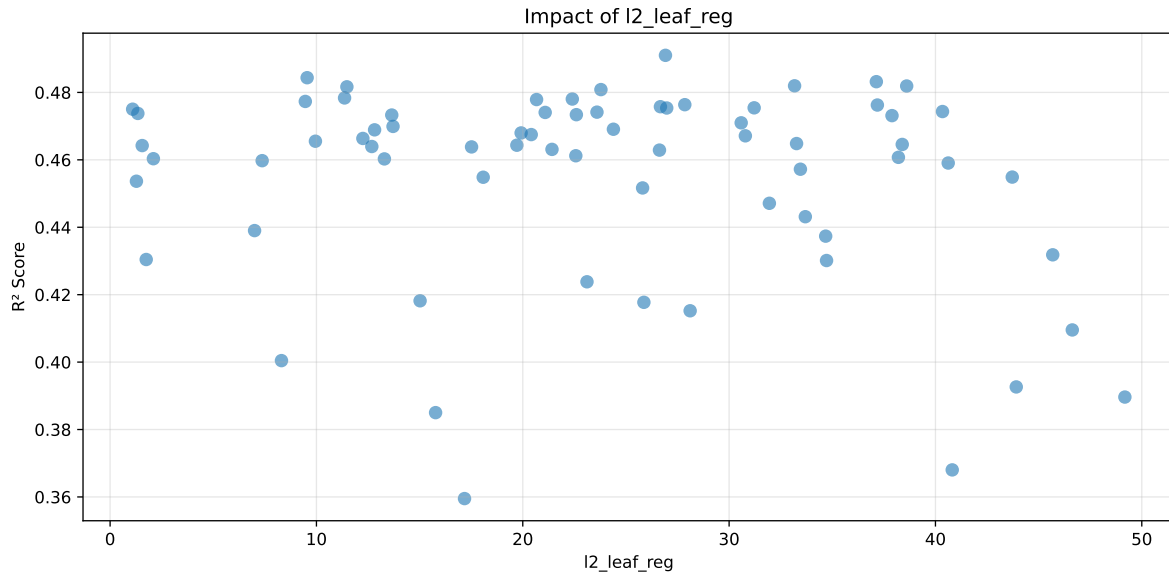
Distribution y_train vs y_test: y_train: mean=1.928, std=0.816, min=0.959, max=6.804
y_test: mean=1.833, std=0.729, min=1.098, max=7.822

===== TOP Importance FEATURES =====
feature importance r_umidity 32.377865 r_NH3 26.505288 r_THI 26.243449 r_temperature
8.407745 r_CO2 6.465653



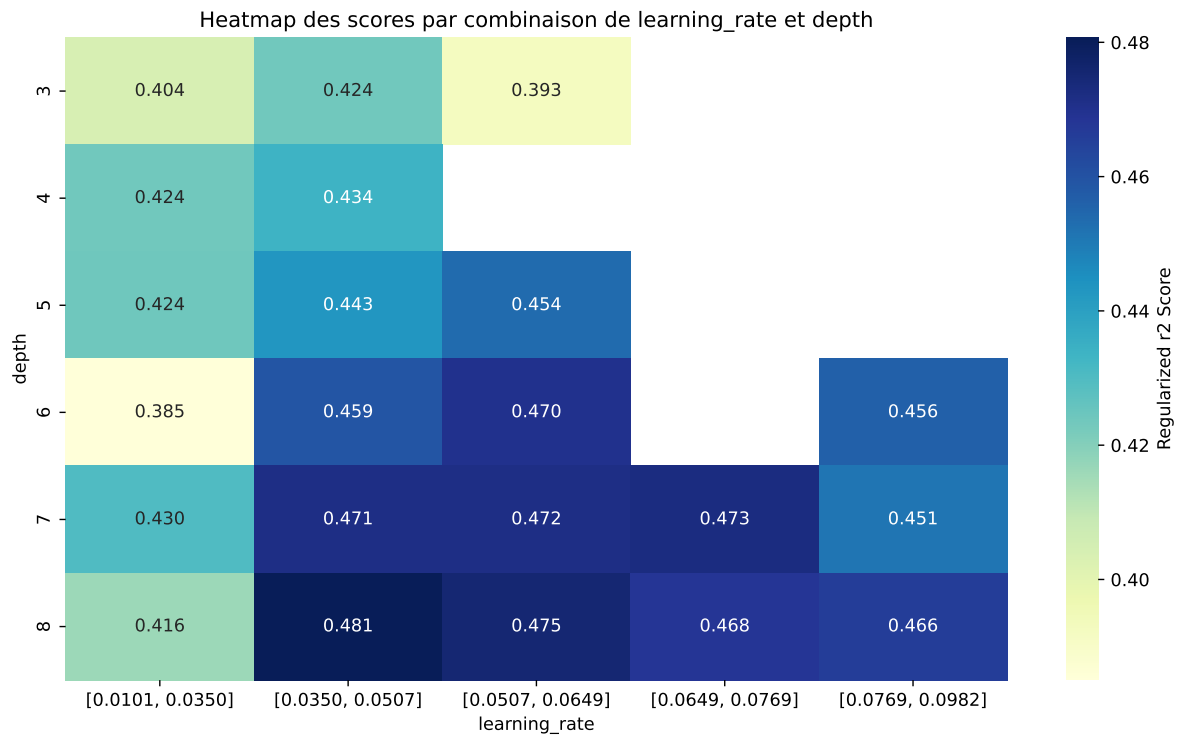
Paramètres avec >5% d'importance : ['learning_rate', 'depth', 'l2_leaf_reg']





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The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.5822	0.3896	0.2141	0.0625	-0.1881	0.244	-0.1456	0.6264	1.4942
2	0.5908	0.385	0.2347	0.0582	-0.2323	0.2791	-0.1059	0.725	1.5344
3	0.5871	0.368	0.2042	0.0625	-0.2892	0.2522	-0.1159	0.6852	1.5953
4	0.7122	0.4304	0.2727	0.0701	-0.4148	0.3202	-0.1102	0.744	1.6546
5	0.6984	0.4177	0.2507	0.0806	-0.4243	0.3122	-0.1056	0.7473	1.6719
6	0.6209	0.3595	0.1929	0.0644	-0.4245	0.2499	-0.1096	0.695	1.727
7	0.68	0.4004	0.2426	0.0671	-0.4381	0.287	-0.1134	0.7168	1.698
8	0.7251	0.4095	0.2166	0.0749	-0.5372	0.261	-0.1485	0.6374	1.7706
9	0.7933	0.4374	0.2612	0.1055	-0.6306	0.3023	-0.135	0.6913	1.8139
10	0.8125	0.4301	0.2663	0.1146	-0.7169	0.2886	-0.1415	0.671	1.8889
---	---	---	---	---	---	---	---	---	---
70	0.9977	0.439	0.2656	0.1077	-1.237	0.3689	-0.0701	0.8403	2.2725

Hyperparameters:

Rank 1: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.03231751268093781, 'reg__l2_leaf_reg': 49.180430849207234, 'reg__bagging_temperature': 3.082467044837591, 'reg__random_strength': 4.046397199964546, 'reg__subsample': 0.8205759688043873, 'reg__min_data_in_leaf': 39, 'reg__leaf_estimation_iterations': 2}

Rank 2: {'reg__iterations': 200, 'reg__depth': 6, 'reg__learning_rate': 0.01793991775310571, 'reg__l2_leaf_reg': 15.774509569989785, 'reg__bagging_temperature': 2.4205717311490087, 'reg__random_strength': 4.785816739028612, 'reg__subsample': 0.5113350540344903, 'reg__min_data_in_leaf': 25, 'reg__leaf_estimation_iterations': 3}

Rank 3: {'reg__iterations': 200, 'reg__depth': 8, 'reg__learning_rate': 0.010923066361442094, 'reg__l2_leaf_reg': 40.81324351354155, 'reg__bagging_temperature': 1.527767539204225, 'reg__random_strength': 2.937531073896061, 'reg__subsample': 0.5124694217641594, 'reg__min_data_in_leaf': 35, 'reg__leaf_estimation_iterations': 9}

Rank 4: {'reg__iterations': 400, 'reg__depth': 5, 'reg__learning_rate': 0.010108532472114894, 'reg__l2_leaf_reg': 1.7502011978865113, 'reg__bagging_temperature': 1.155886547448874, 'reg__random_strength': 3.119905744144321, 'reg__subsample': 0.5128185489455179, 'reg__min_data_in_leaf': 35, 'reg__leaf_estimation_iterations': 8}

Rank 5: {'reg__iterations': 500, 'reg__depth': 5, 'reg__learning_rate': 0.017008868616935884, 'reg__l2_leaf_reg': 25.869483860424147, 'reg__bagging_temperature': 3.9811622549637233, 'reg__random_strength': 3.5241845298411434, 'reg__subsample': 0.6026521071189153, 'reg__min_data_in_leaf': 30, 'reg__leaf_estimation_iterations': 2}

Rank 6: {'reg__iterations': 200, 'reg__depth': 8, 'reg__learning_rate': 0.01566730936256944, 'reg__l2_leaf_reg': 17.176655776108163, 'reg__bagging_temperature': 0.4427220251617203, 'reg__random_strength': 0.2942545728126472, 'reg__subsample': 0.7093048696498693, 'reg__min_data_in_leaf': 48, 'reg__leaf_estimation_iterations': 1}

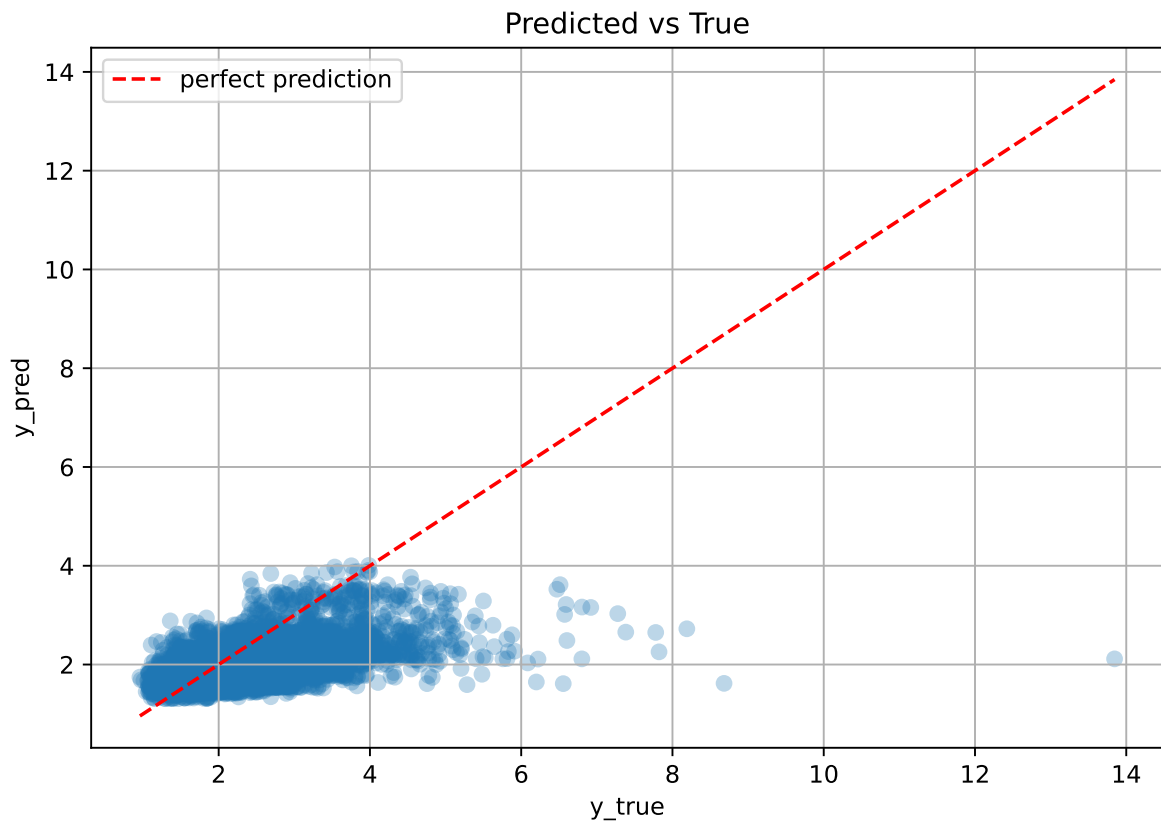
Rank 7: {'reg__iterations': 300, 'reg__depth': 7, 'reg__learning_rate': 0.01625172648192838, 'reg__l2_leaf_reg': 8.305487181507267, 'reg__bagging_temperature': 3.0994733058103066, 'reg__random_strength': 2.422465606543644, 'reg__subsample': 0.7875541181222127, 'reg__min_data_in_leaf': 46, 'reg__leaf_estimation_iterations': 1}

Rank 8: {'reg__iterations': 300, 'reg__depth': 8, 'reg__learning_rate': 0.021107562212317823, 'reg__l2_leaf_reg': 46.631740689946234, 'reg__bagging_temperature': 1.6300054935794905, 'reg__random_strength': 1.7528979085174456, 'reg__subsample': 0.8846137315143563, 'reg__min_data_in_leaf': 10, 'reg__leaf_estimation_iterations': 3}

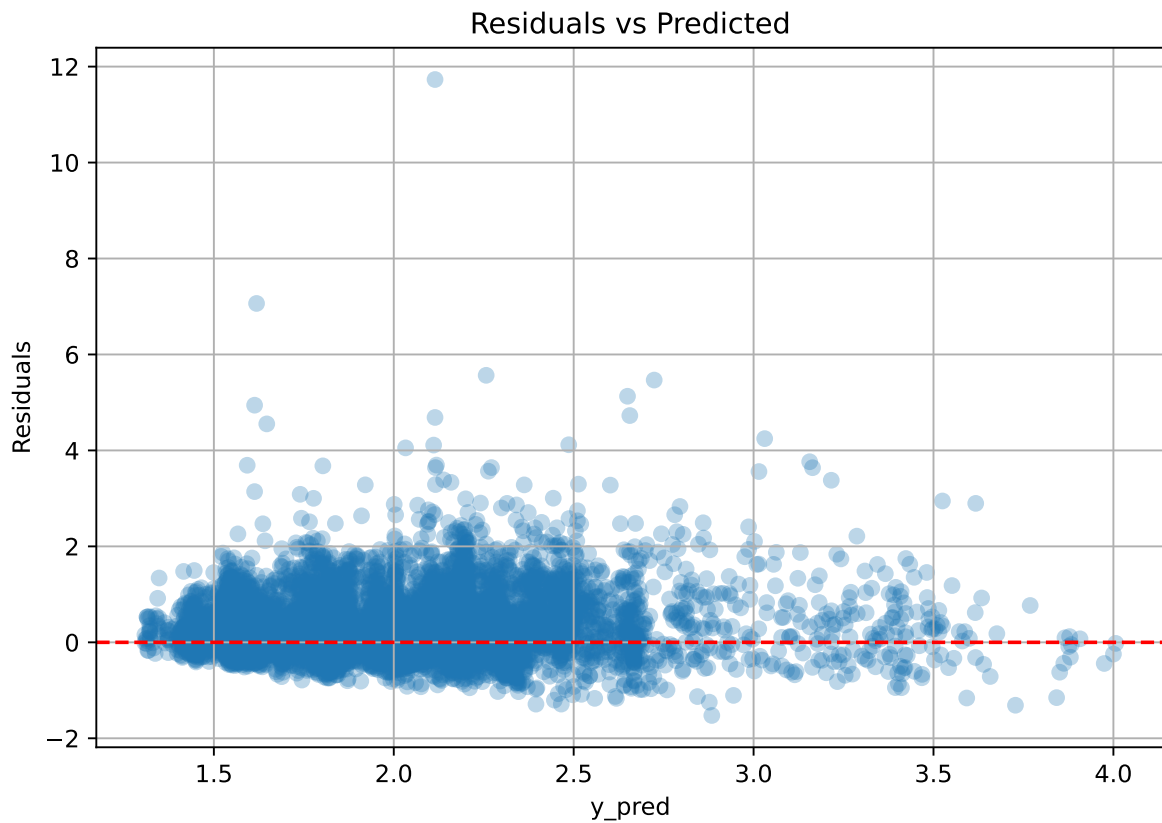
Rank 9: {'reg__iterations': 200, 'reg__depth': 4, 'reg__learning_rate': 0.046911005316887466, 'reg__l2_leaf_reg': 34.67997111998168, 'reg__bagging_temperature': 2.605677987977169, 'reg__random_strength': 2.3490827130820926, 'reg__subsample': 0.5454428709527755, 'reg__min_data_in_leaf': 43, 'reg__leaf_estimation_iterations': 8}

Rank 10: {'reg__iterations': 200, 'reg__depth': 4, 'reg__learning_rate': 0.0485466777458265, 'reg__l2_leaf_reg': 34.722165840593405, 'reg__bagging_temperature': 2.0317850929635846, 'reg__random_strength': 2.2483007059192825, 'reg__subsample': 0.5482343459122715, 'reg__min_data_in_leaf': 43, 'reg__leaf_estimation_iterations': 8}

2.5.10 Scatter



2.5.11 Scatter residuals



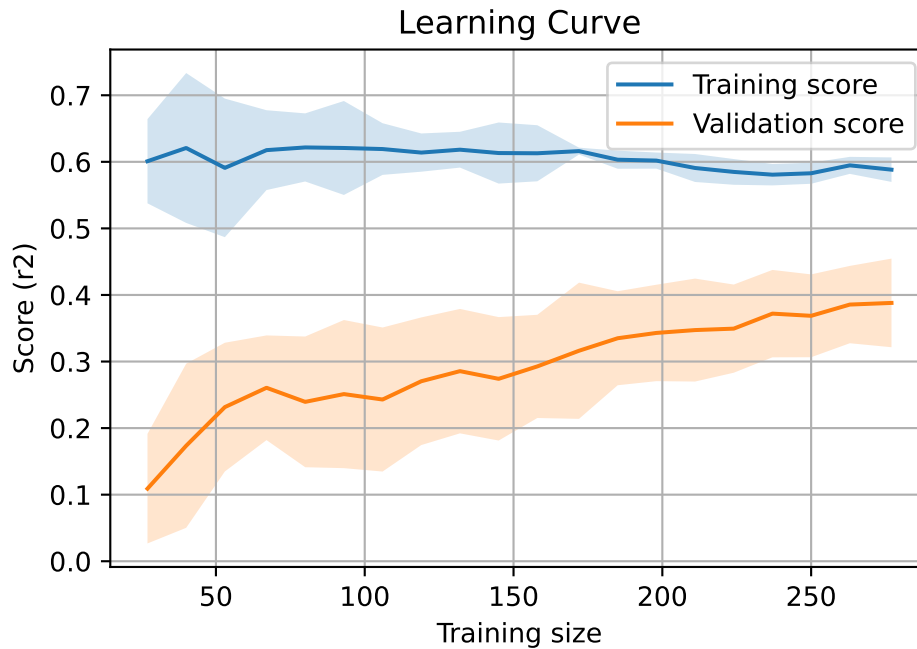
2.5.12 Learning curve — Group: 3_4

`Len(group)` : 568, `Len(training)` : 398,

`Len(training_real)` : 279, `Len(test_of_training)` : 119, `Len(test)` : 170

C:\Users\titou\anaconda3\Lib\site-packages\joblib\externals\loky\process_executor.py:782: UserWarning:

A worker stopped while some jobs were given to the executor. This can be caused by a too short timeout or by an excessive workload.

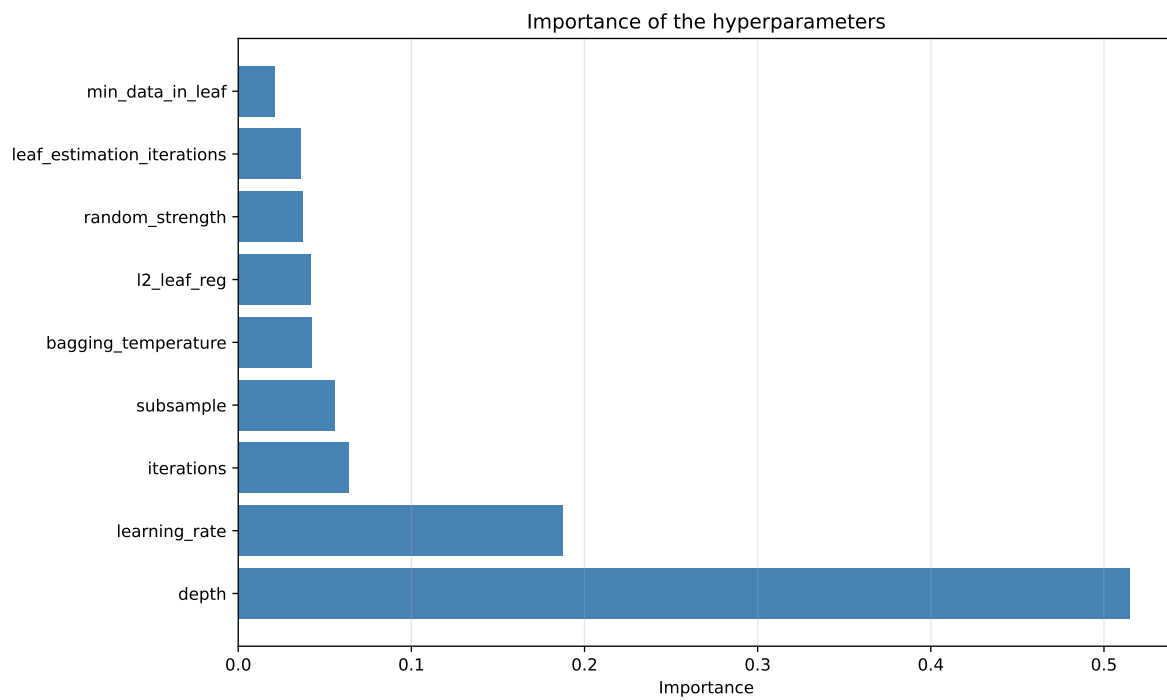
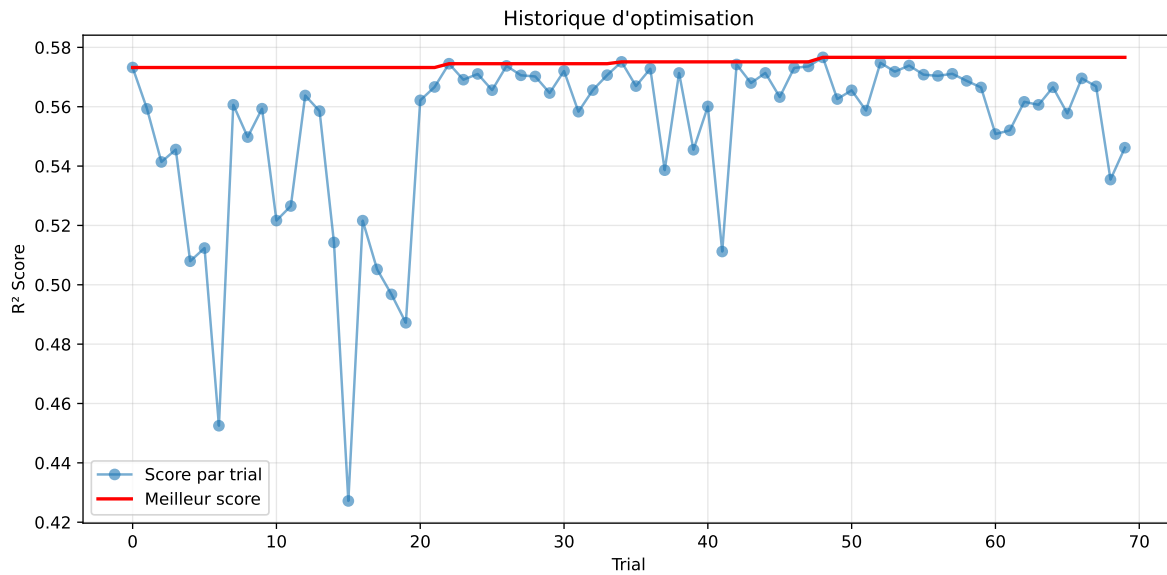


2.5.13 Gridsearch — Group: 4_5

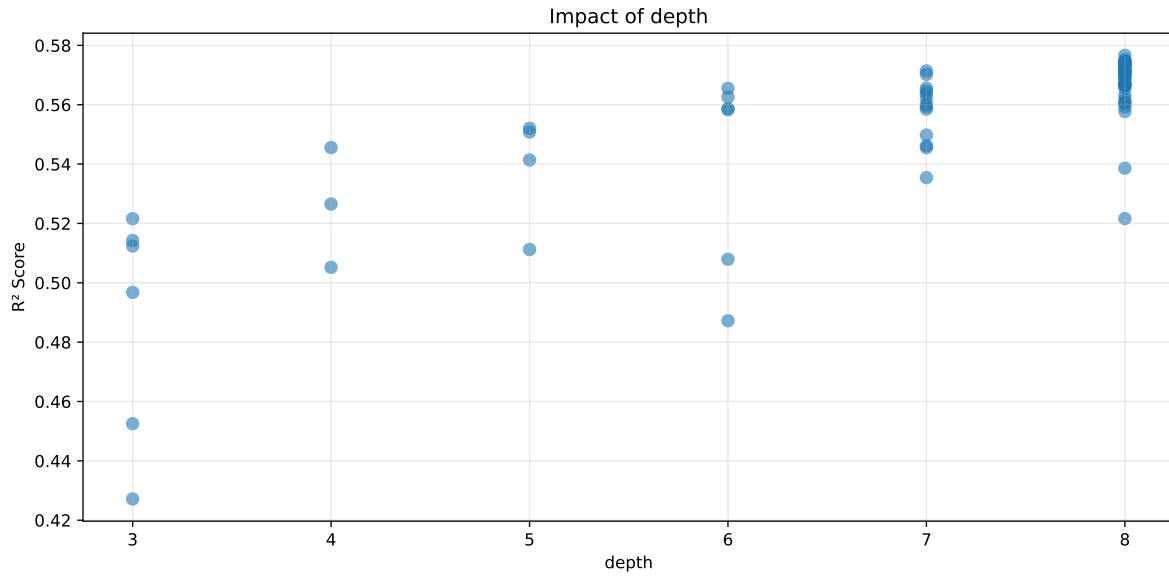
Len(group) : 2726, Len(training) : 1909, Len(test) : 817

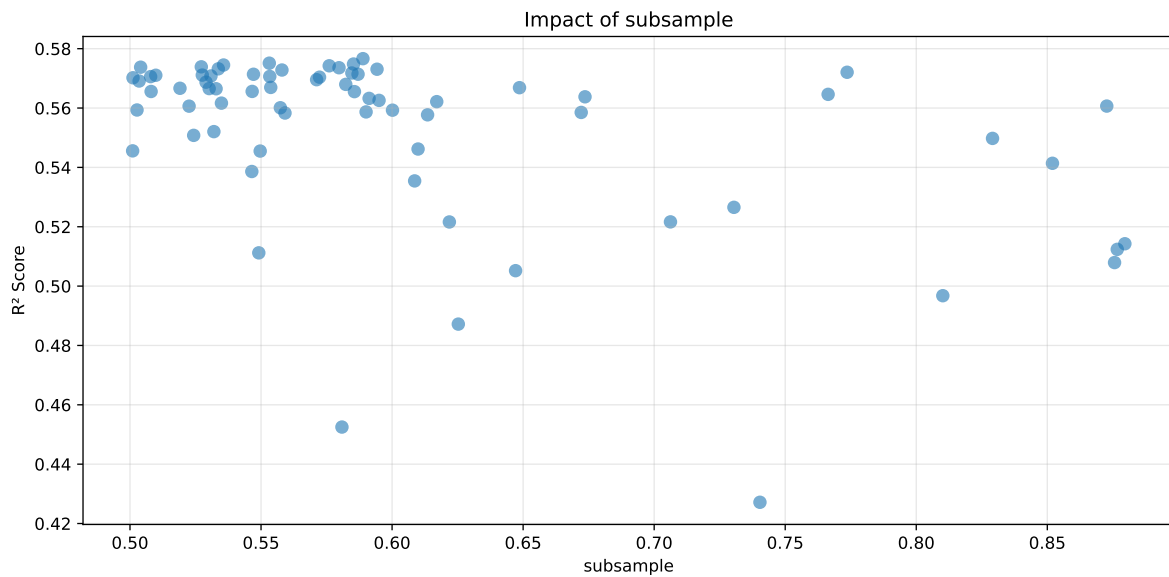
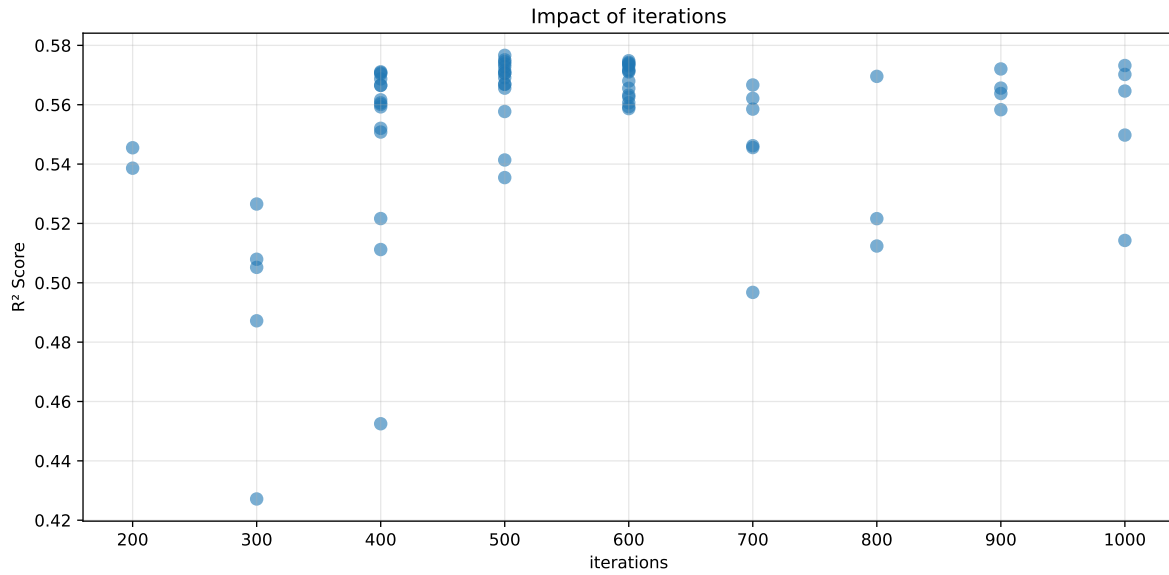
Distribution y_train vs y_test: y_train: mean=2.238, std=0.681, min=0.973, max=5.813
y_test: mean=2.212, std=0.659, min=1.071, max=4.858

===== TOP Importance FEATURES =====
feature importance r_umidity 66.923034 r_temperature 14.296470 r_NH3 7.234930 r_CO2
6.970768 r_THI 4.574798



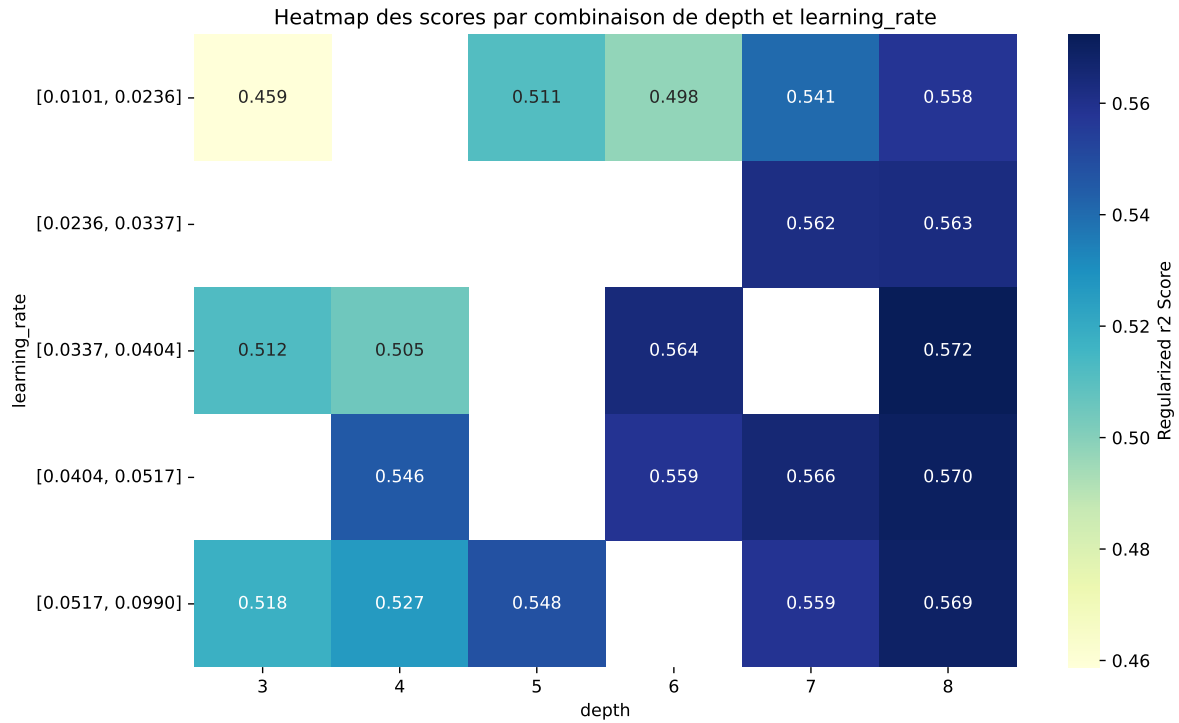
Paramètres avec >5% d'importance : ['depth', 'learning_rate', 'iterations', 'subsample']





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The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.4585	0.4272	0.4892	0.0206	0.333	0.4948	0.0677	1.1584	1.0735
2	0.4939	0.4525	0.5187	0.0241	0.3284	0.5227	0.0702	1.1551	1.0914
3	0.5508	0.4872	0.5496	0.0232	0.2964	0.5569	0.0697	1.1431	1.1306
4	0.5907	0.5079	0.5693	0.0232	0.2597	0.5777	0.0698	1.1373	1.1629
5	0.6015	0.5112	0.5722	0.0254	0.2404	0.5776	0.0664	1.1298	1.1766
6	0.5977	0.5052	0.5713	0.0253	0.2278	0.5738	0.0686	1.1359	1.183
7	0.6201	0.5216	0.5816	0.0213	0.2263	0.5925	0.0708	1.1358	1.1887
8	0.5923	0.4968	0.5637	0.025	0.2102	0.5735	0.0768	1.1545	1.1923
9	0.6533	0.5354	0.592	0.0197	0.1818	0.6014	0.066	1.1232	1.2201
10	0.6711	0.5386	0.5952	0.0188	0.1413	0.6008	0.0621	1.1154	1.2459
---	---	---	---	---	---	---	---	---	---
70	0.9473	0.5498	0.5988	0.0299	-0.643	0.6245	0.0747	1.1359	1.7232

Hyperparameters:

Rank 1: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.013842436384396823, 'reg__l2_leaf_reg': 33.24468183573967, 'reg__bagging_temperature': 0.9064211294270857, 'reg__random_strength': 4.007330629523922, 'reg__subsample': 0.7403573478270629, 'reg__min_data_in_leaf': 49, 'reg__leaf_estimation_iterations': 1}

Rank 2: {'reg__iterations': 400, 'reg__depth': 3, 'reg__learning_rate': 0.013503553830822267,

'reg__l2_leaf_reg': 20.333510844334945, 'reg__bagging_temperature': 3.5159875277697425,
'reg__random_strength': 4.7754569085753324, 'reg__subsample': 0.5808693722506627,
'reg__min_data_in_leaf': 10, 'reg__leaf_estimation_iterations': 2}

Rank 3: {'reg__iterations': 300, 'reg__depth': 6, 'reg__learning_rate': 0.012222296209279027,
'reg__l2_leaf_reg': 17.19238404532058, 'reg__bagging_temperature': 1.9561071861153134,
'reg__random_strength': 3.0351450393179, 'reg__subsample': 0.6252733675980918,
'reg__min_data_in_leaf': 39, 'reg__leaf_estimation_iterations': 3}

Rank 4: {'reg__iterations': 300, 'reg__depth': 6, 'reg__learning_rate': 0.02243172178077319,
'reg__l2_leaf_reg': 36.74539008785682, 'reg__bagging_temperature': 4.33489606095356,
'reg__random_strength': 3.7429685556959607, 'reg__subsample': 0.8756607966394068,
'reg__min_data_in_leaf': 22, 'reg__leaf_estimation_iterations': 5}

Rank 5: {'reg__iterations': 400, 'reg__depth': 5, 'reg__learning_rate': 0.017428886531279345,
'reg__l2_leaf_reg': 13.585776048007897, 'reg__bagging_temperature': 0.6184745565097535,
'reg__random_strength': 2.176962119936025, 'reg__subsample': 0.5491421882101872,
'reg__min_data_in_leaf': 35, 'reg__leaf_estimation_iterations': 6}

Rank 6: {'reg__iterations': 300, 'reg__depth': 4, 'reg__learning_rate': 0.03567002870044463,
'reg__l2_leaf_reg': 22.708659838151316, 'reg__bagging_temperature': 0.9900670699236769,
'reg__random_strength': 4.846353617857982, 'reg__subsample': 0.6471220192266836,
'reg__min_data_in_leaf': 11, 'reg__leaf_estimation_iterations': 9}

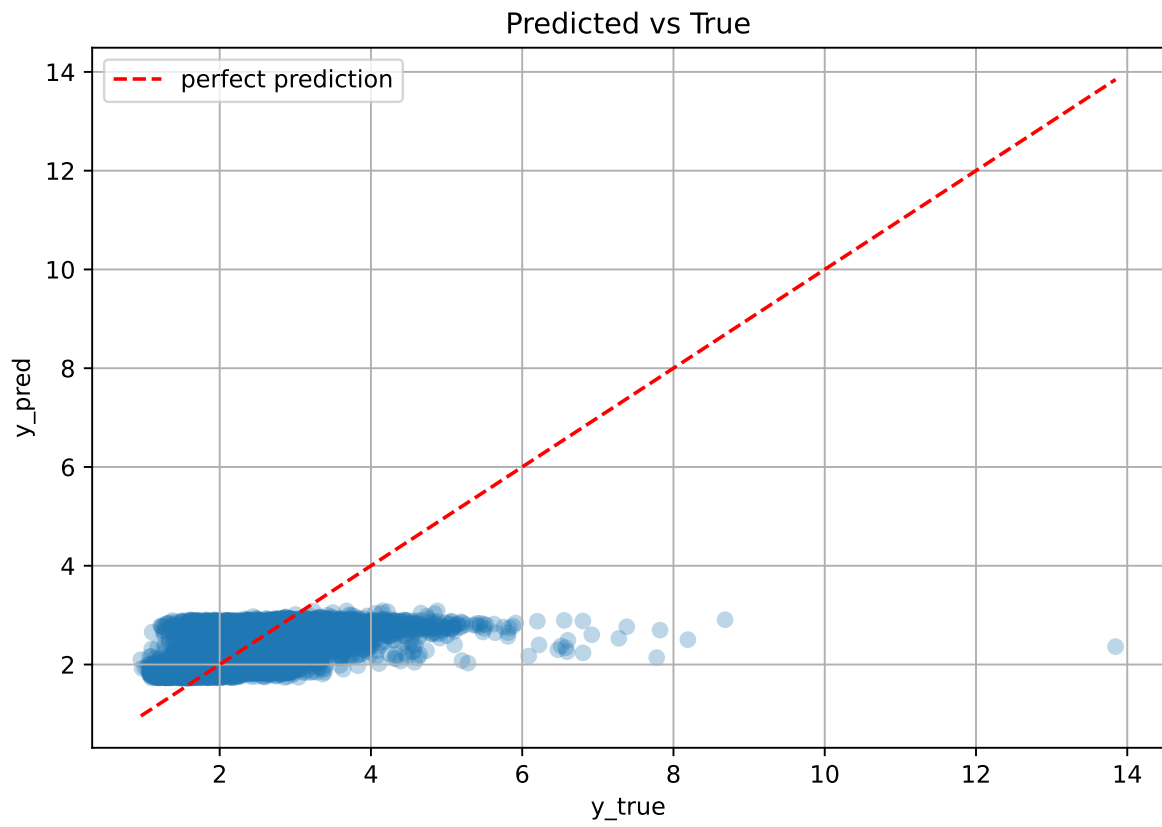
Rank 7: {'reg__iterations': 400, 'reg__depth': 8, 'reg__learning_rate': 0.014617735368505616,
'reg__l2_leaf_reg': 39.90733697556197, 'reg__bagging_temperature': 3.675242214112924,
'reg__random_strength': 3.0920171714060247, 'reg__subsample': 0.7062187839475358,
'reg__min_data_in_leaf': 19, 'reg__leaf_estimation_iterations': 5}

Rank 8: {'reg__iterations': 700, 'reg__depth': 3, 'reg__learning_rate': 0.020741861613994102,
'reg__l2_leaf_reg': 26.391748115342565, 'reg__bagging_temperature': 4.805517262811605,
'reg__random_strength': 3.7305405173238433, 'reg__subsample': 0.8101337714002468,
'reg__min_data_in_leaf': 11, 'reg__leaf_estimation_iterations': 6}

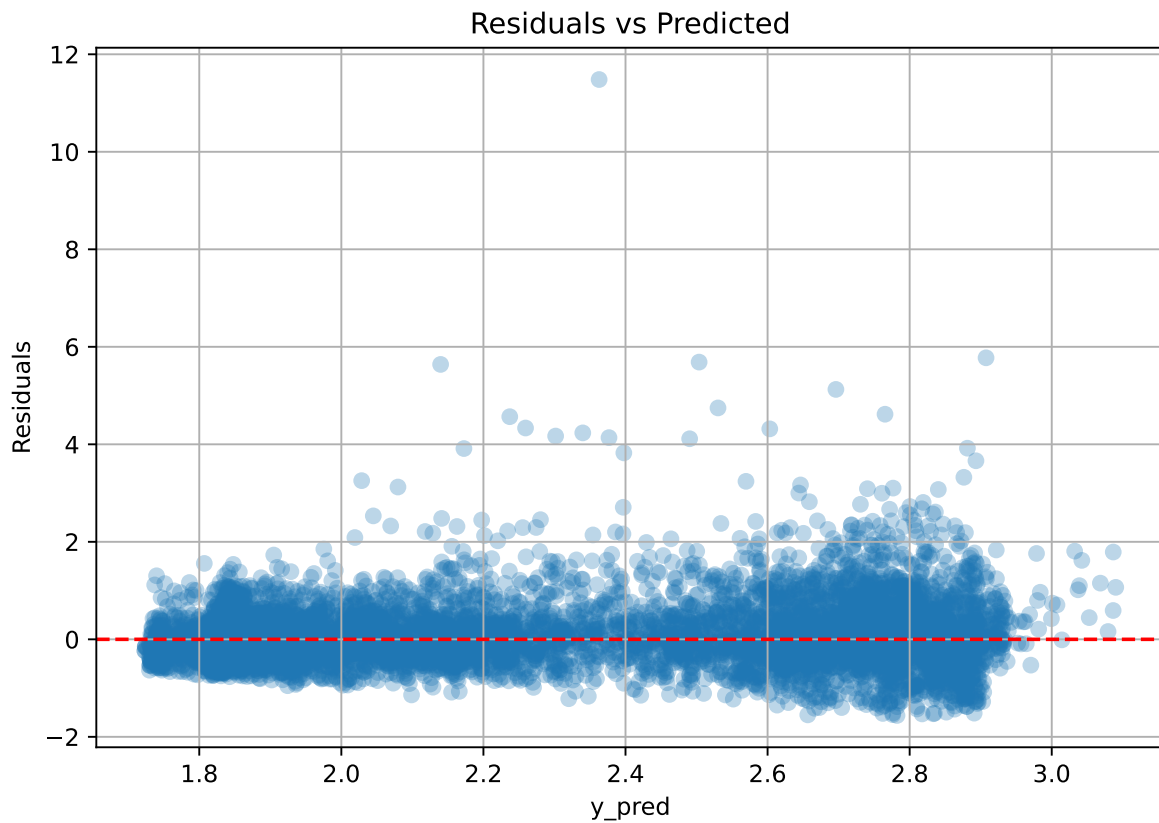
Rank 9: {'reg__iterations': 500, 'reg__depth': 7, 'reg__learning_rate': 0.010364042779103591,
'reg__l2_leaf_reg': 6.09471017377006, 'reg__bagging_temperature': 0.8813098816645939,
'reg__random_strength': 1.4328025946800773, 'reg__subsample': 0.6086101489292756,
'reg__min_data_in_leaf': 29, 'reg__leaf_estimation_iterations': 7}

Rank 10: {'reg__iterations': 200, 'reg__depth': 8, 'reg__learning_rate': 0.030003323064111256,
'reg__l2_leaf_reg': 9.428487745296799, 'reg__bagging_temperature': 0.68393167269498,
'reg__random_strength': 2.3005612395433275, 'reg__subsample': 0.5464813089540255,
'reg__min_data_in_leaf': 34, 'reg__leaf_estimation_iterations': 9}

2.5.14 Scatter



2.5.15 Scatter residuals



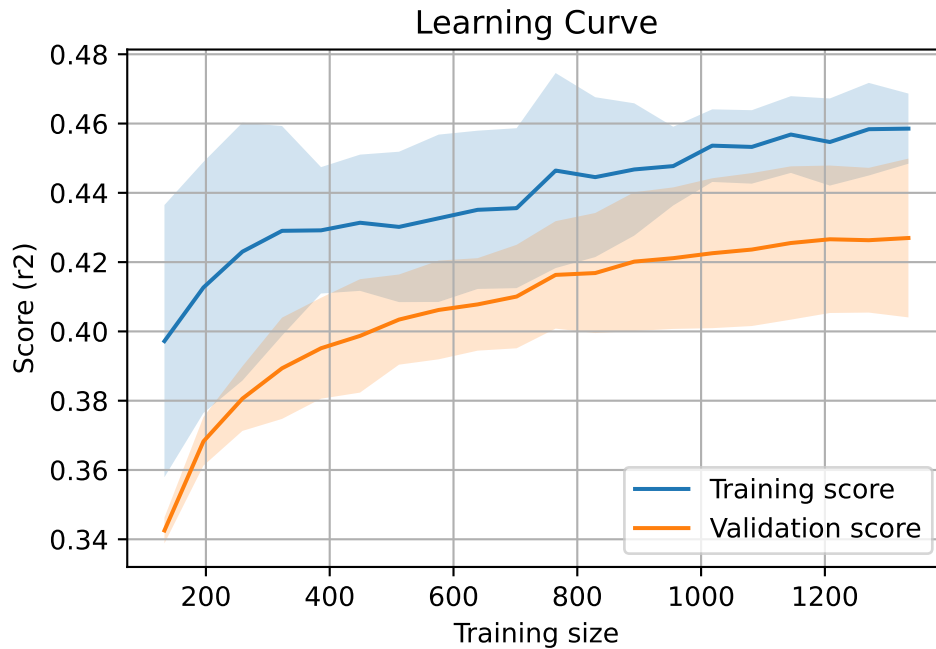
2.5.16 Learning curve — Group: 4_5

`Len(group)` : 2726, `Len(training)` : 1909,

`Len(training_real)` : 1337, `Len(test_of_training)` : 572, `Len(test)` : 817

C:\Users\titou\anaconda3\Lib\site-packages\joblib\externals\loky\process_executor.py:782: Use

A worker stopped while some jobs were given to the executor. This can be caused by a too sho

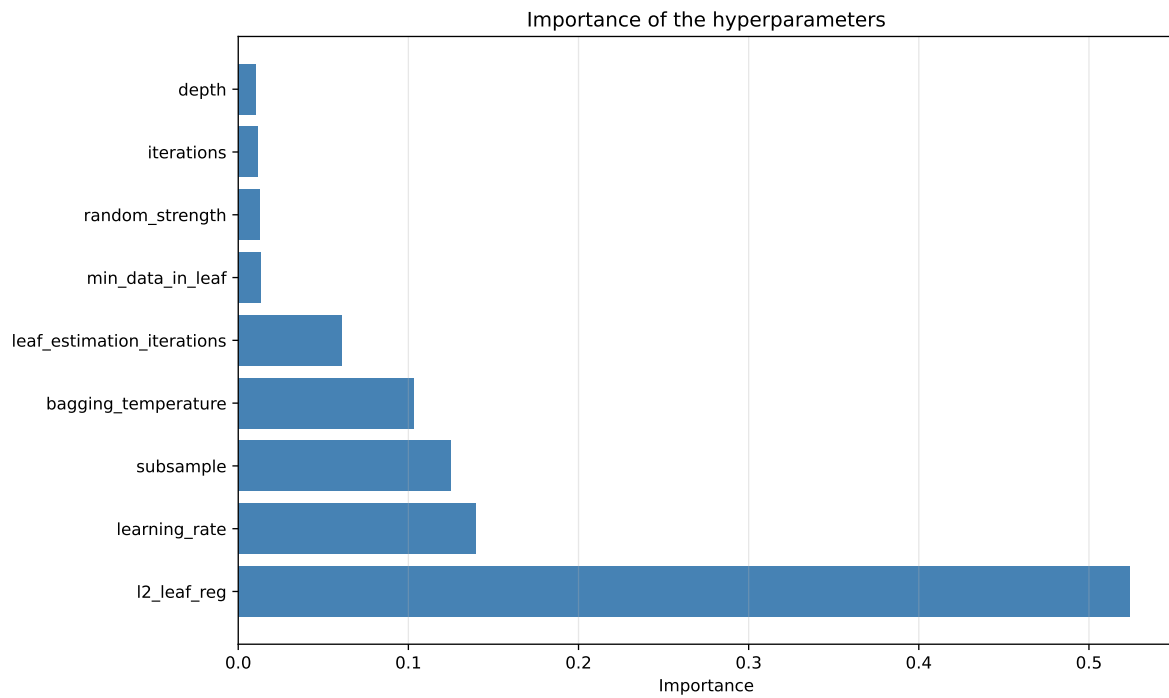
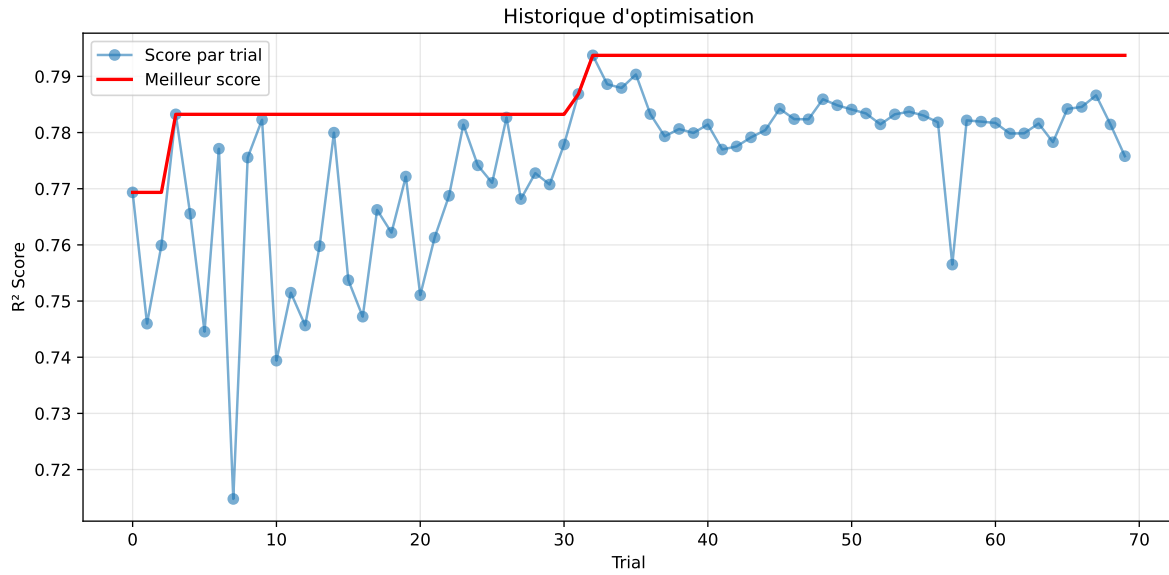


2.5.17 Gridsearch — Group: 6_3

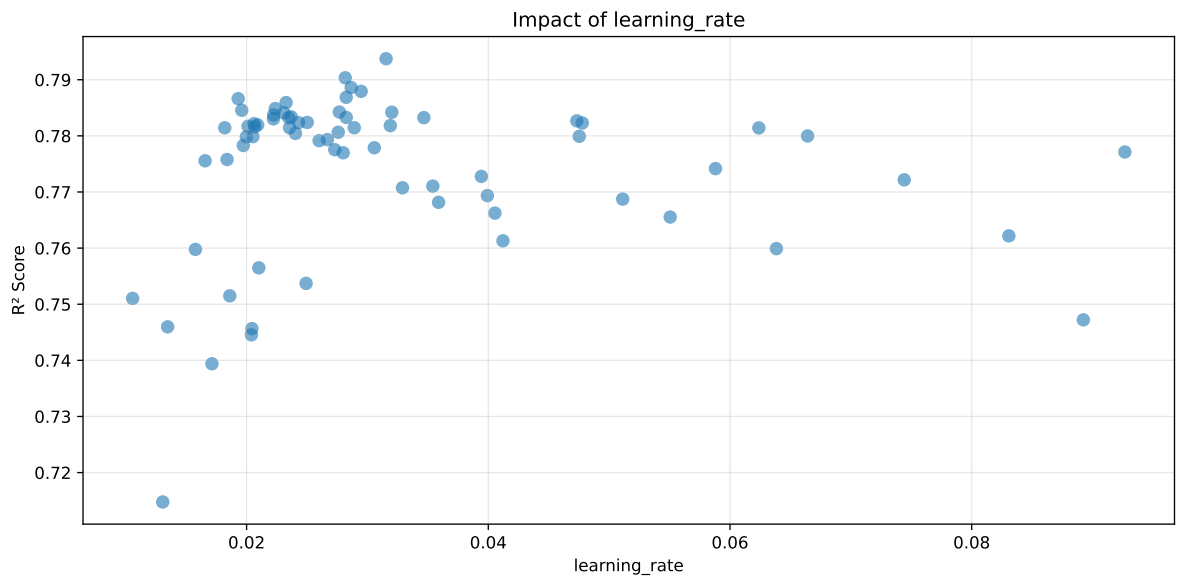
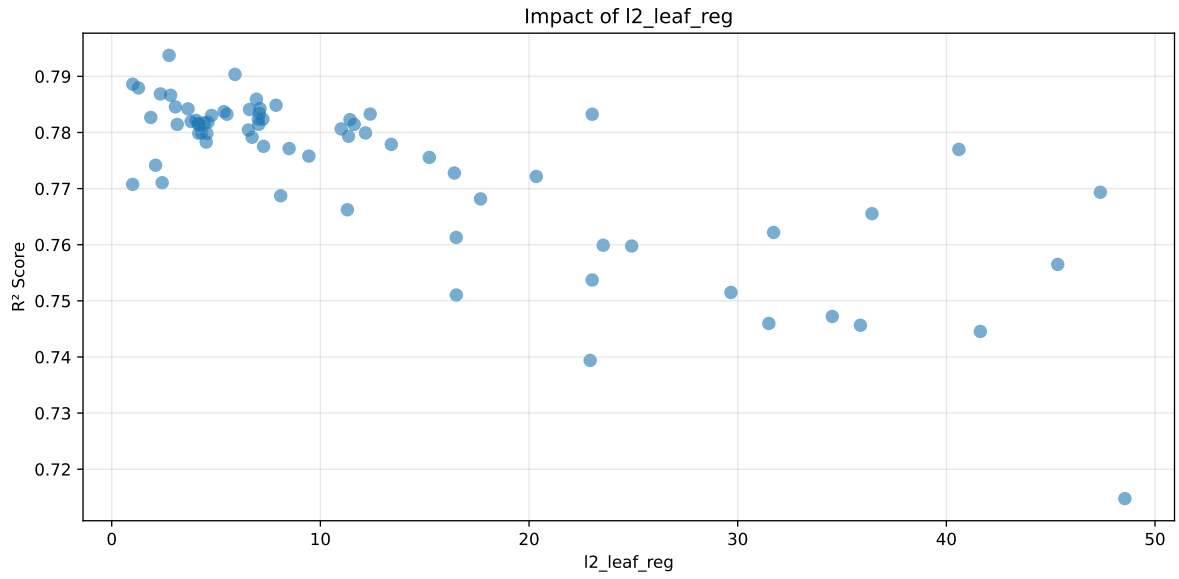
Len(group) : 561, Len(training) : 393, Len(test) : 168

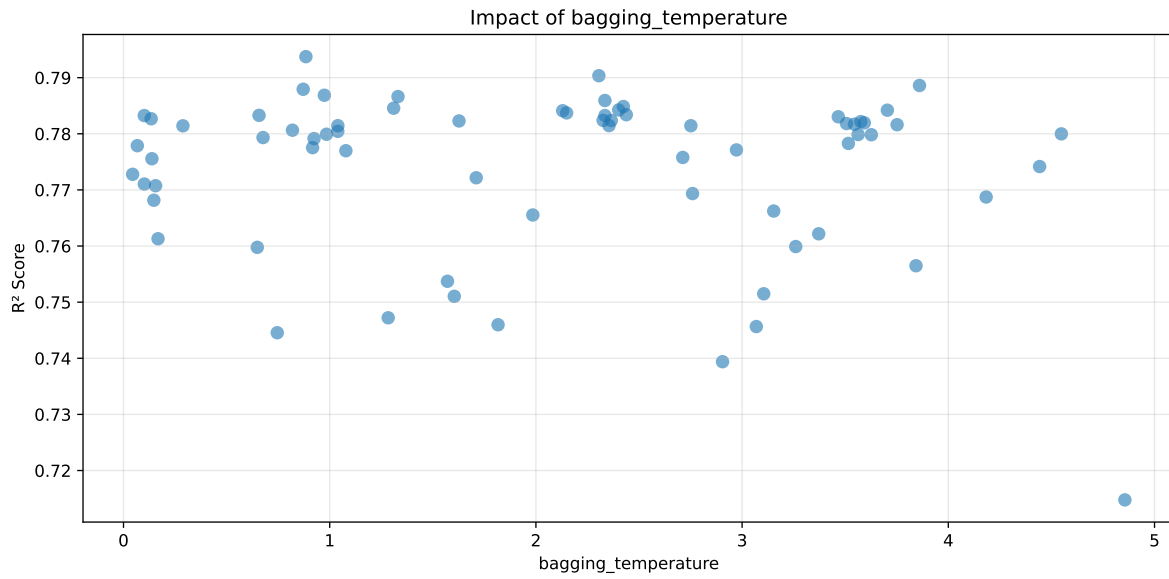
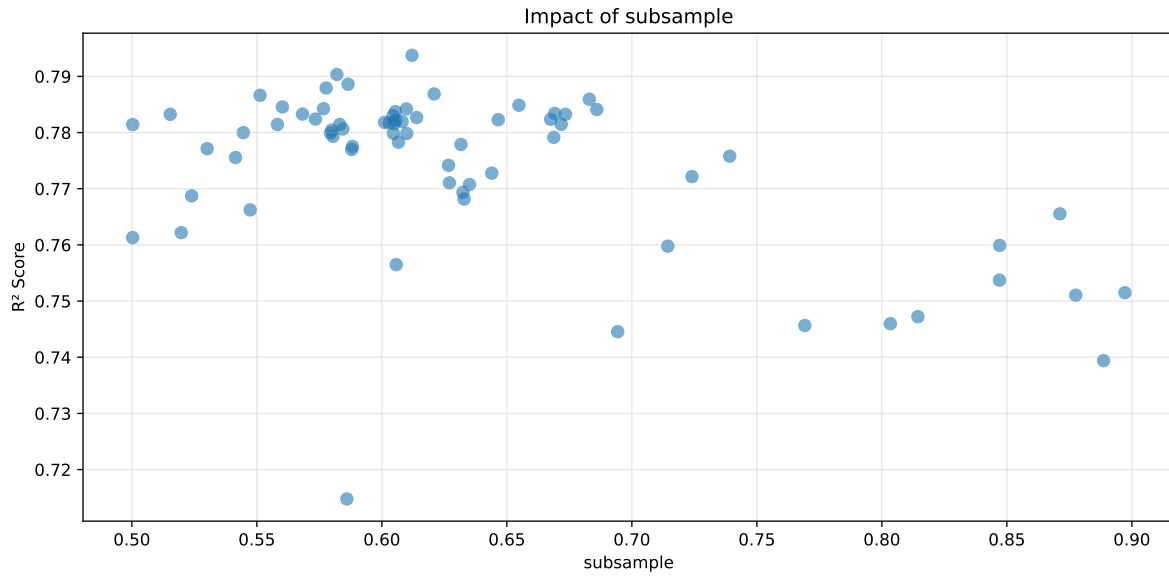
Distribution y_train vs y_test: y_train: mean=1.935, std=0.720, min=1.041, max=6.086
y_test: mean=2.031, std=0.744, min=1.078, max=4.912

===== TOP Importance FEATURES =====
feature importance r_umidity 38.465368 r_NH3 30.870429 r_THI 12.773737 r_temperature
12.573820 r_CO2 5.316647



Paramètres avec >5% d'importance : ['l2_leaf_reg', 'learning_rate', 'subsample', 'bagging_temperature', 'leaf_estimation_iterations']





rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.7707	0.7148	0.7027	0.0615	0.5469	0.7256	0.0109	1.0152	1.0783
2	0.8473	0.7457	0.7623	0.057	0.4406	0.7856	0.0399	1.0535	1.1364
3	0.8684	0.751	0.7646	0.0602	0.399	0.7844	0.0333	1.0444	1.1562
4	0.865	0.7445	0.7615	0.0576	0.3832	0.7902	0.0457	1.0614	1.1618
5	0.889	0.7537	0.7762	0.0573	0.3479	0.7957	0.0419	1.0556	1.1795
6	0.9026	0.7565	0.7783	0.0603	0.3182	0.8104	0.0539	1.0712	1.1931
7	0.8809	0.7394	0.7522	0.0602	0.3147	0.7895	0.0501	1.0678	1.1914
8	0.8972	0.7515	0.7837	0.0565	0.3144	0.8058	0.0543	1.0723	1.1939
9	0.8908	0.746	0.7618	0.0596	0.3114	0.7867	0.0408	1.0546	1.1942
10	0.9508	0.7798	0.8056	0.0568	0.2671	0.8231	0.0433	1.0555	1.2192
---	---	---	---	---	---	---	---	---	---
70	0.9992	0.7599	0.7849	0.0547	0.0419	0.8135	0.0536	1.0705	1.315

Hyperparameters:

Rank 1: {'reg__iterations': 600, 'reg__depth': 5, 'reg__learning_rate': 0.013061033955601441, 'reg__l2_leaf_reg': 48.555722168749746, 'reg__bagging_temperature': 4.856217349128799, 'reg__random_strength': 3.951730436613652, 'reg__subsample': 0.5859816201066739, 'reg__min_data_in_leaf': 30, 'reg__leaf_estimation_iterations': 3}

Rank 2: {'reg__iterations': 500, 'reg__depth': 4, 'reg__learning_rate': 0.02044206270957214, 'reg__l2_leaf_reg': 35.882183869086745, 'reg__bagging_temperature': 3.068887574489776, 'reg__random_strength': 2.9329427255280276, 'reg__subsample': 0.7691752486647574, 'reg__min_data_in_leaf': 45, 'reg__leaf_estimation_iterations': 5}

Rank 3: {'reg__iterations': 1000, 'reg__depth': 5, 'reg__learning_rate': 0.010569907787031315, 'reg__l2_leaf_reg': 16.51643218579123, 'reg__bagging_temperature': 1.60411936608292, 'reg__random_strength': 2.306523535102692, 'reg__subsample': 0.8775600396451624, 'reg__min_data_in_leaf': 10, 'reg__leaf_estimation_iterations': 2}

Rank 4: {'reg__iterations': 600, 'reg__depth': 3, 'reg__learning_rate': 0.020400208857813126, 'reg__l2_leaf_reg': 41.6281663774313, 'reg__bagging_temperature': 0.7456286488527925, 'reg__random_strength': 0.7711303203597092, 'reg__subsample': 0.6943361710159892, 'reg__min_data_in_leaf': 22, 'reg__leaf_estimation_iterations': 6}

Rank 5: {'reg__iterations': 500, 'reg__depth': 3, 'reg__learning_rate': 0.024924534840834286, 'reg__l2_leaf_reg': 23.02596412990948, 'reg__bagging_temperature': 1.5710643636973027, 'reg__random_strength': 1.8901579847367744, 'reg__subsample': 0.8470431739183464, 'reg__min_data_in_leaf': 33, 'reg__leaf_estimation_iterations': 8}

Rank 6: {'reg__iterations': 600, 'reg__depth': 6, 'reg__learning_rate': 0.021009307786336178, 'reg__l2_leaf_reg': 45.33763769168348, 'reg__bagging_temperature': 3.8433271973099266, 'reg__random_strength': 4.964562518269427, 'reg__subsample': 0.6057721616985547, 'reg__min_data_in_leaf': 32, 'reg__leaf_estimation_iterations': 6}

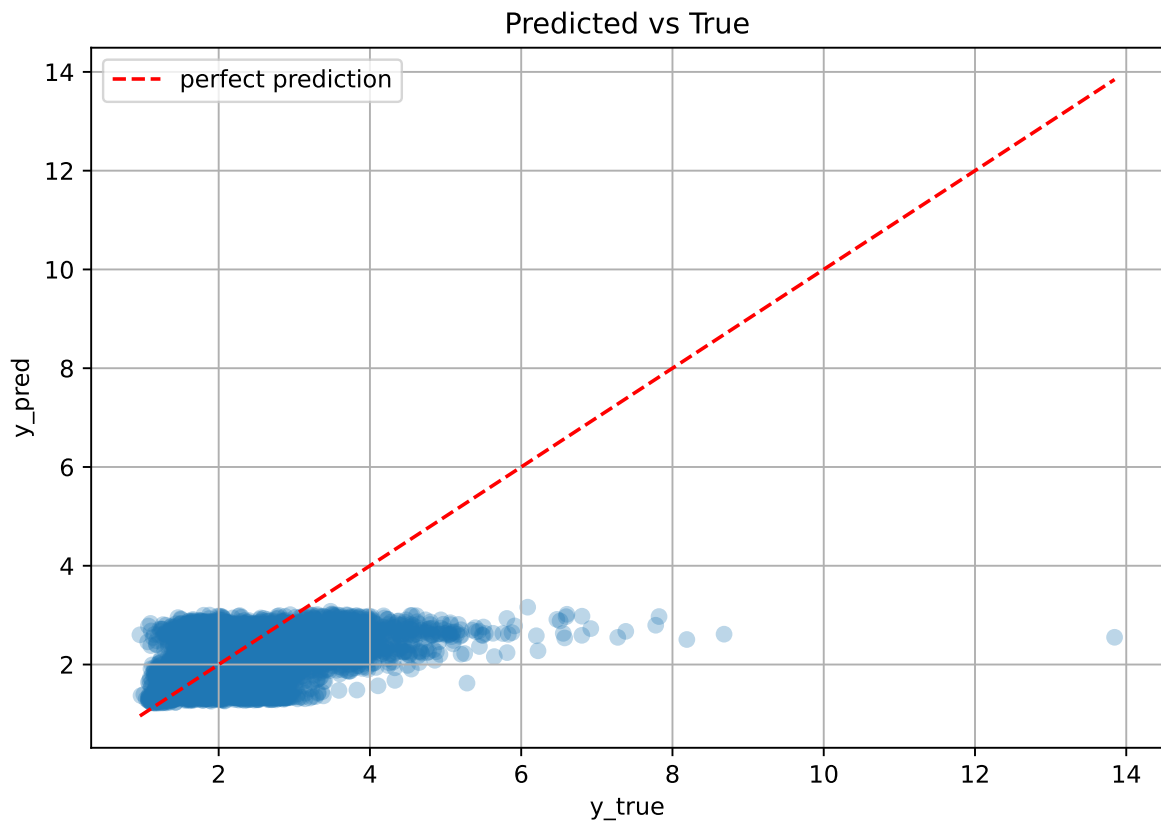
Rank 7: {'reg__iterations': 900, 'reg__depth': 8, 'reg__learning_rate': 0.017131511781344092, 'reg__l2_leaf_reg': 22.929062413138244, 'reg__bagging_temperature': 2.9050626612510704, 'reg__random_strength': 1.2924665161710402, 'reg__subsample': 0.8886653360858625, 'reg__min_data_in_leaf': 40, 'reg__leaf_estimation_iterations': 1}

Rank 8: {'reg__iterations': 800, 'reg__depth': 4, 'reg__learning_rate': 0.018609530639332537, 'reg__l2_leaf_reg': 29.67550853236998, 'reg__bagging_temperature': 3.104866953006973, 'reg__random_strength': 4.708544211319628, 'reg__subsample': 0.8972022855227555, 'reg__min_data_in_leaf': 13, 'reg__leaf_estimation_iterations': 5}

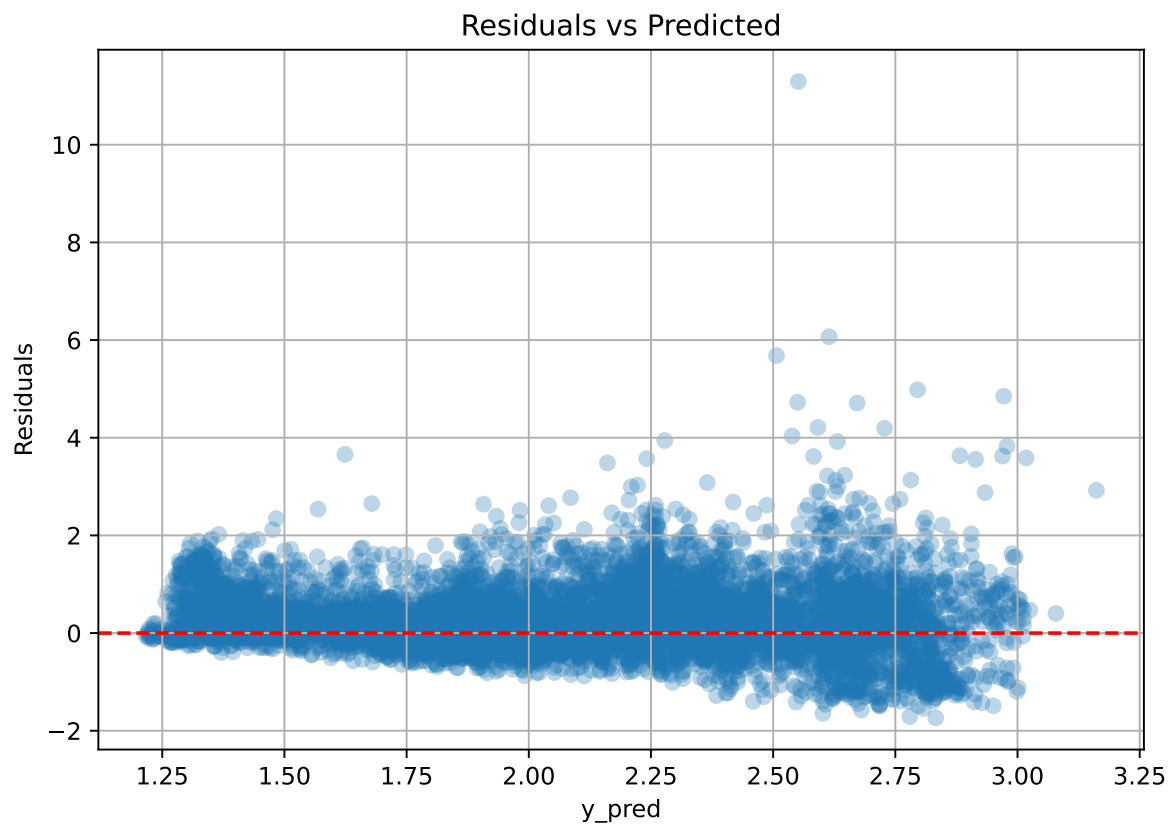
Rank 9: {'reg__iterations': 500, 'reg__depth': 8, 'reg__learning_rate': 0.013467191479435496, 'reg__l2_leaf_reg': 31.49161674218752, 'reg__bagging_temperature': 1.8164842492958906, 'reg__random_strength': 3.4452768681865, 'reg__subsample': 0.8033943888487963, 'reg__min_data_in_leaf': 44, 'reg__leaf_estimation_iterations': 8}

Rank 10: {'reg__iterations': 600, 'reg__depth': 5, 'reg__learning_rate': 0.019977660460264536, 'reg__l2_leaf_reg': 4.560191154671733, 'reg__bagging_temperature': 3.6266501171043597, 'reg__random_strength': 4.851759206756966, 'reg__subsample': 0.6099832764626769, 'reg__min_data_in_leaf': 31, 'reg__leaf_estimation_iterations': 6}

2.5.18 Scatter



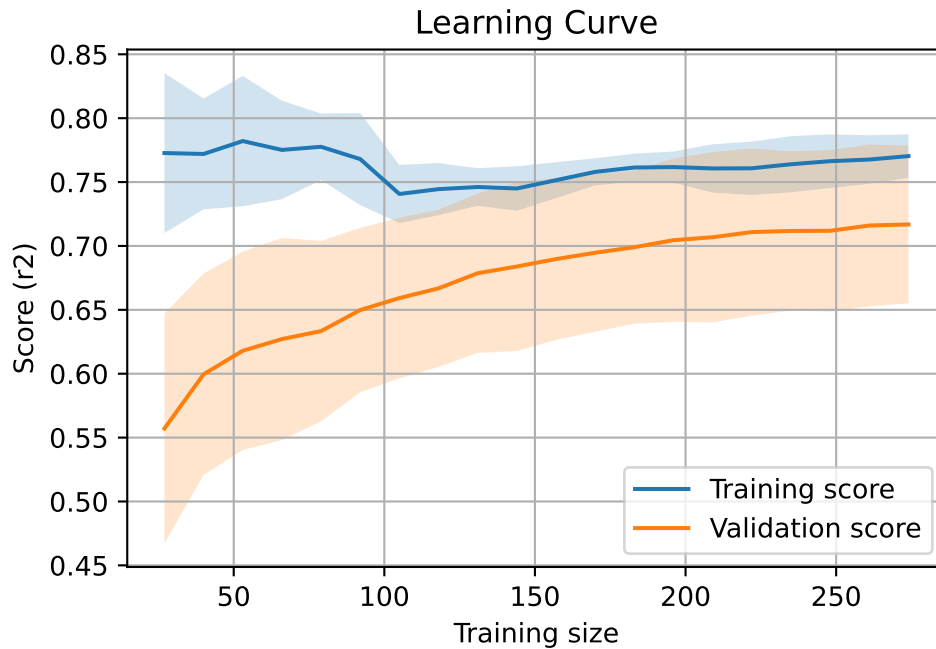
2.5.19 Scatter residuals



2.5.20 Learning curve — Group: 6_3

`Len(group)` : 561, `Len(training)` : 393,

`Len(training_real)` : 276, `Len(test_of_training)` : 117, `Len(test)` : 168

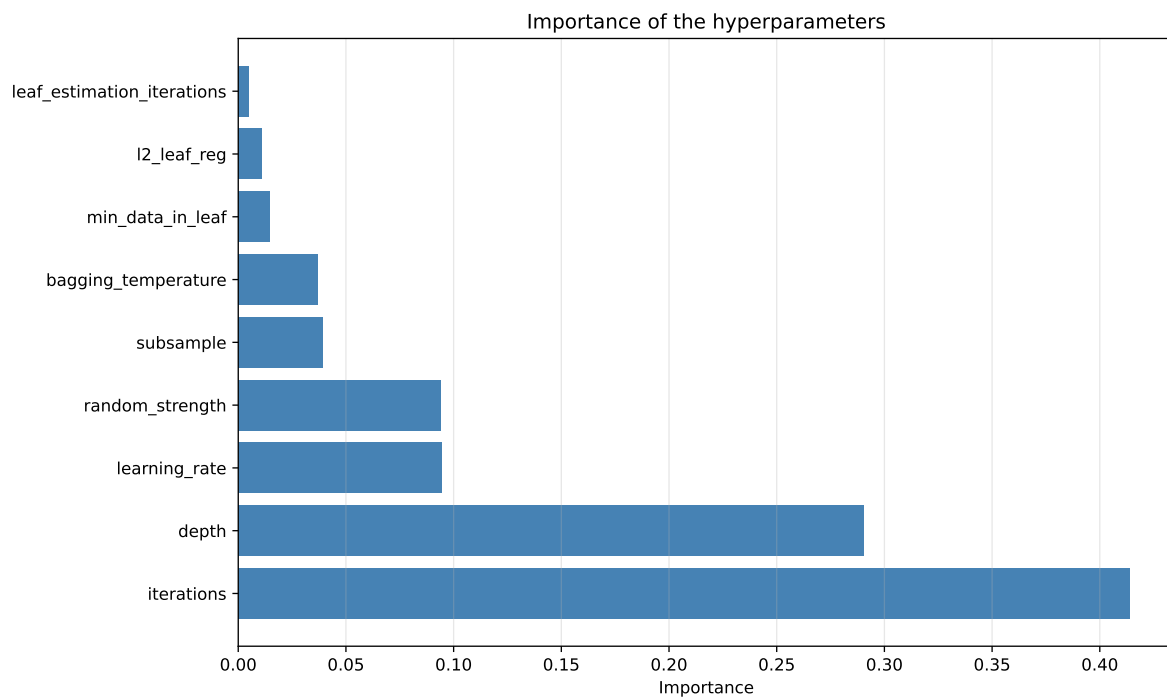
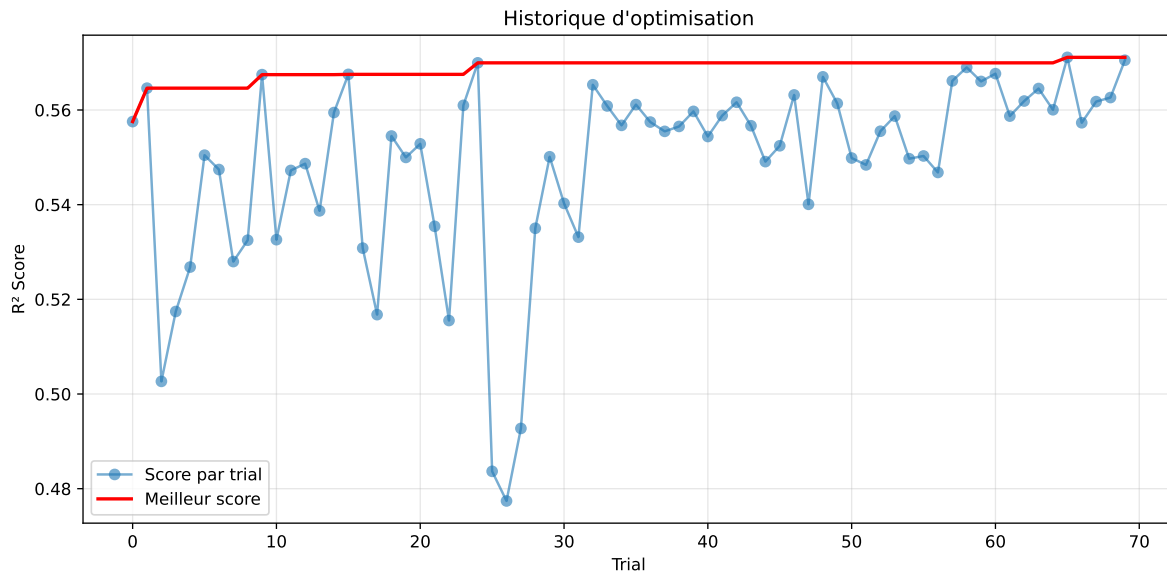


2.5.21 Gridsearch — Group: 6_4

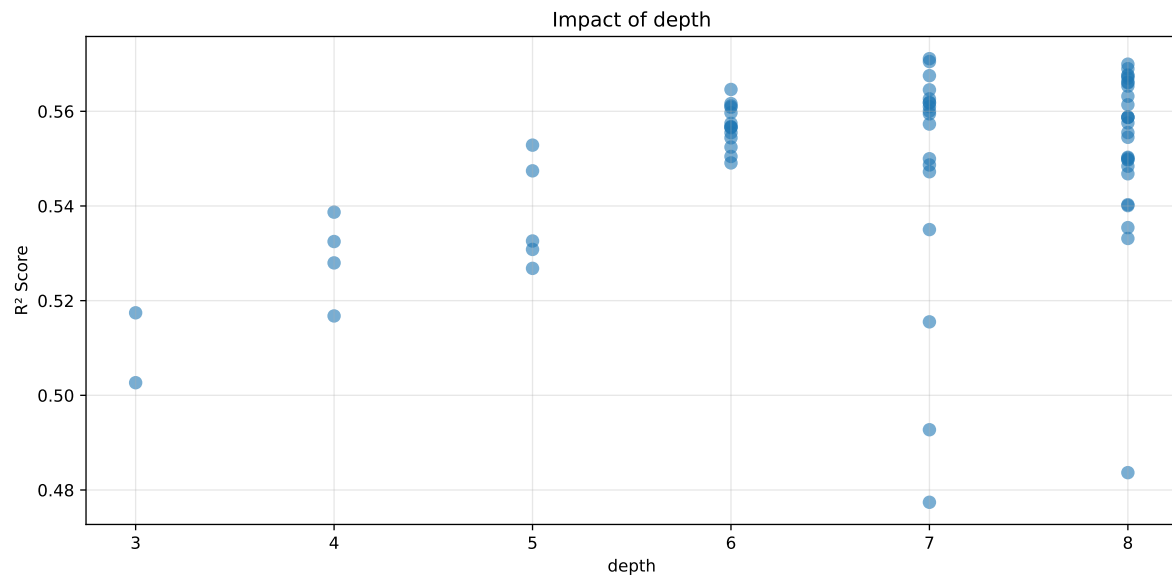
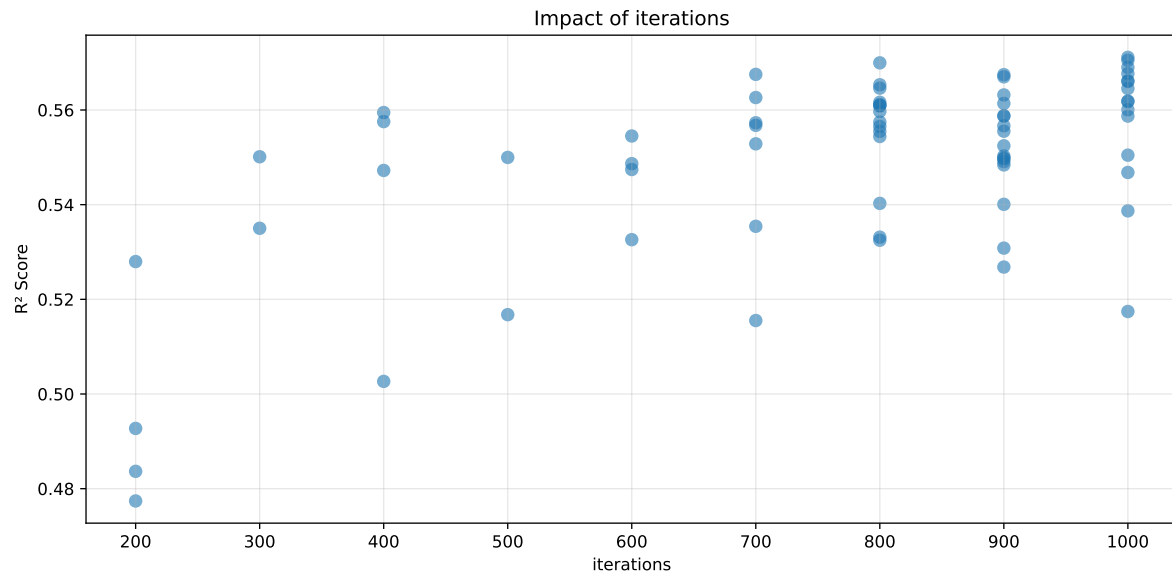
Len(group) : 792, Len(training) : 555, Len(test) : 237

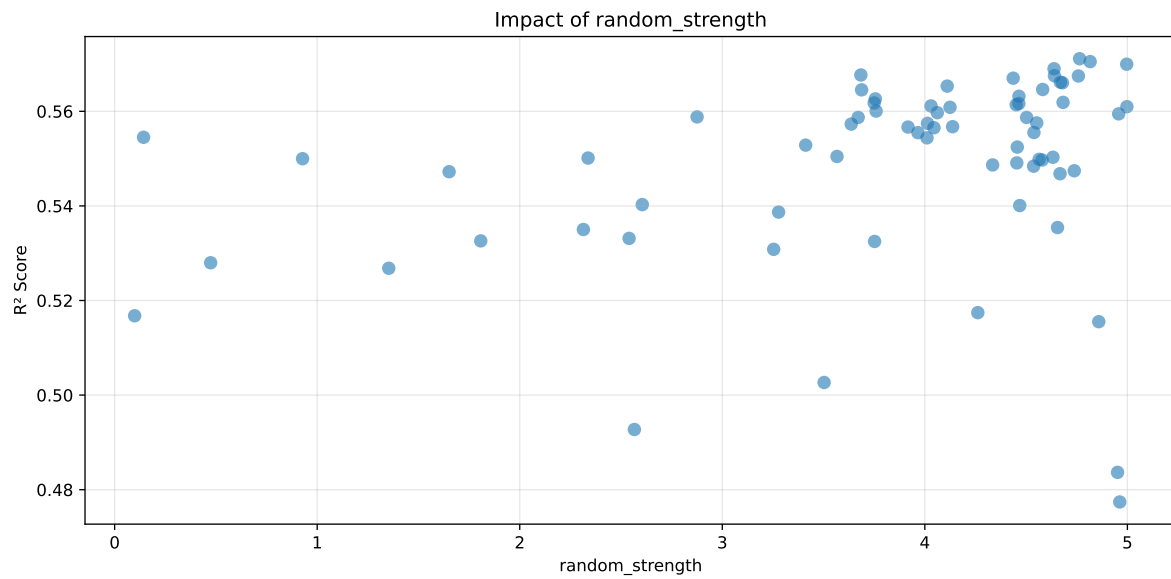
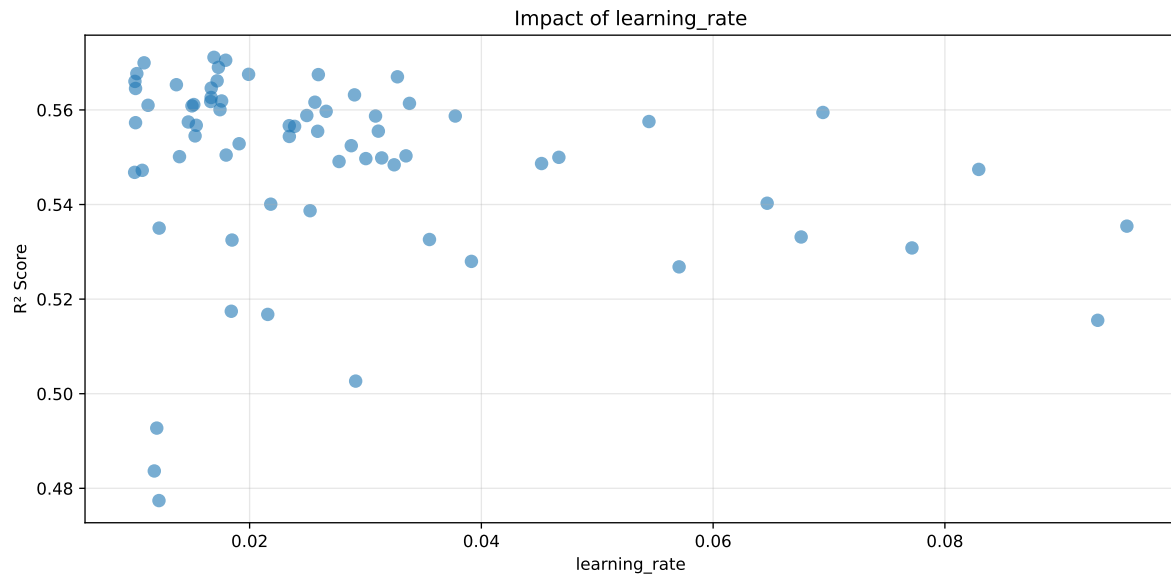
Distribution y_train vs y_test: y_train: mean=2.486, std=0.782, min=1.253, max=7.383
y_test: mean=2.410, std=0.685, min=1.398, max=5.144

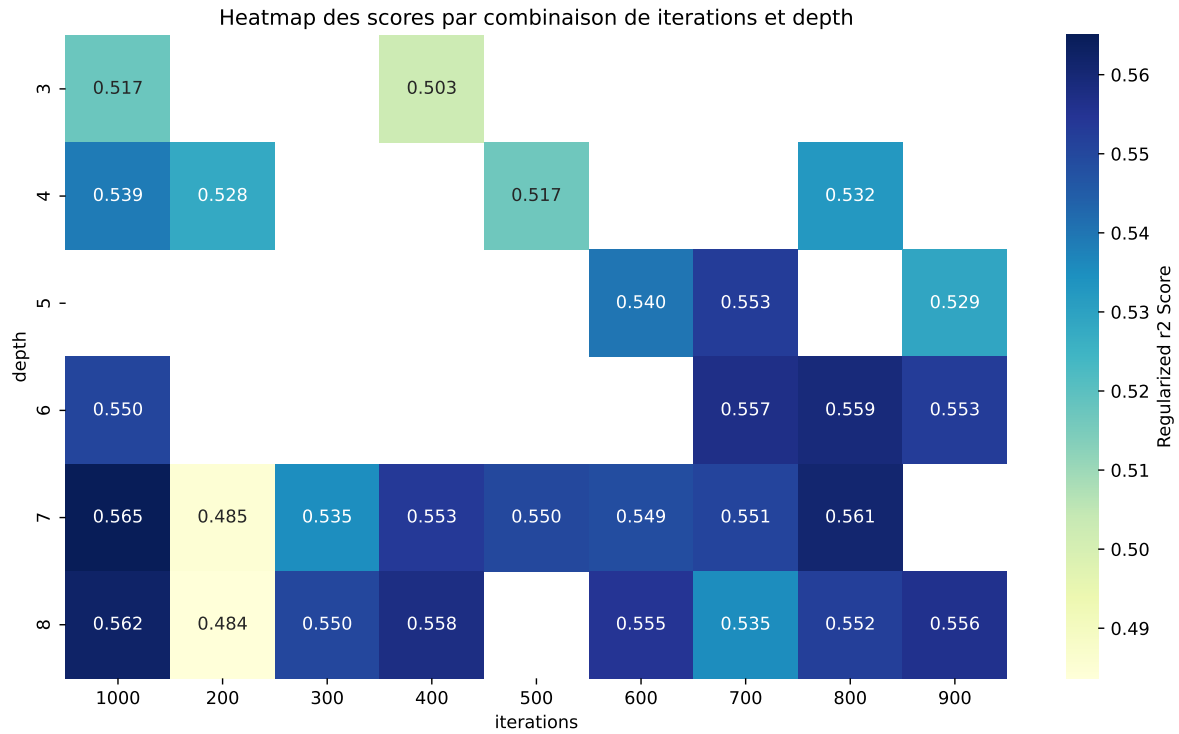
===== TOP Importance FEATURES =====
feature importance r_THI 32.214400 r_NH3 23.037453 r_umidity 19.587706 r_CO2 14.306071
r_temperature 10.854371



Paramètres avec >5% d'importance : ['iterations', 'depth', 'learning_rate', 'random_strength']







rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.5828	0.4774	0.5635	0.0118	0.1614	0.592	0.1146	1.24	1.2207
2	0.601	0.4837	0.5674	0.0166	0.1317	0.5998	0.1161	1.24	1.2425
3	0.6144	0.4927	0.5824	0.0183	0.1279	0.6022	0.1094	1.2221	1.2468
4	0.6936	0.535	0.6333	0.03	0.0591	0.6542	0.1192	1.2228	1.2965
5	0.7492	0.5468	0.6422	0.0282	-0.0605	0.6713	0.1245	1.2276	1.3702
6	0.7551	0.5472	0.6493	0.0354	-0.0765	0.6734	0.1261	1.2305	1.3799
7	0.7385	0.5325	0.6389	0.0416	-0.0854	0.6608	0.1284	1.2411	1.3868
8	0.7799	0.5573	0.6558	0.027	-0.1106	0.684	0.1267	1.2273	1.3995
9	0.767	0.5401	0.6276	0.0325	-0.1408	0.6564	0.1163	1.2154	1.4202
10	0.7993	0.561	0.6564	0.0251	-0.1541	0.6858	0.1248	1.2225	1.4249
---	---	---	---	---	---	---	---	---	---
70	0.9999	0.5155	0.6102	0.0475	-0.9375	0.6473	0.1317	1.2556	1.9395

Hyperparameters:

Rank 1: {'reg__iterations': 200, 'reg__depth': 7, 'reg__learning_rate': 0.012177722201811948, 'reg__l2_leaf_reg': 33.50016334763174, 'reg__bagging_temperature': 3.5162578991536293, 'reg__random_strength': 4.962389798621308, 'reg__subsample': 0.7197664390212782, 'reg__min_data_in_leaf': 30, 'reg__leaf_estimation_iterations': 7}

Rank 2: {'reg__iterations': 200, 'reg__depth': 8, 'reg__learning_rate': 0.01177443834529251, 'reg__l2_leaf_reg': 32.85045841071112, 'reg__bagging_temperature': 3.318130420868385, 'reg__random_strength': 4.951542619497158, 'reg__subsample': 0.7401339290193116, 'reg__min_data_in_leaf': 27, 'reg__leaf_estimation_iterations': 7}

Rank 3: {'reg__iterations': 200, 'reg__depth': 7, 'reg__learning_rate': 0.01198755897267876, 'reg__l2_leaf_reg': 31.893652649505032, 'reg__bagging_temperature': 3.5237792786281124, 'reg__random_strength': 2.565901282319279, 'reg__subsample': 0.737038255229535, 'reg__min_data_in_leaf': 30, 'reg__leaf_estimation_iterations': 7}

Rank 4: {'reg__iterations': 300, 'reg__depth': 7, 'reg__learning_rate': 0.012192980233185506, 'reg__l2_leaf_reg': 34.066667409555905, 'reg__bagging_temperature': 3.5492695810923824, 'reg__random_strength': 2.314345862788802, 'reg__subsample': 0.7234641626440524, 'reg__min_data_in_leaf': 32, 'reg__leaf_estimation_iterations': 8}

Rank 5: {'reg__iterations': 1000, 'reg__depth': 8, 'reg__learning_rate': 0.010077917562115932, 'reg__l2_leaf_reg': 15.119662299694024, 'reg__bagging_temperature': 2.0618994095914775, 'reg__random_strength': 4.6675291418438905, 'reg__subsample': 0.6991569515841174, 'reg__min_data_in_leaf': 25, 'reg__leaf_estimation_iterations': 1}

Rank 6: {'reg__iterations': 400, 'reg__depth': 7, 'reg__learning_rate': 0.010734808912350353, 'reg__l2_leaf_reg': 13.428963454047508, 'reg__bagging_temperature': 2.825964093802523, 'reg__random_strength': 1.6512503588948724, 'reg__subsample': 0.8366747703372376, 'reg__min_data_in_leaf': 50, 'reg__leaf_estimation_iterations': 5}

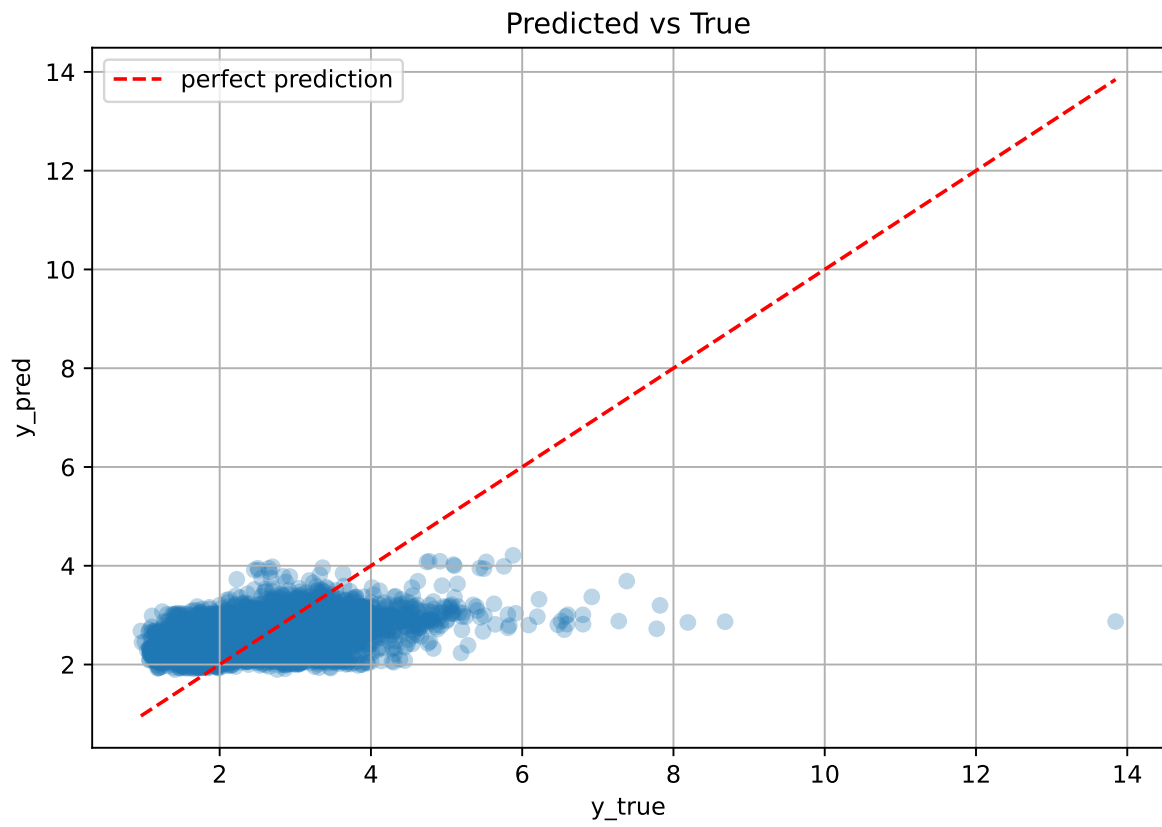
Rank 7: {'reg__iterations': 800, 'reg__depth': 4, 'reg__learning_rate': 0.018481312331534182, 'reg__l2_leaf_reg': 20.049144838968882, 'reg__bagging_temperature': 4.086145751784971, 'reg__random_strength': 3.7518112502462406, 'reg__subsample': 0.7930194074524215, 'reg__min_data_in_leaf': 10, 'reg__leaf_estimation_iterations': 2}

Rank 8: {'reg__iterations': 700, 'reg__depth': 7, 'reg__learning_rate': 0.01015700531859524, 'reg__l2_leaf_reg': 12.006264711250509, 'reg__bagging_temperature': 2.063940463617275, 'reg__random_strength': 3.6365154118743006, 'reg__subsample': 0.6922917576282456, 'reg__min_data_in_leaf': 21, 'reg__leaf_estimation_iterations': 3}

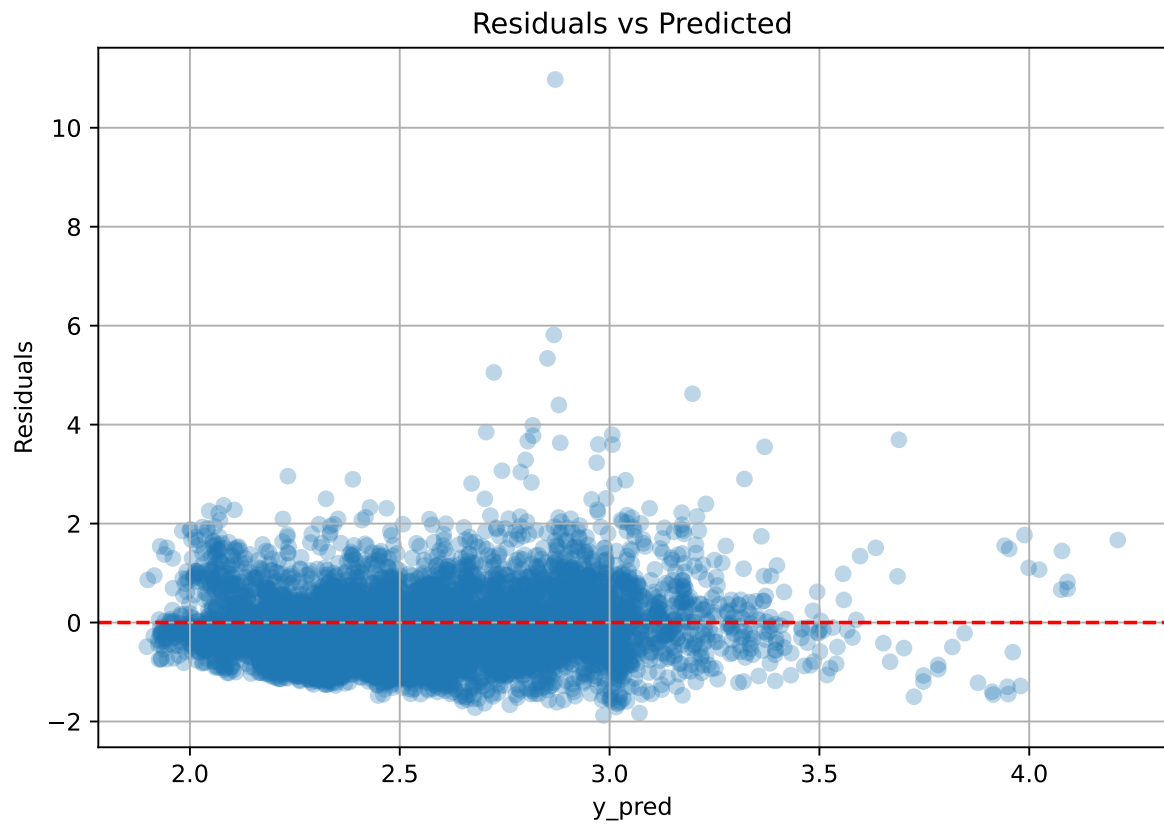
Rank 9: {'reg__iterations': 900, 'reg__depth': 8, 'reg__learning_rate': 0.021823754855618833, 'reg__l2_leaf_reg': 45.300071743481446, 'reg__bagging_temperature': 4.212582479570103, 'reg__random_strength': 4.468558612411265, 'reg__subsample': 0.8124849095591439, 'reg__min_data_in_leaf': 40, 'reg__leaf_estimation_iterations': 1}

Rank 10: {'reg__iterations': 800, 'reg__depth': 7, 'reg__learning_rate': 0.011234594731373736, 'reg__l2_leaf_reg': 35.13380106273488, 'reg__bagging_temperature': 3.245836674349518, 'reg__random_strength': 4.997487079082786, 'reg__subsample': 0.7339885276313184, 'reg__min_data_in_leaf': 28, 'reg__leaf_estimation_iterations': 7}

2.5.22 Scatter



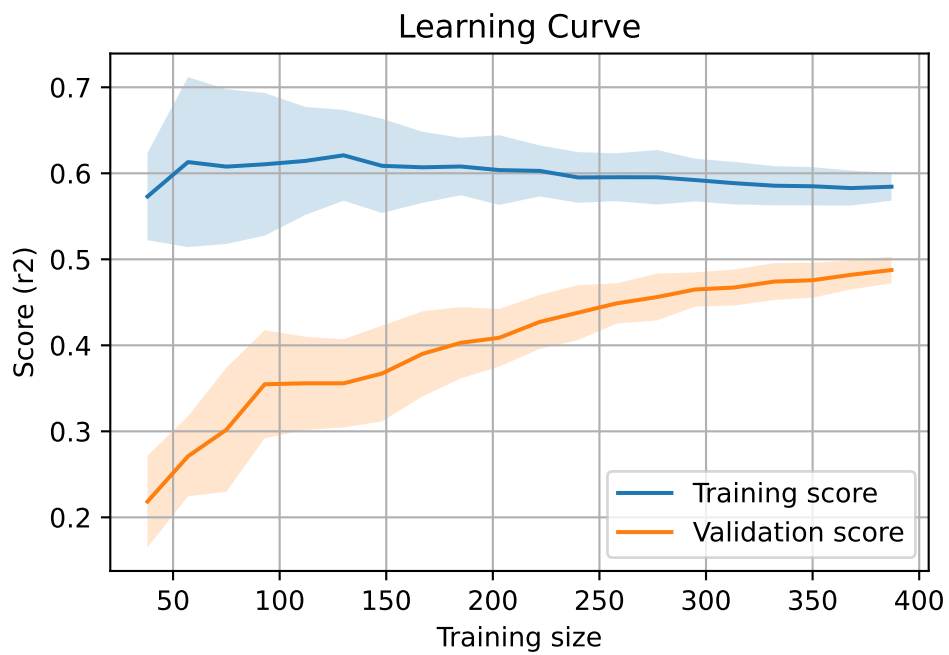
2.5.23 Scatter residuals



2.5.24 Learning curve — Group: 6_4

`Len(group)` : 792, `Len(training)` : 555,

`Len(training_real)` : 389, `Len(test_of_training)` : 166, `Len(test)` : 237



2.6 Sumup-table

model	group	Len training	Len test	Len total	mean_train_cv	mean_test_cv	std_test_cv	test_cv	test	diff	r_test	r_train	pen_score
Linear	1_3	557	238	795	0.563	0.551	0.044	0.498	0.5	-0.05	0.908	1.022	nan
Linear	2_11	1909	817	2726	0.463	0.463	0.017	0.394	0.394	-0.069	0.852	1.0	nan
Linear	3_4	398	170	568	0.307	0.248	0.036	0.187	0.203	-0.045	0.818	1.241	nan
Linear	4_5	1909	817	2726	0.431	0.404	0.026	0.457	0.457	0.054	1.133	1.069	nan
Linear	6_3	393	168	561	0.625	0.677	0.067	0.616	0.623	-0.054	0.921	0.924	nan
Linear	6_4	555	237	792	0.172	0.181	0.042	0.26	0.264	0.083	1.459	0.952	nan
Linear	MEAN	5721	2447	8168	0.427	0.42	0.039	0.402	0.407	-0.014	1.015	1.035	nan
LGBM	1_3	557	238	795	0.612	0.572	0.043	0.54	0.559	-0.014	0.976	1.07	0.4522
LGBM	2_11	1909	817	2726	0.682	0.651	0.014	0.551	0.569	-0.082	0.874	1.048	0.651
LGBM	3_4	398	170	568	0.296	0.246	0.034	0.1	0.144	-0.102	0.584	1.205	0.0948
LGBM	4_5	1909	817	2726	0.453	0.412	0.024	0.471	0.494	0.082	1.199	1.099	0.2897
LGBM	6_3	393	168	561	0.718	0.691	0.054	0.636	0.678	-0.013	0.981	1.038	0.691
LGBM	6_4	555	237	792	0.512	0.433	0.029	0.518	0.558	0.125	1.289	1.184	0.1938
LGBM	MEAN	5721	2447	8168	0.546	0.501	0.033	0.469	0.5	-0.001	0.984	1.107	0.395
CatBoost	1_3	557	238	795	0.655	0.61	0.045	0.583	0.599	-0.011	0.982	1.075	0.4727
CatBoost	2_11	1909	817	2726	0.681	0.663	0.013	0.573	0.582	-0.081	0.878	1.027	0.6632
CatBoost	3_4	398	170	568	0.582	0.39	0.062	0.214	0.244	-0.146	0.626	1.494	-0.1881
CatBoost	4_5	1909	817	2726	0.458	0.427	0.021	0.489	0.495	0.068	1.158	1.074	0.333
CatBoost	6_3	393	168	561	0.771	0.715	0.062	0.703	0.726	0.011	1.015	1.078	0.5469
CatBoost	6_4	555	237	792	0.583	0.477	0.012	0.564	0.592	0.115	1.24	1.221	0.1614
CatBoost	MEAN	5721	2447	8168	0.622	0.547	0.036	0.521	0.54	-0.007	0.983	1.161	0.332

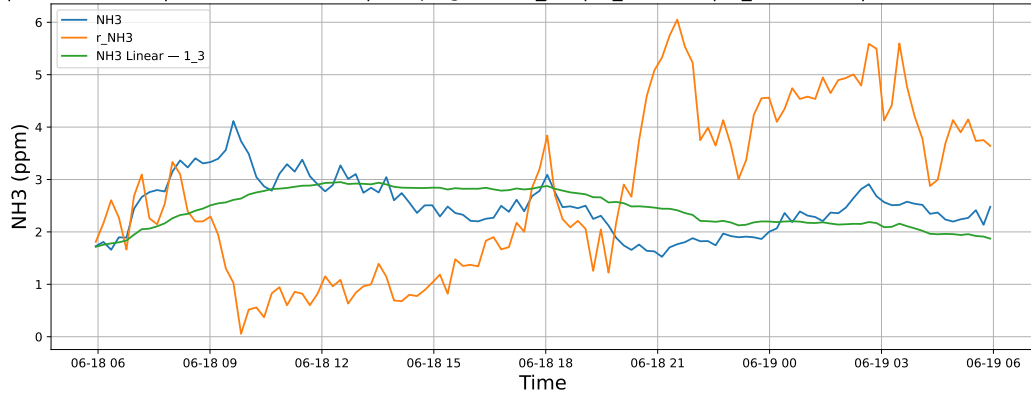
3 Plot

3.1 Standalone plots

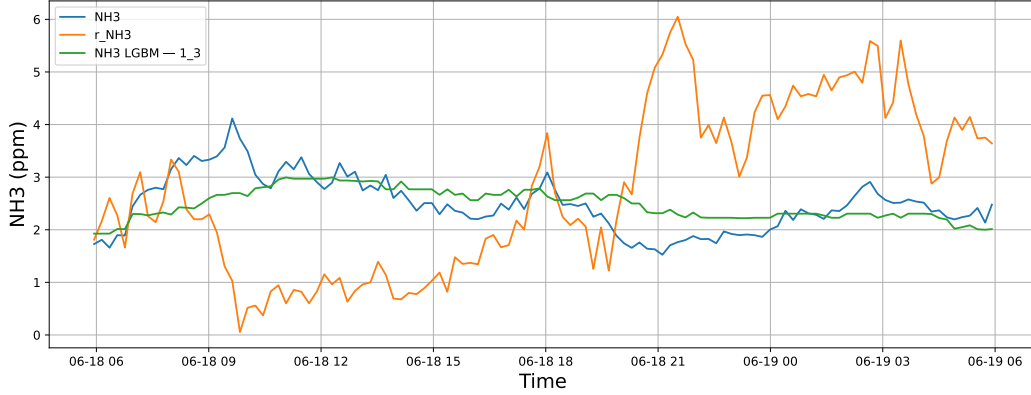
3.1.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

3.1.1.1 Time window : 24

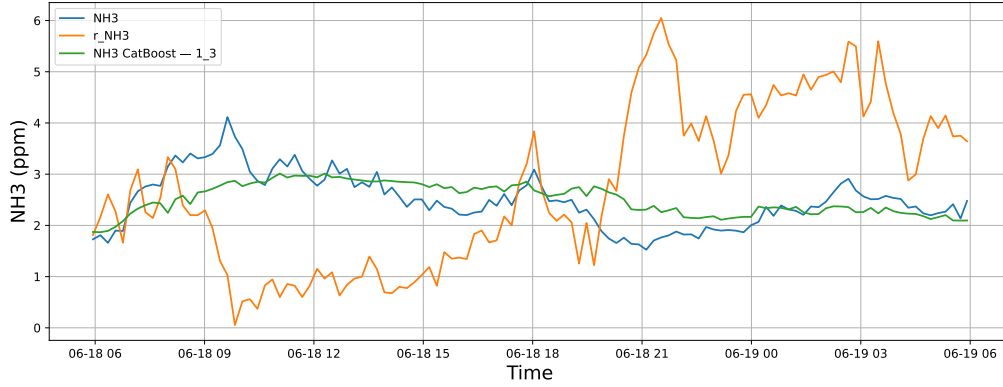
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

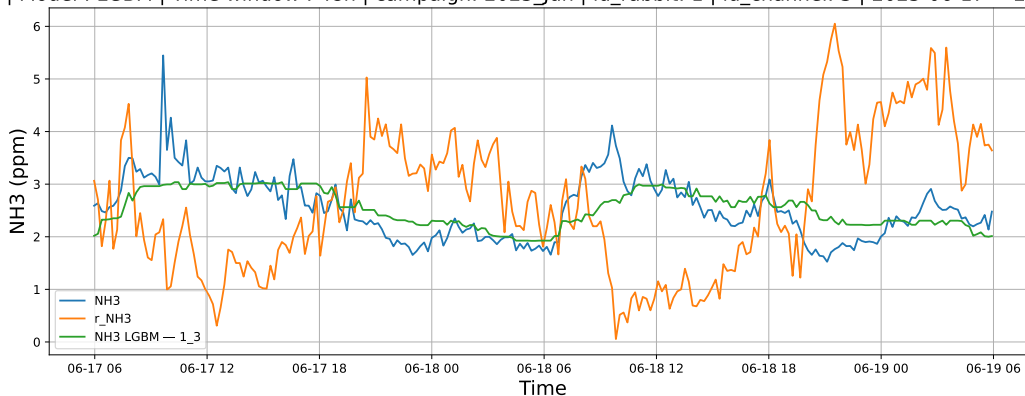


3.1.1.2 Time window : 48

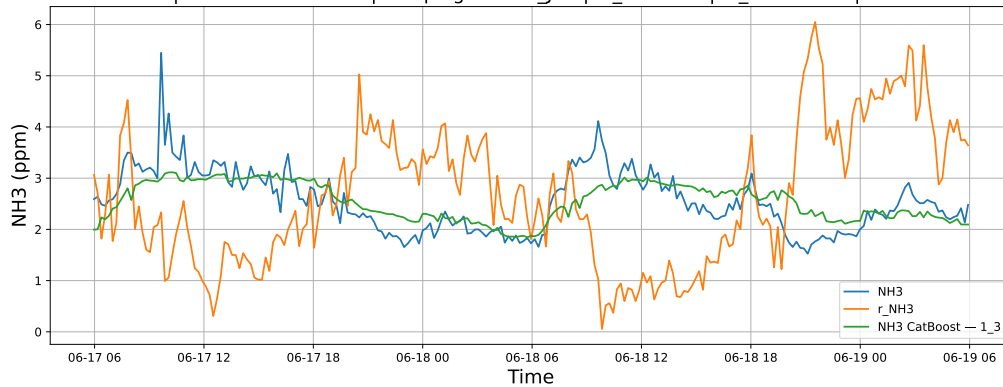
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

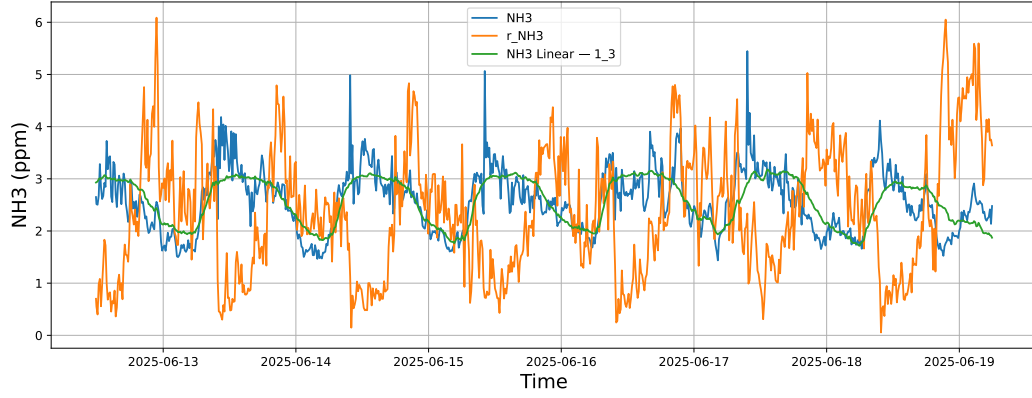


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

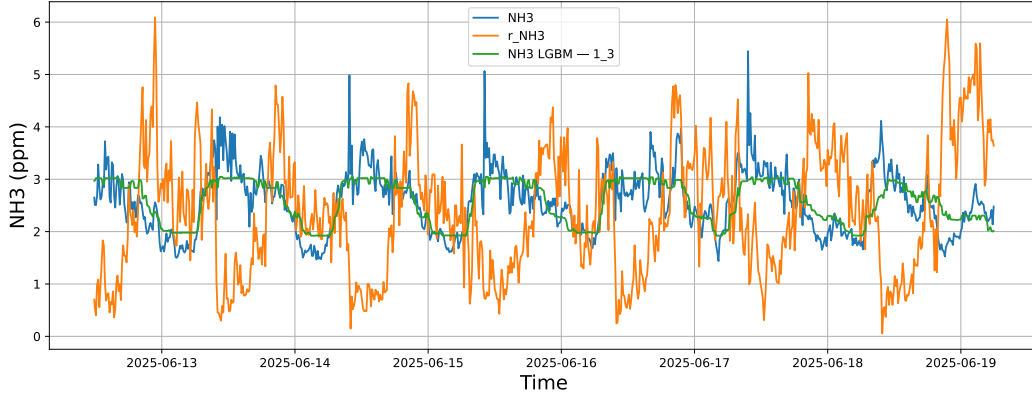


3.1.1.3 Time window : ALL

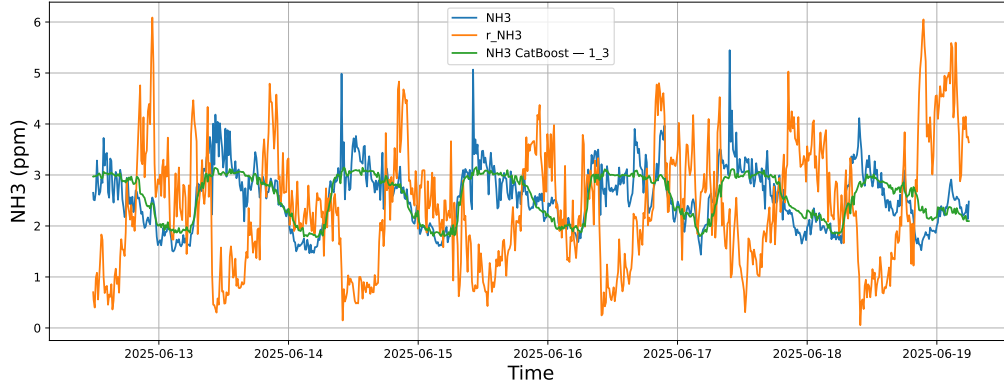
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



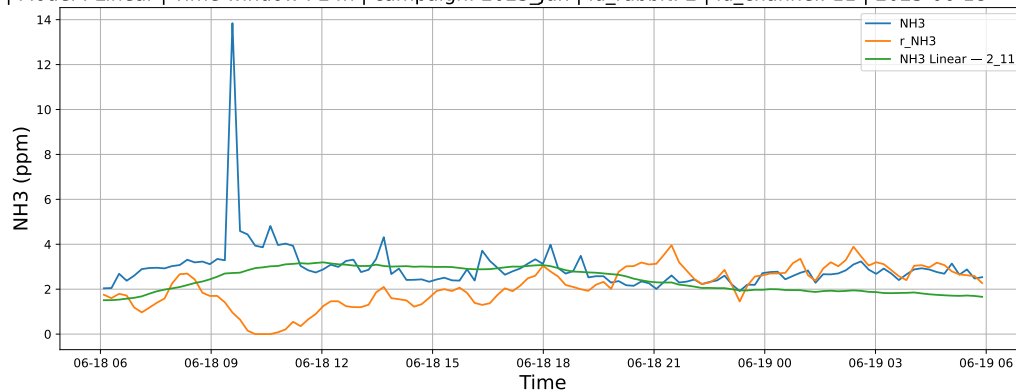
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



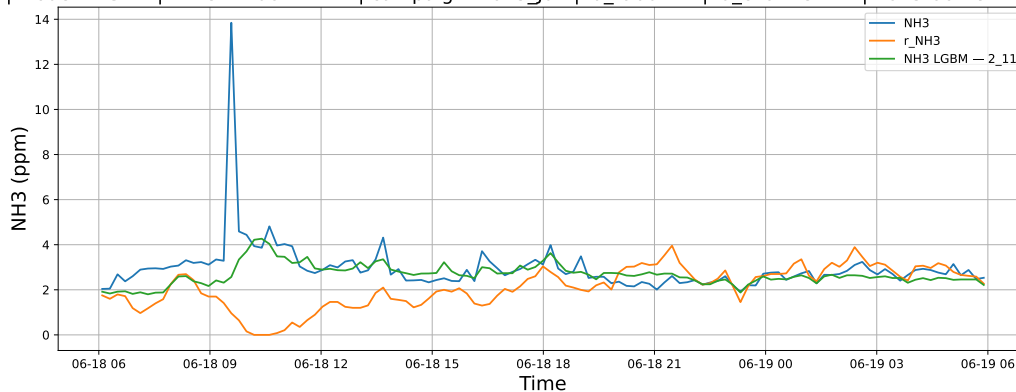
3.1.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

3.1.2.1 Time window : 24

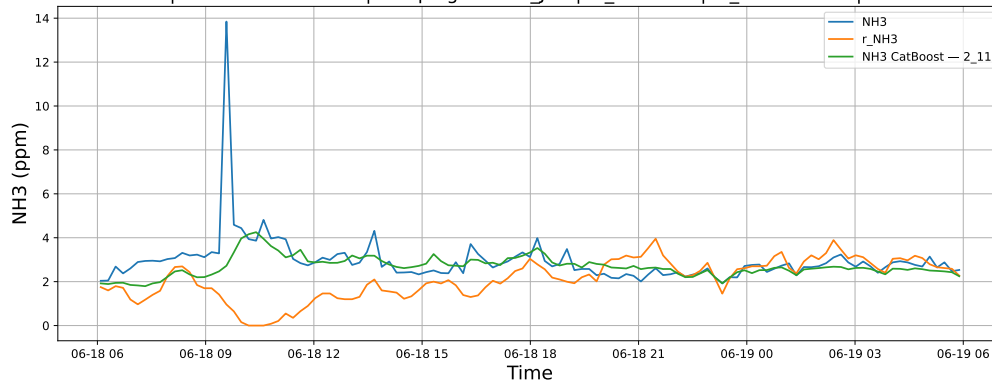
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

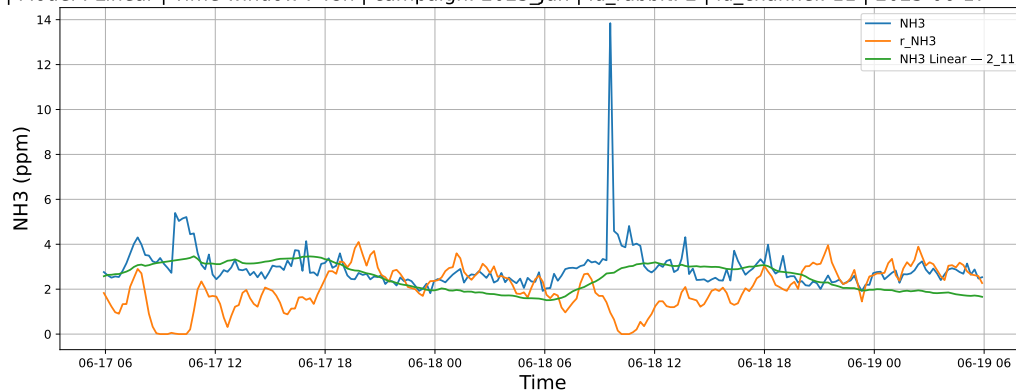


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

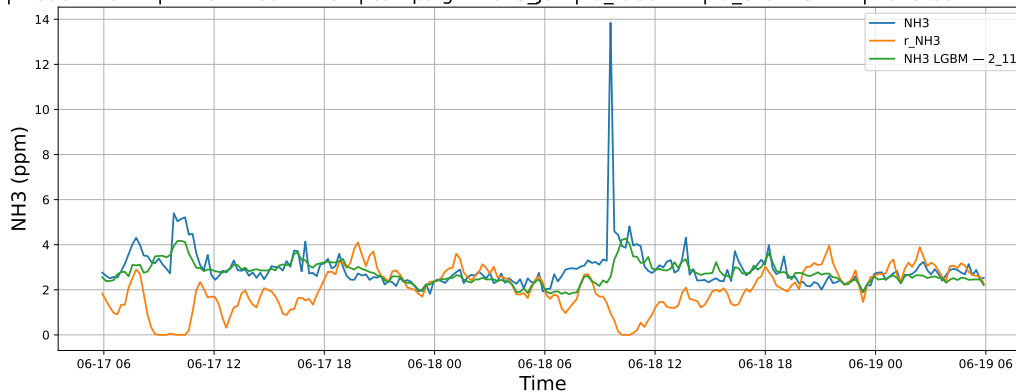


3.1.2.2 Time window : 48

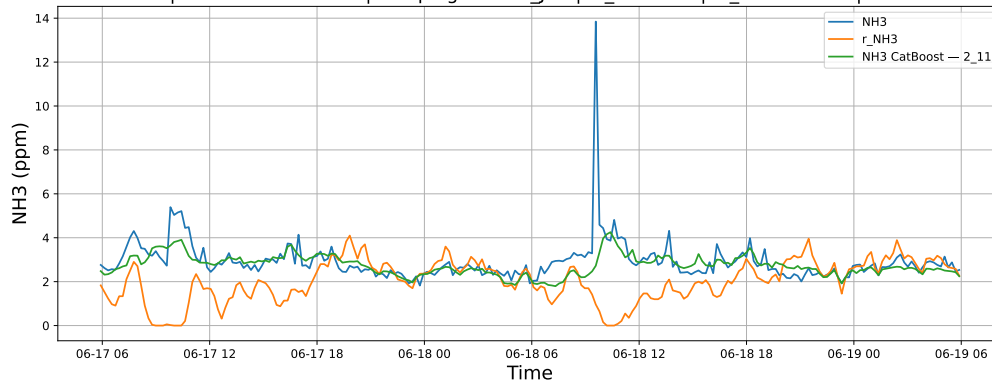
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

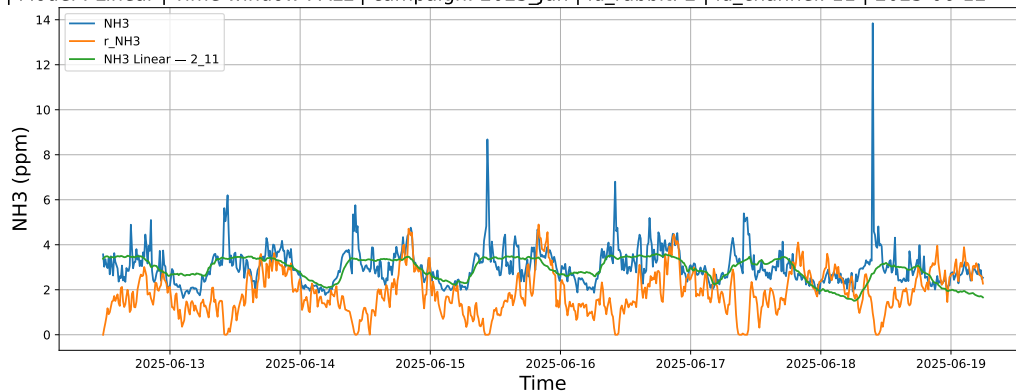


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

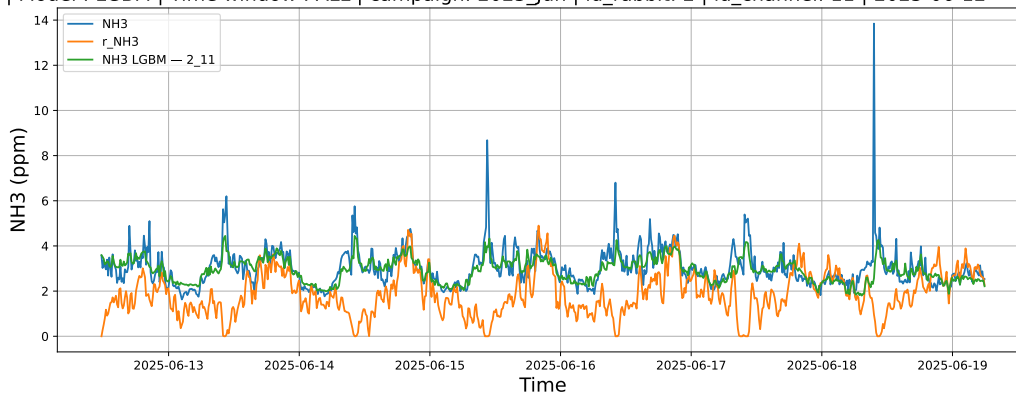


3.1.2.3 Time window : ALL

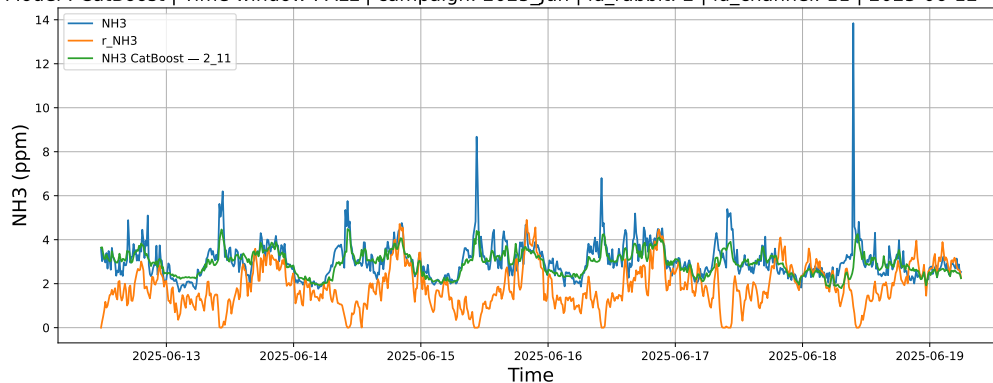
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



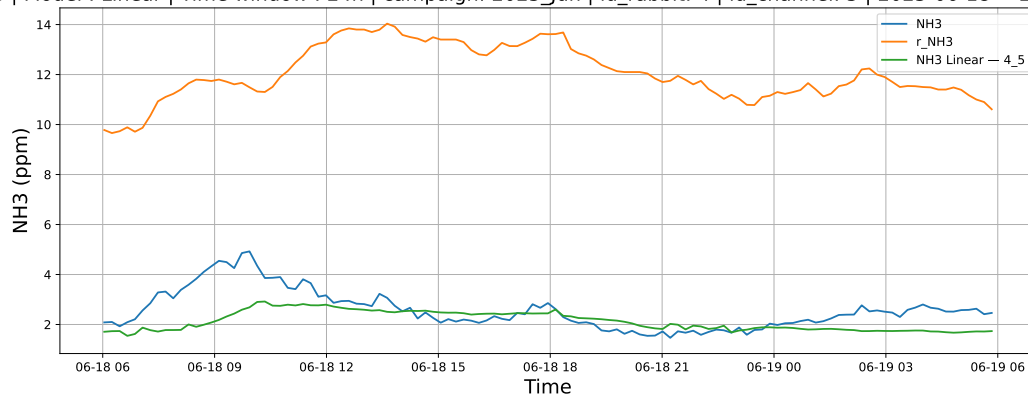
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



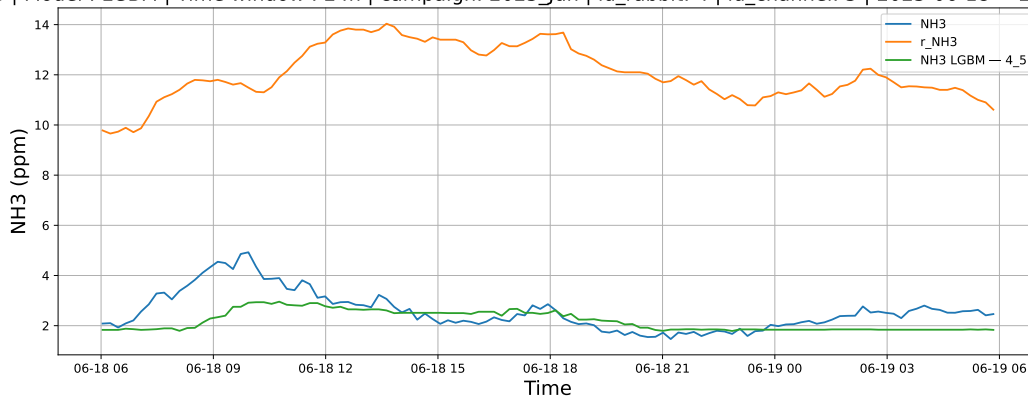
3.1.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

3.1.3.1 Time window : 24

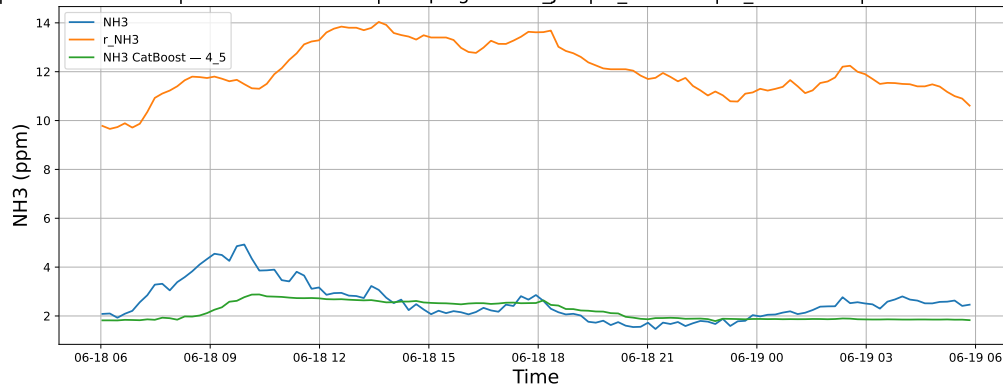
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

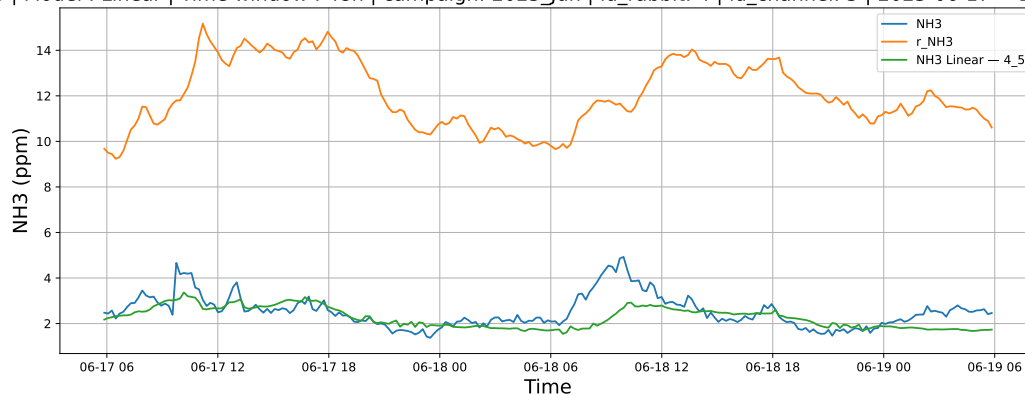


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

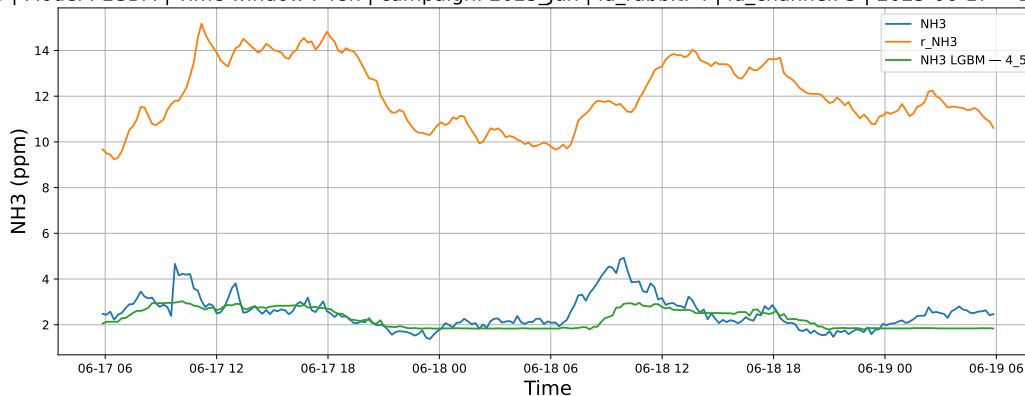


3.1.3.2 Time window : 48

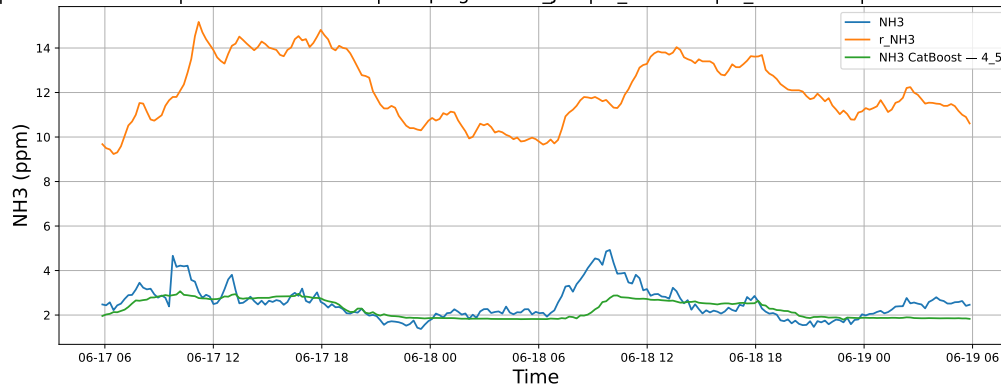
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

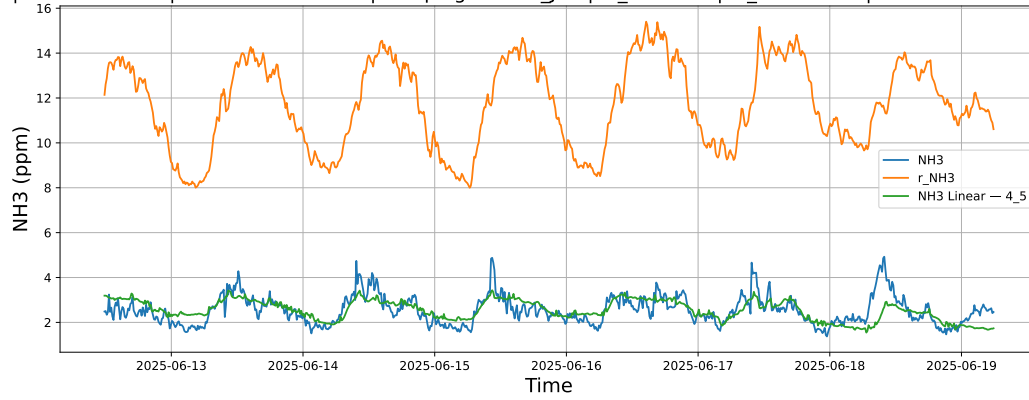


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

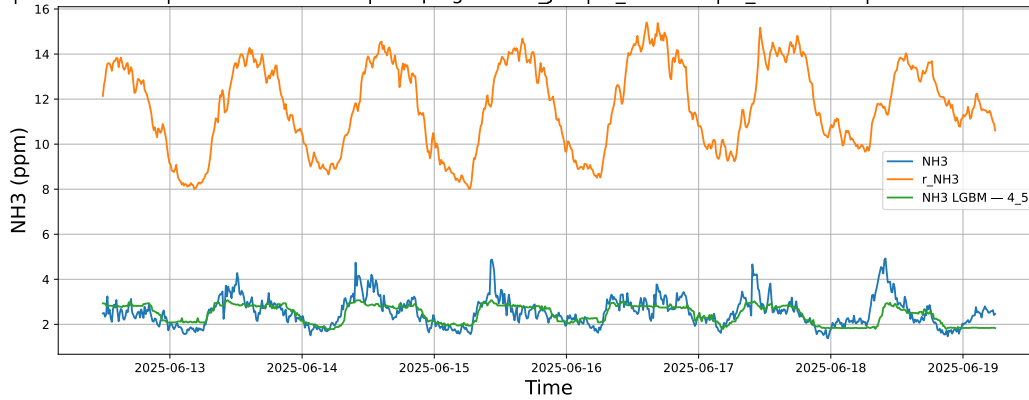


3.1.3.3 Time window : ALL

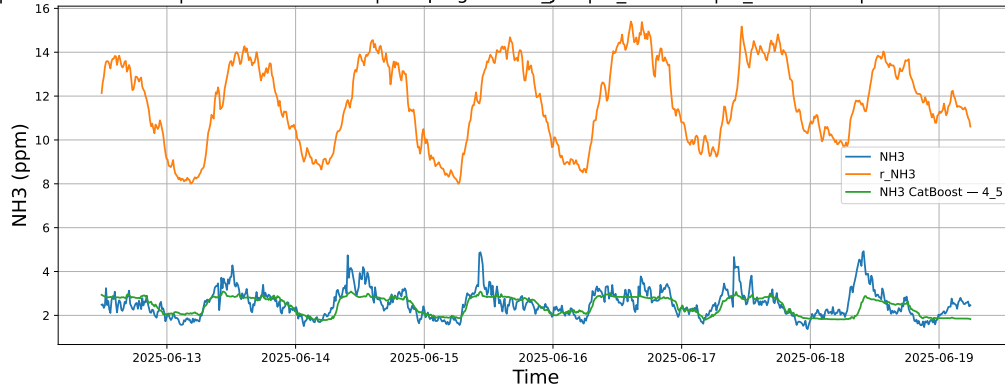
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



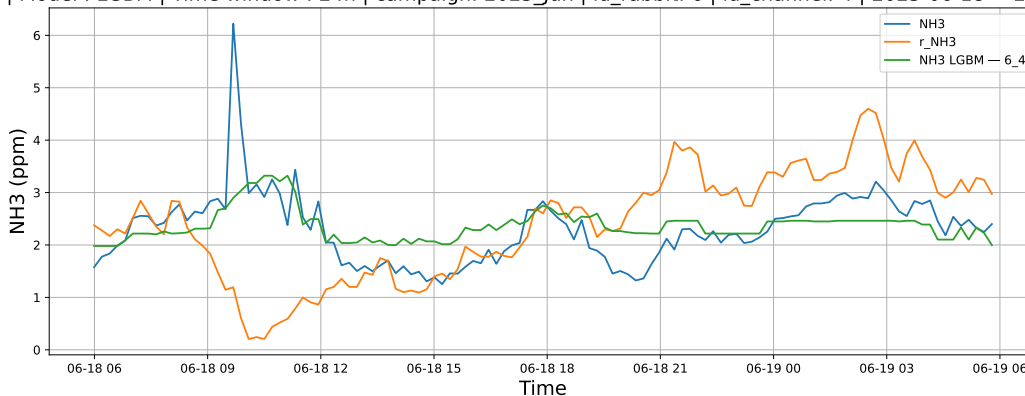
3.1.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

3.1.4.1 Time window : 24

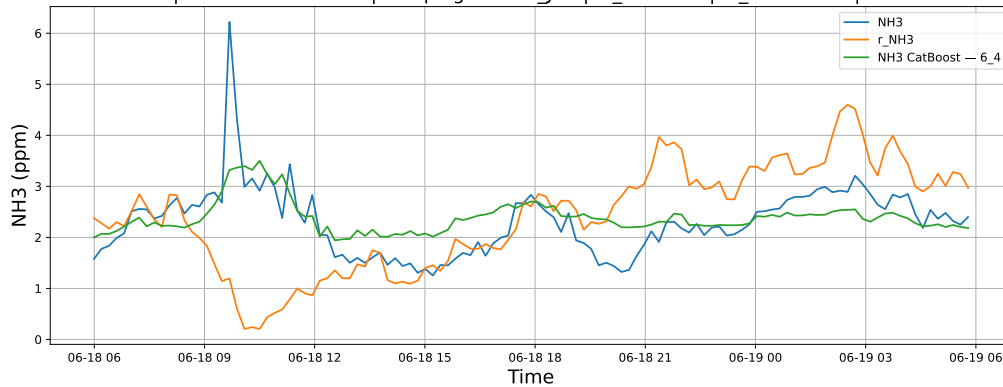
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

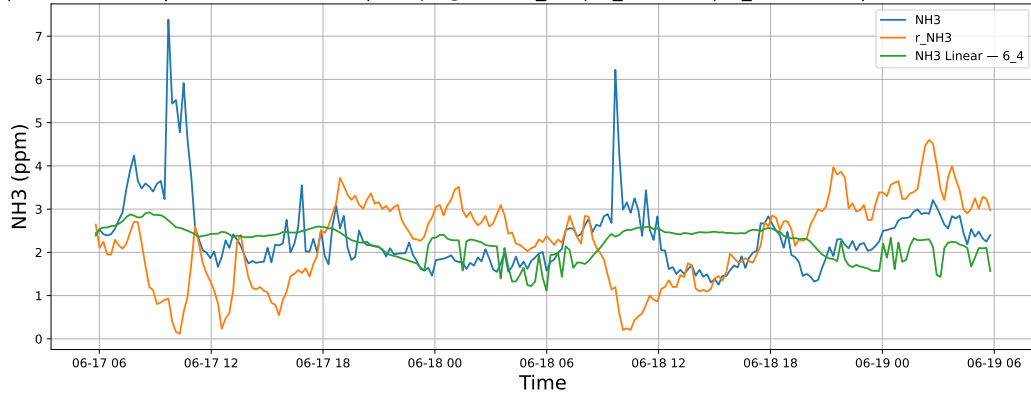


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

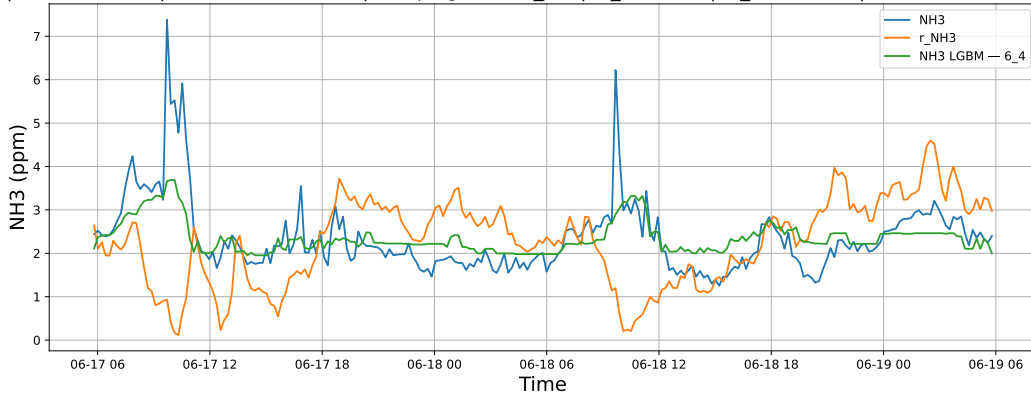


3.1.4.2 Time window : 48

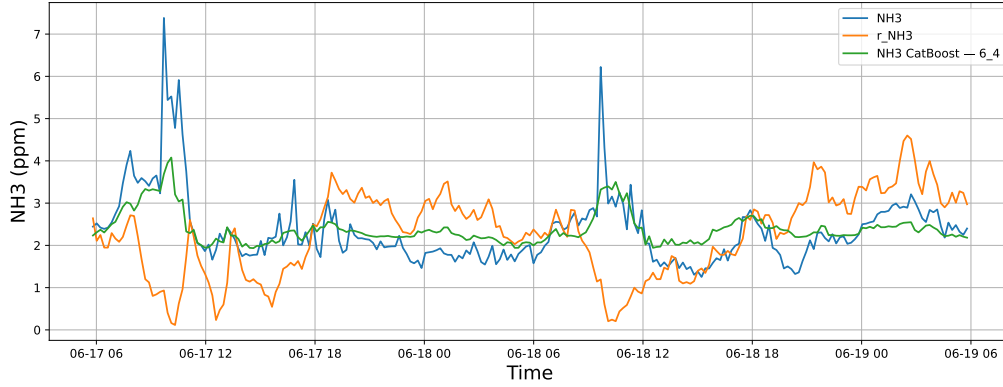
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

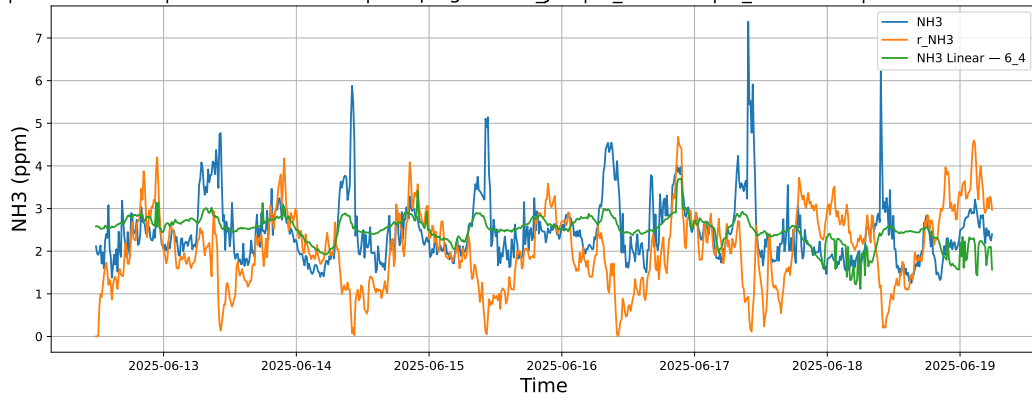


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

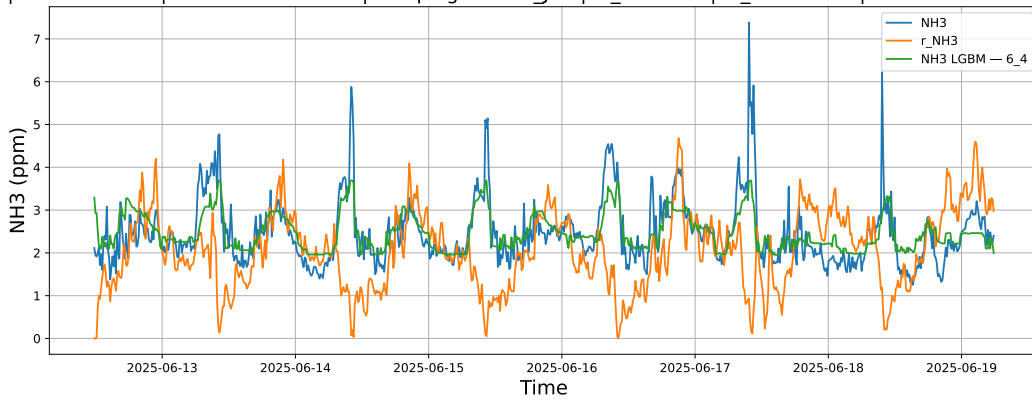


3.1.4.3 Time window : ALL

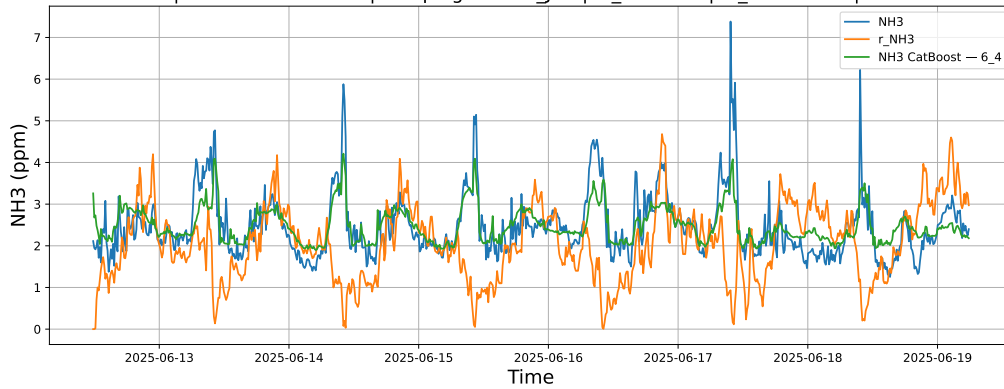
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



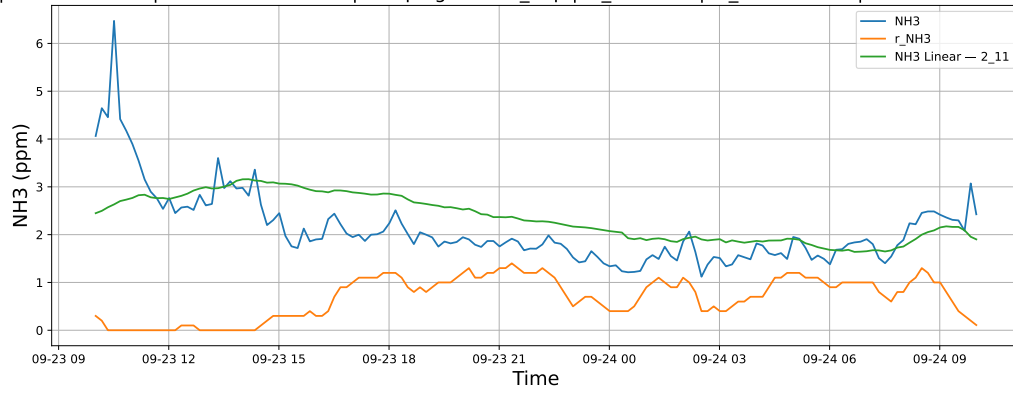
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



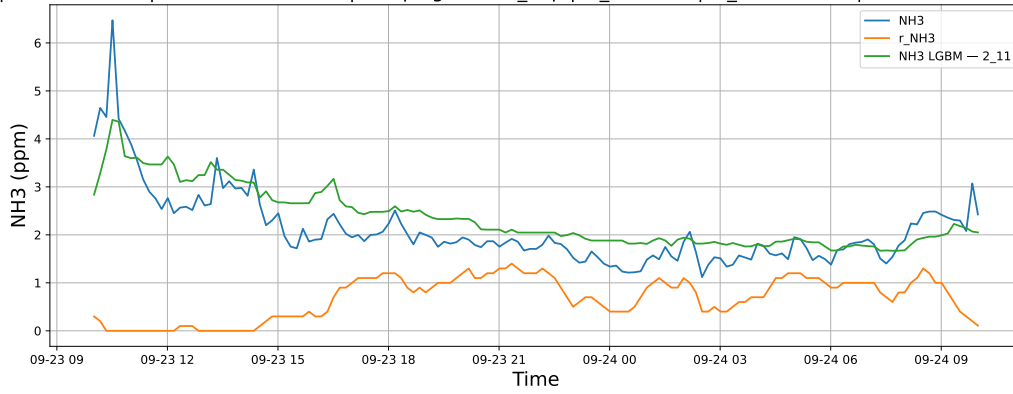
3.1.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

3.1.5.1 Time window : 24

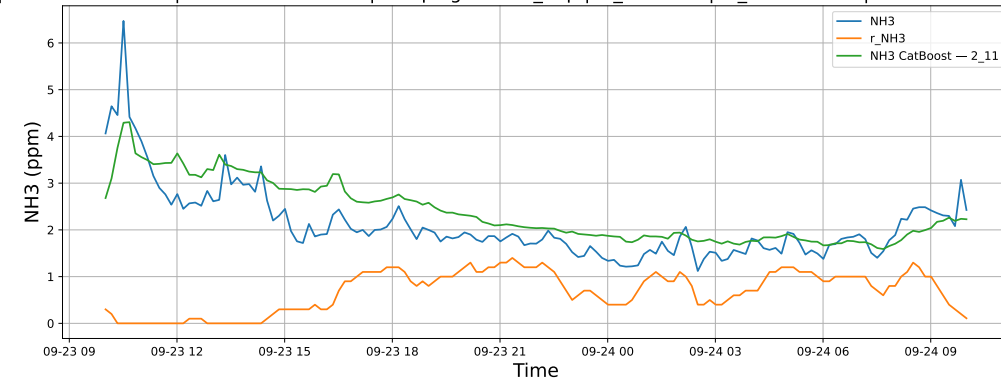
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

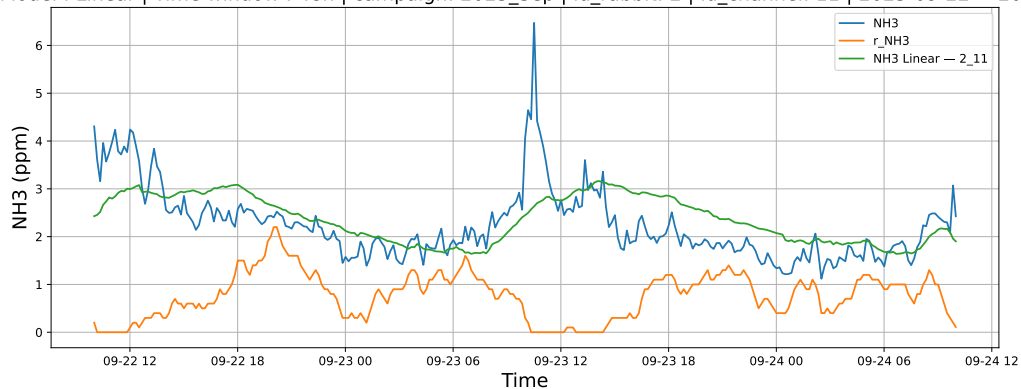


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

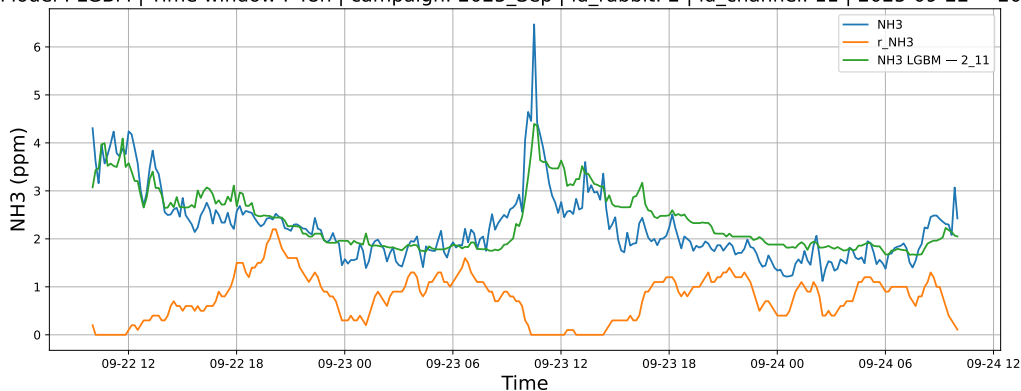


3.1.5.2 Time window : 48

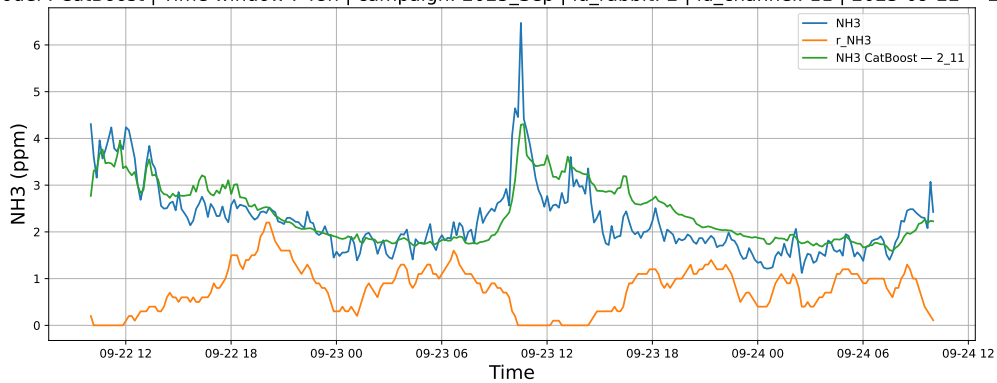
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

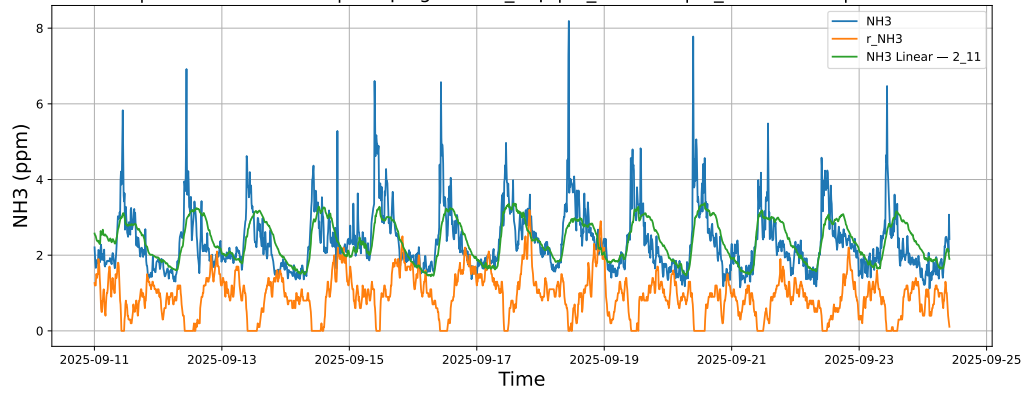


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

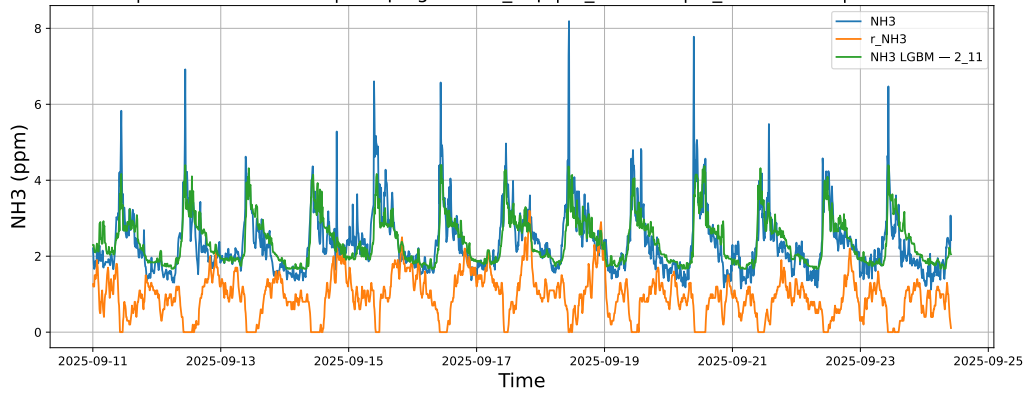


3.1.5.3 Time window : ALL

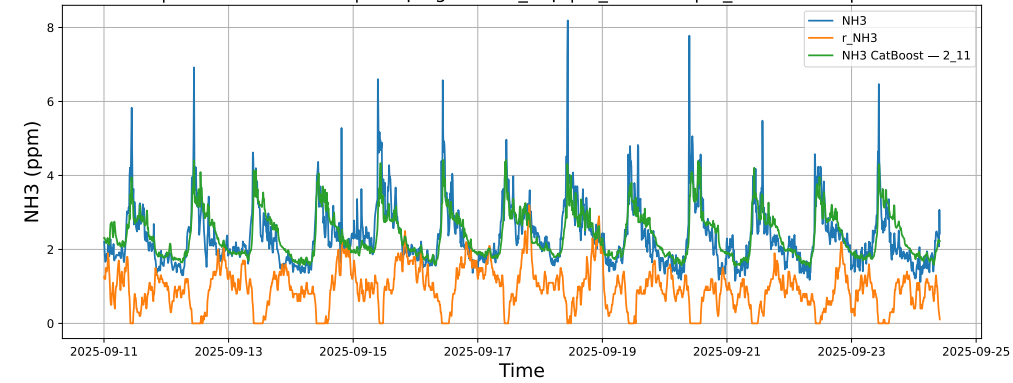
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



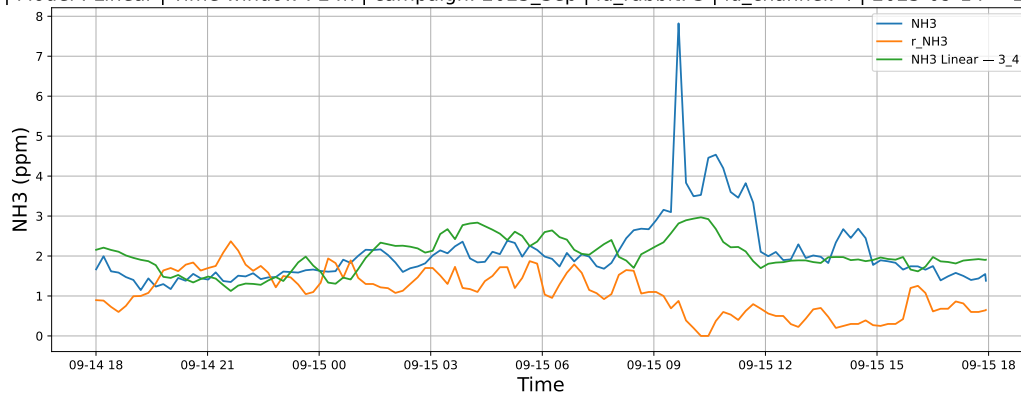
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



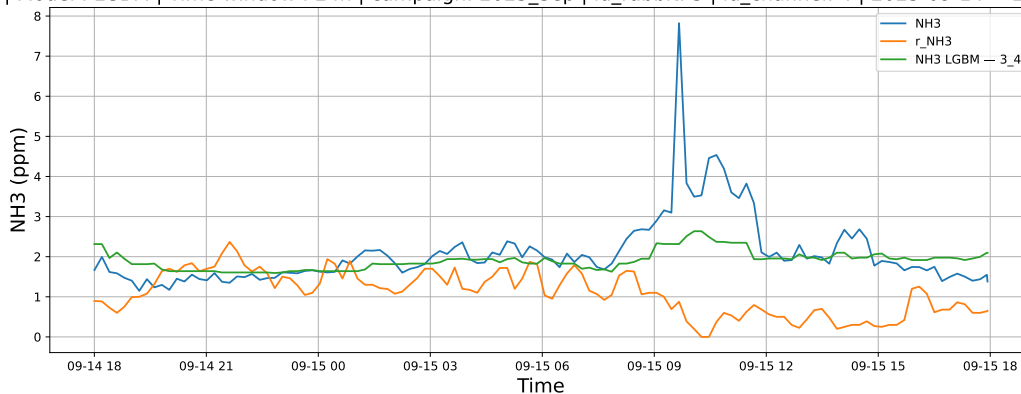
3.1.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

3.1.6.1 Time window : 24

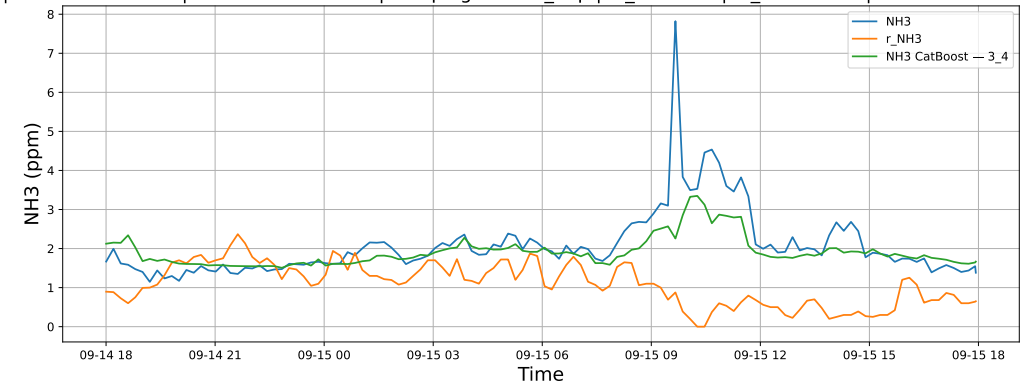
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

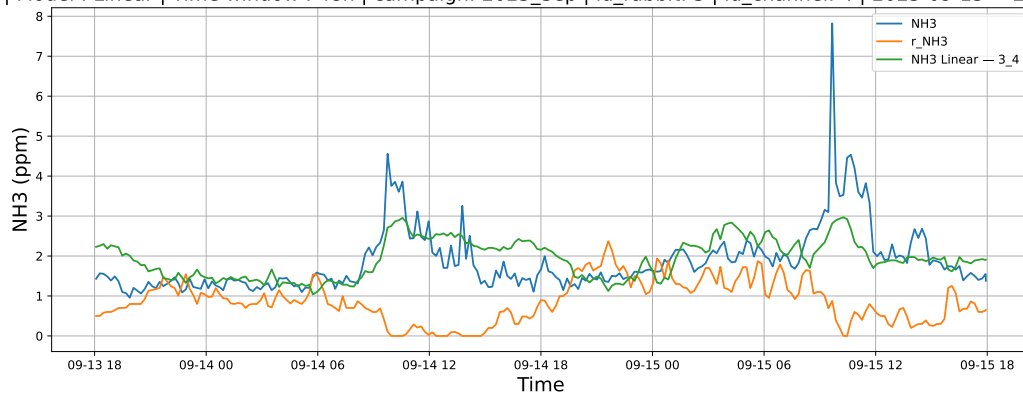


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

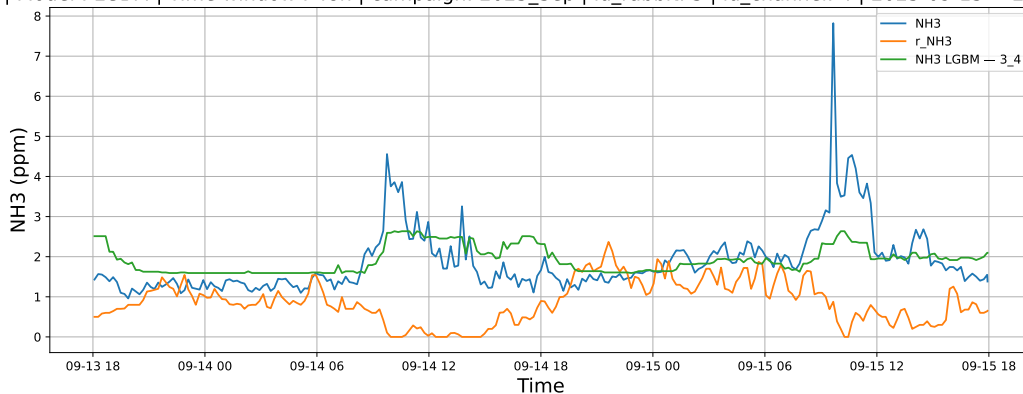


3.1.6.2 Time window : 48

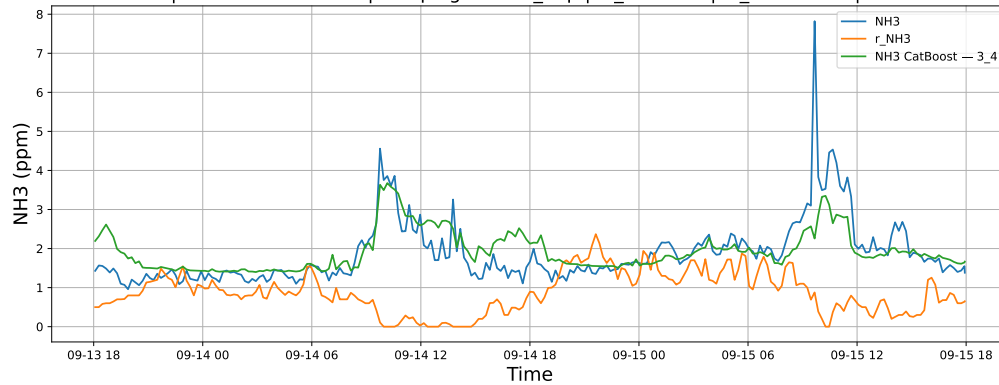
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

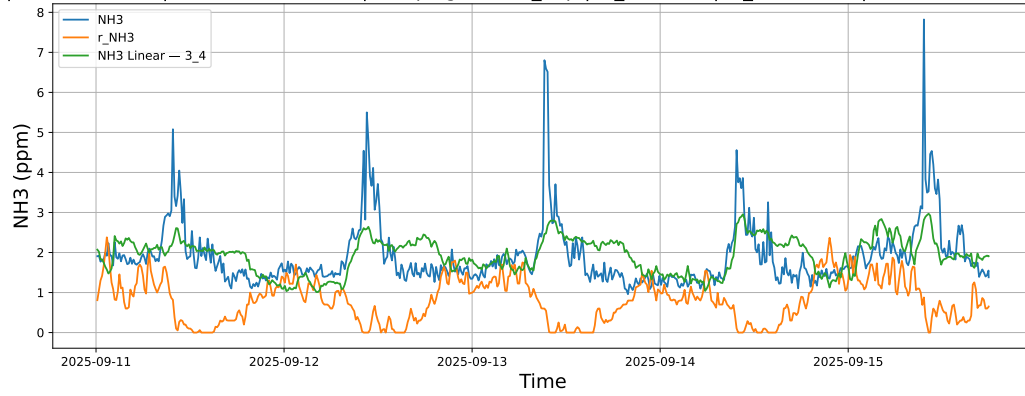


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

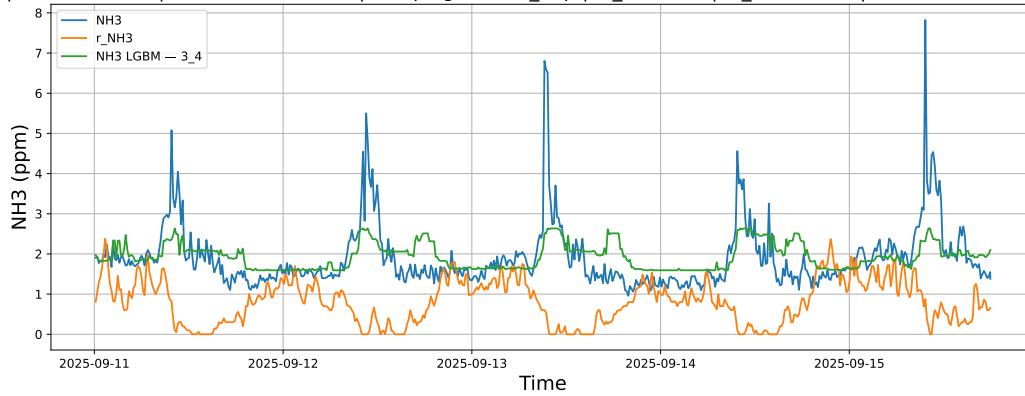


3.1.6.3 Time window : ALL

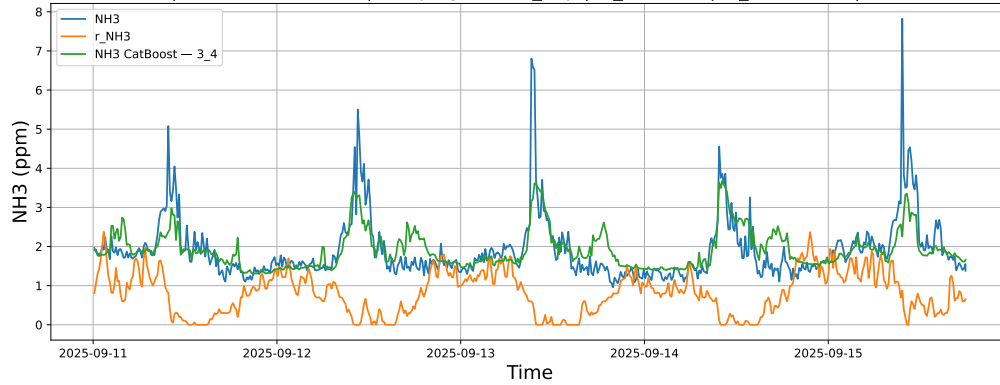
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



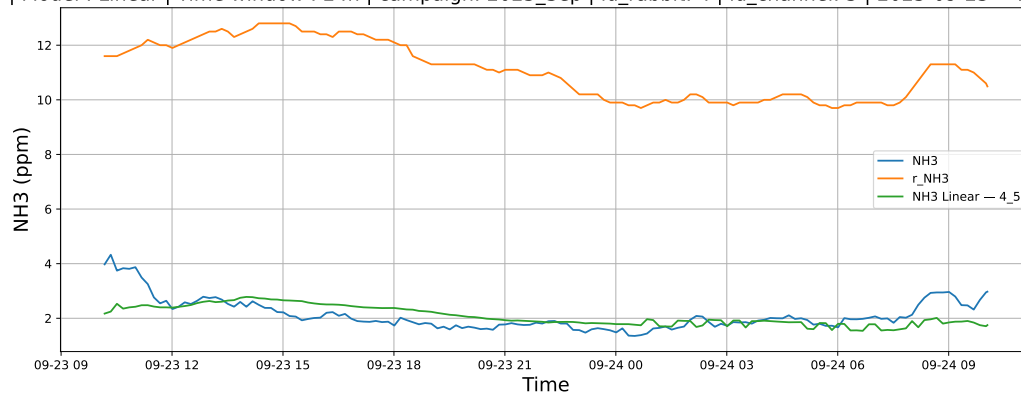
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



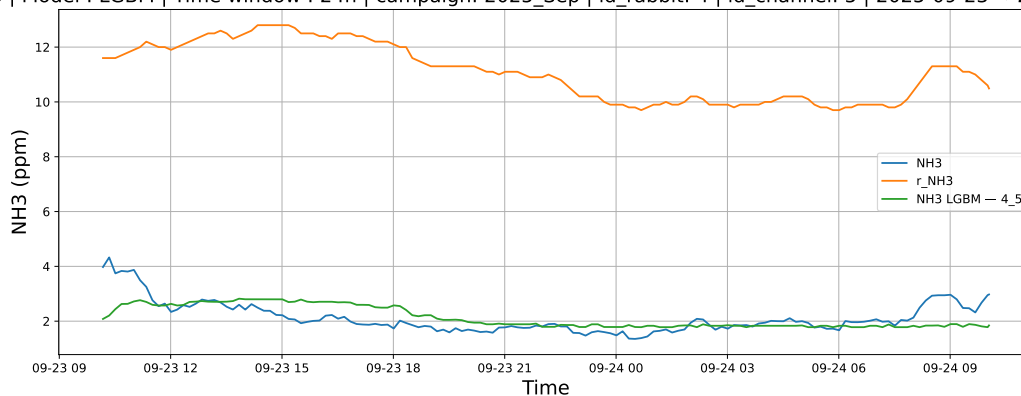
3.1.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

3.1.7.1 Time window : 24

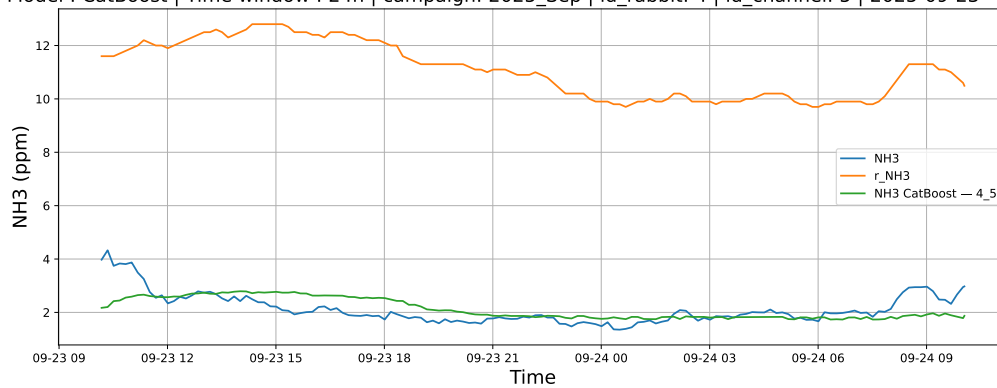
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

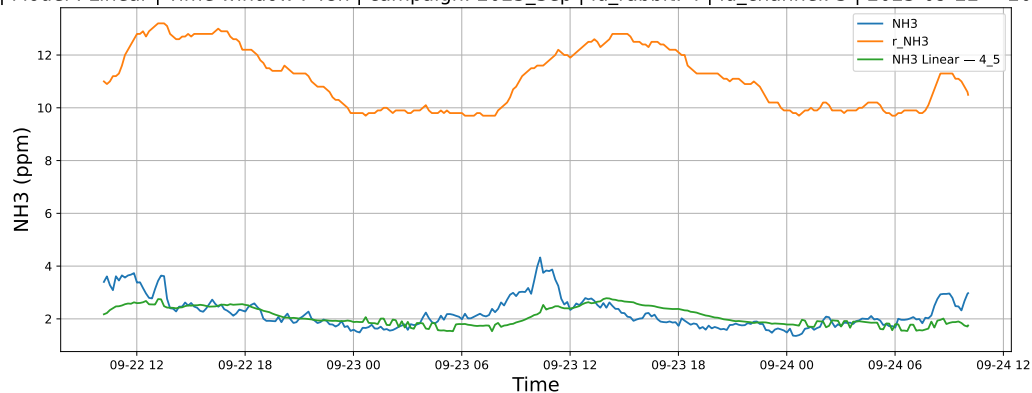


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

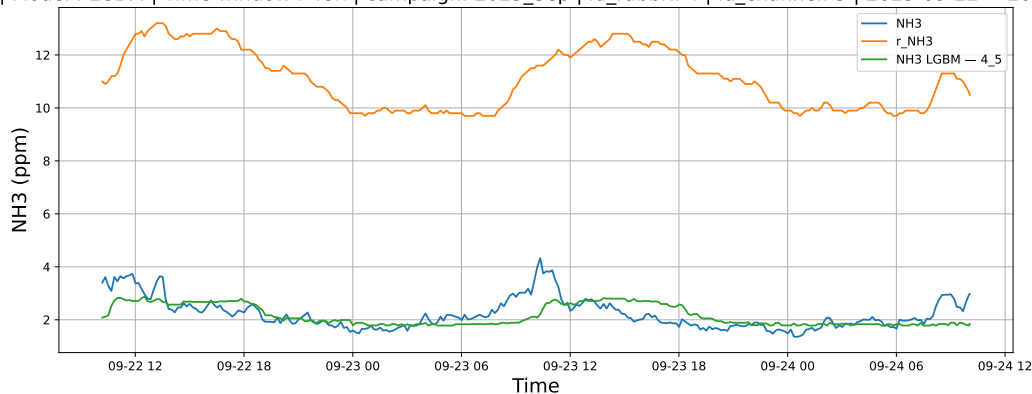


3.1.7.2 Time window : 48

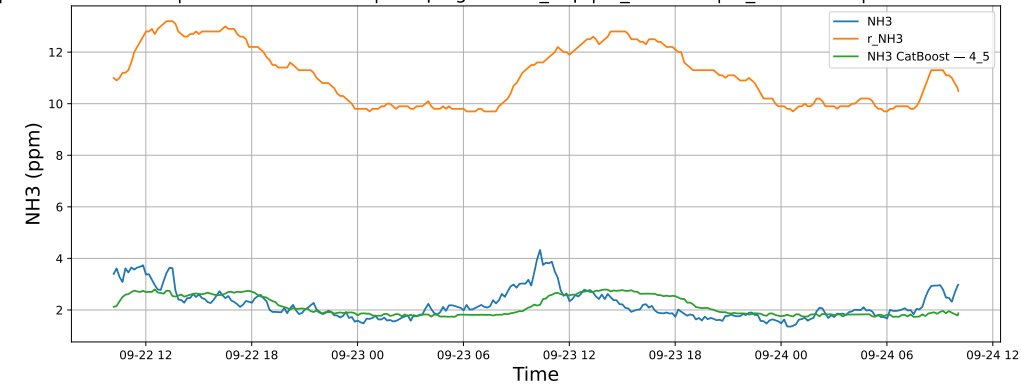
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

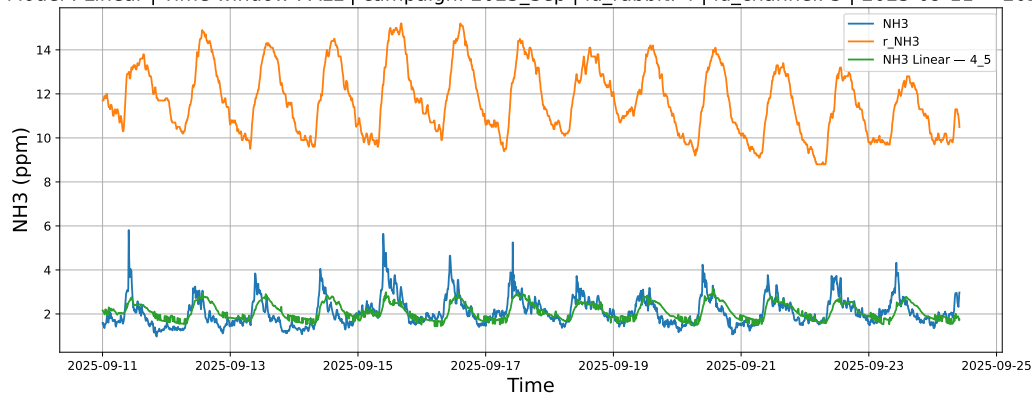


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

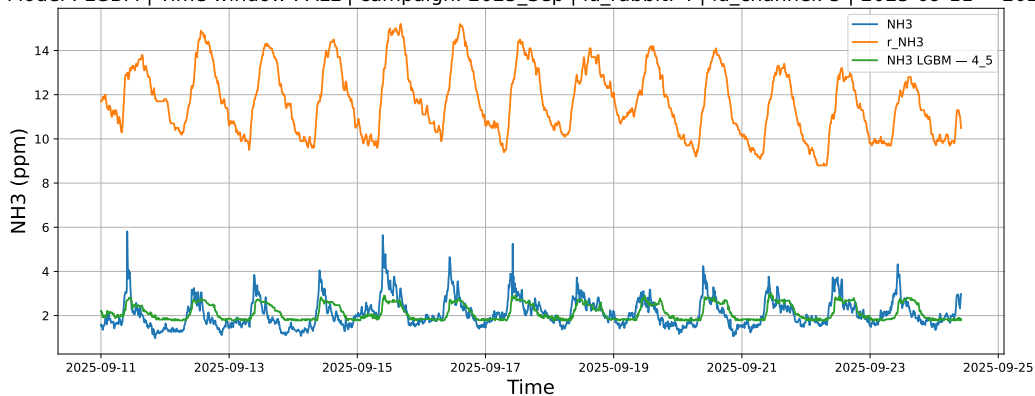


3.1.7.3 Time window : ALL

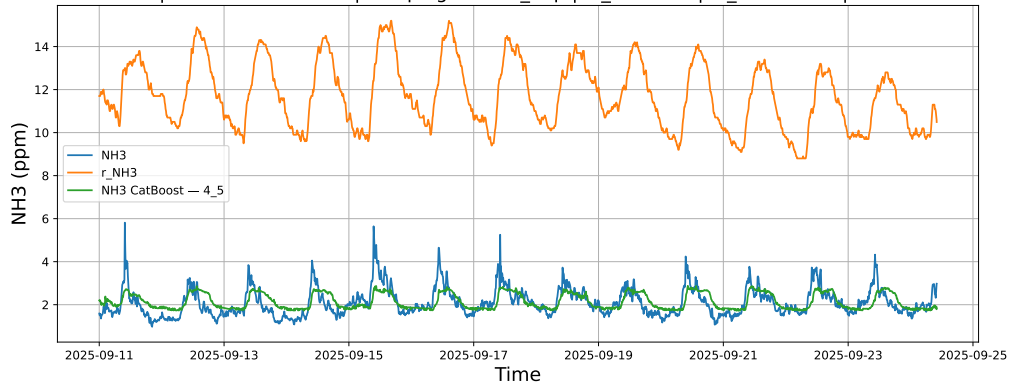
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



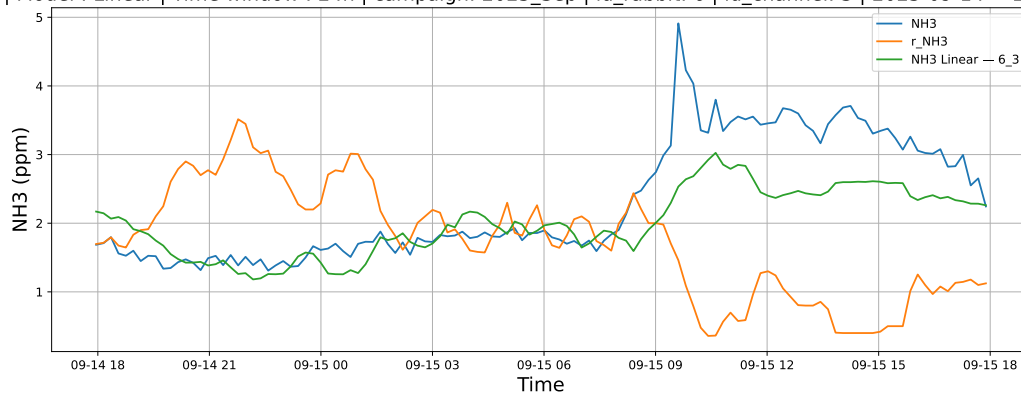
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



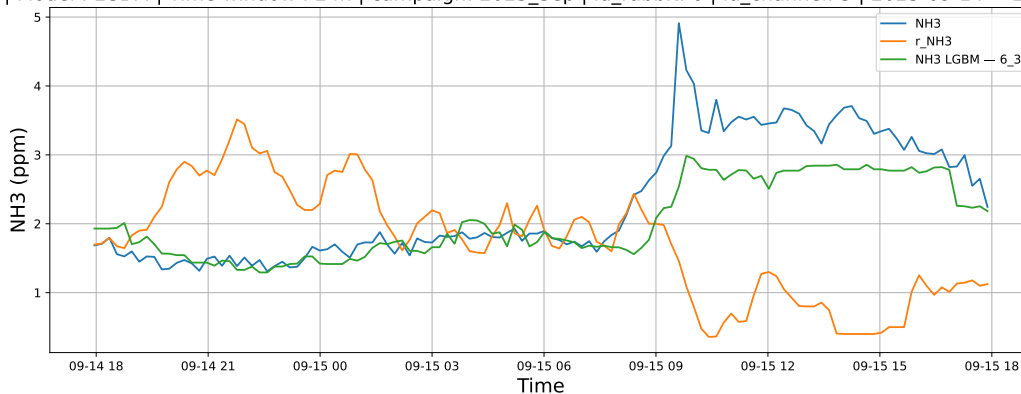
3.1.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

3.1.8.1 Time window : 24

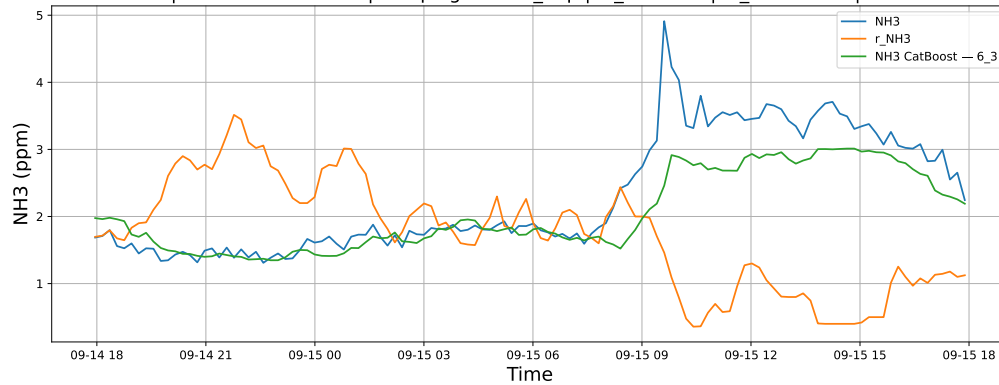
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

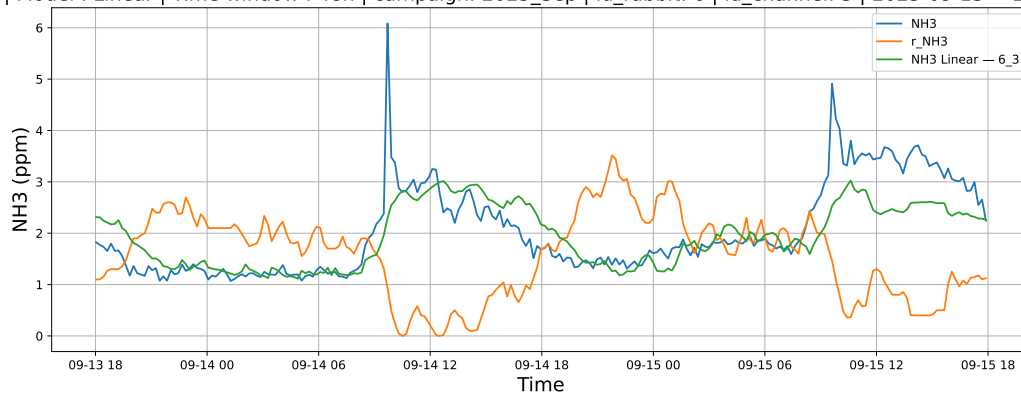


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

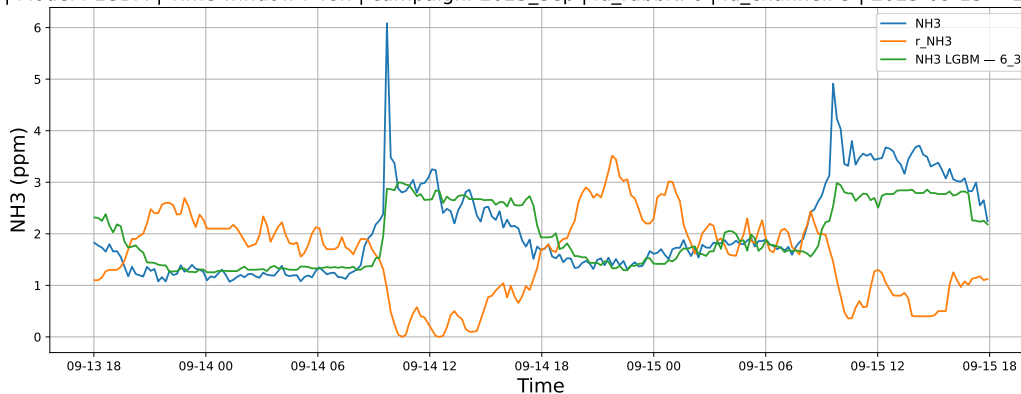


3.1.8.2 Time window : 48

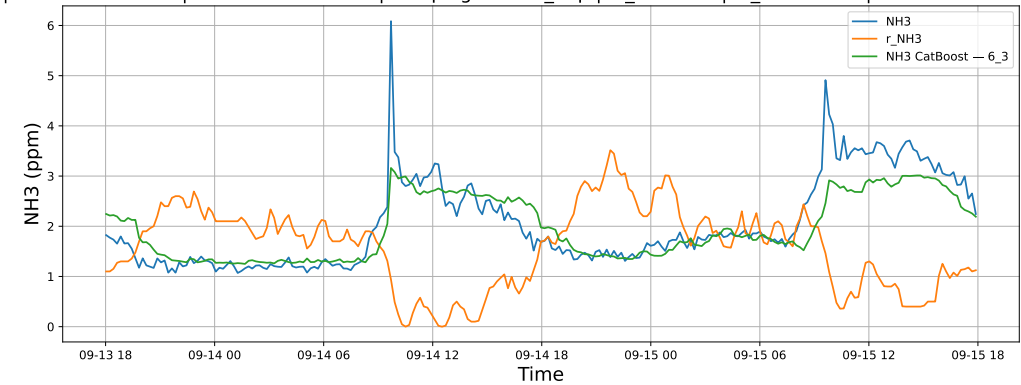
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

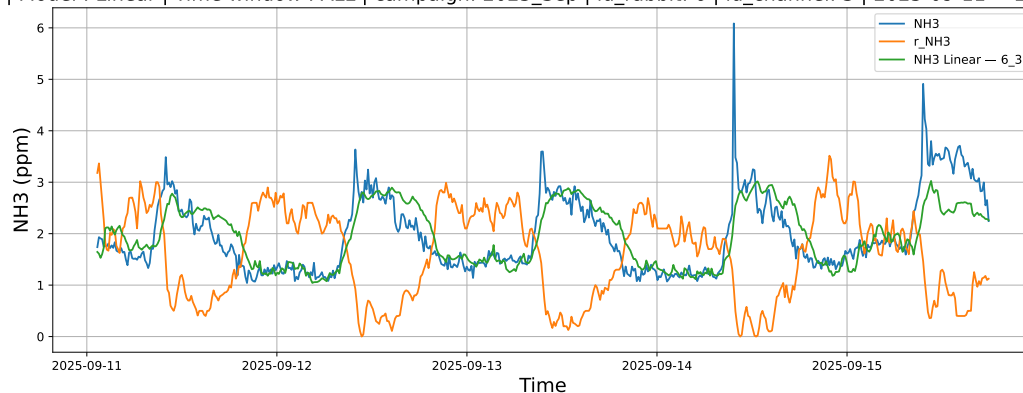


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

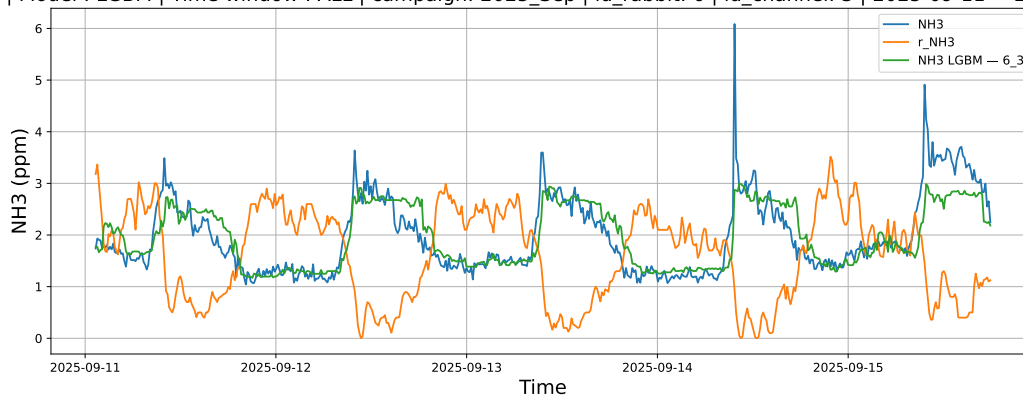


3.1.8.3 Time window : ALL

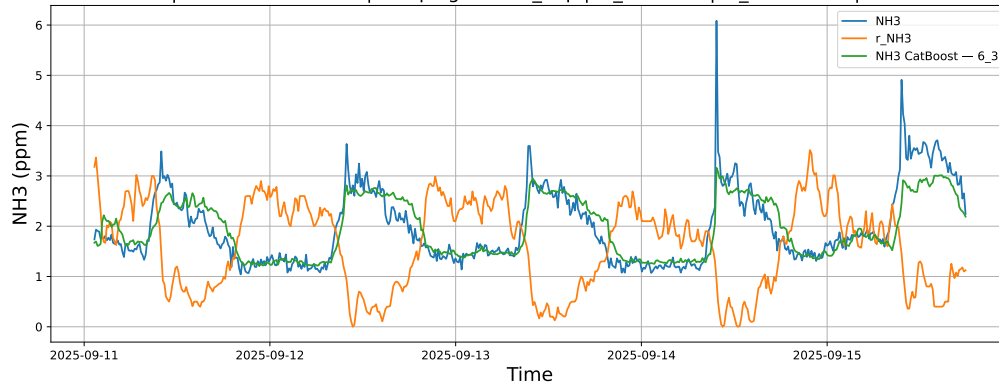
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

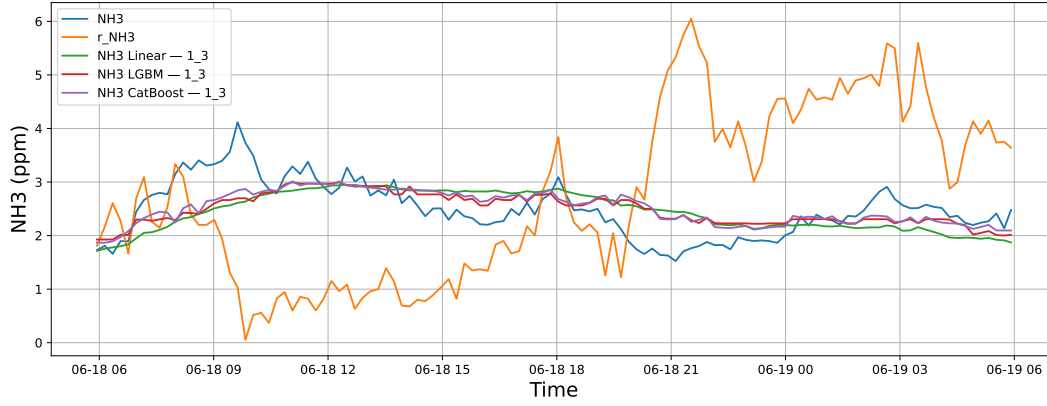


3.2 Compare plots

3.2.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

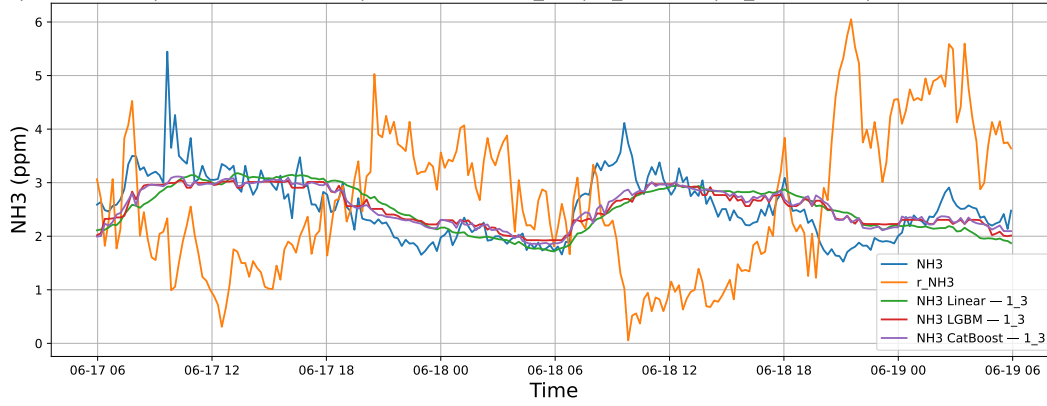
3.2.1.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



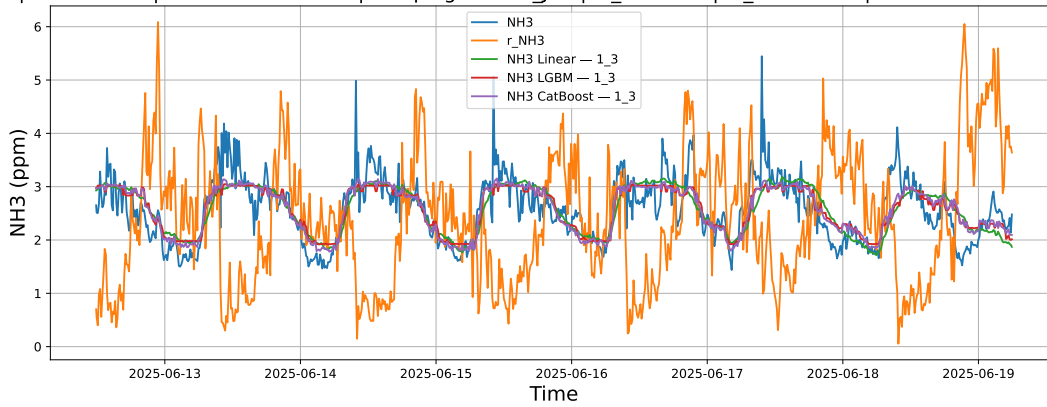
3.2.1.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



3.2.1.3 Time window : ALL

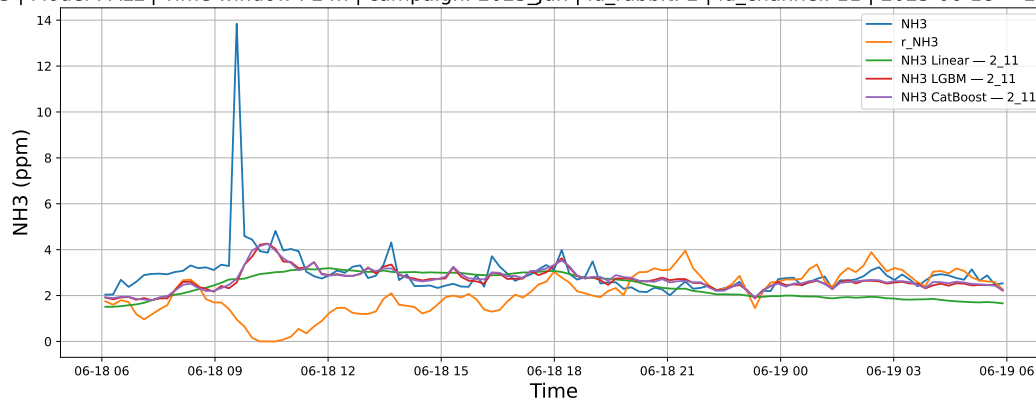
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



3.2.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

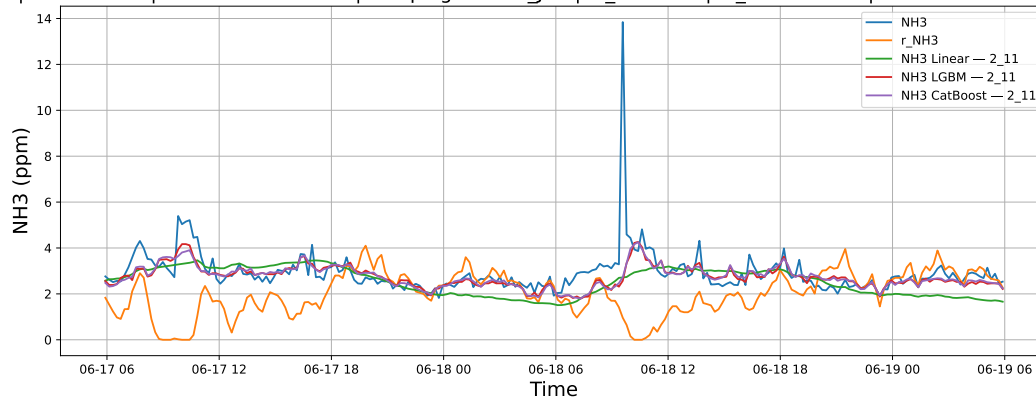
3.2.2.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



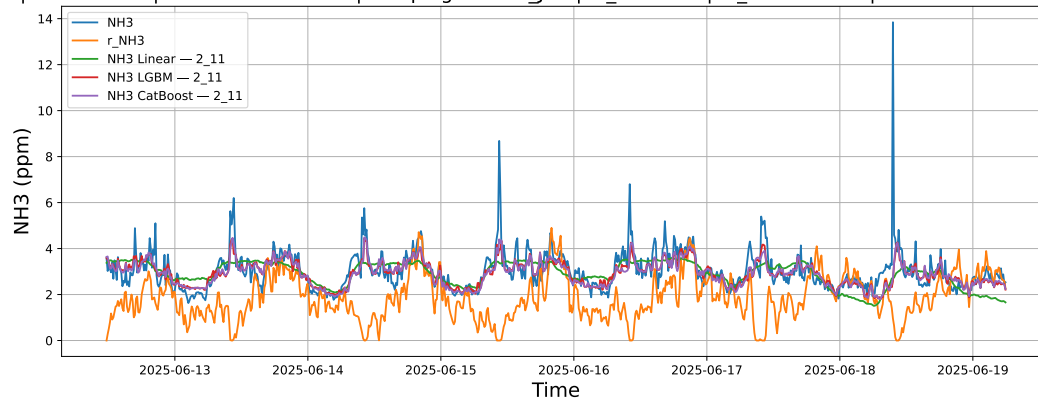
3.2.2.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



3.2.2.3 Time window : ALL

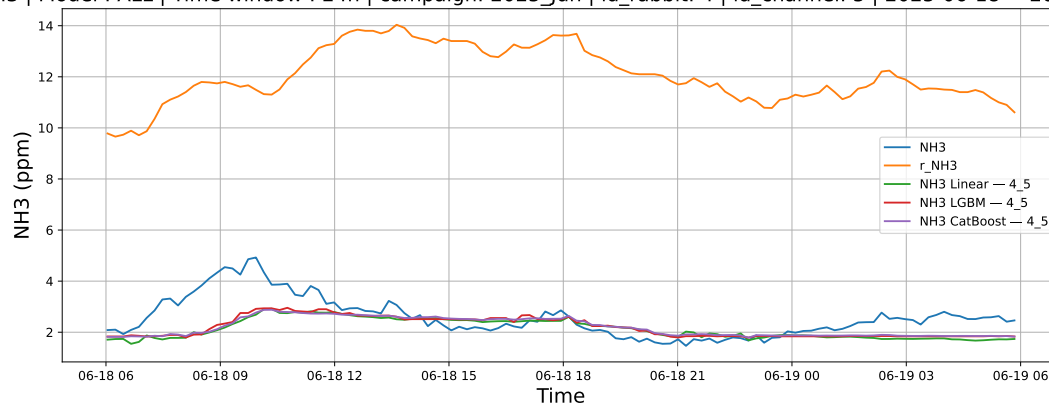
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



3.2.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

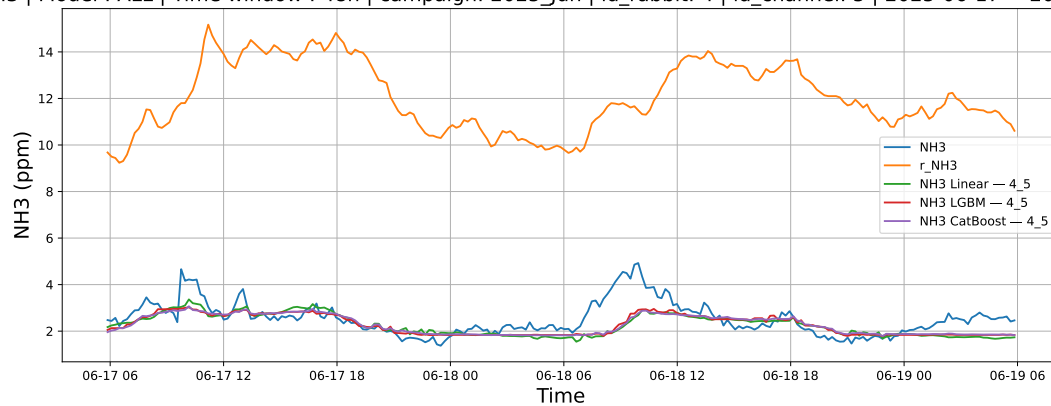
3.2.3.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



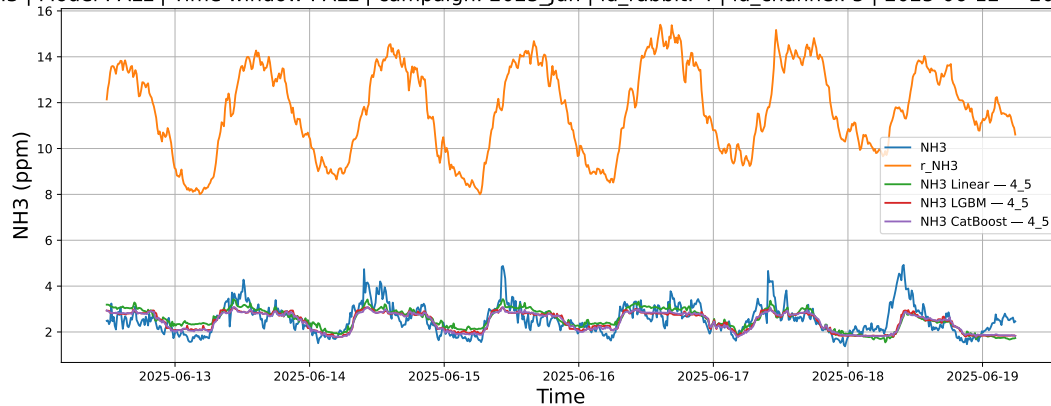
3.2.3.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



3.2.3.3 Time window : ALL

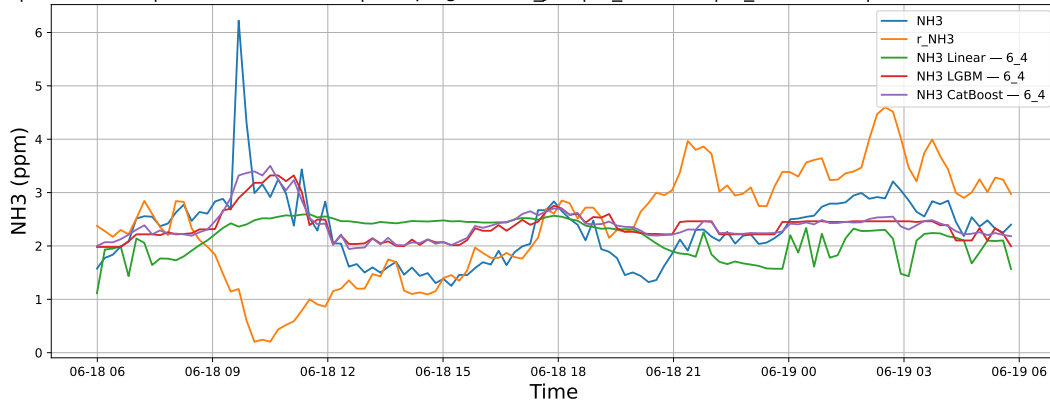
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



3.2.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

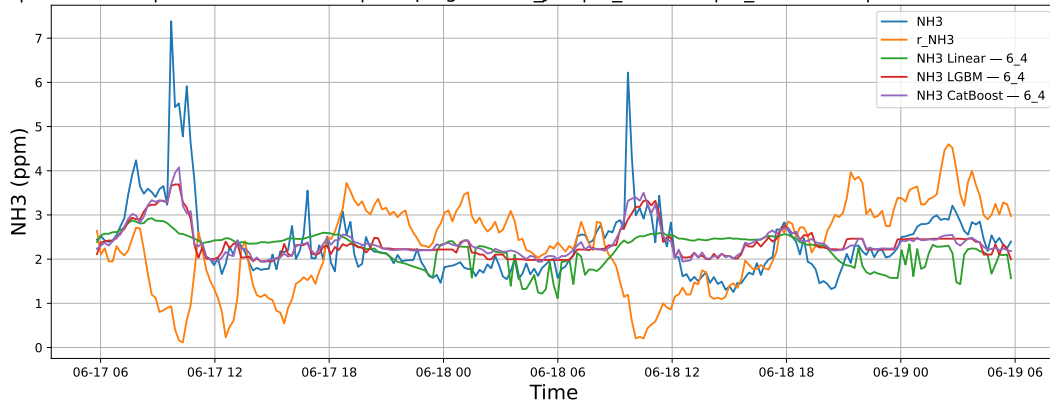
3.2.4.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



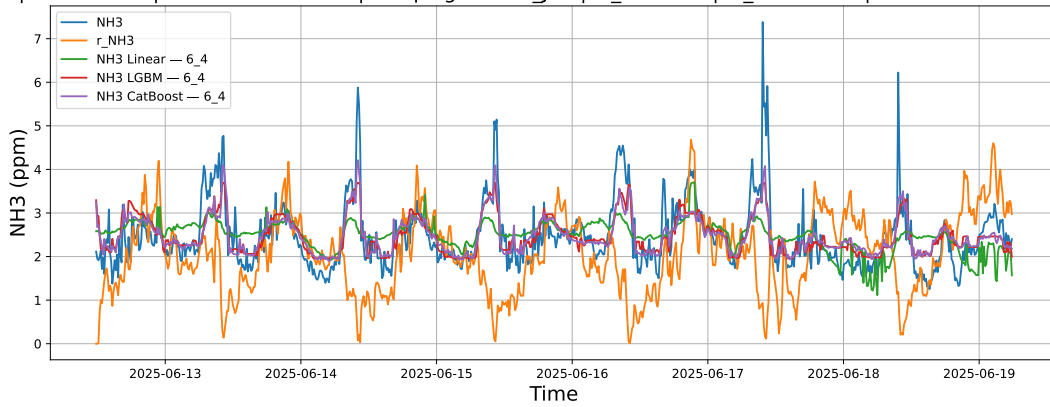
3.2.4.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



3.2.4.3 Time window : ALL

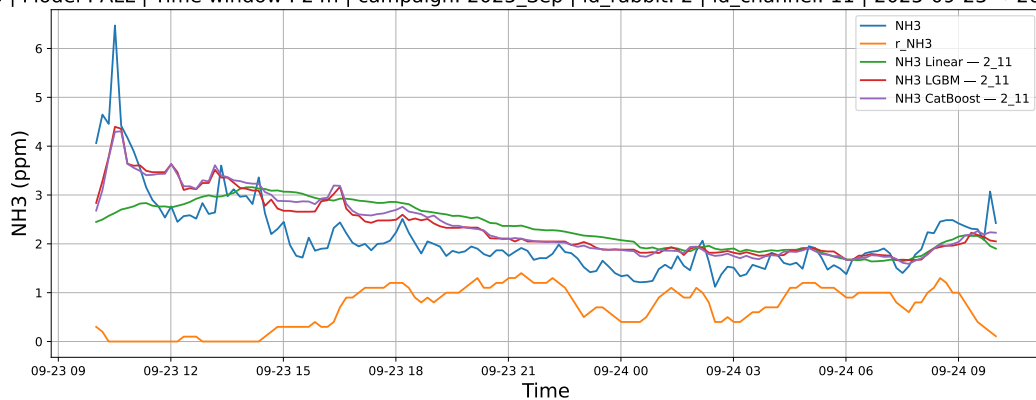
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



3.2.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

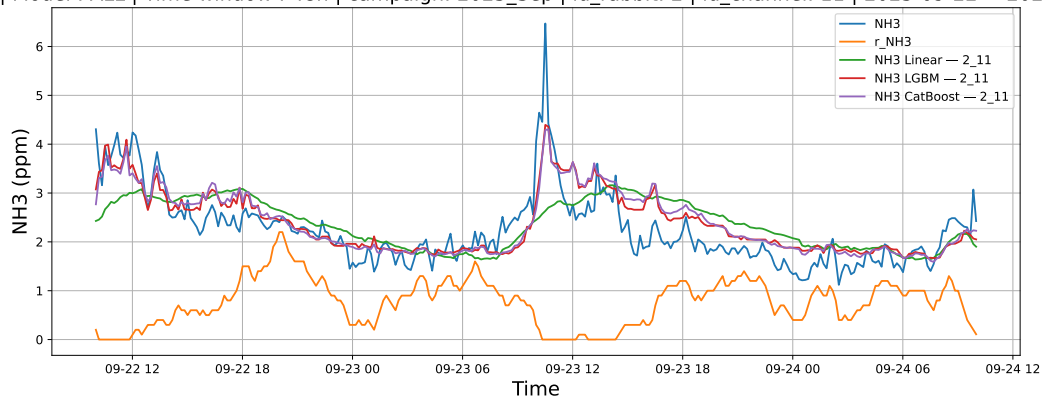
3.2.5.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



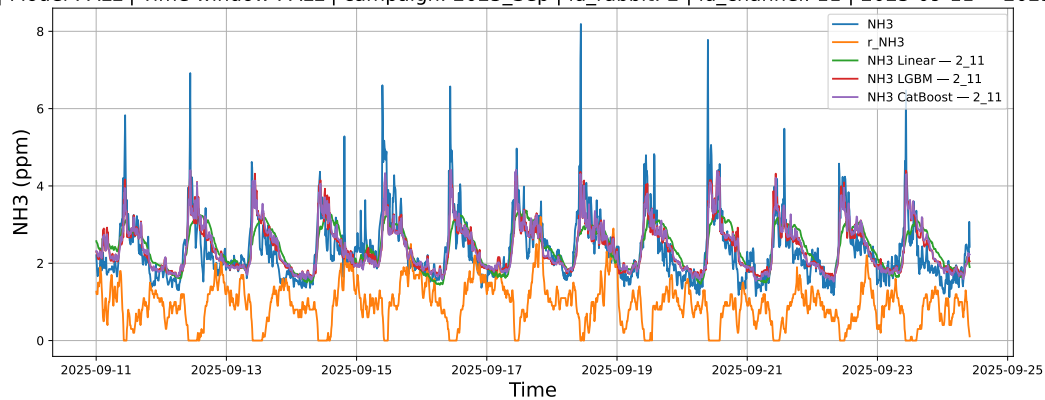
3.2.5.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



3.2.5.3 Time window : ALL

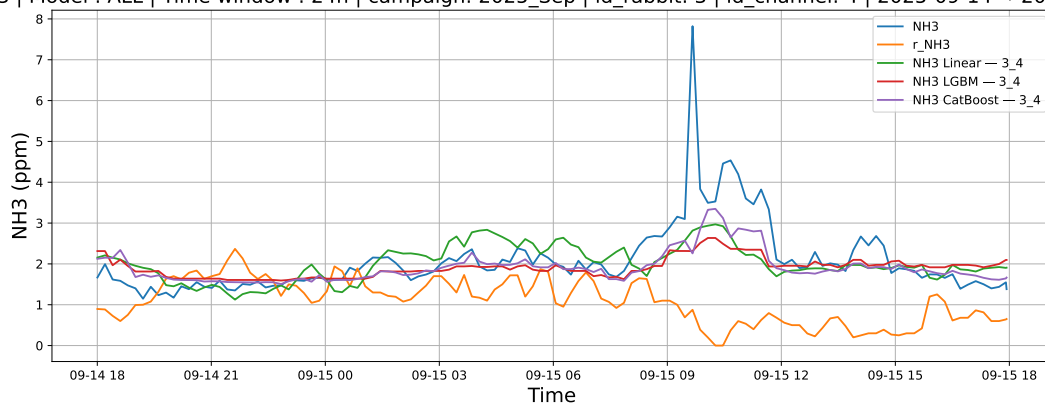
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



3.2.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

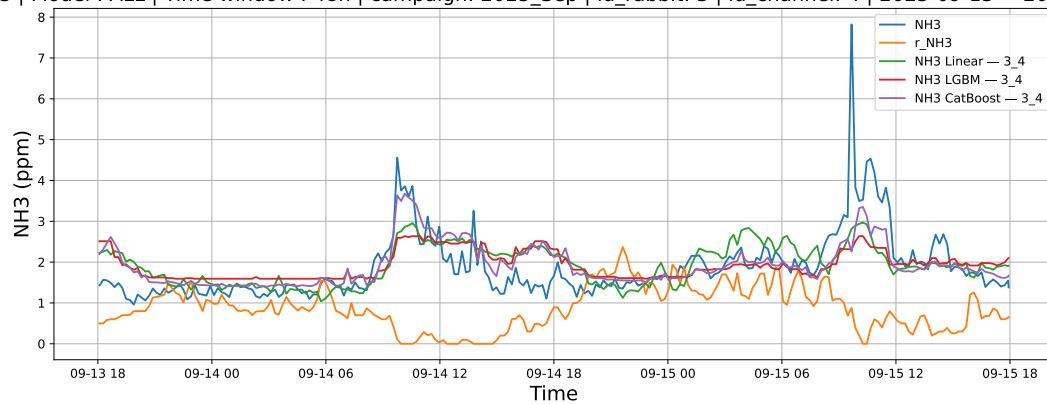
3.2.6.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



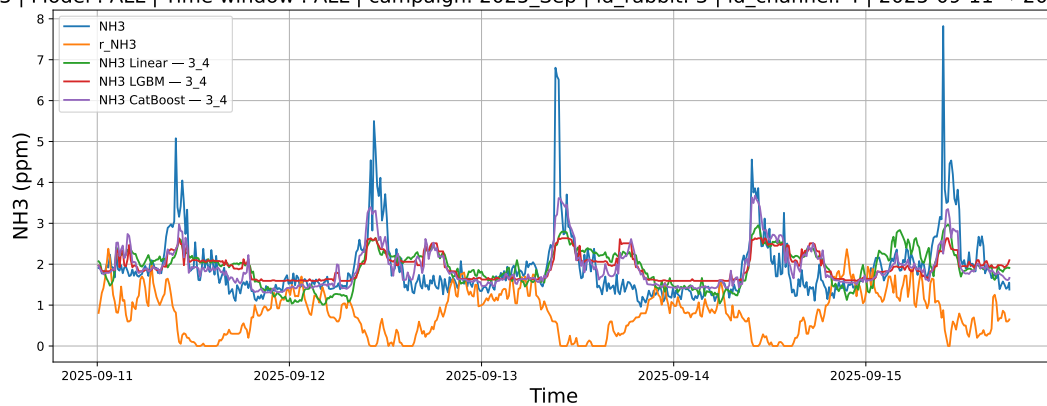
3.2.6.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



3.2.6.3 Time window : ALL

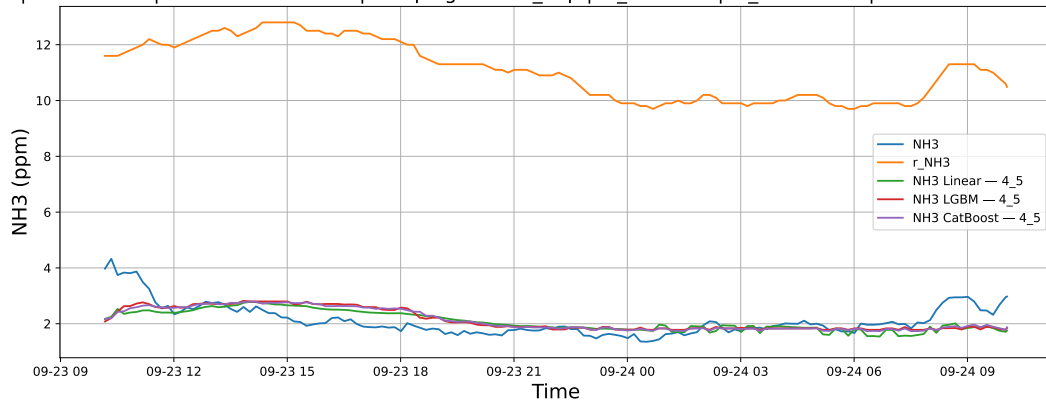
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



3.2.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

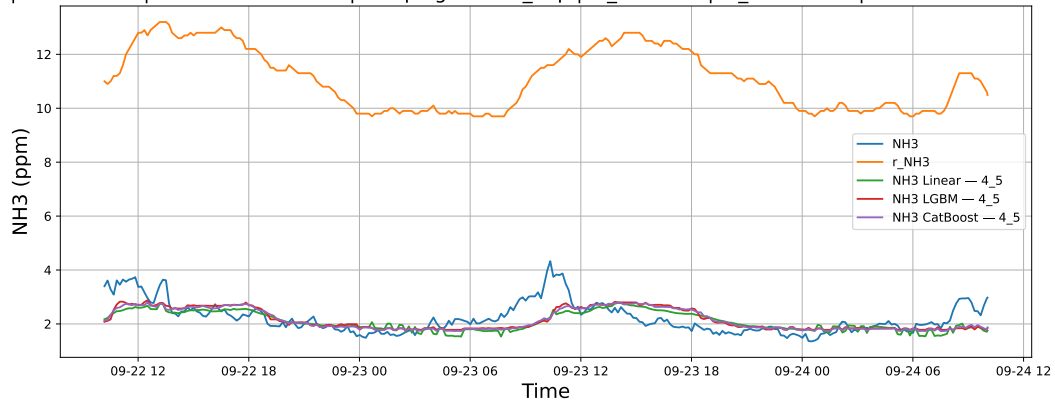
3.2.7.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



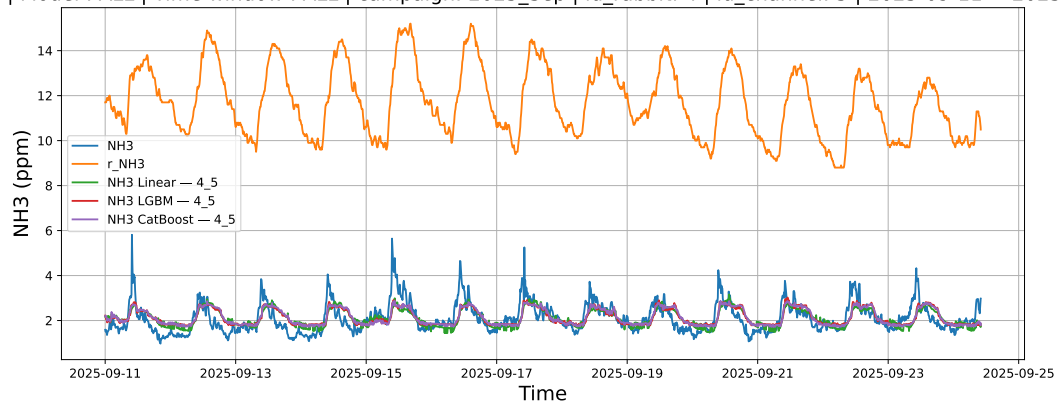
3.2.7.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



3.2.7.3 Time window : ALL

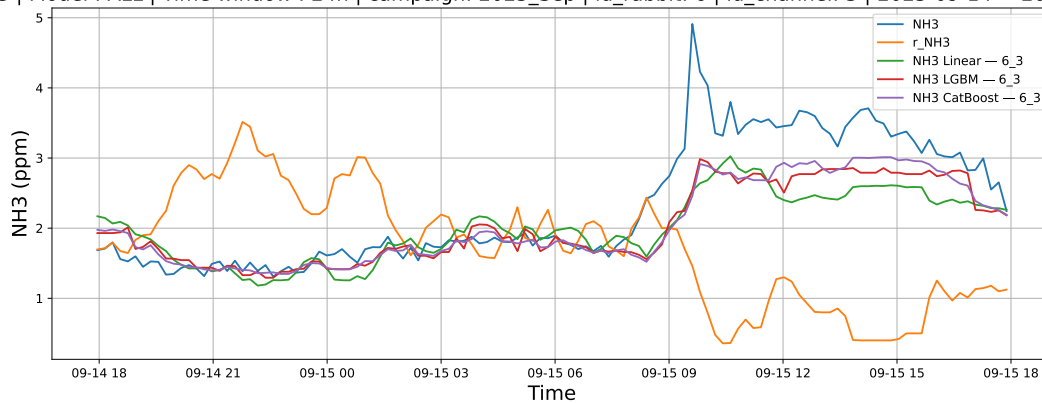
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



3.2.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

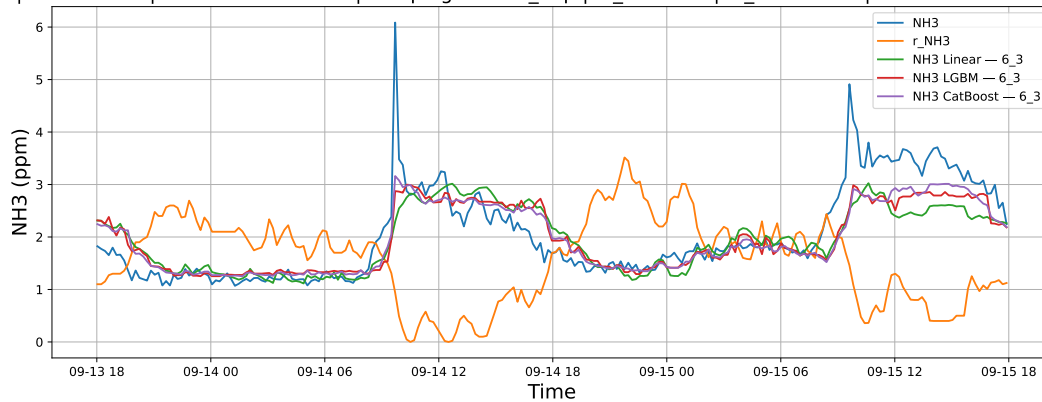
3.2.8.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



3.2.8.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



3.2.8.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

